CHAPTER 8

Axial Compressor Aerodynamics

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8.1 Introduction

In this chapter, we address compressors and turbines in an aircraft gas turbine engine. We first present the fundamental equations that are applicable to all types of turbomachinery and then follow with the flow characteristics of each machine in subsequent Chapters 9 and 10.

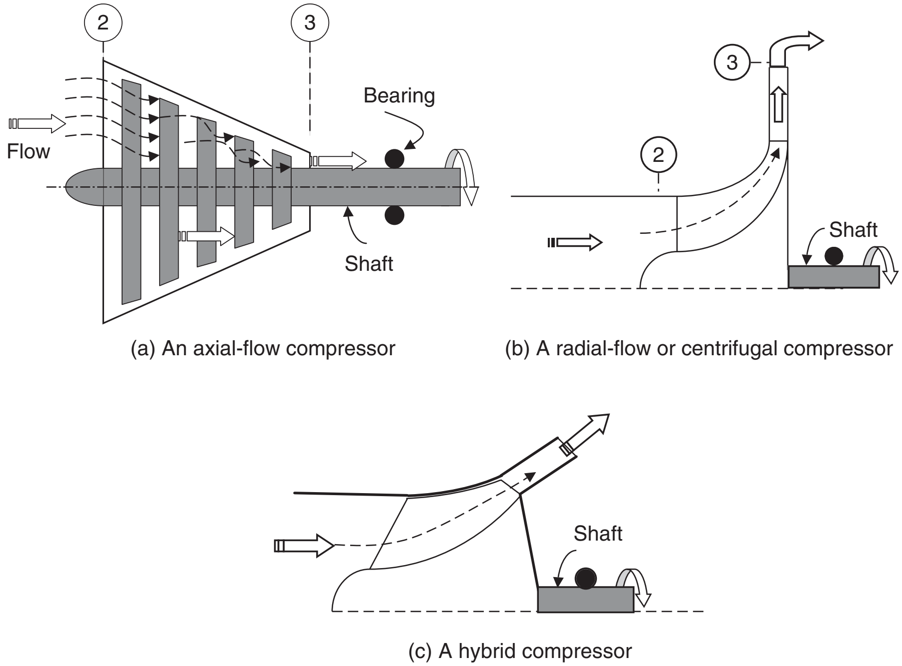
Turbomachinery is at the heart of gas turbine engines. The role of mechanical compression of air in an engine is given to the compressor. The shaft power to drive the compressor typically is produced by expanding gases in the turbine. The machines that exchange energy with a fluid, called the working fluid, through shaft rotation are known as turbomachinery. The machines where the fluid path is predominantly along the axis of the shaft rotation are called axial-flow turbomachinery. In contrast to these machines, in radial-flow turbomachinery the fluid path undergoes a 90° turn from the axial direction. These machines are sometimes referred to as centrifugal machines. A mixed-flow turbomachinery is a hybrid between the axial and the radial-flow machines. In aircraft gas turbine engines, the axial-flow compressors and turbines enjoy the widest application and development (Figure 8.1). The centrifugal compressors and radial-flow turbines are used in small gas turbine engines and automotive turbocharger applications.

8.2 The Geometry

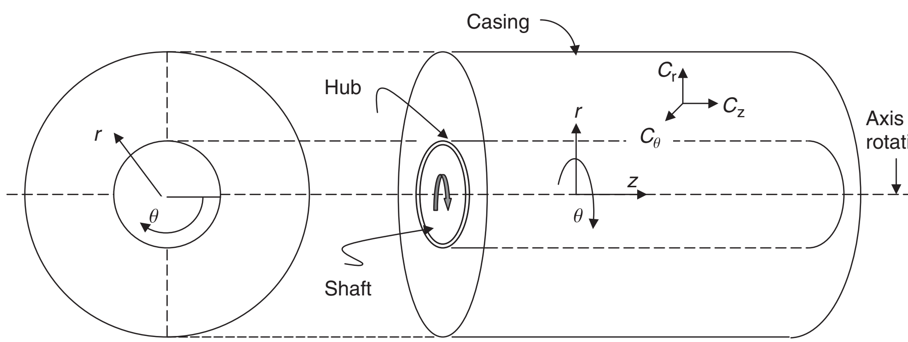
The geometry of rotating blades demands a cylindrical duct with a shaft centric configuration. This in turn leads to the choice of cylindrical coordinates for the analysis of flows in

Aircraft Propulsion, Second Edition. Saeed Farokhi.

Companion Website: www.wiley.com/go/farokhi



■ **FIGURE 8.1** **Schematic drawing of different types of compressors in aircraft gas turbine engines**

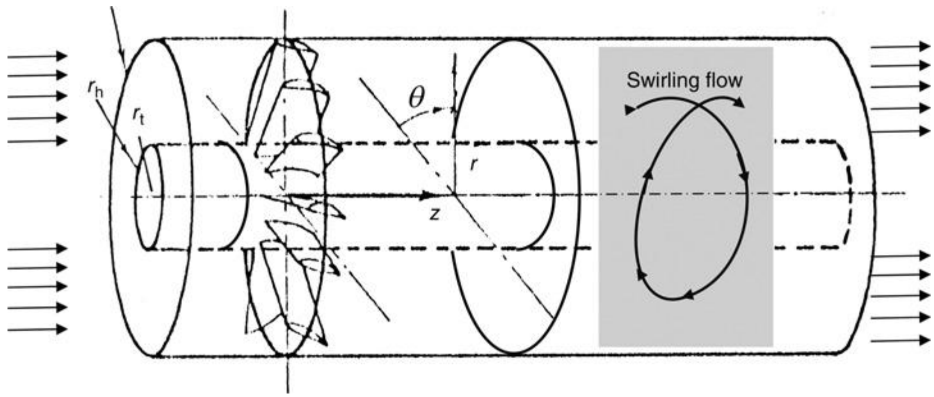


■ **FIGURE 8.2** **Definition sketch for the coordinates and the velocity components of the flow in cylindrical coordinates**

turbomachinery. The coordinates $[r, \theta, z]$ are in radial, tangential or azimuthal, and axial directions, respectively. The velocity components $[C_r, C_\theta, C_z]$ are the radial, tangential (or sometimes referred to as swirl or azimuthal velocity), and axial velocity components, respectively. These are shown in a definition sketch in Figure 8.2.

8.3 Rotor and Stator Frames of Reference

In turbomachinery, energy transfer between the blades and the fluid takes place in an inherently unsteady manner. This is achieved by a set of rotating blades, called the rotor. The rotor blades are three-dimensional aerodynamic surfaces, which experience aerodynamic forces. The rotor blades are cantilevered at the hub and thus feel a root bending moment and a torque. The reaction to the blade forces and moments is exerted on



■ **FIGURE 8.3**
Isolated rotor in a cylinder. Source: Adapted from Marble 1964

the fluid, via the action–reaction principle of Newton. Stationary blades called the stator follow the rotor blades in what is known as a turbomachinery *stage.* The stator blades are three-dimensional aerodynamic surfaces as well. They are cantilevered from the casing and experience forces and moments, like the rotor. The exception is that the stator forces and moments are stationary (in the laboratory frame of reference) and thus perform no work on the fluid. The energy of the fluid is, thus, expected to remain constant in passing through the stator blades. The position of the observer is, however, important in viewing the flowfield and the energy exchange in turbomachinery. An observer fixed at the casing (or laboratory) is called an *absolute* observer. If the observer is attached to the rotor blade and spins with it, then it is called a *relative* observer. The frames of reference are then called the absolute and relative frames of reference, respectively. Consider an isolated rotor in a cylindrical duct, as shown in Figure 8.3. An absolute observer sees the blades' aerodynamic forces are in motion, at an angular rate, that is, the angular velocity w of the shaft. Hence, as measured by this observer, the total enthalpy of the fluid goes up in crossing the rotor row. On the contrary, let us put ourselves in the frame of reference of a relative observer who is spinning with the rotor. According to a relative observer, the blades are not moving! An observer fixed at the rotor measures aerodynamic forces and moments of the blades, however, as the forces are stationary, there is no work done on the fluid according to this observer. Thus, the relative observer measures the same total enthalpy across the blade row.

The flowfield as seen by a relative observer attached to an isolated rotor in a cylinder is thus steady. The absolute observer on the casing, however, sees the passing of the blades and thus experiences an unsteady flowfield. As the rotor blades pass by, a periodic pressure pulse (due to blade tip) is registered at the casing, which signifies an unsteady event with a periodicity of blade passing frequency. To be able to analyze a flowfield in a steady frame of reference offers tremendous advantages in the nature and the solution of governing equations. Consequently, in analyzing the flow within a rotor blade row, we employ the relative observer stance, while the stator flows are viewed from the standpoint of an absolute observer. We need to be mindful, however, that in practice there are no isolated rotors and thus the flowfield in rotating machinery is *inherently* unsteady. We shall present some preliminary discussions on the scale and effect of unsteadiness in axial-flow compressors later in this chapter.

The velocity components as seen by observers in the two frames of reference are related. First, we note that the radial and axial velocity components are identical in the two frames, as the relative observer moves only in the θ, or the angular, direction. Therefore, the swirl or tangential velocity is the only component of the velocity vector field that is affected by the observer rotation. At a radial position r on the rotor, the relative observer rotates with a speed ωr and thus registers a tangential velocity, which is ωr less than the absolute swirl velocity, that is,

$$\text{Relative swirl} = \text{Absolute swirl} - \omega r \tag{8.1}$$

The fluid velocity vectors in the two frames are labeled as $\vec{C}$ and $\vec{W}$ for the absolute and relative observers, respectively. Therefore, the absolute velocity vector is described as

$$\vec{C} = C_r\hat{e}_r + C_\theta\hat{e}_\theta + C_z\hat{e}_z \tag{8.2}$$

The relative velocity vector is

$$\vec{W} = W_r\hat{e}_r + W_\theta\hat{e}_\theta + W_z\hat{e}_z \tag{8.3}$$

The rotor blade spins with an angular velocity ω, hence it describes a solid body rotation as

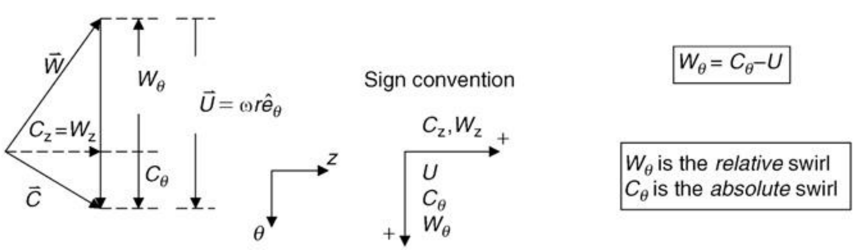
$$\vec{U} = \omega r\hat{e}_\theta \tag{8.4}$$

Comparing the velocity vectors as described by Equations 8.2–8.4, we conclude that the following vector identities hold, namely,

$$\vec{W} = \vec{C} - \vec{U} \tag{8.5a}$$
$$\vec{C} = \vec{W} + \vec{U} \tag{8.5b}$$

These three vectors form a triangle known as the *velocity triangle.*

Figure 8.4 shows a definition sketch of the velocity triangle and the sign convention. The positive tangential or azimuthal angle θ is in the direction of rotor rotation $\vec{U}$. The swirl velocity component is considered positive if in the direction of rotor rotation. For example, we note that the absolute swirl is pointing in the positive θ direction, therefore it is considered positive. The relative swirl velocity component W_θ is in the opposite direction, hence it is a negative quantity. The scalar relation given in Equation 8.1 between the two swirl components is always valid.



■ **FIGURE 8.4**
The velocity triangle

8.4 The Euler Turbine Equation

The Euler turbine equation is called the fundamental equation of turbomachinery. Once we derive this simple yet powerful expression, its significance becomes evident. Let us consider a streamtube that enters a turbomachinery blade row. Figure 8.5 illustrates a generic streamtube with its geometry and velocity components. In general, stream surfaces undergo a radial shift when interacting with a blade row, as depicted in Figure 8.5.

The mass flow rate in the streamtube is constant, by definition, and is labeled as $\dot{m}$. The angular momentum of the fluid in the streamtube is the moment of the tangential momentum of the fluid about the axis of rotation, namely,

$$\text{(Time rate of change of the) Fluid angular momentum} = \dot{m}rC_\theta \qquad \textbf{(8.6)}$$

The change of fluid angular momentum between the exit and inlet of the streamtube is the applied torque exerted by the blade on the fluid, that is,

$$\dot{m}(r_2C_{\theta 2} - r_1C_{\theta 1}) = \tau_{\text{fluid}} \qquad \textbf{(8.7a)}$$

The torque is the product of blade tangential force F_θ and the moment arm r from the axis of rotation. Hence the blade torque is

$$\tau_{\text{blade}} = F_{\theta,\text{blade}} \cdot r = -\tau_{\text{fluid}} \qquad \textbf{(8.7b)}$$

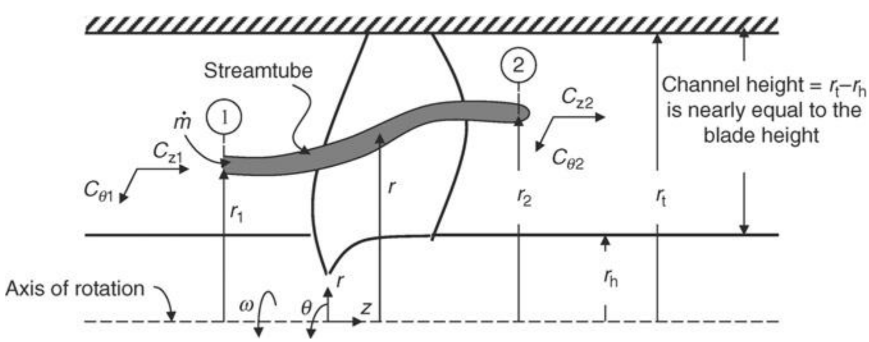
The expression 8.7a (or 8.7b) is valid for the rotor as well as the stator. In case of rotor, there is an angular motion, hence the product of the angular velocity of the blade and the torque provides the power transmitted to the fluid, namely,

$$\wp = \tau_{\text{fluid}} \cdot \omega = \dot{m}\omega(r_2C_{\theta 2} - r_1C_{\theta 1}) = \dot{m}\omega\Delta(rC_\theta) \qquad \textbf{(8.8)}$$

The ratio of shaft power to mass flow rate is called the specific work of the rotor, w_c for the compressor and w_t for the turbine, hence,

$$w_\text{c} \equiv \frac{\wp_\text{c}}{\dot{m}_\text{c}} = \omega\Delta(rC_\theta)_\text{c} \qquad \textbf{(8.9a)}$$

$$w_\text{t} \equiv \frac{\wp_\text{t}}{\dot{m}_\text{t}} = \omega\Delta(rC_\theta)_\text{t} \qquad \textbf{(8.9b)}$$



■ **FIGURE 8.5**
Streamtube interacting with a turbomachinery blade row

This is the Euler turbine equation written for the fluid interacting with a compressor or turbine rotor. As evidenced in Equation 8.9, the rotor (specific) work appears as the change in (specific) angular momentum across a blade row times the shaft speed. The rotor and stator torques are proportional to the change in angular momentum across the rotor and stator blade rows, respectively, via Equation 8.7.

The first law of thermodynamics applied to a steady and adiabatic process demands that the change of total enthalpy of the fluid across the blade row to be equal to the blade-specific work delivered to the fluid, namely,

$$h_{t2} - h_{t1} = \frac{\wp}{\dot{m}} = w_c \quad \textbf{(8.10a)}$$

From the Euler turbine equation, the exit stagnation enthalpy in Equation 8.10a is related to the inlet stagnation enthalpy and the change of the angular momentum across the rotor row,

$$h_{t2} = h_{t1} + \omega\Delta(rC_\theta) \quad \textbf{(8.10b)}$$

The nondimensional total enthalpy change is then

$$\frac{h_{t2}}{h_{t1}} = 1 + \frac{\omega\Delta(rC_\theta)}{h_{t1}} = \frac{T_{t2}}{T_{t1}} \quad \textbf{(8.10c)}$$

Note that we assumed a calorically perfect gas in Equation 8.10c when we replaced the ratio of stagnation enthalpy with the ratio of total temperatures.

8.5 Axial-Flow Versus Radial-Flow Machines

In an axial-flow turbomachinery, the fluid path is predominantly along the axis of rotation. In radial-flow or centrifugal machinery, the fluid path departs from axial and attains a predominantly radial motion at the exit. As a result, the following approximation is typically made for axial-flow machines,

$$r_1 \approx r_2 \approx r \quad \textbf{(8.11)}$$

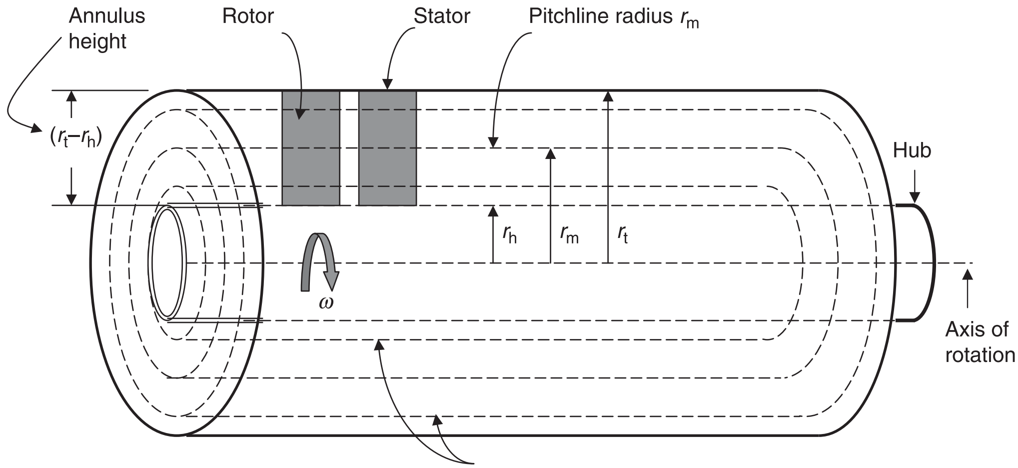
Therefore, the Euler turbine equation may be simplified to

$$w_c \cong \omega r(\Delta C_\theta) = U(C_{\theta 2} - C_{\theta 1}) \quad \textbf{(8.12)}$$

The assumption of constant "r" in axial-flow turbomachinery places the stream surfaces on cylindrical surfaces. This means that stream surfaces do not undergo significant *radial deviation*. Thus, radial deviation or radial shift of stream surfaces is often assumed negligible in axial-flow machines. The simplified flowfield in an axial turbomachinery annulus may be divided into a series of $r=$ constant cylindrical cuts, as shown in Figure 8.6.

The pitchline radius r_m is defined as the mid-radius between the hub and the casing, that is,

$$r_m \equiv \frac{r_h + r_t}{2} \quad \textbf{(8.13)}$$



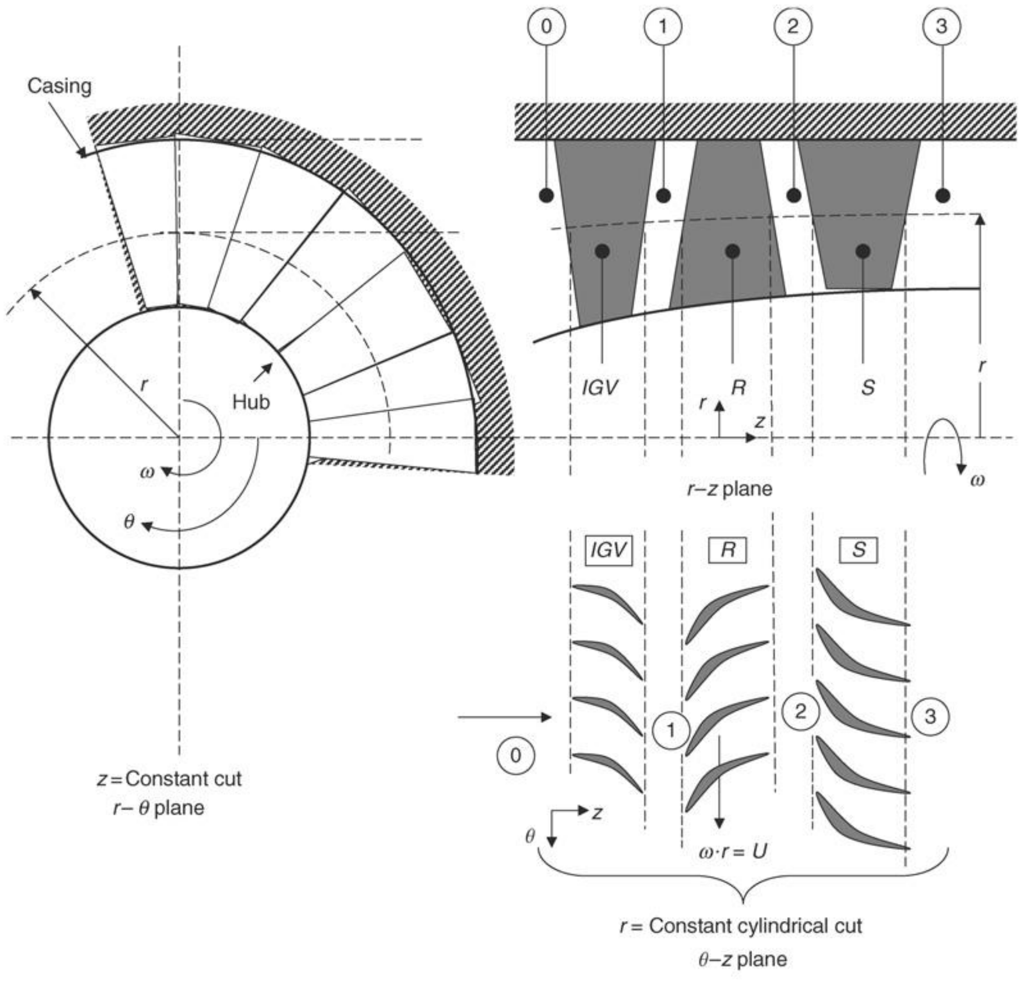
■ **FIGURE 8.6 A simple model of an axial compressor stage (rotor and stator) in a cylindrical annulus showing a mean or pitchline radius r_m along with other cylindrical cuts of the annulus representing simplified stream surfaces**

Often, the pitchline radius of an axial-flow turbomachinery is assumed to represent the mean or average of the flowfield properties in the annulus and thus serves as the first line of attack in a *one-dimensional* flow analysis approach. The hub and the tip radii represent the maximum deviations from the mean and thus are analyzed next. For a more accurate analysis, other cylindrical cuts are introduced in the annulus, as shown in Figure 8.6. To further improve the accuracy of our analysis, we need to incorporate the radial disposition of the stream surfaces interacting with turbomachinery blade rows. In general, the stream surfaces undergo a radial shift interacting with a blade row and thus form *conical surfaces* in the vicinity of a blade. We need to employ a three-dimensional flow theory such as radial equilibrium theory or an actuator disc theory to approximate their exit radius r_2 of stream surfaces that have entered the blade row at the inlet radius of r_1. We will estimate the radial shift of stream surfaces using radial equilibrium theory Section 8.6.5.

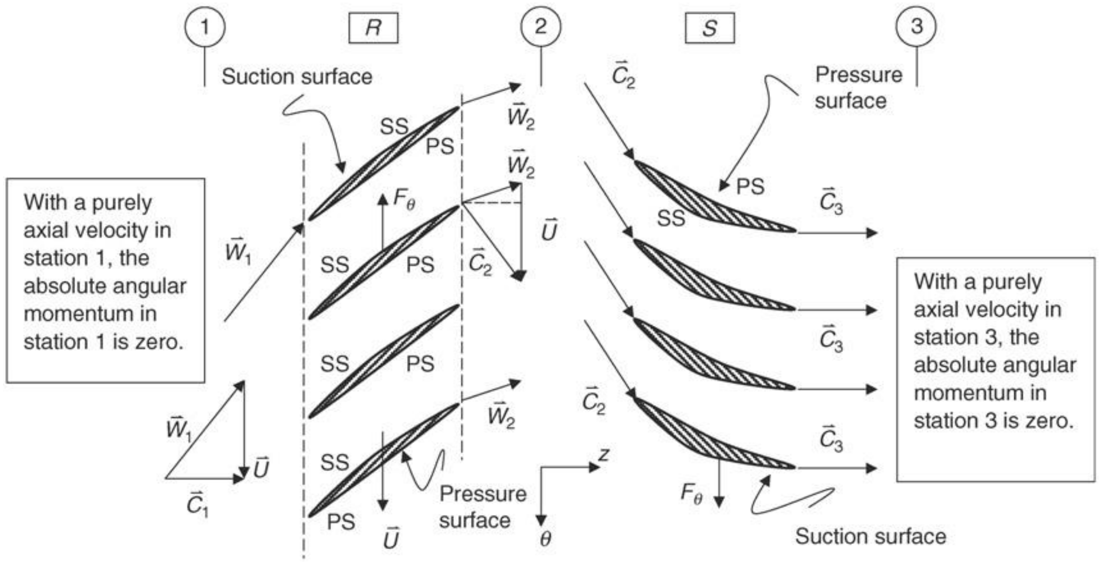
A centrifugal or radial compressor receives the fluid in the axial direction near the axis of rotation and imparts large angular momentum to the fluid by expelling the fluid at a higher radius and with a large swirl kinetic energy. The radial turbines receive the fluid at a large radius and with high swirl kinetic energy and expel the fluid near the axis of rotation with a small angular momentum. In the process of absorbing the swirl kinetic energy, a large blade torque is created, which in a rotor translates into shaft power. Hence, the radial shift of stream surfaces needs to be large for an efficient centrifugal compression/expansion and thus may not be neglected from our analysis. Fortunately, unlike axial-flow turbomachinery that we have to employ three-dimensional flow theories to predict the exit radius of an incoming stream surface, in a centrifugal machine the exit radius for all stream surfaces is fixed by the impeller geometry, that is, the impeller exit radius. We will treat centrifugal compressors in Chapter 9.

8.6 Axial-Flow Compressors and Fans

Axial-flow compressors and fans provide mechanical compression for the air stream that enters a gas turbine engine. Thermodynamically, their function is to increase the fluid pressure, efficiently. Hence, the boundary layers on the blades of compressors and fans, as well as their hub and casing, are exposed to an adverse, or rising, pressure gradient. Boundary layers exposed to an adverse pressure gradient, due to their inherently low momentum, cannot tolerate significant pressure rise. Consequently, to achieve a large pressure rise, axial-flow compressors and fans need to be staged. With this stipulation, a multistage machinery, or compression system, is born. In a stage, the rotor blade imparts angular momentum to the fluid, while the following stator blade row removes the angular momentum from the fluid. A definition sketch of a compressor stage with an inlet guide vane that imparts angular momentum to the incoming fluid is shown in Figure 8.7. A



■ **FIGURE 8.7 Definition sketch for station numbers and three different planes, r–θ, r–z, and θ–z, in a compressor stage with an inlet guide vane (IGV)**



■ **FIGURE 8.8** **Cylindrical (*r* = constant) cut of a compressor stage with velocity triangles showing the rotor imparting and the stator removing the swirl and angular momentum to the fluid**

compressor stage without an inlet guide vane is depicted in Figure 8.8. In either case, the principle of rotor increasing the fluid angular momentum and the stator blade removing the swirl (or angular momentum) is independent of any preswirl in the incoming flow to the stage. We may introduce an inlet guide vane upstream of the rotor blades that imparts a preswirl (in the direction of the rotor motion) to the incoming stream and yet the principle of rotor–stator angular momentum interactions with the fluid remains intact.

Based on the absolute velocity field in regions 1, 2, and 3, for a generic inlet condition that may include a preswirl $C_{\theta 1}$ created by an inlet guide vane, we may write the rotor and stator torques

$$\tau_{\text{rotor}} = -\tau_{\text{fluid}} = -\dot{m} \cdot r(C_{\theta 2} - C_{\theta 1}) \tag{8.14}$$

$$\tau_{\text{stator}} = -\tau_{\text{fluid}} = -\dot{m} \cdot r(C_{\theta 3} - C_{\theta 2}) \tag{8.15}$$

Assuming the absolute swirl and angular momentum across the stage remains the same, that is, $C_{\theta 1} = C_{\theta 3}$ and $r_1 = r_3$ the rotor and stator torques become equal and opposite, namely,

$$\tau_{\text{rotor}} = -\tau_{\text{stator}} \tag{8.16}$$

We observe the suction and pressure surfaces of the rotor and stator blades, as shown in Figure 8.8, and note that the blade aerodynamic forces in the θ-direction F_θ for the rotor and stator blades are in opposite directions. Since the moment of this tangential force, that is, $r{\cdot}F_\theta$ from the axis of rotation, is the blade torque, we conclude that the rotor and stator torques are opposite in direction and nearly equal in magnitude.

EXAMPLE 8.1

The absolute flow at the pitchline to a compressor rotor is swirl free. The exit flow from the rotor has a positive swirl, $C_{\theta 2} = 145$ m/s. The pitchline radius is $r_m = 0.5$ m, and the rotor angular speed is $\omega = 5600$ rpm. Calculate the specific work at the pitchline and the rotor torque per unit mass flow rate.

SOLUTION

Rotor tangential speed at pitchline is

$$U_m = \omega r_m$$
$$= (5600\text{ rev/min})(2\pi\text{ rad/rev})(\text{min}/60\text{ s})(0.5\text{ m})$$
$$= 293.2\text{ m/s}$$

$$w_c \cong \omega r(\Delta C_\theta) = U(C_{\theta 2} - C_{\theta 1}) = 293.2\text{ m/s}(145\text{ m/s})$$
$$\approx 42.516\text{ kJ/kg}$$
$$\tau_{r,m}/\dot{m} = r_m(C_{\theta 2} - C_{\theta 1}) = 0.5\text{ m}(145\text{ m/s}) = 72.5\text{ m}^2/\text{s}$$

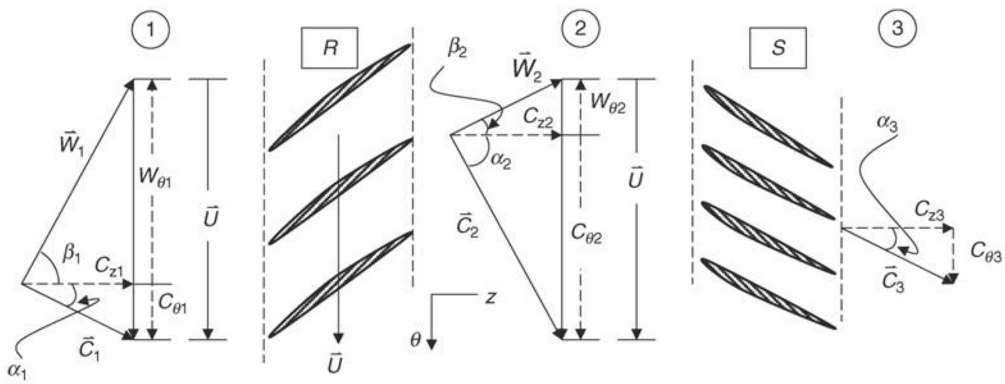
8.6.1 Definition of Flow Angles

The flow angles are measured with respect to the axial direction, or axis of the machine, and are labeled as α and β, which correspond to the absolute and relative flow velocity vectors $\vec{C}$ and $\vec{W}$, respectively. Figure 8.9 is a definition sketch that shows the absolute and relative flow angles in a compressor stage.

We may use these absolute and relative flow angles to express the velocity components in the axial and the swirl direction as

$$C_{\theta 1} = C_{z1} \cdot \tan(\alpha_1) \quad \textbf{(8.17)}$$
$$W_{\theta 2} = C_{z2} \cdot \tan(\beta_1) \quad \textbf{(8.18)}$$
$$C_{\theta 2} = C_{z2} \cdot \tan(\alpha_2) \quad \textbf{(8.19)}$$
$$W_{\theta 2} = C_{z2} \cdot \tan(\beta_2) \quad \textbf{(8.20)}$$

One method of accounting for positive and negative swirl velocities is through a convention for positive and negative flow angles. We observe that the absolute velocity vector upstream



■ FIGURE 8.9 Definition sketch for the absolute and relative flow angles in a compressor stage (a cylindrical cut of the stage, r = constant) of the rotor has a swirl component in the direction of the rotor rotation. Hence, the absolute flow angle α_1 is considered positive. The opposite is true for the relative velocity vector $\vec{W}_1$, which has a swirl component in the opposite direction to the rotor rotation. Both the relative flow angle β_1 and the swirl velocity facing it $W_{\theta 1}$ are thus negative.

A turbomachinery blade row is designed to maintain an attached boundary layer, under normal operating conditions. Hence, the flow angles at the exit of the blades are primarily fixed by the blade angle at the exit plane. For example, the relative velocity vector at the exit of the rotor, $\vec{W}_2$, should nearly be tangent to the rotor suction surface at the trailing edge. Hence, β_2 is fixed by the geometry of the rotor and remains nearly constant over a wide operating range of the compressor. The same statement may be made about α_1 or α_3. These angles remain constant over a wide range of the operation of the compressor as well. Again, remember that the exit flow angle argument is made for an attached (boundary layer) flow. The other flow angles, as in β_1 or α_2, change with rotor speed U (i.e., ωr). Consequently, use is made of the nearly constant exit flow angles α_1, β_2, and α_3 in expressing performance parameters of the compressor stage and the blade row.

The axial velocity components C_{z1}, C_{z2}, or C_{z3} contribute to the mass flow rate through the machine. A common (textbook) design approach in axial-flow compressors and fans maintains a constant axial velocity throughout the stages. We shall make use of this simplifying design approach repeatedly in this chapter. Figure 8.8 shows an example of a constant axial velocity in a compressor stage. Although a simplifying design assumption, the reader needs to be aware that the goal of constant axial velocity is nearly impossible to achieve in practice. This is due to a highly three-dimensional nature of the flowfield, set up by three-dimensional pressure gradients, in a turbomachinery stage. We will address this and other topics related to three-dimensional flow in Section 8.6.5.

A useful concept in turbomachinery design calls for a *repeated stage* (also referred to as *normal stage*). This implies that the velocity vectors at the exit and entrance to a stage are the same, that is,

$$\vec{C}_3 = \vec{C}_1 \tag{8.21a}$$

$$\alpha_3 = \alpha_1 \tag{8.21b}$$

The velocity triangles at the inlet and the exit of the stage shown in Figure 8.8 have made use of the repeated stage concept. Another concept in turbomachinery calls for a *repeated row* design, which leads to flow angle implications that are noteworthy. In a repeated row design, the exit relative flow angle has the same magnitude as the absolute inlet flow angle, and the inlet relative flow angle has the same magnitude as the exit absolute flow angle, that is,

$$|\beta_2| = |\alpha_1| \tag{8.22a}$$

$$|\alpha_2| = |\beta_1| \tag{8.22b}$$

The example of the compressor stage shown in Figure 8.9 has used the concept of repeated row. Note that a repeated row design leads to a repeated stage but the reverse is not necessarily correct. Namely, we may have a repeated stage design that does not use a repeated row concept. Figure 8.8 shows an example of a repeated stage that does not obey a repeated row design.

8.6.2 Stage Parameters

The Euler turbine equation that we derived earlier is the starting point for this section with the assumption of $r_1 \approx r_2 \approx r$,

$$\frac{T_{t2}}{T_{t1}} = 1 + \frac{U\left(C_{\theta 2} - C_{\theta 1}\right)}{c_p T_{t1}}$$

We may replace the swirl velocities by the flow angles and the axial velocity components, namely,

$$\frac{T_{t2}}{T_{t1}} = 1 + \frac{U\left(C_{z2}\tan\alpha_2 - C_{z1}\tan\alpha_1\right)}{c_p T_{t1}} = 1 + \left(\frac{U^2}{c_p T_{t1}}\right)\left(\frac{C_{z1}}{U}\right)\left(\frac{C_{z2}}{C_{z1}}\tan\alpha_2 - \tan\alpha_1\right) \tag{8.23}$$

Expression 8.23 for the total temperature rise across the rotor (or stage) involves nondimensional groups C_{z1}/U and $c_p T_{t1}/U^2$. These groups appear throughout the turbomachinery literature and deserve a special attention. Also, the axial velocity ratio C_{z2}/C_{z1} appears that is often set equal to 1, as a first-order design assumption. Equation 8.23 is expressed in terms of the absolute flow angle at the exit of the rotor, which is not, however, a good choice, since it varies with the rotor speed. A better choice for the flow angle in plane 2, that is, downstream of the rotor, is the relative flow angle β_2. The relative exit flow angle β_2 remains nearly unchanged as long as the flow remains attached to the blades. To express the total temperature rise across the rotor to the flow angles α_1 and β_2, we replace the absolute swirl $C_{\theta 2}$ by the relative swirl speed, namely,

$$C_{\theta 2} = U + W_{\theta 2} \tag{8.24}$$

Therefore,

$$\frac{T_{t2}}{T_{t1}} = 1 + \left(\frac{U^2}{c_p T_{t1}}\right)\left[1 + \frac{C_{z2}}{U}\tan\beta_2 - \frac{C_{z1}}{U}\tan\alpha_1\right] \tag{8.25a}$$

If we assume a constant axial velocity design, that is, $C_{z1} = C_{z2}$, then we get

$$\frac{T_{t2}}{T_{t1}} = 1 + \left(\frac{U^2}{c_p T_{t1}}\right)\left[1 + \left(\frac{C_z}{U}\right)\left(\tan\beta_2 - \tan\alpha_1\right)\right] \tag{8.25b}$$

Here, we have expressed the stage total temperature ratio as a function of two nondimensional parameters. Note that β_2 is a negative angle and α_1 is a positive angle, according to our sign convention. Therefore, the contribution to the stage total temperature ratio falls with increasing (C_z/U) for a given wheel speed and inlet stagnation enthalpy. The ratio of axial-to-wheel speed is called the *flow coefficient* ϕ

$$\phi \equiv \frac{C_z}{U} \tag{8.26a}$$

We may divide both numerator and the denominator of Equation 8.26a by the speed of sound in plane 1, that is, a_1, to get the ratio of axial to blade (rotational) or tangential Mach number, namely,

$$\phi = \frac{C_z/a_1}{U/a_1} = \frac{M_z}{M_T} \tag{8.26b}$$

where M_z is the axial Mach number, and M_T is the blade tangential Mach number based on U and a_1. For example, a stream surface with an axial Mach number of 0.5 that approaches a section of a rotor that is spinning at Mach 1 has a flow coefficient of 0.5. Now, let us interpret the first nondimensional group $U^2/(c_pT_{t1})$. We may divide this expression by the square of the upstream speed of sound a_1^2 to get

$$\frac{U^2/a_1^2}{c_pT_{t1}/(\gamma RT_1)} = \frac{(\gamma-1)\,M_T^2}{1+\left(\frac{\gamma-1}{2}\right)M_1^2} \tag{8.27}$$

Noting that the absolute Mach number M_1 is expressible in terms of the axial Mach number and a constant preswirl angle α_1 as

$$M_1 = \frac{M_z}{\cos\alpha_1} \tag{8.28}$$

We conclude that

$$\frac{U^2}{c_pT_{t1}} = \frac{(\gamma-1)\,M_T^2}{1+\left(\frac{\gamma-1}{2}\right)\frac{M_z^2}{\cos^2\alpha_1}} \tag{8.29}$$

Based on Equations 8.26b and 8.29, we may recast the stage stagnation temperature ratio in terms of blade tangential and axial Mach numbers as

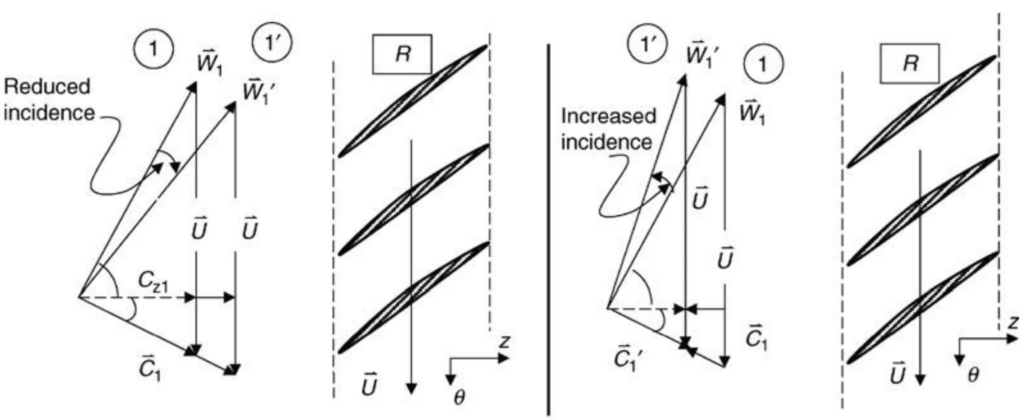
$$\frac{T_{t2}}{T_{t1}} = 1 + \left[\frac{(\gamma-1)\,M_T^2}{1+\left(\frac{\gamma-1}{2}\right)\frac{M_z^2}{\cos^2\alpha_1}}\right]\left[1+\left(\frac{M_z}{M_T}\right)\left(\tan\beta_2-\tan\alpha_1\right)\right] \tag{8.30}$$

Kerrebrock (1992) offers an insightful discussion of compressor aerodynamics based on this equation. Two Mach numbers, that is, the axial and the blade tangential Mach numbers, appear to influence the stage temperature ratio in three ways, namely, via M_z, M_T, and M_z/M_T influence in Equation 8.30. However, if we divide the first bracket on the right-hand side (RHS) of Equation 8.30 by the square of the blade tangential Mach number, we reduce this dependency to two parameters, namely,

$$\frac{T_{t2}}{T_{t1}} = 1 + \left[\frac{\gamma - 1}{\left(\frac{1}{M_T^2}\right) + \left(\frac{\gamma - 1}{2\cos^2\alpha_1}\right)\left(\frac{M_z}{M_T}\right)^2} \right] \left[1 + \left(\frac{M_z}{M_T}\right)(\tan\beta_2 - \tan\alpha_1) \right] \tag{8.31}$$

Consequently, two parameters govern the stage temperature ratio (or pressure ratio) and these are the blade tangential Mach number M_T and the flow coefficient or the ratio of the axial-to-tangential Mach number M_z/M_T. For given flow angles α_1 and β_2, an increase in the flow coefficient reduces the temperature rise in the stage. We may interpret an increase in the flow coefficient as an increase in the axial Mach number or the mass flow rate through the machine. As the mass flow rate increases, keeping the blade rotational Mach number the same, the blade angle of attack, or in the language of turbomachinery the *incidence angle,* decreases, hence the total temperature ratio drops. The opposite effect is observed with a decreasing mass flow rate through the machine where the incidence angle increases and thus blade work on the fluid increases to produce higher temperature or pressure rise. To visualize the effect of throughflow on rotor work, temperature and pressure rise, a rotor blade at different axial Mach numbers (or flow rate) and the same blade tangential Mach number (or wheel speed) are shown in Figure 8.10.

A reduced flow rate leads to an increased compressor temperature (or pressure) ratio. However, there is a limitation on how low the flow rate, or axial Mach number, can sink before the blades stall, for a fixed shaft rotational speed. Consequently, at reduced flow rates we could enter a blade stall flow instability, which marks the lower limit of



(a) The effect of *increased mass flow rate* while maintaining the same rotational speed is to reduce blade incidence angle

(b) The effect of *decreased mass flow rate*, with constant *U*, is to expose the blades to a higher incidence

■ **FIGURE 8.10** **Inlet velocity triangles for a compressor rotor with a changing flow rate** the axial flow speed on the compressor map for a given shaft speed. The increased flow rate that leads to a reduction of the stage total temperature rise has its own limitation. The phenomenon of negative stall could be entered as the inlet Mach number is increased significantly.

The second parameter is the blade tangential Mach number M_T. A higher blade tangential Mach number increases the stage total temperature rise as deduced from Equation 8.31. However, the limitations on the blade tangential Mach number are the appearance of strong shock waves at the tip as well as the structural limitations under centrifugal and vibratory stresses. The rotor shock losses increase (nonlinearly) with the relative tip Mach number; however, the advantage of higher work ($\propto M_T^2$) on the fluid outweigh the negatives of such a design at modest tip Mach numbers. The relative tip Mach number is defined as

$$M_{\text{tip,r}} = \sqrt{M_z^2 + \left(M_{\text{T,tip}}^2\right)} \tag{8.32}$$

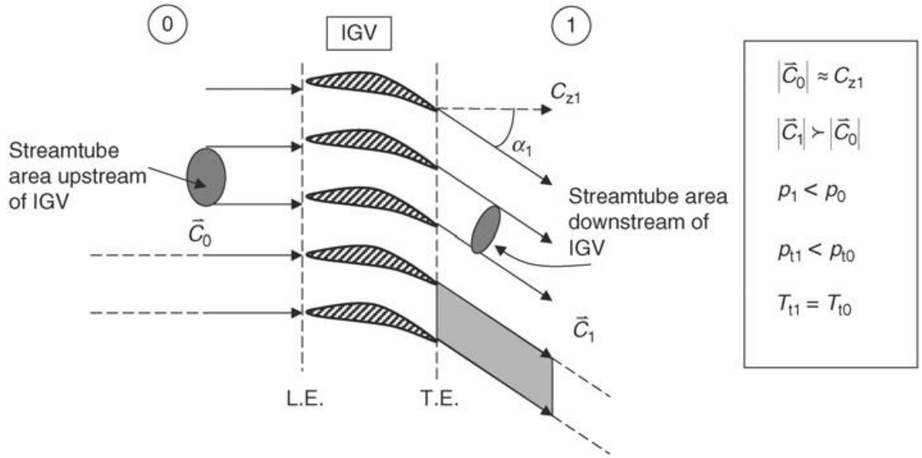
where it represents the case of zero preswirl. The general case that includes a nonzero preswirl, may be written as

$$M_{\text{tip,r}} = \sqrt{M_z^2 + \left(M_{\text{T,tip}} - M_z \tan \alpha_1\right)^2} \tag{8.33}$$

The relative blade Mach numbers at the tip of operational fan blades have been supersonic for the past three decades. By necessity, these blade sections should be thin to avoid stronger bow shocks. A typical value for the relative tip Mach number is ~1.2 but may be designed as high as ~1.7. The thickness-to-chord ratio of supersonic blading may be as low as ~3%. High strength-to-weight ratio titanium alloys represented the enabling technology that allowed the production of supersonic (tip) fans in the early 1970s.

The inlet absolute flow angle to the rotor α_1 is a design choice that has the effect of reducing the rotor relative tip speed. The inlet preswirl is created by a set of inlet guide vanes, known as an IGV. To reduce the relative flow to the rotor tip, the IGV turns the flow in the direction of the rotor rotation, by α_1. The flowfield entering an IGV is swirl-free and thus the function of the IGV is to impart positive swirl (or positive angular momentum) to the incoming fluid. This in effect reduces the rotor blade loading whose purpose is to impart angular momentum to the incoming fluid. An inlet guide vane and its flowfield are shown in Figure 8.11.

We note that the blade passages in the IGV form a subsonic nozzle (i.e., contracting area) and cause flow acceleration across the blade row. The result of the flow acceleration is found in the static pressure drop due to flow acceleration, as well as a total pressure drop due to frictional losses of the blade passages. Hence, if the compressor design could avoid the use of an IGV, then certain advantages, including cost and weight savings, are gained. The advantage of operating at higher tip speeds at times outweighs the disadvantages of an IGV. The inlet guide vane may also be actuated rapidly if a quick response in thrust modulation is needed. For example, consider a lift fan, or a deflected engine exhaust flow, to support a VTOL aircraft in hover mode. The ability to modulate the jet lift (or vertical thrust) for stability and maneuver purposes may not be achieved through a spool up or spool down throttle sequence of the engine. Due to a large moment of inertia of the rotating parts in a turbomachinery, the rapid spooling is not fast enough for the control



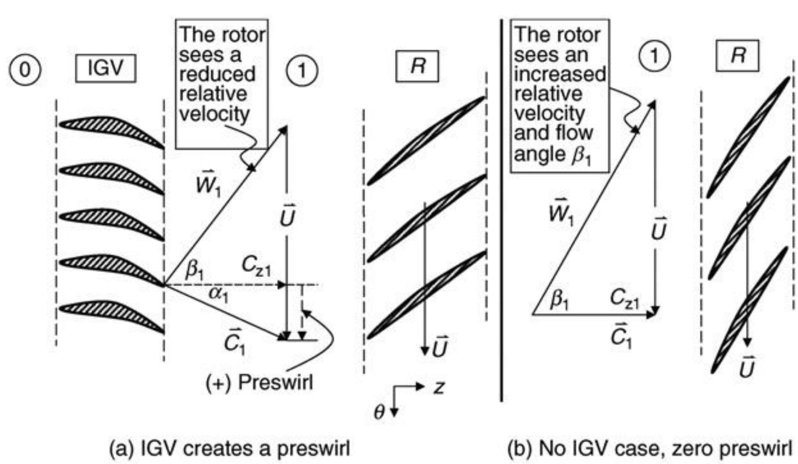
■ **FIGURE 8.11** **An inlet guide vane is seen to impart swirl to the fluid (note the shrinking stream tube area implies flow acceleration and static pressure drop)**

and stability purposes of a VTOL aircraft. In such applications, fast-acting IGV actuators could modulate the inlet flow to the rotor and hence the thrust/lift. An adjustable exit louver at the deflected nozzle end achieves the same goal. The inlet swirl angle α_1 may be zero or be adjustable in the range of $\pm 30°$ or more using a variable geometry IGV to relieve the relative tip Mach number, improve efficiency at compressor off-design operation, and provide for rapid thrust/lift modulation in military aircraft.

An example of a velocity triangle upstream of the rotor blade with and without the inlet guide vane is shown in Figure 8.12. In both cases, we maintain the rotational speed and the mass flow rate, i.e., the axial velocity C_z constant.

The stage total pressure ratio is related to the stage total temperature ratio via a stage efficiency parameter η_s. Recalling from the chapter on cycle analysis,

$$\eta_s \equiv \frac{T_{t3s} - T_{t1}}{T_{t3} - T_{t1}} = \frac{\pi_s^{\frac{\gamma-1}{\gamma}} - 1}{\tau_s - 1} \tag{8.34}$$



■ **FIGURE 8.12** **Impact of IGV on the rotor relative flow for constant axial velocity and rotor speed**

Therefore, we may calculate the stage total pressure ratio from the velocity triangles that are used in the Euler turbine equation to establish τ_s and an efficiency parameter η_s to get

$$\pi_s = \left[1 + \eta_s\left(\tau_s - 1\right)\right]^{\frac{\gamma}{\gamma-1}} \tag{8.35}$$

Since the axial flow compressor pressure ratio per stage is small (i.e., near 1), the stage adiabatic efficiency and the polytropic efficiency are nearly equal. We recall that the polytropic efficiency was also called the "small-stage" efficiency, valid for an infinitesimal stage work. Although the approximation

$$\eta_s \cong e_c \tag{8.36}$$

is valid for low-pressure ratio axial flow compressor stages. The exact relationship between these parameters is derived in the cycle analysis chapter to be

$$\eta_s = \frac{\tau_s - 1}{\tau_s^{\frac{1}{e_c}} - 1} \tag{8.37}$$

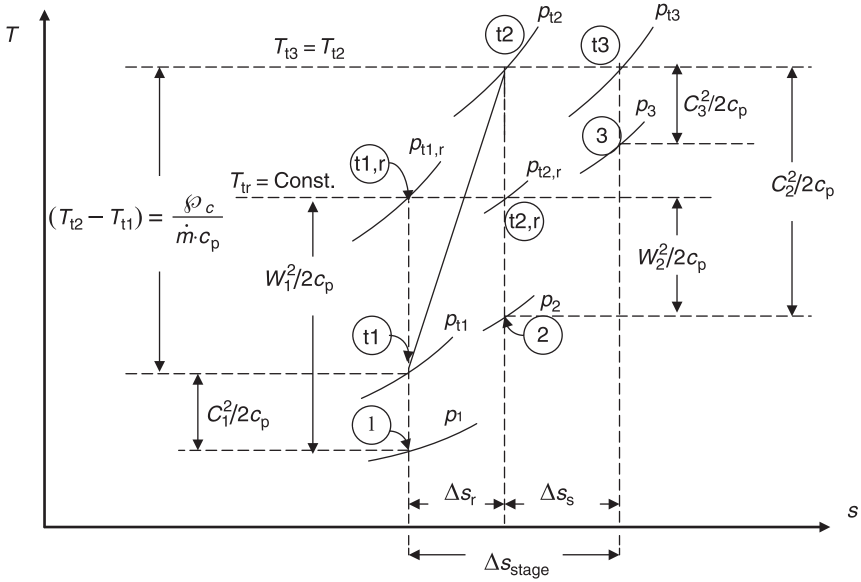
Therefore, by calculating the stage total temperature ratio from the Euler turbine equation and assuming polytropic efficiency, of say 0.90, we may calculate the stage adiabatic efficiency. Otherwise, we may calculate the stage total pressure ratio from the polytropic efficiency e_c and the stage total temperature ratio directly, via

$$\pi_s = \tau_s^{\frac{\gamma \cdot e_c}{\gamma-1}} \tag{8.38}$$

The frictional and shock losses are the main contributors to a total pressure loss in the relative frame of reference. We note that in a relative frame of reference, the blade passage is stationary and thus blade aerodynamic forces perform no work. Examining the fluid total pressure across the blade through the eyes of a relative observer, there are only losses to report. As noted, these losses stem from the viscous and turbulent dissipation of mechanical (kinetic) energy into heat as well as the flow losses associated with a shock. From the vantage point of an absolute observer, the blades are rotating and doing work on the fluid, thus increasing the fluid total temperature and pressure in the absolute frame of reference. These processes may be shown on a *T–s* diagram (Figure 8.13) that is instructive to review. For example, follow the static state of gas across the stage then follow the stagnation states in the absolute and relative frames. Explain the behavior of kinetic energy of gas as the fluid encounters the stationary and rotating blade rows, as seen by absolute and relative observers.

The *relative* total enthalpy is constant across the rotor along a stream surface (in steady flow without a radial shift in the stream surface),

$$h_{t,r} = h_1 + \frac{W_1^2}{2} = h_2 + \frac{W_2^2}{2} \tag{8.39}$$



■ FIGURE 8.13 **Absolute and relative states of gas across a compressor rotor and stator**

We may replace the total kinetic energy terms in Equation 8.39 by the sum of the kinetic energy of the component velocities, namely,

$$h_1 + W_{z1}^2/2 + W_{\theta 1}^2/2 = h_2 + W_{z2}^2/2 + W_{\theta 2}^2/2 \tag{8.40}$$

Let us substitute the absolute swirl minus the wheel speed ($C_\theta - U$) for the relative swirl in Equation 8.40, to get

$$h_1 + C_{z1}^2/2 + C_{\theta 1}^2/2 - C_{\theta 1} \cdot U = h_2 + C_{z2}^2/2 + C_{\theta 2}^2/2 - C_{\theta 2} U \tag{8.41}$$

The sum of the first three terms is the absolute total enthalpy on each side of the above equality, therefore,

$$h_{t1} - UC_{\theta 1} = h_{t2} - UC_{\theta 2} \tag{8.42}$$

This enthalpy constant in the rotor frame is known as "rothalpy," which may be rearranged to help us arrive at the Euler turbine equation, that is,

$$h_{t2} - h_{t1} = \frac{\wp}{\dot{m}} = w_c = U(C_{\theta 2} - C_{\theta 1}) \tag{8.43}$$

The rotor specific work on the fluid in nondimensional form is written as

$$\psi \equiv \frac{\Delta h_t}{U^2} = \frac{C_{\theta 2}}{U} - \frac{C_{\theta 1}}{U} = \frac{W_{\theta 2} + U}{U} - \frac{C_{\theta 1}}{U} = 1 + \left(\frac{C_z}{U}\right)\left(\tan\beta_2 - \tan\alpha_1\right) \tag{8.44}$$

The function ψ is the nondimensional stage work parameter, which is called the *stage-loading factor or parameter* defined in Equation 8.44. The ratio of axial-to-wheel speed was called the flow coefficient ϕ, which forms another two-parameter family for the stage characteristics, namely,

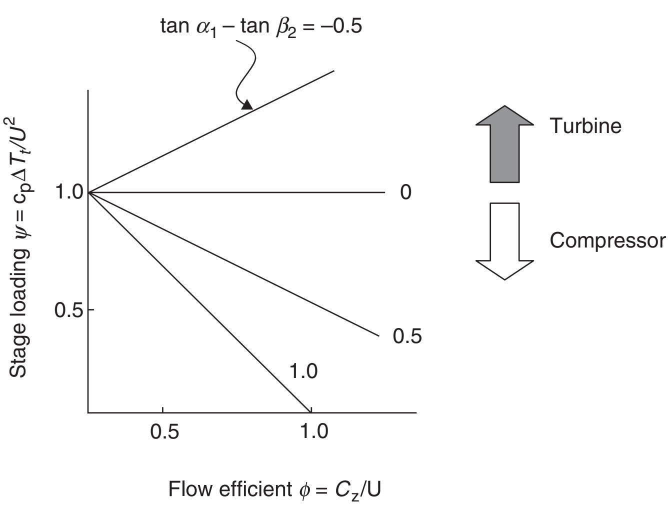
$$\psi = 1 + \phi\left(\tan\beta_2 - \tan\alpha_1\right) \tag{8.45}$$

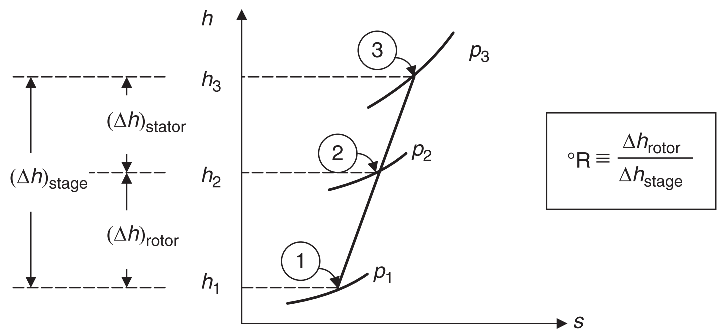
We note that the rotor exit relative flow angle is a negative quantity that leads to a stage-loading factor that is less than 1. In the case of zero preswirl, or no inlet guide vane, the stage loading factor increases with a decreasing rotor exit flow angle β_2. In the limit of zero relative swirl at the rotor exit and no inlet guide vane, the stage loading factor approaches unity. However, a rotor relative exit flow angle of zero implies significant turning in the rotor blade passage, which may lead to flow separation. The stage-loading factor is an alternate form of expressing the stage characteristics and, in essence, takes the place of the rotor tangential Mach number M_T, which was presented earlier. Horlock (1973) takes advantage of the linear dependence of the stage loading and the flow coefficient in Equation 8.45 to explore the off-design behavior of ideal turbomachinery stages. We shall discuss the off-design behavior of turbomachinery later in this chapter but for now show the linear dependence of the two parameters in Figure 8.14 (adapted from Horlock, 1973).

We define a stage *degree of reaction* °R as the fraction of static enthalpy rise across the stage that is accomplished by the rotor. Although the stator does no work on the fluid, it still acts as a diffuser that decelerates the fluid and thus causes an increase in fluid temperature, or static enthalpy. The stator takes out the swirl (kinetic energy) put in by the rotor and thus converts it to a static pressure rise. The degree of reaction measures the rotor share of the stage enthalpy rise as compared with the burden on the stator. This process is shown in an *h–s* diagram for the static states in a compressor stage. Remember that the static states are independent of the motion of the observer, hence they carry no subscript labels besides the station number (Figure 8.15).

$$°\mathrm{R} \equiv \frac{h_2 - h_1}{h_3 - h_1} = \frac{h_{t2} - h_{t1} - (C_2^2 - C_1^2)/2}{h_{t3} - h_{t1} - (C_3^2 - C_1^2)/2} \tag{8.46a}$$

■ FIGURE 8.14 Linear dependence of stage loading and flow coefficient parameters in ideal turbomachinery stages. Source: Adapted from Horlock 1973





■ **FIGURE 8.15** **Static states of gas in a compressor stage and a definition sketch for the stage degree of reaction °R**

The stagnation enthalpy across the stator remains constant as the stator blades do no work on the fluid, hence $h_{t3} = h_{t2}$ and if we assume a repeated stage, with $C_1 = C_3$, we simplify the above expression to get

$$°R \cong 1 - \frac{C_2^2 - C_1^2}{2(h_{t2} - h_{t1})} = 1 - \frac{C_{z2}^2 + C_{\theta 2}^2 - C_{z1}^2 + C_{\theta 1}^2}{2U(C_{\theta 2} - C_{\theta 1})} \tag{8.46b}$$

Now, for a constant axial velocity across the rotor, $C_{z2} = C_{z1}$, we get a simple expression for the stage degree of reaction, as

$$°R \cong 1 - \frac{C_{\theta 2} + C_{\theta 1}}{2U} = 1 - \frac{C_{\theta,\text{mean}}}{U} \tag{8.46c}$$

where $C_{\theta,\text{mean}}$ is the average swirl across the rotor. Since the flow has to fight an uphill battle with an adverse pressure gradient throughout a compressor stage, it stands to reason to expect/design an equal burden of the static pressure rise in the rotor as that of a stator. Consequently, a 50% degree of reaction stage may be thought of as desirable. We may express the swirl velocity components across the rotor in terms of the absolute in flow and relative exit flow angles, α_1 and β_2, respectively,

$$°R = 1 - \frac{W_{\theta 2} + U + C_{\theta 1}}{2U} = \frac{1}{2} - \frac{C_z \tan\beta_2 + C_z \tan\alpha_1}{2U} = \frac{1}{2} - \left(\frac{C_z}{U}\right)\left(\frac{\tan\beta_2 + \tan\alpha_1}{2}\right) \tag{8.47}$$

For a 50% degree of reaction stage (at some spanwise radius r), the rotor exit flow angle has to be equal in magnitude and opposite to the inlet absolute flow angle, which is dictated by an IGV, namely,

$$\beta_2 = -\alpha_1$$

This is the condition for a repeated row design, as noted in Equations 8.22a and 8.22b. Therefore a purely axial inflow (with no inlet guide vane) demands a purely axial relative outflow in order to produce a 50% degree of reaction. Again, we need to examine the net turning angle across the blade and assess the potential for flow separation. We shall introduce another parameter that will shed light on the state of the boundary layer at the blade exit and that is the diffusion factor.

On the degree of reaction, there is a body of experimental results that support the proposal that a boundary layer on a spinning blade, that is, the rotor, is more *stable* than a corresponding boundary layer on a stationary blade, that is, the stator, hence allocating a slightly higher burden of static pressure rise to the rotor. Here the word *stable* is used in the context of resistant to adverse pressure gradient or higher stalling pressure rise. Based on this, a degree of reaction of 60% may be a desirable split between the two blade rows in a compressor stage. As we shall see in the three-dimensional flow section in this chapter, the desirable degree of reaction is often compromised at different radii along a blade span to satisfy other requirements, namely, a healthy state of boundary layer flow.

EXAMPLE 8.2

An axial-flow compressor stage has a pitchline radius of $r_m = 0.5$ m. The rotational speed of the rotor at pitchline is $U_m = 212$ m/s. The absolute inlet flow to the rotor is described by $C_{zm} = 155$ m/s and $C_{\theta 1m} = 28$ m/s. The stage degree of reaction at pitchline is $°R_m = 0.60$, $\alpha_3 = \alpha_1$, and C_{zm} remains constant. Calculate

(a) rotor angular speed ω in rpm
(b) rotor exit swirl $C_{\theta 2m}$
(c) rotor specific work at pitchline w_{cm}
(d) relative velocity vector at the rotor exit
(e) rotor and stator torque per unit mass flow rate
(f) stage loading parameter at pitchline ψ_m
(g) flow coefficient ϕ_m.

SOLUTION

$\omega = U_m/r_m = (212 \text{ m/s})/0.5 \text{ m} = (424 \text{ rad/s}) (\text{rev}/2\pi \text{ rad}) (60 \text{ s/min}) \approx 4{,}049$ rpm

From Equation 8.46c, written at the pitchline

$$°R_m \cong 1 - \frac{C_{\theta 2m} + C_{\theta 1m}}{2U_m}$$

We isolate $C_{\theta 2m}$ to be

$$\begin{aligned} C_{\theta 2m} &= 2U_m(1 - °R_m) - C_{\theta 1m} \\ &= 2(212 \text{ m/s})(0.4) - 28 \text{ m/s} \approx 141.6 \text{ m/s} \end{aligned}$$

Euler turbine equation describes the rotor specific work,

$$w_c \equiv \frac{\wp_c}{\dot{m}_c} = \omega\Delta\left(rC_\theta\right)_c, \quad \text{therefore,}$$

$$\begin{aligned} w_{cm} &= U_m(C_{\theta 2m} - C_{\theta 1m}) \\ &= 212 \text{ m/s } (141.6 - 28)\text{m/s} \approx 24.08 \text{ kJ/kg} \end{aligned}$$

Rotor relative swirl at the exit is

$$W_{\theta 2m} = C_{\theta 2m} - U_m = (141.6 - 212)\text{m/s} = -71 \text{ m/s}$$

Since the axial component of velocity remains constant, we can write the vector

$$\vec{W}_{2m} = 155\hat{k} - 71\hat{e}_\theta$$

Since $\alpha_3 = \alpha_1$, the rotor and stator torques are equal and opposite to eachother, i.e.,

$$\begin{aligned} \tau_{rm}/\dot{m} &= r_m(C_{\theta 2m} - C_{\theta 1m}) = 0.5\text{m}(141.6 - 28) \text{ m/s} \\ &= 56.8\text{m}^2/\text{s} = -\tau_{sm}/\dot{m} \end{aligned}$$

The stage loading parameter and flow coefficients are

$$\begin{aligned} \psi_m &= \Delta C_\theta/U_m = (141.6 - 28)/212 \approx 0.5358 \\ \phi_m &= C_{zm}/U_m = 155/212 = 0.731 \end{aligned}$$

Another figure of merit for a compressor blade section that addresses the health of a boundary layer is, as noted earlier, the *Diffusion Factor*, or the *D*-factor. Its definitions for rotor and stator blades are, respectively

$$D_r \equiv 1 - \frac{W_2}{W_1} + \frac{|W_{\theta 2} - W_{\theta 1}|}{2\sigma_r W_1} \qquad \text{(rotor } D\text{-Factor)} \tag{8.48}$$

$$D_s \equiv 1 - \frac{C_3}{C_2} + \frac{|C_{\theta 3} - C_{\theta 2}|}{2\sigma_s C_2} \qquad \text{(stator } D\text{-Factor)} \tag{8.49}$$

where σ defines the blade solidity, that is, the ratio of blade chord c to spacing s

$$\sigma_r \equiv \frac{c_r}{s_r} \qquad \text{(rotor solidity)} \tag{8.50}$$

$$\sigma_s \equiv \frac{c_s}{s_s} \qquad \text{(stator solidity)} \tag{8.51}$$

Modern compressor design utilizes a high solidity ($\sigma_m \geq 1$) blading at the pitchline radius (i.e., r_m). Since the blade spacing increases linearly with radius, the solidity of a constant chord blade also decreases linearly with the blade span, that is,

$$s(r) = \frac{2\pi \cdot r}{N_b} \qquad \text{(blade spacing)} \tag{8.52}$$

where the N_b is the number of blades in the rotor or the stator row. We can see the variation of blade row solidity with blade span by substituting Equation 8.52 for the blade spacing, as

$$\sigma(r) = \frac{N_b \cdot c}{2\pi \cdot r} \qquad \text{(blade solidity)} \tag{8.53}$$

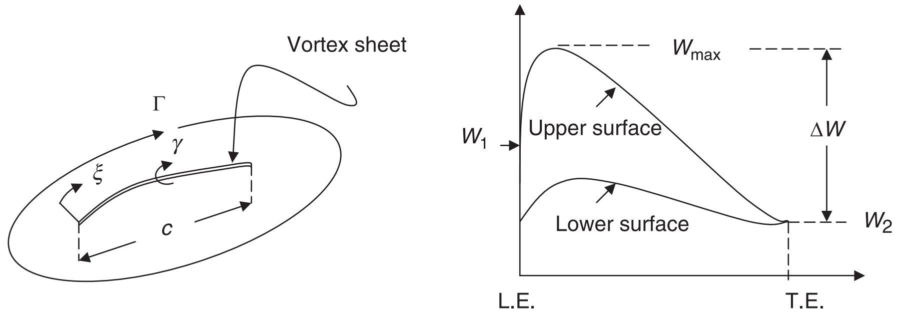
The hub section (r_h) has thus the highest solidity and the tip section (r_t) the lowest. The rationale for the definition of D-factor and its link to blade stall is made by Lieblein (1953, 1959, 1965). We review it here for its physical importance. First note that the definition of rotor diffusion factor is the same as the stator D-factor, except the parameters for the rotor are all in relative frame of reference. Next note that the change in swirl velocity across the rotor is the same regardless of the frame of reference, that is, the change in absolute swirl is the same in magnitude as the change in relative swirl, namely,

$$|W_{\theta 2} - W_{\theta 1}| = |C_{\theta 2} - C_{\theta 1}| \tag{8.54}$$

The blade circulation Γ is the integral of the vortex sheet strength γ over the chord, namely,

$$\Gamma \equiv \int_0^c \gamma(\xi)\mathrm{d}\xi \approx \bar{\gamma} \cdot c \tag{8.55}$$

Figure 8.16 shows an element of a blade section represented by a vortex sheet of local strength $\gamma(\xi)$. The local strength is equal to the local tangential velocity jump across the



■ **FIGURE 8.16**
Blade airfoil section is represented by a vortex sheet and a velocity distribution

blade. The average vortex sheet strength is thus the average velocity jump across the sheet, namely,

$$\bar{\gamma} \approx (\Delta W)_{avg} \approx \frac{W_{max} - W_1}{2} \tag{8.56}$$

Therefore, the average (positive) circulation around the blade section is

$$\Gamma_{avg} \approx \frac{c}{2}\left(W_{max} - W_1\right) \tag{8.57}$$

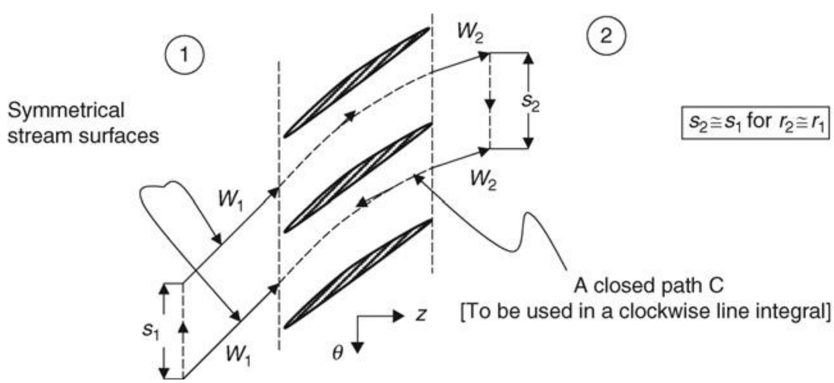
Now, let us consider a control volume symmetrically wrapped around a blade section, as shown in Figure 8.17.

By performing a closed line integral in the clockwise direction around the path C (Figure 8.18), we calculate the magnitude of blade circulation Γ as

$$\Gamma = s|W_{\theta 1} - W_{\theta 2}| \tag{8.58}$$

Therefore equating the two expressions for the (magnitude of) blade circulation, we get

$$W_{max} \approx W_1 + \frac{|W_{\theta 2} - W_{\theta 1}|}{2\sigma} \tag{8.59}$$



■ **FIGURE 8.17**
Control volume for determining the blade circulation Γ along a stream surface

The maximum adverse pressure gradient on the blade suction surface leads to a maximum flow diffusion, which may be measured by the following parameter

$$\text{Maximum flow diffusion} \approx \frac{W_{\max} - W_2}{W_2} \approx \frac{W_{\max} - W_2}{W_1} \tag{8.60}$$

This parameter is called the *Diffusion Factor* D

$$D \equiv \frac{W_{\max} - W_2}{W_1} = \frac{W_1 + (|\Delta C_\theta|/2\sigma) - W_2}{W_1} = 1 - \frac{W_2}{W_1} + \frac{|\Delta C_\theta|}{2\sigma W_1} \tag{8.61}$$

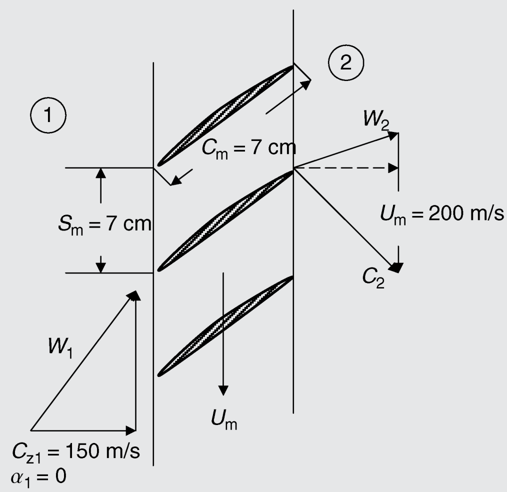
EXAMPLE 8.3

A rotor blade row is cut at pitchline, r_m. The velocity vectors at the inlet and exit of the rotor are shown.

Assuming that $U_{m1} = U_{m2} = 200$ m/s and $C_{z1} = C_{z2} = 150$ m/s

And $\beta_2 = -35°$, calculate

(a) W_{1m} and W_{2m}
(b) *D*-factor D_{rm}
(c) circulation Γ_m



SOLUTION

$$W_{1m} = [(150)^2 + (200)^2]^{1/2} = 250 \text{ m/s}$$

$$W_{\theta 2m} = C_{z2} \tan\beta_2 = (150 \text{ m/s}) \tan(-35°)$$

$$W_{\theta 2m} \approx -105 \text{ m/s}$$

$$W_{2m} = [(150)^2 + (105)^2]^{1/2} = 183.1 \text{ m/s}$$

The *D*-factor for the rotor at the pitchline is

$$D_{rm} \equiv 1 - \frac{W_{2m}}{W_{1m}} + \frac{|W_{\theta 2m} - W_{\theta 1m}|}{2\sigma_{rm} W_{1m}}$$

The rotor solidity at the pitchline is $\sigma_{rm} = c/s = 7/7 = 1.0$ The relative tangential speed at the inlet to the rotor is equal in magnitude to the rotor speed U_m, but it has a negative value, as it points in the opposite direction to the rotor rotation

$$W_{\theta 1m} = -200 \text{ m/s}$$

Therefore, we have

$$D_{rm} = 1 - (183.1/250) + |(-105 + 200)|/(2.1.250) \approx 0.457$$

Since *D*-factor is less than ~0.5, the rotor boundary layer at the pitchline is expected to be attached.

Circulation around the rotor at pitchline is

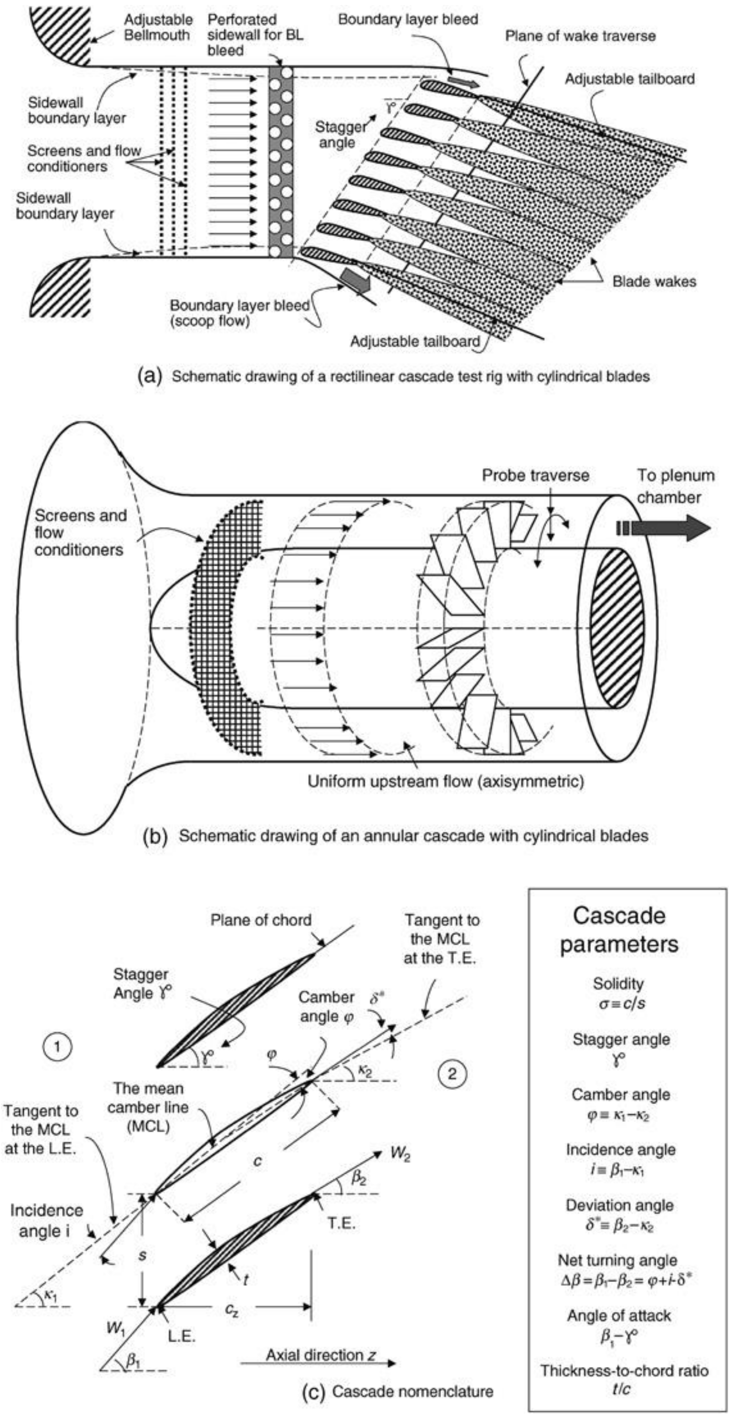
$$\Gamma_m = s_m |W_{\theta 1m} - W_{\theta 2m}| = (0.07\text{m})|(-200 + 105)| \approx 6.65 \text{ m}^2/\text{s}$$

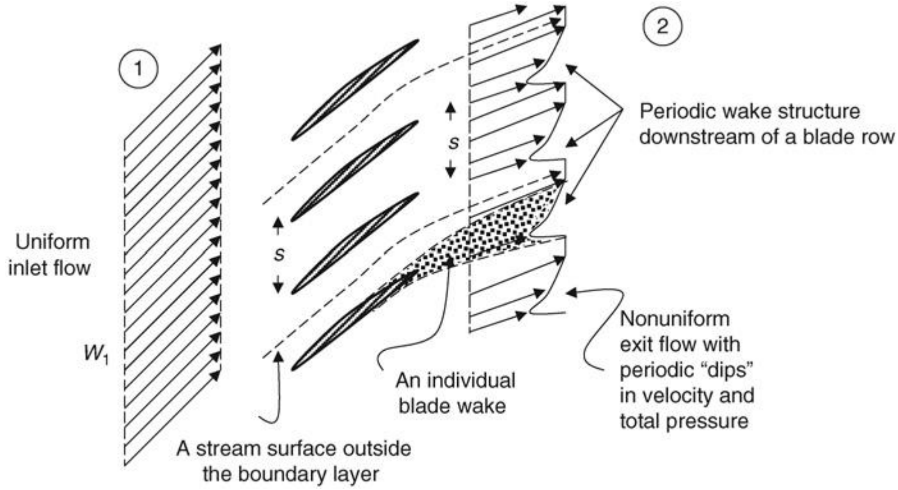
8.6.3 Cascade Aerodynamics

A large body of experimental data supports a correlation between the blade stall and the diffusion factor. Let us first examine the attributes of a stalled flow in a compressor blade row. A separated boundary layer will create a thick wake, which may be characterized by its momentum deficit length scale θ^*. Equivalently, a wake is a region of deficit total pressure, hence in the blade frame of reference stall creates a rapid rise in the total pressure loss across the blade row. In either case, we need to examine the wake profile downstream of a blade row to quantify momentum and total pressure loss. To concentrate on the study of compressor blades, NACA (the predecessor of NASA) developed the so-called 65 series of airfoil profiles in early 1950s. To test the aerodynamic behavior of the newly developed blade shapes, NACA performed an extensive series of cascade experiments, concentrating on the impact of the 65-series compressor airfoil shapes on two-dimensional compressor loss and stall characteristics. Airfoil shapes are defined around a mean camber line, typically of circular arc or parabolic shape and a prescribed thickness distribution. A cascade is composed of a series of two-dimensional or cylindrical blades placed in a uniform flow to simulate the two-dimensional flowfield about a *spanwise section* of a compressor blade row. The goal of the investigation is to quantify the profile loss of two-dimensional blade elements as a function of the profile geometry and cascade parameters such as blade chord-to-spacing ratio, blade stagger angle, and other parameters. To construct the three-dimensional loss characteristics of compressor blades, two-dimensional cascade loss data are *stacked up* to represent the 3D picture. This approach neglects the cross interaction of the stream surfaces that is created through 3D pressure gradients. We learned in wing theory that three-dimensional pressure gradients lead to the formation of the streamwise vortices in the wake, which causes a 3D induced velocity (and induced drag) along the blade span. We shall address three-dimensional losses that are overlooked by the cascade data later in this chapter. Let us return to two-dimensional cascade studies performed at NACA. The wake profile with its momentum deficit and total pressure loss holds the key to characterizing the behavior of a compressor blade section. Figure 8.18a shows a rectilinear cascade, Figure 8.18b an annular cascade, and Figure 8.18c defines the geometric parameters of the blade section and the cascade. Figure 8.19 shows periodic blade wakes downstream of a cascade where the momentum deficit and the total pressure loss are concentrated. The thickness of the wake is exaggerated in Figure 8.19 for visualization purposes, but a thicker suction surface boundary layer than the pressure surface is intentionally graphed to show the behavior of these boundary layers that merge to form the blade wake in turbomachinery.

Let us review the cascade parameters as shown in Figure 8.18c. An important geometrical cascade parameter is the solidity σ, which is defined as the ratio of chord-to-spacing. A high solidity blading is capable of a higher net turning angle than a low solidity blading. Consequently, a high solidity cascade is less susceptible to stall. Experimental evidence to support this assertion will be presented as cascade test results. As noted earlier, the modern compressor and fan design utilizes a high solidity blading ($\sigma_t \cong 1$). The angle of the mean camber line at the leading and trailing edge is used as a reference where we measure the inlet flow incidence angle i and the exit deviation angle δ^*. The incidence angle is defined as the flow angle between the tangent to the mean camber line and relative velocity vector at the inlet. In compressors, incidence angle takes the place of angle of attack in external aerodynamics. An optimum, incidence angle is defined as the incidence that causes minimum (total pressure) loss across a given cascade. A

■ **FIGURE 8.18**
Cascade types and nomenclature

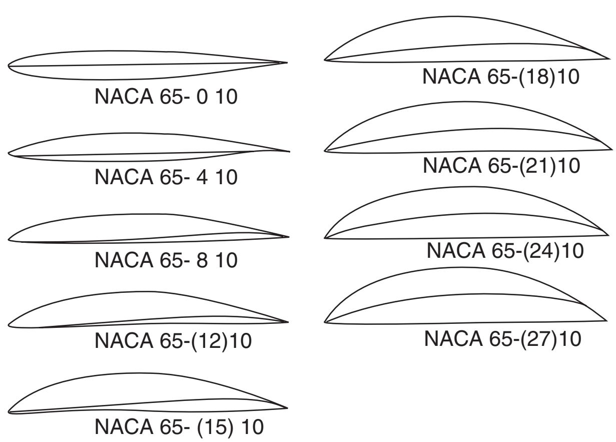




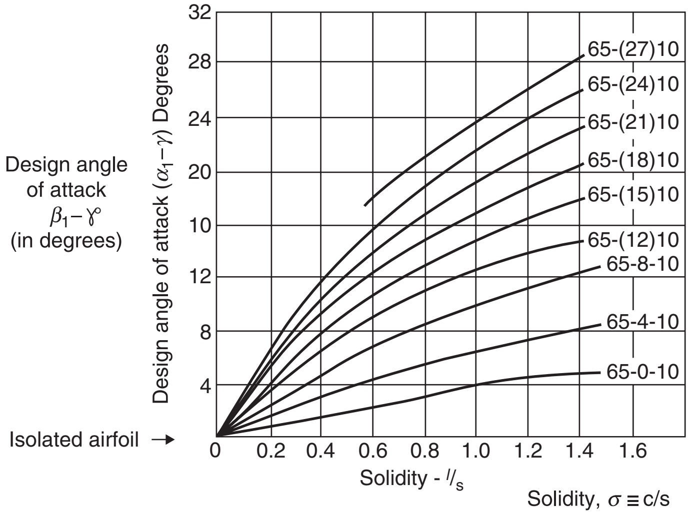
■ **FIGURE 8.19** **Survey of the wake-velocity profile downstream of a cascade**

typical value of the optimum incidence angle for a subsonic flow cascade is $i_{opt} \sim 2°$. Cascade experimental results support this rule of thumb. The camber angle φ is defined as the angle formed at the intersection of the two tangents to the mean camber line at the leading and trailing edges, as shown. A large camber means a large flow turning, which may lead to flow separation. Hence, compressor blade camber is small compared with a turbine blade, which is capable of large turning due to a favorable pressure gradient. The stagger angle $\gamma°$ is sometimes referred to as the *blade-setting angle* is the blade chord angle with respect to the axial direction. The stagger angle increases with radius as the blade rotational speed (ωr) increases linearly with radius. The difference between the stagger angle and the relative inflow angle is called the angle of attack, as in external aerodynamics. Finally, the net turning angle refers to the difference between the inlet and exit flow angles, respectively.

The 65-series compressor cascade airfoil shapes are shown in Figure 8.20. The design (theoretical) lift coefficient for an isolated airfoil shape (at zero angle of attack) is listed in parenthesis (times ten) following the 65-series designation. Note that the lift



■ **FIGURE 8.20** **The 65-series cascade airfoil shapes with 10% thickness. Source: Herrig, Emery, and Erwin 1951**



■ **FIGURE 8.21**
Design angle of attack, $\beta_1 - \gamma°$, for the 65-series airfoils with 10% thickness as a function of solidity. Source: Herrig, Emery, and Erwin 1951

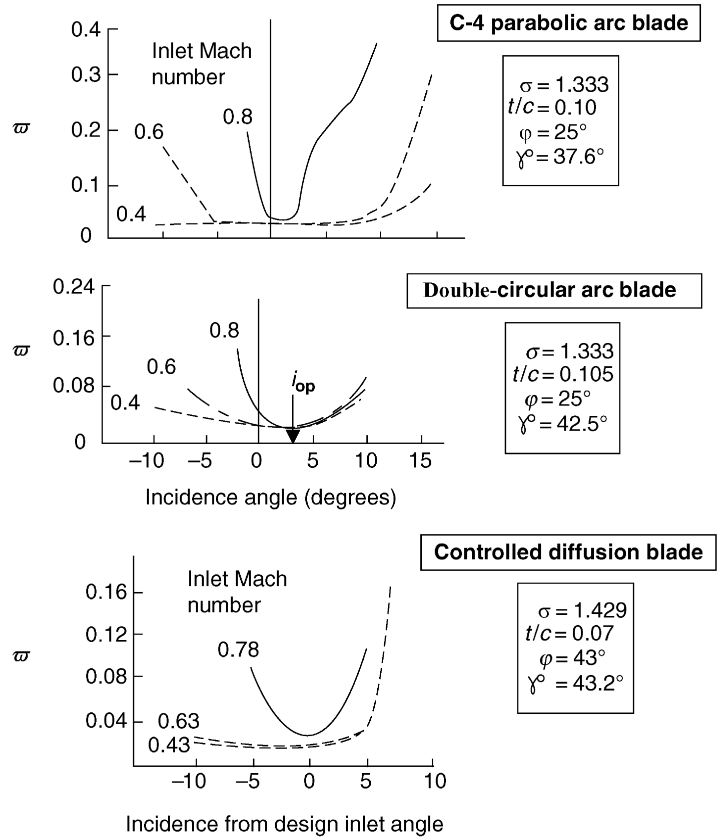
coefficient at zero angle of attack is entirely due to camber. The *isolated* means that the airfoil is not in a cascade configuration, that is, solidity is zero. The thickness-to-chord ratio (in percent) comprises the last two digits of the series designation.

For example, a NACA 65-(18)10 represents the 65-series camber shape that produces an isolated theoretical (i.e., inviscid) lift coefficient of 1.8 (at zero angle of attack) and has a 10% thickness-to-chord ratio. The cascade parameters, namely, solidity and stagger, are included in the individual loss characteristic (bucket) curves that are produced for different cascades. NACA defined a design angle of attack for the 65-series airfoils based on the smoothness of the pressure distributions on the airfoils. Figure 8.21 shows the design angle of attack for the 65-series airfoils arranged in a cascade of varying solidity. The zero solidity refers to an isolated airfoil case. Note that the combination of high solidity and high camber that leads to a high angle of attack (Figure 8.21) does not mean that the incidence angle is very large too. The blade inlet angle κ_1 makes a large angle with respect to the blade chord for highly cambered airfoils (see Figure 8.20 for airfoil shapes with high camber). The message from Figure 8.21 is that higher solidity allows for larger turning, which translates into an increase in inlet flow angle (keeping the exit angle ~ fixed).

A survey of total pressure downstream of cascade reveals the presence of periodic wakes. Defining an *average* total pressure loss parameter for a cascade as

$$\varpi \equiv \frac{p_{t1} - \bar{p}_{t2}}{\rho_1 W_1^2/2} \tag{8.62}$$

where the average downstream total pressure may be taken as the area-average of the total pressure survey. The denominator of Equation 8.62 is the familiar dynamic pressure ($\rho_1 W_1^2/2$) in the cascade frame of reference. The cascade loss “bucket curves” are shown in Figure 8.22 for a parabolic arc, a double-circular arc, and a controlled diffusion blading at different inlet Mach numbers (adapted from Kerrebrock, 1992). The works of Hobbs and Weingold (1984) and Hechert, Steinert and Lehmann (1985) on development of



■ **FIGURE 8.22**
Variation of cascade total pressure loss parameter with inlet Mach number and flow incidence angle. Source: Adapted from Kerrebroch 1992

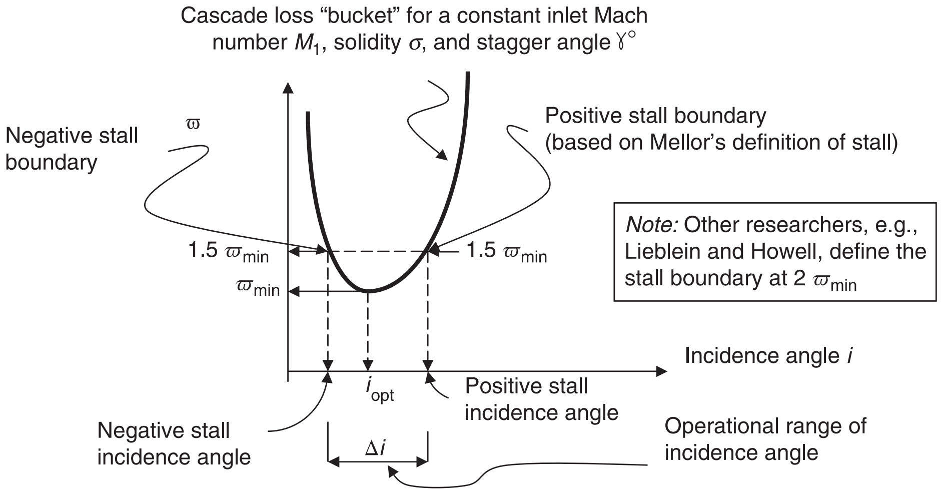
controlled diffusion airfoils and their comparison to NACA-65 airfoils for compressors should be consulted for further reading.

The minimum loss incidence angle is referred to as ϖ_{min} that corresponds to the optimum incidence angle i_{opt} We further define an acceptable operational range for the incidence angle that corresponds to 150% of the minimum loss ϖ_{min}, which Mellor referred to as the positive and negative stall boundaries. A definition sketch is shown in Figure 8.23.

Mellor presents the operational range of each 65-series airfoils in various cascade arrangements that are very useful for preliminary design purposes. Mellor's unpublished graphical correlations (originated at MIT Gas Turbine Laboratory) were published by Horlock (1973), Hill and Peterson (1992), among others.

The correlation between the diffusion factor and the wake momentum deficit thickness θ^* is presented in Figure 8.24 (Lieblein, 1965). The cascade data of Lieblein present 10% thick blades with the mean camber line shapes described by the NACA 65-series profiles and the British C-4 parabolic arc profiles, as shown in Figure 8.24. The Reynolds number for the cascade tests is ~250,000. Blade stall leads to a thickening of the wake that results in a rapid increase in profile drag and hence momentum deficit thickness. From Figure 8.24, we note this behavior occurs around a D-factor of 0.6. Hence, the maximum

■ **FIGURE 8.23**
Definition sketch for the stall boundaries based on the cascade loss curve

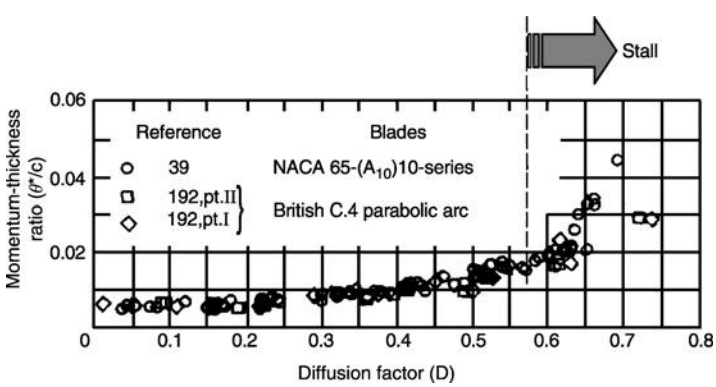


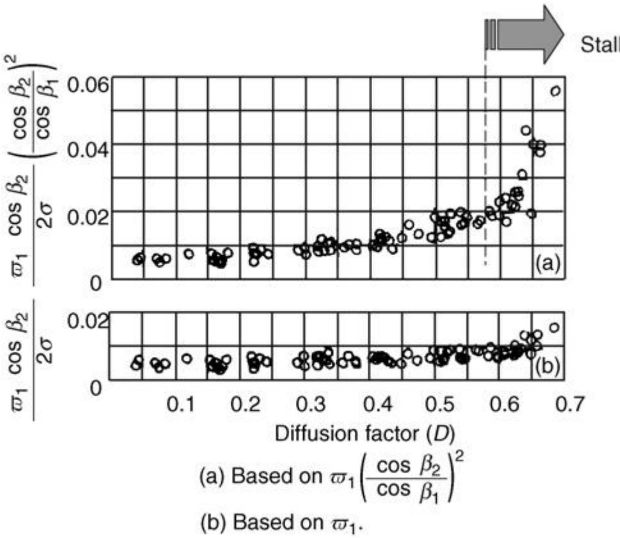
diffusion factor associated with a well-behaved boundary layer on these classical blade profiles is $D_{max} \sim 0.6$. A higher D-factor (of ∼0.7) may be achieved in cascades of modern controlled-diffusion profiles.

How does the momentum deficit thickness θ^*/c relate to the total pressure loss parameter ϖ in a cascade? By averaging the total pressure downstream of the cascade assuming a periodic momentum deficit thickness with zero momentum, we can correlate these two parameters. Figure 8.25 (Lieblein, 1965) shows two total pressure loss functions and their correlation with D-factor.

To demonstrate the functional form of the correlation (see the ordinate of Figure 8.25), we approximate the cascade exit flow as being composed of uniform flow with periodic gaps of zero momentum, each with θ^*/c thickness corresponding to wakes. The static pressure is assumed to be uniform in the measuring station downstream of the cascade. Figure 8.26 shows a definition sketch of this simplified cascade exit flow model. Although the original contribution to this derivation is due to Lieblien-Roudebush (1956), this author has benefited from Kerrebrock's (1992) treatment of this and other turbomachinery subjects.

■ **FIGURE 8.24**
Correlation between the diffusion factor and the wake momentum deficit thickness. Source: Lieblein 1965 (reference numbers are in Lieblein). Courtesy of NASA





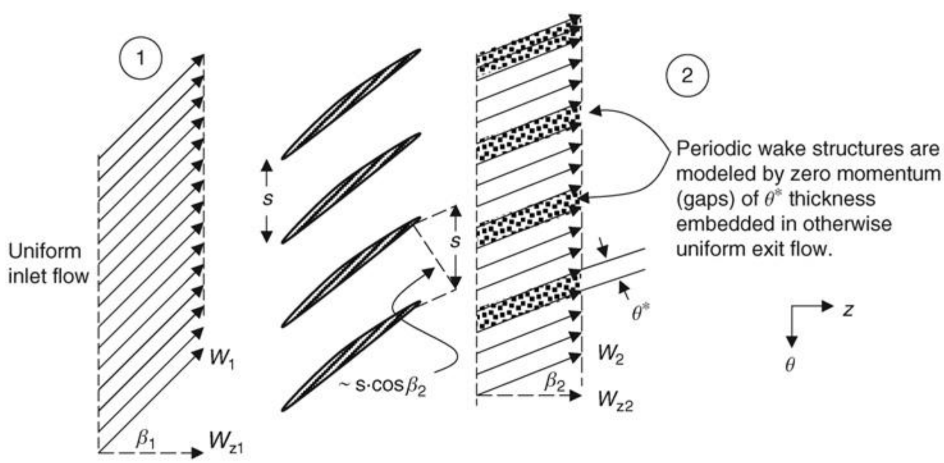
■ **FIGURE 8.25**
Total pressure loss correlation with *D*-factor. Source: Lieblein 1965. Courtesy of NASA

The total pressure downstream of the cascade is the sum of static and dynamic pressures, according to (incompressible) Bernoulli equation. The dynamic pressure is zero in the wake region of θ^* thickness. We may area-average the total pressure in the downstream region according to

$$p_{t2} \cdot (s \cdot \cos\beta_2 - \theta^*) + p_2 \cdot \theta^* = \bar{p}_{t2} \cdot s \cdot \cos\beta_2 \tag{8.63a}$$

$$\bar{p}_{t2} = p_{t1} - \frac{p_{t2} - p_2}{s \cdot \cos\beta_2} \cdot \theta^* = p_{t1} - \frac{\rho \cdot W_2^2}{2s \cdot \cos\beta_2} \cdot \theta^* \tag{8.63b}$$

Note that we have used the concept of constant total pressure in an inviscid fluid by using $p_{t1} = p_{t2}$ outside the wake region. Therefore, in the blade frame of reference, the total pressure remains conserved in the inviscid limit, assuming there are no shocks in



■ **FIGURE 8.26**
Definition sketch for a cascade exit flow model with periodic wakes

the blade passage. In this context, we have applied the Bernoulli equation, which strictly speaking holds for an incompressible fluid. All these approximations are reasonable for our purposes here of demonstrating subsonic cascade data correlation of total pressure loss with D-factor. Now, in terms of the total pressure loss parameter ϖ we get

$$\varpi = \frac{p_{t1} - \bar{p}_{t2}}{\rho_1 W_1^2/2} \approx \frac{\rho\left(\dfrac{W_z^2}{\cos^2\beta_2}\right)\cdot\theta^*}{2s\cdot\cos\beta_2\cdot\rho\left(\dfrac{W_z^2}{\cos^2\beta_1}\right)/2} = \left(\frac{\cos\beta_1}{\cos\beta_2}\right)^2 \frac{\sigma}{\cos\beta_2}\left(\frac{\theta^*}{c}\right) \tag{8.64}$$

Here we made a simple approximation of constant axial flow across the cascade. Finally, we conclude that since θ^*/c correlated with the diffusion factor, then the RHS of

$$\frac{\theta^*}{c} = \varpi\cdot\left(\frac{\cos\beta_2}{\sigma}\right)\left(\frac{\cos\beta_2}{\cos\beta_1}\right)^2 = f(D, t/c, Re_c, M, \sigma, \text{ camber shape}) \tag{8.65}$$

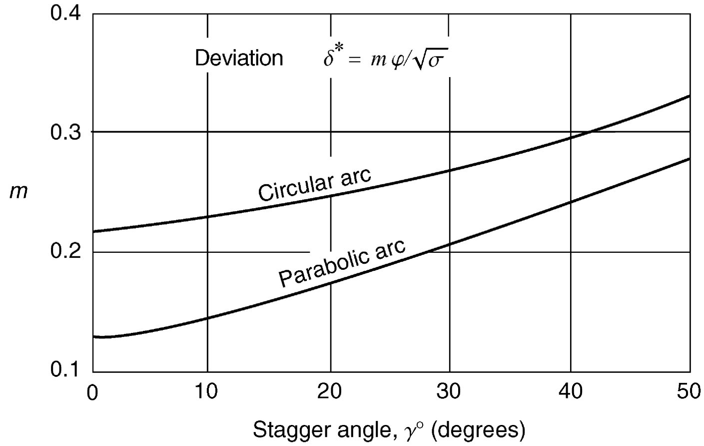
correlates with the D-factor as well, keeping other cascade parameters constant. This functional dependence of the total pressure loss parameter on D-factor is demonstrated in Figure 8.25.

The exit flow angle deviates from the exit blade angle by what is called a deviation angle δ^* due to boundary layer buildup at the trailing edge. Cascade experiments have revealed a correlation between the deviation angle and the geometric parameters of a cascade, namely, the camber angle, the solidity, and the stagger angle. The following correlation is due to Carter (1955).

$$\delta^* = \frac{m\varphi}{\sigma^n} \quad \text{(Carter's rule for deviation angle)} \tag{8.66}$$

where φ is the camber angle, σ is the solidity, m is a function of cascade stagger angle, and n is $1/2$ for a compressor cascade and 1 for an inlet guide vane (or an accelerating passage such as turbines). The higher exponent of solidity for a turbine ($n = 1$) than the compressor ($n = 1/2$) leads to higher exit deviation angles in a compressor, as expected. The boundary layer build up in a compressor in adverse pressure gradient is greater than its counterpart in the turbine. The dependence of m on stagger angle and the profile shape is presented in Figure 8.27, following Carter (1955).

The double-circular arc blade has a circular arc mean camber line shape where the point of maximum camber and thickness occur at mid-chord. The parabolic arc mean camber line blade that is shown in Figure 8.27 has a maximum camber location at 40% chord. The location of the maximum camber and even the thickness affect the pressure distribution about the blade and hence the exit flow deviation angle. It is instructive to know that in turbomachinery blading the chordwise location of the maximum camber moves farther aft with an increasing inlet relative Mach number. For example, a multiple-circular arc blade may be used in the supersonic section of a transonic compressor with the maximum camber placed at 75% chord. We shall discuss multiple-circular arc and other profile shapes of suitable blades for supersonic applications later in this chapter.



■ **FIGURE 8.27** **Compressor cascade deviation angle rule following Carter (1955). Source: Reproduced with permission from Elsevier**

A more general definition of the diffusion factor should involve the radial shift of stream surfaces across compressor blade rows. The constant radius, cylindrical cut approximation of the flowfield is rather restrictive. The method allows for a "pitchline," one-dimensional analysis of a compressor, but falls short of a realistic "mean" flow modeling. A general D-factor is defined as

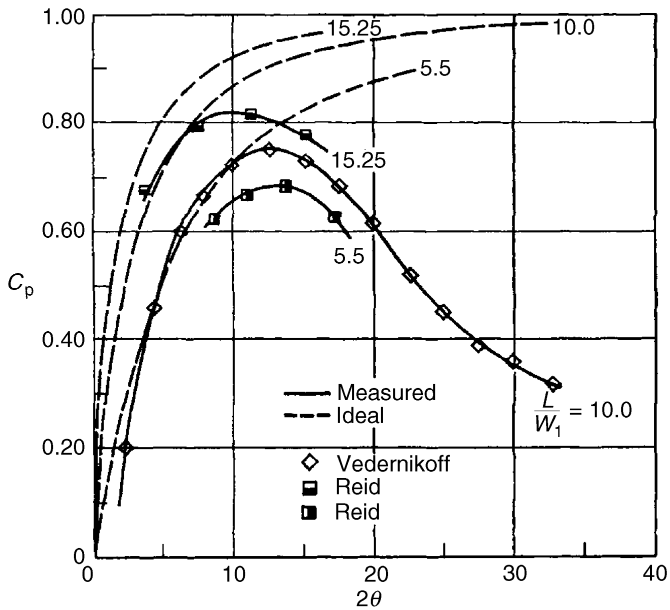
$$D \equiv 1 - \frac{W_2}{W_1} + \frac{|r_2 W_{\theta 2} - r_1 W_{\theta 1}|}{2\sigma r_m \cdot W_1} = 1 - \frac{W_2}{W_1} + \frac{|r_2 W_{\theta 2} - r_1 W_{\theta 1}|}{(r_1 + r_2)\,\sigma \cdot W_1} \tag{8.67}$$

Note that the change in angular momentum has replaced the change in tangential momentum in the definition of a general D-factor. We will address the general diffusion factor in the three-dimensional flow section in this chapter.

Another performance limiting parameter in a compressor in addition to the D-factor is the static pressure rise coefficient, as in a diffuser. The pressure rise coefficient and the D-factor are two different ways of measuring the same thing, namely, the tendency of the boundary layer to stall. However, the D-factor examines the stall behavior of a compressor blade suction surface, whereas the stalling pressure rise coefficient analogy with a diffuser examines the end wall stall that dominates the multistage compressor aerodynamics. The new limiting parameter is defined as

$$C_p \equiv (p_2 - p_1)/\left(\rho_1 W_1^2/2\right) \tag{8.68}$$

The static pressure rise across a blade row, in the numerator of Equation 8.68, is independent of the observer frame of reference. The dynamic pressure at the inlet to the blade row is in the relative frame, that is, as in a cascade parameter. From our studies of boundary layers in adverse pressure gradient we have learned that $C_{p,\max} \approx 0.6$. Interestingly, the numerical values of $C_{p,\max}$ and the limiting D-factor are both ~0.6, a coincidence. The nature of the compressor end wall boundary layer and its stalling characteristics is more complicated than a simple diffuser. Figure 8.28 shows the pressure rise data for a two-dimensional diffuser, where a working range of C_p within ~0.4 to ~0.8 is observed (from Kline *et al.*, 1959).



■ **FIGURE 8.28** **Static pressure rise limitation in a 2D diffuser (data from Kline, Abbott and Fox 1959). Source: Kline, Abbott and Fox 1959. Fig. 4, p. 326. Reproduced with permission from ASME**

The main differences between a diffuser flow and a flow in a compressor blade row are

(a) Compressor boundary layer is highly skewed due to large streamwise and radial pressure gradients, that is, leading to (a highly) three-dimensional flow separation
(b) Inherent unsteadiness in turbomachinery due to relative motion of neighboring blade rows are absent in a diffuser
(c) Compressibility effects through the passage shock interaction with the boundary layer is absent in diffuser performance charts such as Figure 8.28
(d) Upstream wake transport in downstream blade rows is also absent in diffusers
(e) Blade tip clearance flow is unique to turbomachinery
(f) End wall regions create secondary vortex formations as in corner and scraping vortex structures in compressors with no counterpart in a diffuser

In light of these complicating factors, to suggest a single value for the maximum (sometimes called *stalling)* pressure rise coefficient for all compressor blade sections is oversimplistic. For example, the stalling pressure rise coefficient near the end walls may be ~0.48 (first stipulated by de Haller, 1953) and at the pitchline radius ~0.6. A stalling pressure rise correlation for multistage axial-flow compressors is successfully derived by Koch (1981). We will present Koch's correlation in the context of compressor stall margin later in this chapter. A simple application of de Haller pressure rise criterion in the incompressible limit restricts the maximum flow deceleration (W_2/W_1) in a blade row to 0.72. The shock in passage also changes the abruptness of the static pressure rise, hence the boundary layer separation. Despite these shortcomings, the concept of a stalling pressure rise in a compressor blade row is very attractive and useful. We shall examine some limiting features of stalling pressure rise coefficient in a compressor blade. First, let us use Bernoulli equation in recasting the static pressure rise coefficient C_p as

$$C_p = 1 - \frac{W_2^2}{W_1^2} - \frac{\Delta p_t}{(\rho_1 W_1^2/2)} = 1 - \frac{W_2^2}{W_1^2} - \varpi_{\text{cascade}} \tag{8.69}$$

The cascade total pressure loss parameter ϖ_{min} is ~2% for unstalled blades (Figure 8.21). For a maximum pressure rise coefficient of ~0.6, the maximum flow deceleration (expressed in terms of W_2/W_1) is

$$W_2 \geq 0.62\ W_1$$

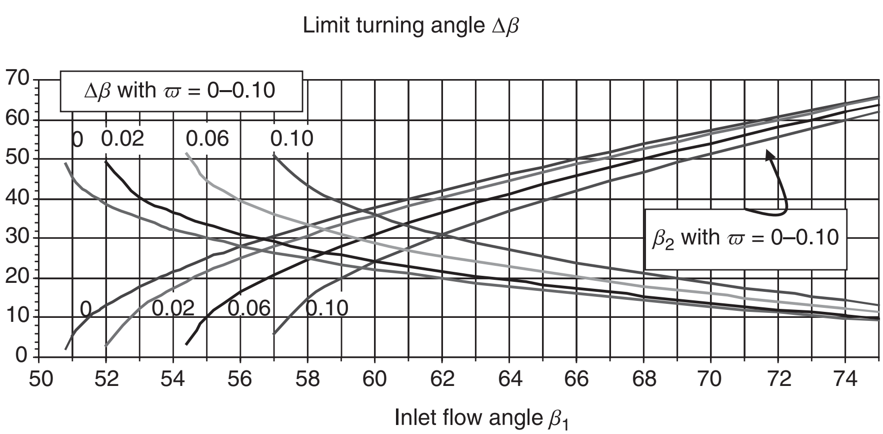
The implication of this (maximum) flow deceleration may be found in the turning angle limitation, namely, for a constant axial velocity W_z, we have $W_1 = W_z/\cos\beta_1$ and $W_2 = W_z/\cos\beta_2$,

$$C_p = 1 - \left(\cos^2\beta_1/\cos^2\beta_2\right) - \varpi_{\text{cascade}} \tag{8.70}$$

We may solve Equation 8.20 for the exit flow angle β_2 for a given cascade total pressure loss parameter, inlet flow angle, and the maximum pressure rise coefficient. The maximum turning angle ($\Delta\beta$) associated with a limiting pressure rise coefficient is graphed in Figure 8.29. The cases of zero total pressure loss, that is, ideal flow, and a cascade with a total pressure loss parameter of 2, 6, and 10% are shown. The effect of total pressure loss increase is seen in Figure 8.29 as a reduction in exit flow angle and thus an increase in flow turning in a compressor blade row. Also, from Equation 8.70, we observe that the maximum static pressure rise coefficient $C_{p,\text{max}}$ for a given inlet flow angle β_1 and total pressure loss occurs at the exit flow angle $\beta_2 = 0$, or axial flow direction at the exit, as expected.

Another aspect of the pressure rise coefficient may be found in its relation to the relative inlet Mach number M_{1r}. From the definition of the static pressure rise coefficient,

■ **FIGURE 8.29**
The exit flow and the limit turning angles (in degrees) based on a $C_{p,\text{max}}$ of 0.6



we may express the dynamic pressure in terms of the inlet static pressure and the inlet relative Mach number, as

$$C_p = \frac{p_2 - p_1}{\gamma \cdot p_1 M_{1r}^2/2} \quad \textbf{(8.71a)}$$

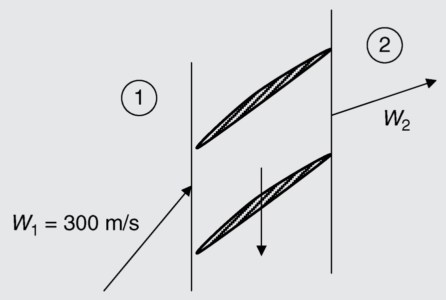
$$\frac{p_2}{p_1} = 1 + C_p \cdot \gamma \cdot M_{1r}^2/2 \quad \textbf{(8.71b)}$$

EXAMPLE 8.4

A rotor blade row at pitchline radius is shown. The rotor total pressure loss coefficient at this radius is $\varpi_{\mathrm{rm}} = 0.03$.

(a) How much deceleration is allowed in the rotor under de Haller criterion? i.e., What is the minimum W_2?

(b) What is the static pressure rise coefficient, assuming incompressible flow and $W_{2\min}$ from de Haller criterion?



SOLUTION

de Haller criterion states that W_2/W_1 should be >0.72. Therefore, the minimum W_2 is 0.72 W_1 or

$$W_{2,\min} = 216 \text{ m/s}$$

The incompressible flow static pressure rise is given by Equation 8.69

$$C_p = 1 - \frac{W_2^2}{W_1^2} - \varpi_{\text{cascade}} = 1 - (0.72)^2 - 0.03 \approx 0.4516$$

In a modern transonic fan or compressor stage, the relative Mach number to the rotor varies from hub-to-tip of ~0.7 to ~1.4, respectively. The variation of feasible static pressure ratio along the blade span with relative Mach number is shown in Figure 8.30.

We observe that the static pressure ratio p_2/p_1 is nearly limited by ~1.15 near the hub and may increase to ~1.6 for a supersonic relative tip Mach number for a $C_{p,\max}$ of ~0.45 near the tip. The static pressure distribution downstream of a turbomachinery blade row strongly depends on the swirl distribution and the axial velocity profile in the spanwise direction. The most common approach in preliminary analysis of three-dimensional flows in turbomachinery is the radial equilibrium theory that we shall discuss in this chapter.

It is useful to correlate the compressor stage efficiency, at a given radius, with the cascade loss coefficients at the corresponding radius. Let us consider the cascade total pressure loss coefficients of ϖ_{r} and ϖ_{s} to represent the relative total pressure loss in the rotor and the stator blade sections, respectively, nondimensionalized with respect to the inlet dynamic pressure.

$$\varpi_{\mathrm{r}} \equiv \frac{p_{\mathrm{t1r}} - \bar{p}_{\mathrm{t2r}}}{\rho_1 W_1^2/2} = \frac{2p_{\mathrm{t1r}}}{\rho_1 W_1^2}\left(1 - \frac{\bar{p}_{\mathrm{t2r}}}{p_{\mathrm{t1r}}}\right) = \frac{2p_1\left(1 + \frac{\gamma-1}{2}M_{\mathrm{1r}}^2\right)^{\frac{\gamma}{\gamma-1}}}{\rho_1 W_1^2}\left(1 - \frac{\bar{p}_{\mathrm{t2r}}}{p_{\mathrm{t1r}}}\right) \quad \textbf{(8.72)}$$

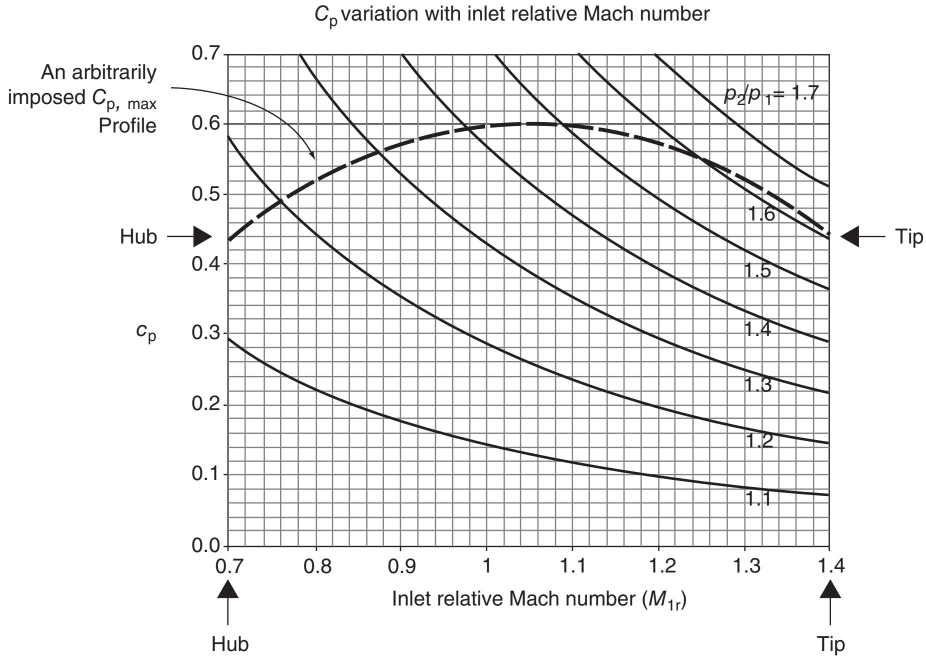


FIGURE 8.30 **Variation of static pressure rise coefficient C_p with inlet relative Mach number (for constant pressure ratio p_2/p_1) with an arbitrarily imposed $C_{p,\max}$ profile**

We may also express the ratio of static pressure to dynamic pressure in terms of the relative Mach number to simplify the above expression as

$$\frac{\bar{p}_{t2r}}{p_{t1r}} = 1 - \varpi_r \left[\frac{\gamma M_{1r}^2/2}{\left(1 + \frac{\gamma - 1}{2} M_{1r}^2\right)^{\frac{\gamma}{\gamma-1}}} \right] \tag{8.73}$$

The stagnation pressure ratio across the stator blade sections may be written in an analogous manner to Equation 8.73, as

$$\frac{\bar{p}_{t3}}{p_{t2}} = 1 - \varpi_s \left[\frac{\gamma M_2^2/2}{\left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\frac{\gamma}{\gamma-1}}} \right] \tag{8.74}$$

The compressor stage adiabatic efficiency η_s is

$$\eta_s = \frac{\pi_s^{\frac{\gamma-1}{\gamma}} - 1}{\tau_s - 1} \tag{8.75}$$

We may use the following chain rule to express the stage total pressure ratio as

$$\pi_s \equiv \frac{p_{t3}}{p_{t1}} = \left(\frac{p_{t3}}{p_{t2}}\right)\left(\frac{p_{t2}}{p_{t2r}}\right)\left(\frac{p_{t2r}}{p_{t1r}}\right)\left(\frac{p_{t1r}}{p_{t1}}\right) \tag{8.76}$$

Equations 8.73 and 8.74 describe two of the parentheses in Equation 8.76, and we may relate the remaining two expressions to flow Mach numbers in the absolute and relative frame, namely

$$\frac{p_{\mathrm{t1r}}}{p_{\mathrm{t1}}} = \left[\frac{1 + \frac{\gamma - 1}{2} M_{1\mathrm{r}}^2}{1 + \frac{\gamma - 1}{2} M_1^2}\right]^{\frac{\gamma}{\gamma-1}} \tag{8.77}$$

$$\frac{p_{\mathrm{t2}}}{p_{\mathrm{t2r}}} = \left[\frac{1 + \frac{\gamma - 1}{2} M_2^2}{1 + \frac{\gamma - 1}{2} M_{2\mathrm{r}}^2}\right]^{\frac{\gamma}{\gamma-1}} \tag{8.78}$$

From the inlet conditions, we can calculate M_1 and $M_{1\mathrm{r}}$ that are needed in Equation 8.77. The absolute and relative Mach numbers in station 2, that is, downstream of the rotor, may be calculated from the velocities C_2 and W_2 from the velocity triangles and the speed of sound a_2. The speed of sound may be related to fluid total enthalpy and kinetic energy according to

$$h_{\mathrm{t}} = a^2/(\gamma - 1) + C^2/2 \tag{8.79}$$

Therefore,

$$a_2^2 = (\gamma - 1)\left[h_{\mathrm{t2}} - \frac{C_2^2}{2}\right] = (\gamma - 1)\left\{\left[h_{\mathrm{t1}} + \omega\left(r_2 C_{\theta 2} - r_1 C_{\theta 1}\right)\right] - \frac{(C_{z2}^2 + C_{\theta 2}^2)}{2}\right\} \tag{8.80}$$

Hence, the absolute Mach number downstream of the rotor, in terms of known quantities, is

$$M_2 = \sqrt{\frac{(C_{z2}^2 + C_{\theta 2}^2)}{(\gamma - 1)\left[h_{\mathrm{t1}} + \omega(r_2 C_{\theta 2} - r_1 C_{\theta 1}) - (C_{z2}^2 + C_{\theta 2}^2)/2\right]}} \tag{8.81}$$

The relative Mach number $M_{2\mathrm{r}}$ is

$$M_{2\mathrm{r}} = \sqrt{\frac{\left[C_{z2}^2 + (C_{\theta 2} - \omega \cdot r_2)^2\right]}{(\gamma - 1)\left[h_{\mathrm{t1}} + \omega(r_2 C_{\theta 2} - r_1 C_{\theta 1}) - \left(C_{z2}^2 + C_{\theta 2}^2\right)/2\right]}} \tag{8.82}$$

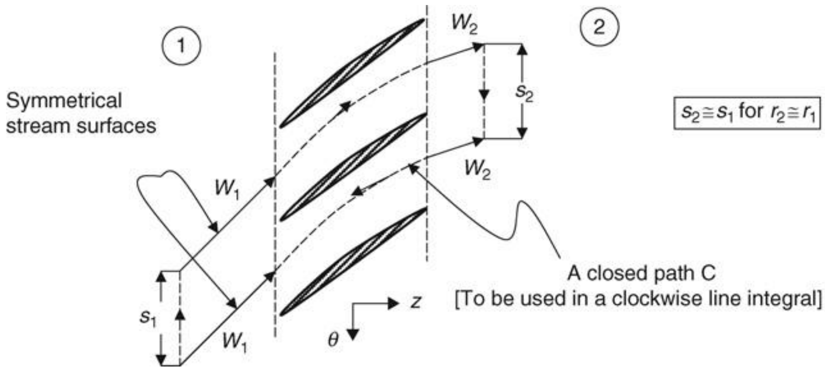
Now, we have established all the parameters that enter the stage adiabatic efficiency Equation 8.75 in terms of the basic velocity triangles and the cascade total pressure loss coefficients. Our analysis is valid for compressible flows in turbomachinery. However, by limiting the analysis to incompressible fluids, we may relate the stage efficiency to the stagnation pressure loss in the rotor and stator blade rows, via

$$\eta_s \approx 1 - \frac{\Delta p_{t-\text{Rotor}} + \Delta p_{t-\text{Stator}}}{\rho\left(h_{t2} - h_{t1}\right)} \tag{8.83}$$

The derivation of Equation 8.83 is straightforward and is shown in Hill and Peterson (1992). The cascade total pressure loss data directly feed into Equation 8.83 for the stage efficiency estimation. In general, we have to supplement the cascade total pressure loss coefficient by shock losses in supersonic flow. In addition, we need to account for the effect of the shock-boundary layer interaction, which increases the blade profile losses. These effects belong to what is known as the *compressibility effects.* A host of other losses in a turbomachinery stage occur that are not usually simulated in a cascade experiment. For example, the end-wall effects, including the tip clearance flow, the streamwise vortices in the blades' wakes due to spanwise variation of blade circulation, as well as flow unsteadiness that result in vortex shedding in the wakes, are not simulated in a typical cascade experiment. The cascade total pressure loss parameter ϖ describes the profile loss of a two-dimensional blade in a steady flow. We will use it as a foundation to construct a more elaborate blade loss model, only.

8.6.4 Aerodynamic Forces on Compressor Blades

The conservation principles that we learned and applied to wings and bodies in external aerodynamics apply to internal aerodynamics as well. There are some complicating factors in a turbomachinery flow that are absent in external aerodynamics. The most dominant feature is the presence of strong swirl in turbomachinery that is either completely absent or weak in external flows. Second, the net turning in the flow is large in turbomachinery and zero in external aerodynamics. We may characterize the flow in a turbomachinery as lacking a unique flow direction, unlike external aerodynamics. As an example, the Kutta–Joukowski theorem on lift predicts a magnitude for an ideal two-dimensional lift (on a body that creates circulation, Γ) as $\rho_\infty V_\infty \Gamma$ and a direction for the lift that is normal to V_∞. In turbomachinery blading, there are two distinct flow speeds, one upstream and the other downstream of the blade, that is, W_1 and W_2 and not just a single V_∞. Also, on the question of the direction of this force, is it normal to W_1 or W_2? In turbomachinery, we will define a "mean" flow angle and a "mean" velocity that help us describe the blade lift and its direction. There are other complicating factors in internal flows that deal with flow distortion (e.g., upstream wake transport and interaction with downstream blades) and unsteadiness (i.e., from neighboring blade rows in relative rotation) that are inherent in a turbomachinery stage and have either no or weak counterparts in external aerodynamics. In a compressor, the flow is continually subjected to an adverse pressure gradient, as the static pressure rises along the flow direction. On the contrary, in external aerodynamics the flow is only *locally* subjected to adverse pressure gradient. Therefore, we may distinguish the two adverse pressure gradients experienced by the internal and external flows as one having a "global" and the second a "local" character, respectively. Another subtle, yet important, difference between external aerodynamics and internal flows with swirl (as in turbomachinery) deals with the stability of such flows to external disturbances. The distinguishing feature of having a mean swirl profile in a compressor leads to centrifugal instability waves that may grow and cause compressor stall. The



■ **FIGURE 8.17**
Suitable control surface surrounding a blade along a stream surface for the application of conservation laws

closest external aerodynamic experience to a centrifugal instability comes from the vortex breakdown on a delta wing at high angle of attack. Kerrebrock (1977) has illuminated the behavior of instabilities in a swirling flow (in an annulus) that provides for a new understanding of disturbances in swirling flows in turbomachinery.

Figure 8.17 is reproduced here to help with the control surface definition and the application of conservation principles to the flow in a compressor blade row.

Continuity demands:

$$\rho_1 W_{z1} s_1 = \rho_2 W_{z2} s_2 \tag{8.84}$$

Making an assumption of negligible radial shift in the stream surface from entrance to exit of the blade, we may conclude that $r_2 \approx r_1$ and therefore $s_2 \approx s_1$, hence,

$$\rho_1 W_{z1} \cong \rho_2 W_{z2} \tag{8.85}$$

The momentum equation in the axial (or z-) direction demands

$$\dot{m}(W_{z2} - W_{z1}) = (p_1 - p_2) \cdot s + F_z|_{\text{fluid}} \tag{8.86}$$

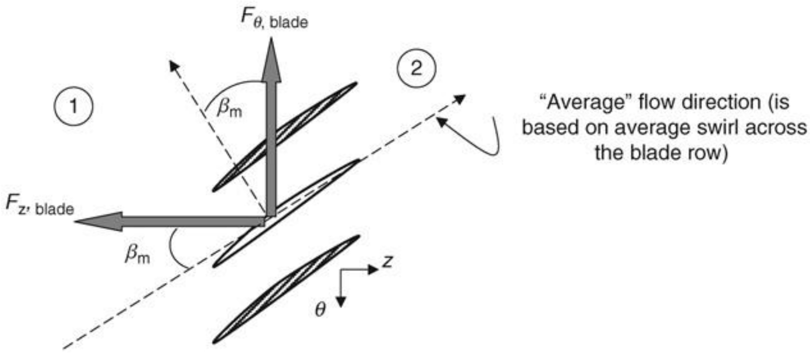
Assuming a constant axial throughflow speed, that is, $W_{z1} = W_{z2}$, we get

$$F_z|_{\text{fluid}} = -F_z|_{\text{blade}} = (p_2 - p_1)s \tag{8.87}$$

Therefore, we conclude that the axial force *acting on the blade* is in negative z-direction, that is, it points in the upstream direction. The conservation of tangential momentum requires

$$\dot{m}(W_{\theta 2} - W_{\theta 1}) = F_\theta|_{\text{fluid}} = -F_\theta|_{\text{blade}} \tag{8.88}$$

Therefore, we conclude that the tangential force on the blade is also in the negative θ-direction. These two blade forces are shown in Figure 8.31.



■ **FIGURE 8.31** **The axial and tangential blade forces acting on a compressor blade**

We may resolve the axial and tangential blade forces into a lift and drag components but first we have to define an average flow direction through the blade row. We define a mean flow direction β_m based on the average swirl in the blade row, namely,

$$\tan \beta_m \equiv W_{\theta m}/W_z \tag{8.89}$$

where the average swirl, $W_{\theta m}$ is defined as

$$W_{\theta m} \equiv \left(W_{\theta 1} + W_{\theta 2}\right)/2 \tag{8.90}$$

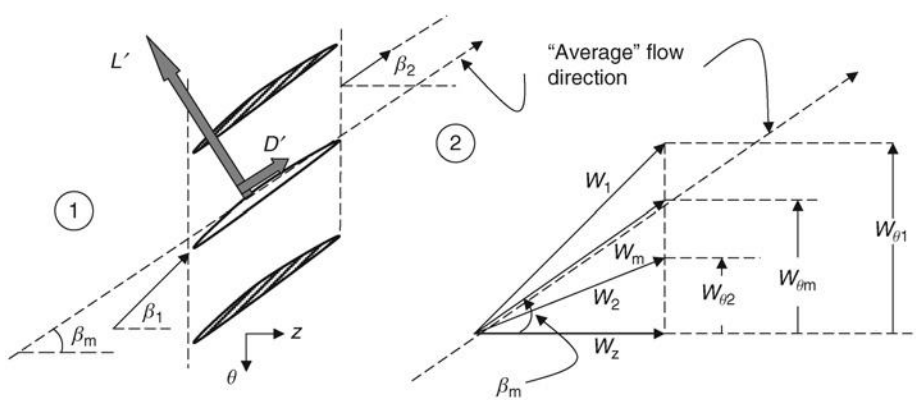
The lift is the sum of the projections of the axial and tangential forces in a direction normal to the "average" flow direction defined in Figure 8.31, that is,

$$L' = F_{\theta,\text{blade}} \cos \beta_m + F_{z,\text{blade}} \sin \beta_m \tag{8.91}$$

Also the drag is the sum of the projections of the two forces in the direction of "average" flow, namely,

$$D' = F_{\theta,\text{blade}} \sin \beta_m - F_{z,\text{blade}} \cos \beta_m \tag{8.92}$$

The lift and drag forces acting on a compressor blade are shown in Figure 8.32.



■ **FIGURE 8.32** **Blade lift and drag forces based on a definition of average flow direction β_m**

Therefore the lift force (per unit span) is

$$L' = \rho \cdot W_z \cdot s(W_{\theta 1} - W_{\theta 2}) \cos \beta_m + (p_2 - p_1) s \cdot \sin \beta_m \tag{8.93}$$

Assuming an incompressible fluid and applying the Bernoulli equation to the static pressure rise term in Equation 8.93, we get

$$L' = \rho \cdot W_z \cdot s(W_{\theta 1} - W_{\theta 2}) \left(\frac{W_z}{W_m}\right) + \left(p_{t2} - \rho \cdot W_2^2/2 - p_{t1} + \rho \cdot W_1^2/2\right) \cdot s \cdot \left(\frac{W_{\theta m}}{W_m}\right) \tag{8.94a}$$

This equation simplifies to

$$L' = \rho \cdot s(W_{\theta 1} - W_{\theta 2}) \left(\frac{W_z^2}{W_m}\right) + \frac{\rho \cdot s}{2} \left(\frac{W_{\theta m}}{W_m}\right) \left(W_z^2 + W_{\theta 1}^2 - W_z^2 - W_{\theta 2}^2\right) - \Delta p_t \cdot s \cdot \left(\frac{W_{\theta m}}{W_m}\right) \tag{8.94b}$$

$$L' = \rho \cdot s(W_{\theta 1} - W_{\theta 2}) \left(\frac{W_z^2}{W_m}\right) + \rho \cdot s \left(\frac{W_{\theta m}}{W_m}\right) \left(\frac{W_{\theta 1} + W_{\theta 2}}{2}\right) (W_{\theta 1} - W_{\theta 2}) - s \cdot \Delta p_t \left(\frac{W_{\theta m}}{W_m}\right) \tag{8.94c}$$

Finally, the blade sectional lift force is expressed as

$$L' = \rho \cdot W_m \cdot s(W_{\theta 1} - W_{\theta 2}) - s \cdot \Delta p_t \left(\frac{W_{\theta m}}{W_m}\right) \tag{8.94d}$$

From Equation 8.58, we may replace the product of blade spacing and the change in swirl velocity by the blade circulation Γ, i.e. $\Gamma \equiv s(W_{\theta 1} - W_{\theta 2})$, to get the familiar Kutta–Joukowski theorem counterpart in turbo-machinery flow, namely,

$$L' = \rho \cdot W_m \cdot \Gamma - s \cdot \Delta p_t \left(W_{\theta m}/W_m\right) \tag{8.95}$$

The first term is the familiar Kutta–Joukowski lift, and the second term is the effect of boundary layer formation and viscous losses on lift.

From Equation 8.92, we may write the blade drag force per unit span as

$$D' = \rho \cdot W_z \cdot s(W_{\theta 1} - W_{\theta 2}) \left(\frac{W_{\theta m}}{W_m}\right) - \rho \cdot s \cdot W_{\theta m}(W_{\theta 1} - W_{\theta 2}) \left(\frac{W_z}{W_m}\right) + s \cdot \Delta p_t \cdot \left(\frac{W_z}{W_m}\right) \tag{8.96a}$$

Upon cancellation of the first two terms, we get a simple expression, entirely based on profile losses, for the blade drag force as

$$D' = s \cdot \Delta p_t \left(W_z/W_m\right) \tag{8.96b}$$

The lift and drag coefficients are

$$C_l \equiv \frac{L'}{\left(\frac{\rho \cdot W_m^2}{2}\right) \cdot c} = \frac{2\Gamma}{W_m \cdot c} - \frac{\varpi}{\sigma} \cdot \sin \beta_m \tag{8.97}$$

where the cascade total pressure loss is nondimensionalized based on the inlet relative dynamic pressure and the σ is the blade solidity.

$$C_d \equiv \frac{D'}{\left(\frac{\rho \cdot W_m^2}{2}\right) \cdot c} = \frac{\varpi}{\sigma} \cdot \cos \beta_m \tag{8.98}$$

As expected, the blade two-dimensional drag coefficient in the limit of low-speed flow is related to the total pressure loss in the cascade, that is, the momentum deficit thickness in the blade wake. Furthermore, the drag coefficient is *linearly* proportional to the cascade total pressure loss measured in the wake, as expected.

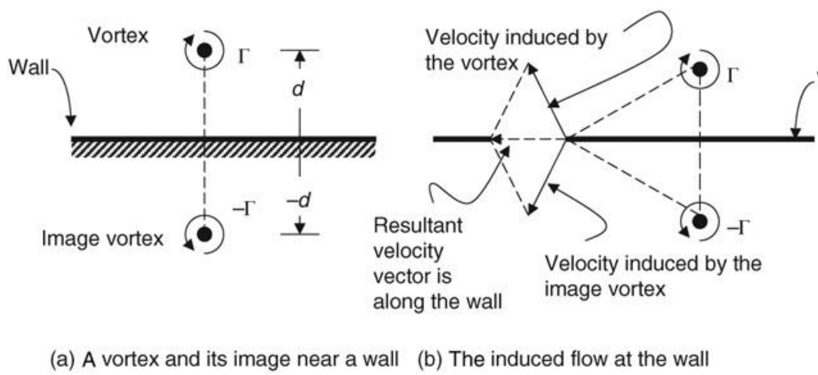
We note that the wake-related loss in lift, in Equation 8.97, is proportional to the blade drag coefficient, Equation 8.98, therefore, the relation between the two-dimensional lift and drag coefficients may be written as

$$C_l = \frac{2\Gamma}{W_m \cdot c} - C_d \cdot \tan \beta_m \tag{8.99}$$

The first term in Equation 8.98 is the ideal lift coefficient, following the thin airfoil theory, and the second term accounts for the loss of lift due to boundary layer/wake formation. These expressions for the lift and drag coefficients, however, lack compressibility effects, three-dimensional or spanwise effects, and the end wall effects as in a real compressor. In addition, we remember that the actual flow is unsteady and involves wake transport and vortex shedding, which are not modeled here. However, we may take additional steps in correcting for some of the cited shortcomings. From Prandtl's classical wing theory, we remember that the drag due to lift, or expressed as the induced drag coefficient C_{Di}, was related to the wing lift coefficient following

$$C_{Di} = \frac{C_L^2}{\pi \cdot e \cdot AR} \tag{8.100}$$

where the span efficiency factor e in the denominator represents the nonelliptic lift contribution/penalty and the term AR represents the wing aspect ratio. A distinguishing feature of the turbomachinery wake is its spiral shape rather than the flat wake of the Prandtl's lifting line theory, where Equation 8.100 is derived. Second, the presence of solid walls acts as mirrors for the vortices (i.e., they create images), and the total induced flow has to account for the image vortices as well. A simple diagram of a vortex next to a wall and its image that renders the wall a stream surface (i.e., flow tangency condition on the wall is satisfied) is shown in Figure 8.33. The vortex strength is shown as Γ and the image vortex strength is, thus ($-\Gamma$) and is equally disposed on the opposite side of the wall. If we calculate the induced velocity at the wall due to these two vortices, the normal component



■ **FIGURE 8.33** **Solid wall near a vortex is modeled by an image vortex, which helps satisfy the flow tangency condition on the wall**

to the wall vanishes, due to symmetry and the tangential component to the wall is doubled, again due to symmetry (Figure 8.33b).

The presence of the *image vortex* with a *counter swirl* is to reduce the effective induced velocity at the blades, as compared with a vortex filament in an unbounded space. Therefore, the simple expression for the drag polar for a wing, which is the sum of the profile drag and the induced drag

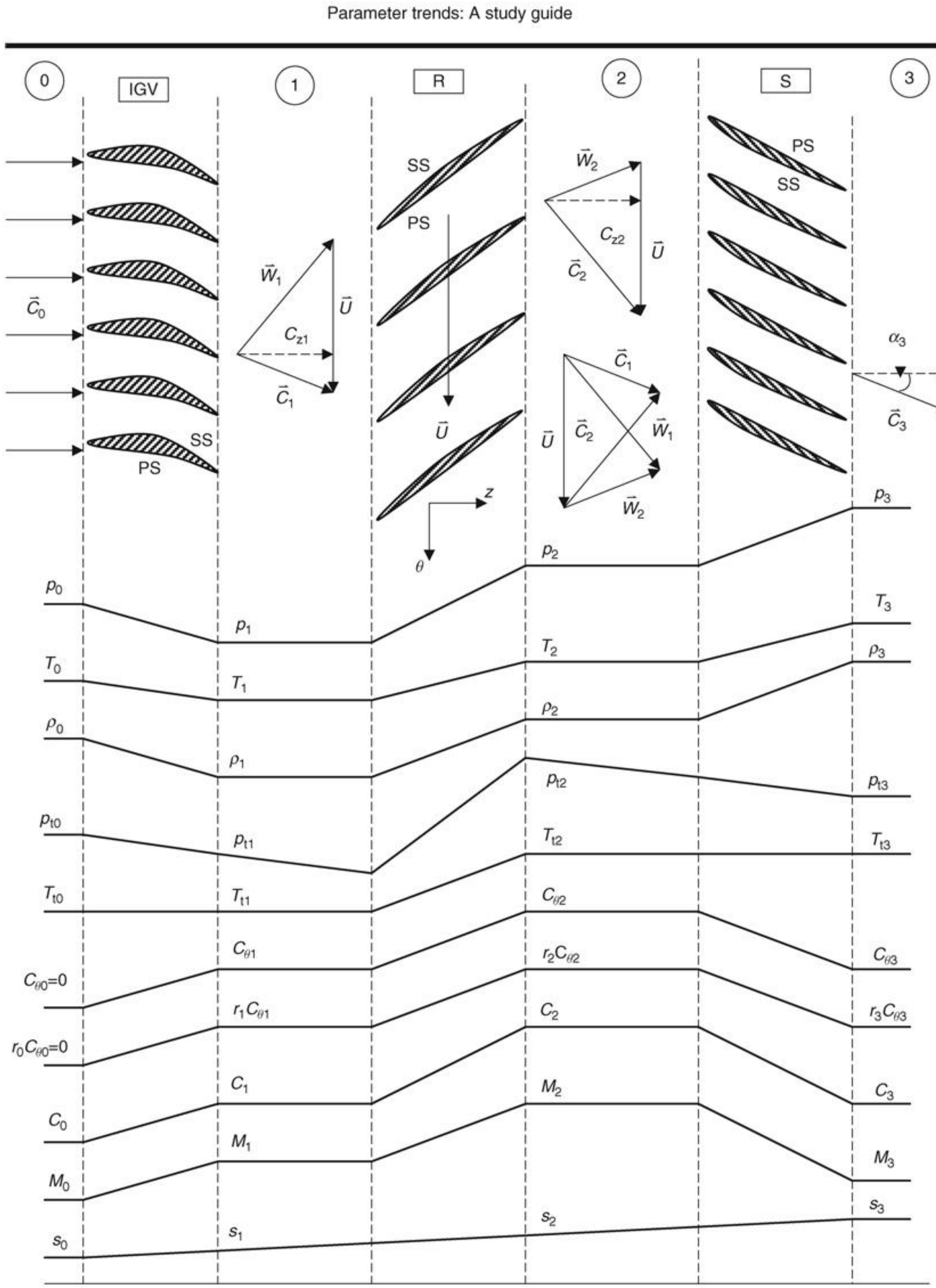
$$C_D = C_{d\cdot 2D} + \frac{C_L^2}{\pi \cdot e \cdot AR} \tag{8.101}$$

should be replaced by a more complex structure in a turbomachinery blading such as

$$C_D = \underbrace{C_{d,\text{cascade}}}_{2D} + \underbrace{\kappa \cdot \left(\frac{C_L^2}{AR}\right) + C_{D,\text{End Wall}} + C_{D,\text{shock}} + C_{D,\text{turbulent-mixing}} + C_{D,\text{rms,Vortex-shedding}}}_{3D} \tag{8.102}$$

In Equation 8.102, κ represents the influence factor on the induced drag due to the presence of solid walls (i.e., the annulus) and helical wakes, the end wall drag coefficient is due to viscous losses and the vortex formation (tip clearance and scraping vortex) near the wall, the drag due to shock is accounted for through $C_{D,\text{shock}}$ Turbulent mixing and the unsteady drag due to vortex shedding (the root-mean square) are reflected in the last term of Equation 8.102. Despite its rather complete look, Equation 8.102 is just an *empirical model*, which requires an extensive experimental database in order to be useful as a predictive tool. However, breaking down the blade loss to its basic constituent levels helps our understanding of the complex flow phenomena in turbo-machinery. We will address three-dimensional losses, compressibility effects, and unsteadiness further in this chapter.

Our discussion of compressor aerodynamics has been limited to 2D flows. The behavior trend of thermodynamic and flow variables in a compressor stage with an IGV in two dimensions are reviewed in Figure 8.34. For additional reading on cascade aerodynamics, Gostelow (1984) should be consulted.



■ **FIGURE 8.34** **Behavior trend of thermodynamic and flow variables (along a streamsurface) in a compressor stage with an IGV**

EXAMPLE 8.5

A compressor stage develops a total pressure ratio of $\pi_s = 1.5$. Its stage adiabatic efficiency is $\eta_s = 0.90$. Calculate the stage total temperature ratio τ_s and compressor polytropic efficiency e_c. Assume $\gamma = 1.4$.

SOLUTION

Stage pressure and temperature ratio are related to stage adiabatic efficiency via

$$\eta_s = \frac{\pi_s^{\frac{\gamma-1}{\gamma}} - 1}{\tau_s - 1}$$

Therefore,

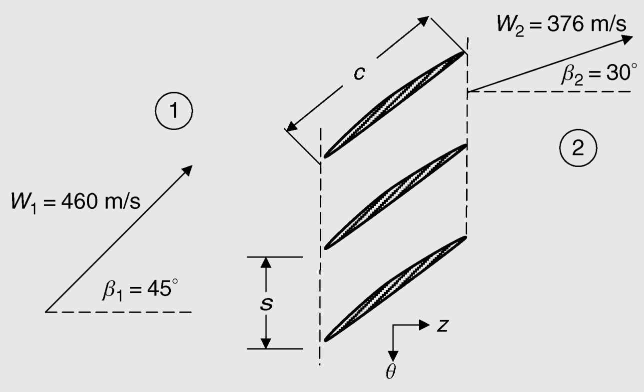
$$\tau_s = 1 + \frac{1}{\eta_s}\left(\pi_s^{\frac{\gamma-1}{\gamma}} - 1\right) = 1 + [1.5^{0.2857} - 1]/0.90 \approx 1.1365$$

Since $\pi_s = \tau_s^{\frac{\gamma \cdot e_c}{\gamma-1}}$, $e_c = \dfrac{\gamma-1}{\gamma}\dfrac{\ln \pi_s}{\ln \tau_s} = 0.2857\dfrac{\ln(1.5)}{\ln(1.1365)} \approx 0.905$

EXAMPLE 8.6

The profile loss parameter for a compressor blade rotor section is $\varpi = 0.05$. The rotor blade chord and spacing at this section are $c = 5.25$ cm and $s = 3.5$ cm, respectively. The relative velocity vectors up- and downstream of the rotor are shown. Calculate

(a) mean relative flow angle β_m
(b) the rotor sectional (2D) drag coefficient C_d and
(c) the rotor circulation Γ
(d) the rotor sectional (2D) lift coefficient



SOLUTION

The inlet swirl is $W_{\theta 1} = W_1 \cdot \sin(45°) = 3253$ m/s
The exit swirl is $W_{\theta 2} = W_2 \cdot \sin(30°) = 188$ m/s
Therefore the average swirl across the rotor is $W_{\theta m} = (W_{\theta 1} + W_{\theta 2})/2 = 256.6$ m/s
Based on the definition of the mean flow angle β_m we have

$$\beta_m \equiv \tan^{-1}\left(\frac{W_{\theta m}}{W_z}\right)$$

The axial velocity upstream of the rotor blade is $W_{z1} = W_1 \cdot \cos(45°) = 325.4$ m/s

The axial velocity downstream of the rotor is $W_{z2} = W_2 \cdot \cos(30°) = 325.6$ m/s

Therefore, $\beta_m \approx \tan^{-1}(256.6/325.5) \approx 38.2°$

The rotor blade solidity at this radius is $\sigma = c/s = 5.25/3.5 = 1.5$

The drag coefficient is related to profile loss, solidity, and mean relative flow angle according to

$$C_d = \frac{\varpi}{\sigma} \cdot \cos\beta_m = \frac{0.05}{1.5}\cos(38.2°) \approx 0.026$$

The magnitude of circulation is

$$\Gamma = s\left|W_{\theta 1} - W_{\theta 2}\right| = (3.5 \times 10^{-2}\text{m})(325.3 - 188)\ \text{m/s} \approx 4.8\text{m}^2/\text{s}$$

To calculate the sectional lift coefficient, we need to calculate $W_m = (W_z^2 + W_{\theta m}^2)^{0.5}$ $W_m \approx 414.5$ m/s

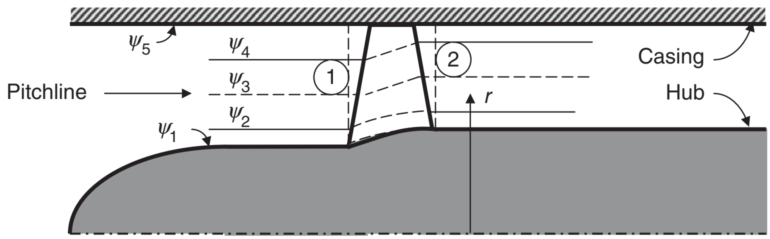
$$C_l = \frac{2\Gamma}{W_m \cdot c} - C_d . \tan\beta_m \approx 0.4416 - 0.0206 \approx 0.4209$$

8.6.5 Three-Dimensional Flow

Our discussion of the flowfield so far has been limited to two-dimensional flows represented by the "pitchline" radius. The next level of complexity brings us to the three-dimensional flow analysis with a host of simplifying assumptions. These assumptions are necessary to render the analysis tenable. To relieve any of the assumptions requires numerical integration of the governing equations, which are coupled and nonlinear. The method of approach, in that case, is the computational fluid dynamics applied to a single or multiple stages in a compressor. Simplifying assumptions for our three-dimensional flow analysis are

1. Flow is steady, that is, $\partial/\partial t \to 0$
2. Flow is axisymmetric, that is, $\partial/\partial\theta \to 0$
3. Flow is adiabatic, that is, $q_{\text{wall}} = 0$
4. Fluid is inviscid and nonheat conducting, that is, $\mu = 0$ and $\kappa = 0$, respectively
5. Blade geometry has zero tip clearance, that is, blades stretch between hub and casing

In addition, we may choose to limit the radial shift of the stream surfaces to within the blade rows and hence require *radial equilibrium* both upstream and downstream of the blade rows. By limiting the radial shift of the stream surfaces to within the blade rows, we are thus creating cylindrical stream surfaces outside the blade rows, that is, $C_r = 0$ both upstream and downstream of the blade row. By assuming an axisymmetric flow, we are smearing out the blade-to-blade flow and thus lose blade periodicity for a finite number of blades. This is akin to having infinitely many blades. For the steady flow, we are placing the



■ **FIGURE 8.35**
Isolated blade row with cylindrical stream surfaces ψ_1 to ψ_5

coordinate system with the blade row that operates in isolation in a cylindrical duct. The implication is that having a neighboring blade row in relative rotation with respect to the blade row of interest would render the flowfield unsteady. In the limit of adiabatic flow, the only mechanism for energy transfer to/from the fluid is the mechanical shaft power. The assumption of inviscid and nonheat conducting fluid eliminates viscous and thermal boundary layer formations. In essence, the only mechanism for energy exchange and momentum transfer is the fluid pressure and not the fluid molecular viscosity and thermal conductivity. The combination of adiabatic and inviscid flow creates a fictitious isentropic environment for the flow interaction with the blades. Figure 8.35 shows cylindrical stream surfaces entering and leaving an isolated blade row in a cylindrical duct.

All parameters in this problem become a function of r only. Note that by taking a cylindrical duct downstream of the blade row, we are in essence decoupling the flow in the vicinity of the blade row from axial pressure gradients. As noted earlier, the axisymmetric flow takes out the θ-dependency; therefore the only independent parameter left is the radial position r. The velocity field in station 2, that is, downstream of the rotor, has zero radial velocity and a general swirl distribution C_θ(r), as well as an axial velocity distribution $C_z(r)$ that needs to be determined from the conservation laws. It is customary to specify a desired swirl distribution downstream of a blade row and calculate the corresponding axial velocity distribution that is supported by the swirl profile. This method of approach is called the vortex design of turbomachinery blades. The pressure distribution $p(r)$ is established by imposing a radial equilibrium condition on the stream surfaces, which simplifies to a balance of pressure gradient and the centrifugal force on a swirling fluid stream, that is,

$$\frac{\partial p}{\partial r} = \frac{\mathrm{d}p}{\mathrm{d}r} = \rho\frac{C_\theta^2}{r} \tag{8.103}$$

From a known swirl distribution $C_\theta(r)$, we get the spanwise blade work distribution following the Euler turbine equation, namely,

$$w_\mathrm{c} = h_{\mathrm{t}2} - h_{\mathrm{t}1} = \omega\left(r_2 C_{\theta 2} - r_1 C_{\theta 1}\right) \tag{8.104}$$

However, the only unknown in the Equation 8.104 is the downstream radius r_2 corresponding to an upstream stream surface radius r_1. This is the crux of the problem of radial shift within the blade row, which needs to be determined. To establish the radial shift of the stream surfaces, Δr, we need to satisfy the continuity equation between the stream surfaces, from hub to tip or in our case from ψ_1 to ψ_5 upstream and downstream of the blade row. For swirl profiles that produce a constant axial velocity downstream of a blade row, the problem becomes very simple. In other cases, we need to integrate a nonuniform axial velocity and density profile in the spanwise direction to calculate the mass flow rates between the stream surfaces and thus establish the radial shift Δr.

8.6.5.1 Blade Vortex Design. The swirl velocity downstream of a blade at a given radius, namely pitchline, may be defined via a two-dimensional approach, as outlined in the previous sections. Anchoring the swirl profile at the pitchline, we may describe a variety of swirl profiles in the spanwise direction that could be analyzed and assessed for their practicality and utility. Some common swirl profiles assume a combination of a potential vortex, a solid body rotation, or, in general, a power profile for the swirl distribution, $C_\theta(r)$. The stagnation enthalpy in the absolute frame of reference may be written as

$$h_{\mathrm{t}} \equiv h + \frac{C_z^2}{2} + \frac{C_\theta^2}{2} \tag{8.105}$$

The radial derivative of the above equation yields,

$$\frac{\mathrm{d}h_{\mathrm{t}}}{\mathrm{d}r} = \frac{\mathrm{d}h}{\mathrm{d}r} + C_z\frac{\mathrm{d}C_{\mathrm{z}}}{\mathrm{d}r} + C_\theta\frac{\mathrm{d}C_\theta}{\mathrm{d}r} \tag{8.106}$$

Also, from Gibbs equation we relate the enthalpy gradient to pressure and entropy gradients as

$$T\frac{\mathrm{d}s}{\mathrm{d}r} = \frac{\mathrm{d}h}{\mathrm{d}r} - \frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r} \tag{8.107}$$

Now, let us substitute Equation 8.107 into Equation 8.106, to get an interim result, that is,

$$\underbrace{\frac{\mathrm{d}h_t}{\mathrm{d}r}}_{\text{blade–work–profile}} = \underbrace{T\frac{\mathrm{d}s}{\mathrm{d}r}}_{\text{blade–loss–profile}} + \underbrace{\frac{1}{\rho}\frac{\mathrm{d}p}{\mathrm{d}r}}_{\text{static–pressure–profile}} + \underbrace{C_z\frac{\mathrm{d}C_z}{\mathrm{d}r}}_{\text{axial–velocity–profile}} + \underbrace{C_\theta\frac{\mathrm{d}C_\theta}{\mathrm{d}r}}_{\text{swirl–profile}} \tag{8.108}$$

A constant-work blade will have the first term in Equation 8.108 vanish, otherwise the work distribution along the blade depends on the angular momentum distribution upstream and downstream of the blade according to Euler turbine equation. The entropy gradient term on the RHS of Equation 8.108 is the blade loss profile in the radial direction. This parameter has to be an input to the problem based on the accumulated loss data published in the literature or the proprietary data of the industry. It is common to ignore the radial loss profiles for the first attempt in establishing the density, velocity gradients downstream of a blade row. The static pressure gradient follows the radial equilibrium theory, namely, Equation 8.103. The axial velocity profile is the third term on the RHS of Equation 8.108, which is the goal of our analysis. The last term is the swirl profile that we consider an assumed function of radial position. Therefore, to establish an axial velocity profile $C_z(r)$, we need to integrate

$$\frac{1}{2}\frac{\mathrm{d}C_z^2}{\mathrm{d}r} = \frac{\mathrm{d}h_{\mathrm{t}}}{\mathrm{d}r} - T\frac{\mathrm{d}s}{\mathrm{d}r} - \frac{C_\theta^2}{r} - C_\theta\frac{\mathrm{d}C_\theta}{\mathrm{d}r} \tag{8.109}$$

The last two terms in Equation 8.109 may be combined into a single derivative involving angular momentum to get

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}(C_z^2) = \frac{\mathrm{d}h_\mathrm{t}}{\mathrm{d}r} - T\frac{\mathrm{d}s}{\mathrm{d}r} - \frac{C_\theta}{r}\frac{\mathrm{d}(rC_\theta)}{\mathrm{d}r} \tag{8.110}$$

The special cases of swirl profile that are considered in compressor blade design are described in the next section.

Case 1: Free-vortex design

$$rC_\theta = \text{constant} \tag{8.111}$$

This is known as the *potential vortex* or *free-vortex* design. The name association "potential vortex" is due to the similarity between this swirl profile and that of a vortex filament ($C_\theta \sim 1/r$). We also remember that the flowfield about a vortex filament is irrotational, hence a *potential* vortex. The irrotational flow around a vortex filament has zero vorticity, hence the name *free-vortex* design.

Let us consider two scenarios here. First, consider the case of zero preswirl upstream of the rotor, that is, the case of no inlet guide vane, and the second scenario with an inlet guide vane that induces a solid-body rotation swirl profile of its own upstream of the rotor.

With no IGV, the flow upstream of the rotor is swirl free, that is,

$$C_{\theta 1} = 0 \tag{8.112}$$

The rotor inducing a free vortex swirl distribution in plane 2, gives

$$(rC_\theta)_2 = \text{constant} \tag{8.113}$$

The Euler turbine equation demands

$$h_{\mathrm{t}2} - h_{\mathrm{t}1} = \omega\left[\left(rC_\theta\right)_2 - \left(rC_\theta\right)_1\right] = \omega\left(rC_\theta\right)_2 = \text{constant} \tag{8.114}$$

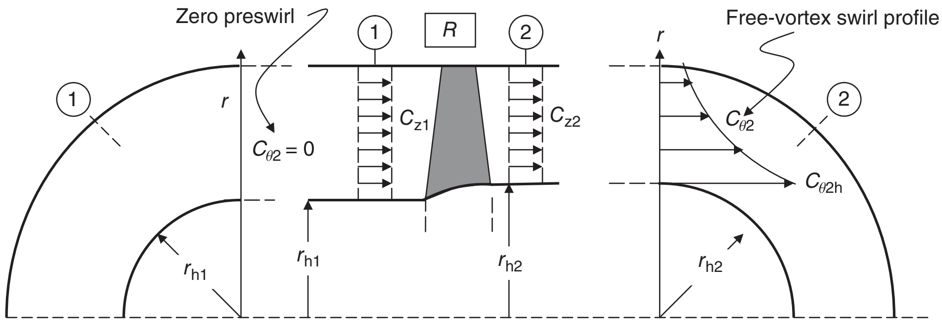
The swirl profile associated with a free-vortex design blade produces a constant work along the blade span. This is known as a *constant-work* rotor. Therefore, the blade work profile and the angular momentum profile terms in Equation 8.110 vanish, that is,

$$\frac{\mathrm{d}h_\mathrm{t}}{\mathrm{d}r} = \frac{\mathrm{d}(rC_\theta)}{\mathrm{d}r} = 0 \tag{8.115}$$

The axial velocity profile equation reduces to

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(C_z^2/2\right) = -T\frac{\mathrm{d}s}{\mathrm{d}r} \tag{8.116}$$

Here, we have a choice and that is to either specify a radial loss profile from the available data in the literature or neglect the loss term in this preliminary phase of establishing an axial velocity profile. Here, we take the second route, although we shall present



■ **FIGURE 8.36** **Velocity field across a rotor blade row with zero preswirl (i.e., no IGV) and free-vortex design for the rotor blade**

and discuss radial distribution of losses associated with modern fan blades later in the chapter.

In the limit of a loss-free rotor and the free-vortex design, we conclude that the axial velocity C_z remains constant, that is, Equation 8.116 yields,

$$C_{z2} = \text{constant} \tag{8.117}$$

Figure 8.36 shows the velocity pattern across the rotor.

In Figure 8.36, we have kept the casing radius constant and have increased the hub radius to account for the density rise across the blade row. There are other choices that we could take, for example, the casing could have been tapered and the hub remained cylindrical or both hub and casing could have been tapered. Since the calculation method is identical, we shall stay with our choice of constant tip radius and an increasing hub radius. It is also customary to design turbomachinery blades based on a constant throughflow speed, that is, axial velocity remains constant. Therefore, based on the continuity equation, the channel area develops inversely proportional to density ratio, assuming a uniform density profile upstream and downstream of the rotor, that is,

$$A_2/A_1 = \rho_1/\rho_2 \tag{8.118a}$$

Therefore, the ratio of r_{h2} to r_t is related to the density ratio and the inlet hub-to-tip radius ratio,

$$\frac{r_{h2}}{r_t} = \sqrt{1 - \frac{\rho_1}{\rho_2}\left[1 - \left(\frac{r_{h1}}{r_t}\right)^2\right]} \tag{8.118b}$$

We tackle the density ratio problem after we examine other swirl profiles of interest. The swirl put in by the rotor needs to be taken out by the stator; therefore, we demand

$$C_{\theta 3} = 0 \tag{8.119}$$

which may be expressed in the shorthand form

$$C_\theta = \pm a/r \tag{8.120}$$

where a is a constant and the plus sign describes the rotor swirl input to the flow and the minus sign designates the stator withdrawal of the swirl put in by the rotor. The exit flow in station 3 downstream of the stator is thus swirl free just as in the flow entering the stage.

The degree of reaction for this design is rewritten from Equation 8.46c, with a free vortex swirl distribution substituted for rotor exit swirl as

$$^{\circ}R = 1 - \frac{C_{\theta 1} + C_{\theta 2}}{2\omega \cdot r} = 1 - \frac{0 + a/r}{2\omega \cdot r} = 1 - \frac{a/\omega}{2r^2} \quad \textbf{(8.121)}$$

To cast this equation relative to the pitchline radius, we divide and multiply the second term in Equation 8.121 by r_{m}^2 to get

$$^{\circ}R(r) = 1 - \frac{\left(a/r_{\mathrm{m}}\right) / \left(\omega \cdot r_{\mathrm{m}}\right)}{2\left(r/r_{\mathrm{m}}\right)^2} \quad \textbf{(8.122)}$$

The term a/r_{m} is the rotor exit swirl velocity at the pitchline radius and the second term, that is, ωr_{m}, is the wheel speed at the pitchline radius. From Equation 8.122, we conclude that the product of these two terms is related to the degree of reaction at the pitchline via

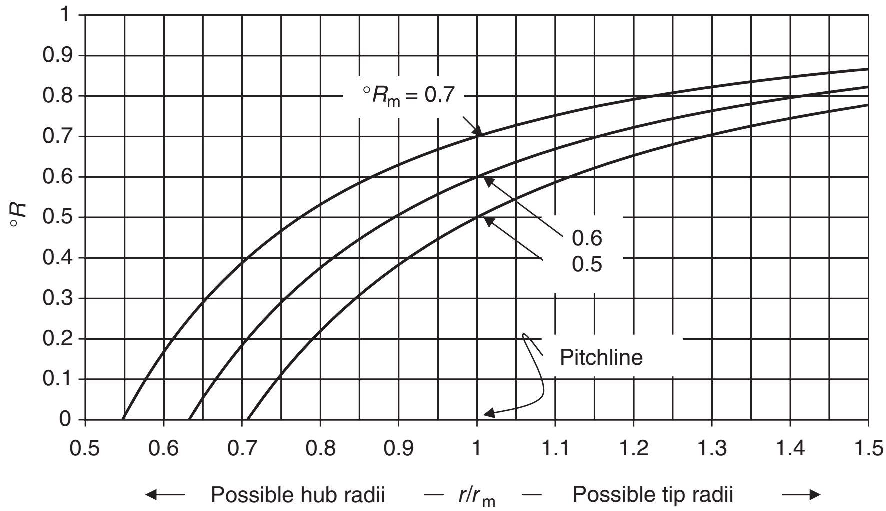
$$\left(\frac{a}{r_{\mathrm{m}}}\right) / \left(\omega \cdot r_{\mathrm{m}}\right) = 2\left(1 - {}^{\circ}R_{\mathrm{m}}\right) \quad \textbf{(8.123)}$$

Therefore, the degree of reaction for a compressor stage with no inlet guide vane and a free vortex swirl distribution produced by its rotor is

$$^{\circ}R(r) = 1 - \frac{1 - {}^{\circ}R_{\mathrm{m}}}{\left(r/r_{\mathrm{m}}\right)^2} \quad \textbf{(8.124)}$$

We have graphed this equation for three stage designs with values of pitchline degree of reaction specified at 0.5, 0.6, or 0.7. We immediately note that the rotor hub is in danger of very low degrees of reaction. Since a negative degree of reaction implies that the flow in that section is not compressing (i.e., that section behaves like a turbine!), we have limited our graphical presentation in Figure 8.37 to positive degree of reaction cases at the hub. For example, in the case of 50% degree of reaction at the pitchline the hub radius should be at ~0.71 r_{m}, which calls for a minimum rotor with hub-to-tip radius ratio of ~0.55. Any lower hub-to-tip radius ratio leads to a negative hub degree of reaction. For a typical value of a modern fan hub-tip radius ratio of ~0.5, we need to raise the pitchline degree or reaction beyond 50%. A second source of concern is the rotor tip region, which is being asked to bear an unusually high share of the stage pressure rise.

The hub section of the rotor is to produce the highest swirl according to Figure 8.36. The tip section is imparting the lowest swirl. Thus, the free vortex swirl profile demands excessive turning of the flow near the hub and very small turning at the tip, which causes the free-vortex blades to be highly twisted. The attractiveness of the free-vortex design is in its simplicity of analysis, which leads to a constant-work rotor and constant axial velocity distribution downstream of blade rows.



■ **FIGURE 8.37** **Spanwise variation of the stage degree of reaction for a free-vortex rotor design and no inlet guide vane**

Now, let us study the case of a compressor stage *with an IGV* that induces a preswirl of solid body rotation upstream of the rotor. The purpose of an inlet guide vane is to reduce the relative Mach number at the tip; hence, a free-vortex design that attains the lowest swirl velocity at the tip would make less sense than a solid body rotation type swirl distribution downstream of the IGV.

Therefore,

$$C_{\theta 1} = b \cdot r \qquad \text{(swirl profile)} \tag{8.125}$$

where b is a constant. Combining this swirl profile with a free-vortex distribution of the rotor, we get

$$C_{\theta 2} = \frac{a}{r} + b \cdot r \tag{8.126}$$

The rotor work distribution is

$$h_{t2} - h_{t1} = \omega \left[(rC_\theta)_2 - (rC_\theta)_1 \right] \tag{8.127a}$$

Substitute swirl profiles 8.125 and 8.126 in the rotor work input (Equation 8.127a) to get

$$h_{t2} - h_{t1} = \omega \left[a + br^2 - br^2 \right] = \omega \cdot a = \text{constant} \tag{8.127b}$$

We conclude that the rotor work distribution along the rotor span is constant.

Now, let us substitute the swirl profile downstream of the rotor and the constant work distribution in Equation 8.110 to get an equation involving the axial velocity distribution,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}(C_z^2) = \frac{\mathrm{d}h_\mathrm{t}}{\mathrm{d}r} - T\frac{\mathrm{d}s}{\mathrm{d}r} - \frac{C_\theta}{r}\frac{d(rC_\theta)}{\mathrm{d}r} = 0 - 0 - \left(\frac{a}{r^2} + b\right)\frac{\mathrm{d}}{\mathrm{d}r}\left(a + br^2\right) \tag{8.128}$$

This equation simplifies to

$$\frac{\mathrm{d}}{\mathrm{d}r}(C_z^2/2) = -\frac{2ab}{r} - 2b^2 r \tag{8.129}$$

We may integrate this equation from a reference radius, for example, the pitchline radius r_m, where the axial velocity is known as $C_{z\mathrm{m}}$, to any radius r to get

$$C_z^2 - C_{z\mathrm{m}}^2 = 2\left[-2ab\ln\frac{r}{r_\mathrm{m}} + b^2 r_\mathrm{m}^2\left(1 - \frac{r^2}{r_\mathrm{m}^2}\right)\right] \tag{8.130}$$

or in nondimensional form the axial velocity profile downstream of the rotor is

$$\frac{C_z(r)}{C_{z\mathrm{m}}} = \sqrt{1 + 2\left(\frac{br_\mathrm{m}}{C_{z\mathrm{m}}}\right)^2\left(1 - \frac{r^2}{r_\mathrm{m}^2}\right) - \left(\frac{4ab}{C_{z\mathrm{m}}^2}\right)\ln\left(\frac{r}{r_\mathrm{m}}\right)} \tag{8.131}$$

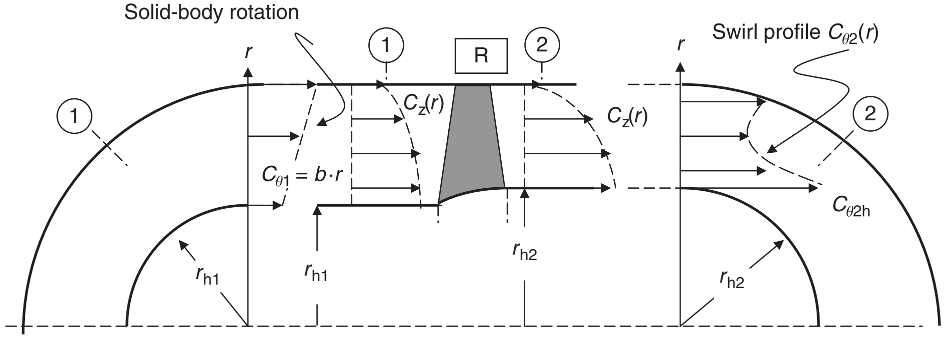
In order to calculate the downstream hub radius after the rotor, we need to integrate the product of density and axial velocity over the inlet and exit planes of the rotor, which are set equal via continuity equation. We need to ask whether the axial velocity downstream of the IGV, which is the rotor upstream, is uniform. We may use the same equation of radial equilibrium theory that we applied to the rotor and note that the IGV does no work on the fluid, hence the fluid total enthalpy remains constant. Consistent with our assumption of no radial loss in the preliminary stage of our blade row calculations, Equation 8.110 simplifies to

$$\frac{\mathrm{d}}{\mathrm{d}r}(C_z^2/2) = -\frac{C_\theta}{r}\frac{\mathrm{d}(rC_\theta)}{\mathrm{d}r} = -b\frac{\mathrm{d}}{\mathrm{d}r}\left(br^2\right) = -\frac{\mathrm{d}}{\mathrm{d}r}\left(b^2r^2\right) \tag{8.132}$$

Upon integration, we get a parabolic axial velocity distribution upstream of the rotor, that is,

$$\frac{C_z(r)}{C_{z\mathrm{m}}} = \sqrt{1 + 2\left(\frac{br_\mathrm{m}}{C_{z\mathrm{m}}}\right)^2\left(1 - \frac{r^2}{r_\mathrm{m}^2}\right)} \quad \text{(axial velocity profile)} \tag{8.133}$$

Let us compare the two axial velocity profiles upstream and downstream of the rotor by examining Equations 8.131 and 8.133. The axial velocity profile upstream of the rotor is parabolic, as noted and the profile downstream of the rotor has an extra logarithmic term, which tends to exacerbate the nonuniformity in the axial profile. This



■ FIGURE 8.38 A schematic drawing of axial and swirl profiles across a compressor rotor with an IGV that induces a solid-body rotation and the rotor is of free-vortex design

implies that the axial velocity at the tip is further reduced downstream of the rotor as compared with the upstream value. In contrast, the axial velocity near the hub shows an increase downstream of the rotor, hence an increased nonuniformity of the axial velocity profile. The swirl and axial velocity profiles across a compressor rotor that induces a free-vortex swirl distribution to an IGV flowfield with a solid-body rotation swirl profile are shown in Figure 8.38. The choice of solid-body rotation for the inlet guide vane is consistent with the desire to reduce the rotor tip Mach number; however, combining it with a free-vortex rotor leads to a highly nonuniform axial velocity profile, which is undesirable. What happens to the spanwise distribution of the degree of reaction in this case?

To answer this question, we recall that the swirl profiles across the rotor are

$$C_{\theta 1}(r) = b \cdot r$$

and

$$C_{\theta 2}(r) = b \cdot r + \frac{a}{r}$$

The equation for the degree of reaction is

$$^\circ R = 1 - \frac{C_{\theta 1} + C_{\theta 2}}{2\omega \cdot r} = 1 - \frac{1}{2}\left[(2b/\omega) + \left(\frac{a/\omega}{r^2}\right)\right] \tag{8.134}$$

We may cast this equation in terms of the pitchline radius, similar to Equation 8.124 as

$$^\circ R = \left(1 - \frac{b}{\omega}\right) - \frac{(a/r_\mathrm{m})/(\omega \cdot r_\mathrm{m})}{2(r/r_\mathrm{m})^2} \tag{8.135}$$

In terms of the degree if reaction at the pitchline, $^\circ R_\mathrm{m}$, we may write Equation 8.135 as

$$^\circ R = (1 - b/\omega) - \frac{(1 - b/\omega) -^\circ R_\mathrm{m}}{(r/r_\mathrm{m})^2} \quad (^\circ R \text{ profile}) \tag{8.136}$$

EXAMPLE 8.7

Compare the degree-of-reaction profile of a compressor stage with and without an IGV for a range of hub-tip radii that result in a positive degree of reaction.

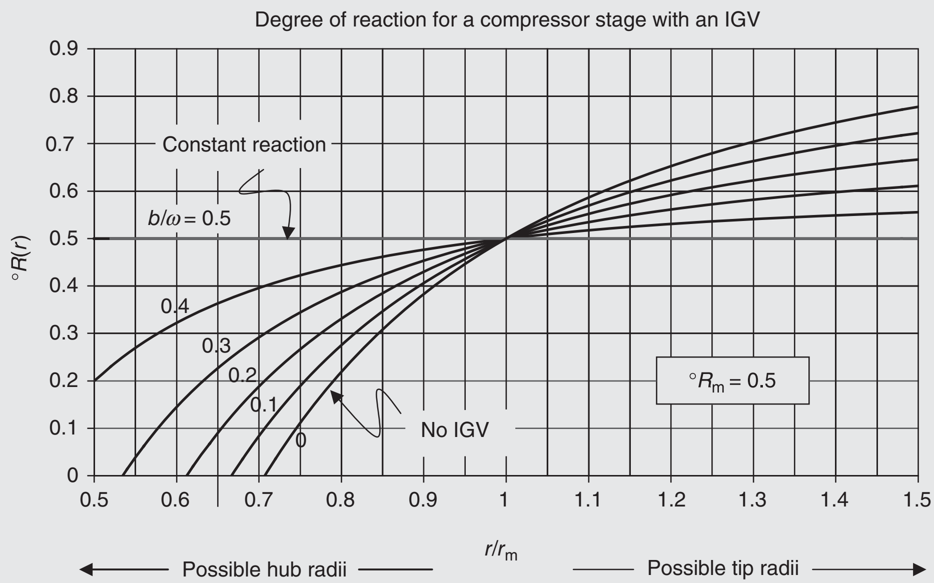
SOLUTION

We make a spreadsheet based on Equations 8.136 and 8.124 for a 50% degree of reaction at the pitchline radius, with varying IGV solid-body rotation swirl profile b/ω. The graph of the spreadsheet table is shown in Figure 8.39. We note that a positive solid-body rotation (i.e., $b > 0$) causes a change in the degree of reaction in the positive direction. The case of $b = 0$ recovers the no IGV stage of Figure 8.37.

We also note that for a special case of

$$1 - \frac{b}{\omega} = {}^\circ R_m \tag{8.137}$$

the reaction along the blade span becomes constant. This view is shown for the case of $b/\omega = 0.5$ in Figure 8.39, which is plotted for a 50% degree of reaction at the pitch-line.



■ **FIGURE 8.39** **IGV with solid-body rotation helps a free-vortex rotor in °*R*(*r*)**

To get the density profile downstream of an IGV, we differentiate the perfect gas law in the radial direction and substitute the centrifugal force for the radial pressure gradient to get

$$\frac{dp}{dr} = R\frac{d}{dr}(\rho \cdot T) = \rho\frac{C_\theta^2}{r} \tag{8.138}$$

This equation may be solved for the density variation as

$$RT\,d\rho + R\rho \cdot dT = \rho\frac{C_\theta^2}{r}dr \tag{8.139}$$

We may isolate $d\rho/\rho$ in terms of the known functions of r, namely

$$\frac{d\rho}{\rho} = \left(\frac{1}{RT}\right)\frac{C_\theta^2}{r}dr - \frac{dT}{T} \tag{8.140}$$

Now, we need to use the conservation of total enthalpy to write an expression for the static temperature distribution downstream of the IGV, that is,

$$T(r) = T_{t0} - \frac{C_z^2 + C_\theta^2}{2c_p} \tag{8.141}$$

The RHS of Equation 8.141 is a known function of the radial coordinate r via Equations 8.126 and 8.133. The differential of the static temperature may be written as

$$c_p dT = -d\left(C_z^2/2\right) - d\left(C_\theta^2/2\right) = 2b^2 r dr - b^2 r dr = b^2 r dr \tag{8.142}$$

The logarithmic derivative of the static temperature is the combination of Equations 8.141 and 8.142 as

$$\frac{dT}{T} = \frac{b^2 r dr}{c_p T_{t0} - (C_z^2 + C_\theta^2)} \tag{8.143}$$

Substituting Equation 8.143 in 8.140, we may calculate the static density profile downstream of the IGV. This process is more accurate, but rather tedious, and is often simplified by making the following assumptions, namely,

$$C_{z1}(r) = C_{zm} = \text{constant} \tag{8.144}$$

$$\rho_1(r) = \rho_m = \text{constant} \tag{8.145}$$

Thus, to establish the hub radius downstream of the rotor, we satisfy the continuity equation via

$$\rho_m C_z \cdot \pi \cdot r_t^2 \left(1 - \left(\frac{r_{h1}}{r_t}\right)^2\right) = \int_{r_{h2}}^{r_t} \rho_2(r) \cdot C_{z2}(r) \cdot 2\pi \cdot r dr \tag{8.146}$$

The density profile downstream of the rotor is a function of temperature and velocity profile using the total enthalpy expression, similar to Equation 8.141. Again, the process is rather tedious and a simplification is in order. We know that the density and temperature are related via a polytropic exponent, namely,

$$\rho \sim T^{\frac{1}{n-1}} \tag{8.147}$$

where n is the polytropic exponent. For an isentropic flow, $n = \gamma$, and for irreversible adiabatic flows, which are encountered in turbomachinery $n < \gamma$. The density distribution is a function of the temperature distribution and the polytropic exponent, that is,

$$\frac{\rho}{\rho_m} = \left(\frac{T}{T_m}\right)^{\frac{1}{n-1}} \tag{8.148}$$

We may use the small stage or polytropic efficiency in a compressor, e_c, to relate pressure and density ratio to temperature ratio across a blade row as

$$\frac{p_2}{p_1} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma \cdot e_c}{\gamma-1}} \quad \textbf{(8.149)}$$

$$\frac{\rho_2}{\rho_1} = \left(\frac{T_2}{T_1}\right)^{\frac{1-\gamma(1-e_c)}{\gamma-1}} \quad \textbf{(8.150)}$$

Therefore the strategy is as follows. We first establish all parameters at the pitchline and then expand the results to other radii by using a blade vortex design choice, for example, free vortex. Therefore, we first establish the swirl at the pitchline using a criterion based on the diffusion factor or the degree of reaction at the pitchline. Then by using the Euler turbine equation, we get the absolute total temperature at the pitchline radius, T_{t2m}. The definition of total temperature at the pitchline and the velocity components leads to the static temperature downstream of the rotor at r_m, that is, T_{2m}. By using a value for compressor polytropic efficiency (say 0.90 or 0.91), we determine the static pressure and the density at r_m, that is, p_{2m} and ρ_{2m}. Now, all parameters are established at the pitchline radius. We are now ready to expand our results to other radii. We choose a blade vortex design type, which fits a swirl profile along the rotor span to the pitchline swirl, $C_{\theta m}$. Therefore, the blade vortex design choice establishes the radial distribution of swirl, that is, $C_\theta(r)$. With the knowledge of swirl downstream of the rotor, we repeat the above procedure at other radii to establish all thermodynamic and flow parameters. It is customary to repeat the procedure for an odd number of stream surfaces (5 or 7), which yield a pitchline stream surface as well. The minimum is thus three, that is, the hub, pitchline, and the tip stream surfaces. We calculate the axial velocity distribution downstream of the rotor, using Equation 8.110. With assumed compressor polytropic efficiency, say 0.91, we calculate the density ratio across the blade row using Equation 8.150. Finally, we conserve the mass flow rate on the two sides of the rotor to establish the annulus area downstream of the rotor. Repeat the same process for the stator. As the process of design is iterative by nature, we need to return to our initial choices of vortex design, D-factor at the pitchline, the degree of reaction at the pitchline, or simply the blade hub-to-tip radius ratio in order to achieve an acceptable preliminary design for the stage. For the choice of blades that produce the desired flow turning at a given radius without flow separation and with an adequate margin of safety, we need to rely on cascade data presented in Figure 8.22.

Case 2: Rotor with forced vortex design (no IGV)
The rotor induced swirl increases linearly with the radius, that is, of solid-body rotation type, hence,

$$C_{\theta 2} = b \cdot r \quad \textbf{(8.151)}$$

The stator removes the swirl put in by the rotor, hence the stator exit plane is swirl free, that is, $C_{\theta 1} = C_{\theta 3} = 0$.

The work distribution along the rotor span is

$$w_c(r) = \omega \left[\left(rC_\theta\right)_2 - \left(rC_\theta\right)_1\right] = \omega \cdot b \cdot r^2 \quad \textbf{(8.152)}$$

Therefore, the rotor loading increases proportional to r^2, that is, the rotor tip is overworked and the hub is not sufficiently loaded. The overloading of the rotor tip could result in flow separation at the tip with excessive flow turning and high diffusion. Usually, the tip region in a modern fan operates in the supersonic range, which limits the amount of flow turning and the appearance of shocks at the blade trailing edge. We shall address the issues of supersonic blade sections in a transonic compressor in a later section of this chapter. The degree of reaction for this vortex profile is

$$^{\circ}R(r) = 1 - \frac{C_{\theta 1} + C_{\theta 2}}{2\omega \cdot r} = 1 - \frac{b}{2\omega} = \text{constant} \tag{8.153}$$

The axial velocity profile may be derived from Equation 8.110 as

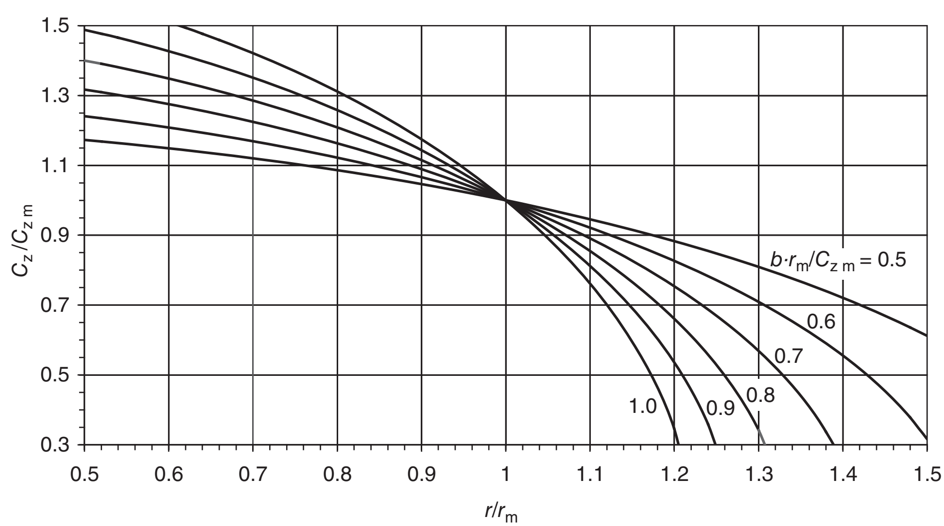
$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}(C_z^2) = \frac{\mathrm{d}h_t}{\mathrm{d}r} - T\frac{\mathrm{d}s}{\mathrm{d}r} - \frac{C_\theta}{r}\frac{\mathrm{d}(rC_\theta)}{\mathrm{d}r} = 2\omega br - 0 - b\frac{\mathrm{d}}{\mathrm{d}r}\left(br^2\right) = 2\omega br - 2b^2 \cdot r \tag{8.154}$$

This equation is integrated with reference to the pitchline to produce

$$\frac{C_z(r)}{C_{z\mathrm{m}}} = \sqrt{1 + 2\left(1 - \frac{\omega}{b}\right)\left(\frac{b \cdot r_\mathrm{m}}{C_{z\mathrm{m}}}\right)^2 \left[1 - \left(\frac{r}{r_\mathrm{m}}\right)^2\right]} \tag{8.155}$$

For $\omega = b$, $^{\circ}R = 0.5$ and the axial velocity remains uniform.

Figure 8.40 shows the spanwise distribution of the axial velocity downstream of the inlet guide vane with a solid-body rotation swirl profile. The spanwise maldistribution of axial velocity is of concern for the same reason as with the excessive tip flow turning, namely, large flow deceleration at the tip could exceed allowable diffusion limits (either through D-factor or C_p) and lead to stall.



■ **FIGURE 8.40**
Axial velocity distribution downstream of an inlet guide vane, which induces a solid-body rotation swirl

Case 3: A general stage vortex design (with IGV)
Let us consider a blend of the free and forced vortex swirl distributions that may be generalized by

$$C_{\theta 1} = C_{\theta 3} = -\frac{a}{r} + br^n \tag{8.156}$$

$$C_{\theta 2} = \frac{a}{r} + br^n \tag{8.157}$$

For $n = 1$, a combination of solid body rotation and a free-vortex swirl distribution are superimposed upstream and downstream of a compressor rotor. The function of stator, which is stated as to remove the swirl put in by the rotor, is seen from Equations 8.156 and 8.157.

The degree of reaction variation along the span may be written as

$$^{\circ}R(r) = 1 - \frac{C_{\theta 1} + C_{\theta 2}}{2\omega \cdot r} = 1 - \frac{br^n}{\omega \cdot r} \quad (^{\circ}R \text{ profile}) \tag{8.158}$$

Obviously, the reaction distribution depends on the exponent n. For $n = 1$, the reaction along the stage height remains constant. This is desirable since the pressure rise along the blade becomes constant. For $n > 1$, the reaction decreases with span and for $n < 1$, the reaction increases with span. Any desired variation, including a constant, may be tailored through our choice of exponent, n of the swirl profile.

The rotor work distribution is

$$w_c(r) = \omega\left[\left(rC_\theta\right)_2 - \left(rC_\theta\right)_1\right] = \omega \cdot \left[(a + br^{n+1}) - (-a + br^{n+1})\right] = 2\omega \cdot a = \text{constant} \tag{8.159}$$

This result is independent of the exponent n in Equation 8.156. Therefore, we create a constant work rotor by the above swirl distribution, which is desirable. The combination of constant work, which means the total enthalpy rise across the rotor is uniform, and the constant reaction (for $n = 1$), which implies a constant static enthalpy rise across the rotor, leads to a constant kinetic energy downstream of the rotor. Since swirl profile is a function of r, then the axial velocity has to attain a profile with the spanwise direction. We made this argument to show that the axial velocity may not be uniform in this case.

The axial velocity profile follows the same approach as the previous cases, namely,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}\left(C_z^2\right) = \frac{\mathrm{d}h_t}{\mathrm{d}r} - T\frac{\mathrm{d}s}{\mathrm{d}r} - \frac{C_\theta}{r}\frac{\mathrm{d}(rC_\theta)}{\mathrm{d}r} = 0 - 0 - \frac{\left(br^n \mp \frac{a}{r}\right)}{r}\frac{\mathrm{d}\left(br^{n+1} \mp a\right)}{\mathrm{d}r} \tag{8.160}$$

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}r}\left(C_z^2\right) = -(n+1)\cdot r^{n-1}\cdot\left(b^2 r^n \mp \frac{ab}{r}\right) \tag{8.161}$$

The case of $n = 1$ is of particular interest since it leads to a constant reaction. The axial velocity profile, for $n = 1$, is

$$C_z(r) = \sqrt{C_{zm}^2 - 2b^2 r_m^2\left(\frac{r^2}{r_m^2} - 1\right) \pm 4ab\ln\left(\frac{r}{r_m}\right)} \tag{8.162}$$

The plus and minus in Equation 8.162 signify the upstream and downstream of the rotor, respectively. In general, the axial velocity decreases with radius with this choice of swirl profile.

8.6.5.2 Three-Dimensional Losses. The factors that render the flow process in a turbomachinery stage irreversible are

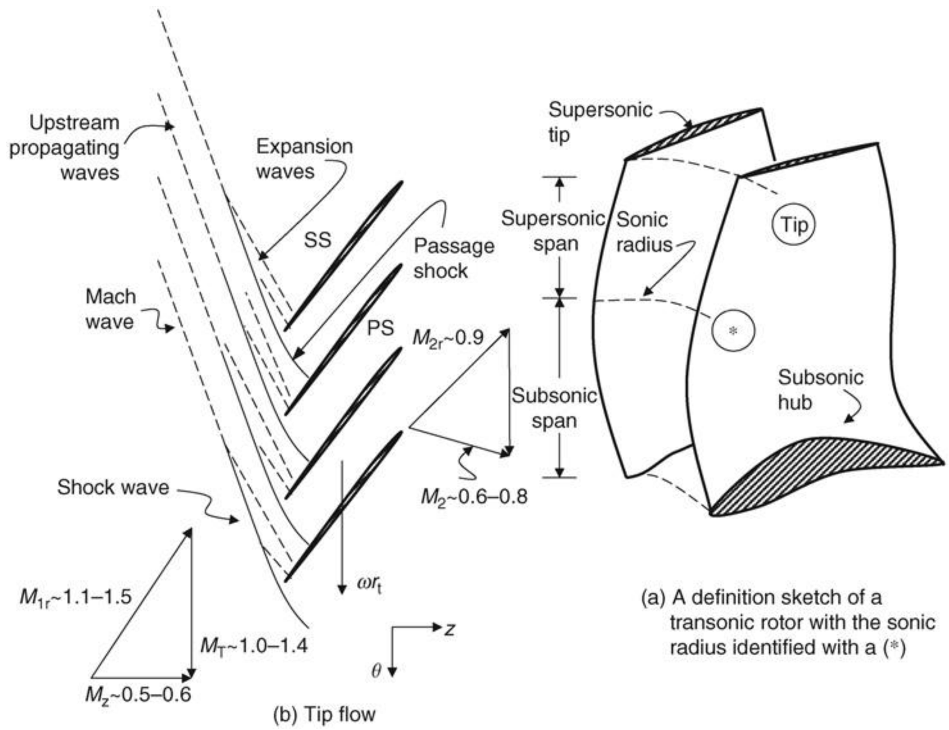
- End wall losses
  - Secondary flow losses
  - Tip clearance loss
  - Labyrinth seal and leakage flow losses
- Shock losses
  - Total pressure loss
  - Shock-boundary layer interaction
- Blade wake losses
  - Viscous profile drag (from cascade experiments)
  - Induced drag losses
  - Radial flow losses
- Unsteady flow losses
  - Upstream wake interaction
  - Vortex shedding in the wake
- Turbulent mixing.

Although the above list seems like a formidable assortment of flow losses, how many of them are totally decoupled from the rest? In a practical sense, the answer has to be none. Is there the possibility of duplication, that is, double bookkeeping, in the above list? The answer is, of course, there is a possibility of counting a loss (or a part of it) twice or three times. Therefore, we conclude that any broken down list of factors contributing to loss is artificial at best and needs to be viewed with caution. The "list" is made only as a tool to help our understanding of complex flow phenomena in broad categories, such as

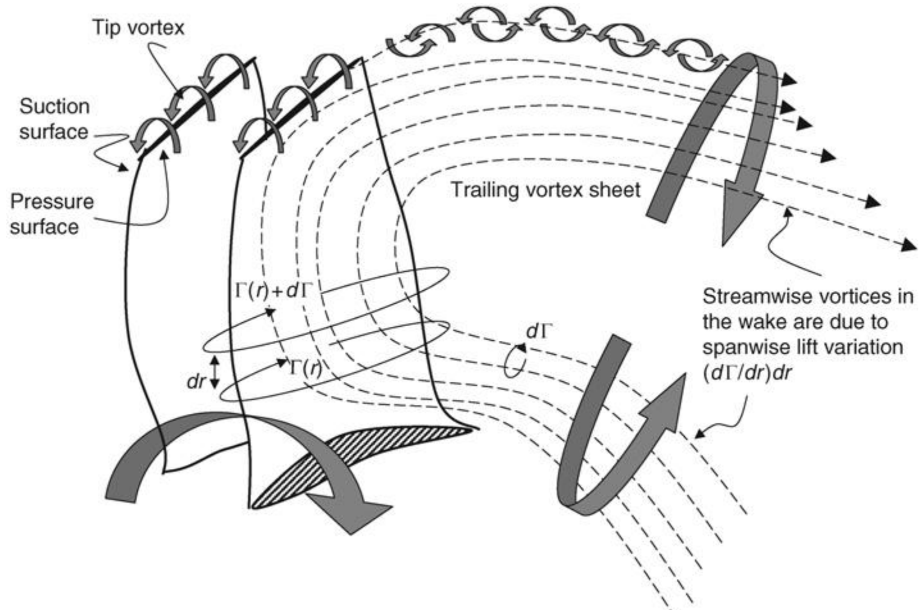
- Compressibility effects
- Viscous and turbulent dissipation
- Unsteadiness
- Three dimensionality.

A transonic rotor creates a shock, which is weakened by suction surface expansion Mach waves and propagates upstream of the rotor row (Kerrebrock, 1981). Note that since the upstream flow is subsonic in the absolute frame of reference, there is no zone of silence preventing the propagation of the rotor created waves. Figure 8.41 shows a typical flowfield.

The broad category of "end wall" losses encompasses the annulus boundary layer, corner vortex, tip clearance flow, and the seal leakage flow. The tip clearance flow is a pressure-driven phenomenon that relieves the fluid on the pressure side towards the suction surface, as shown in Figure 8.42.

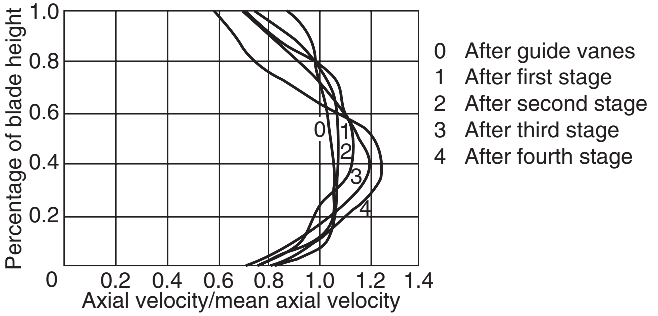


■ FIGURE 8.41 Typical transonic rotor and its tip flowfield/wave pattern (bow shock and expansion waves)

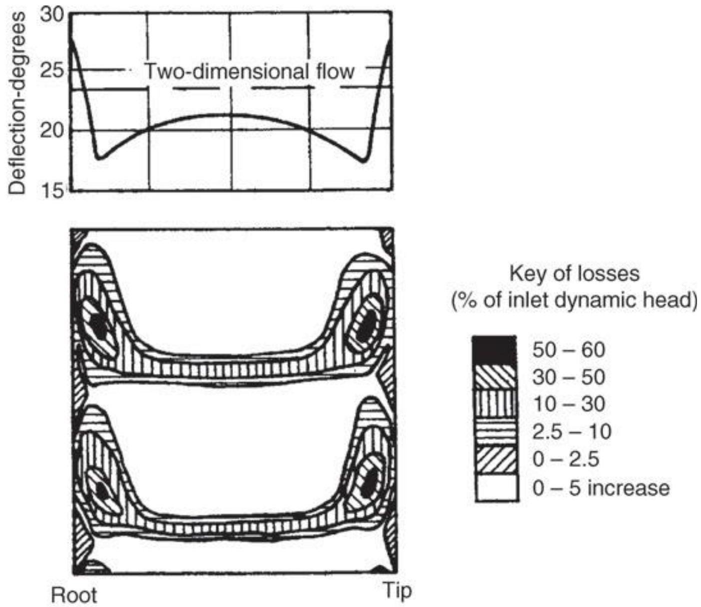


■ FIGURE 8.42 Schematic drawing of tip vortex rollup and the inviscid wake composed of streamwise vortices due to spanwise lift variation on the blade

■ **FIGURE 8.43**
An aspect of end wall flow losses studied by. Source: Howell 1945. Reproduced with permission from SAGE



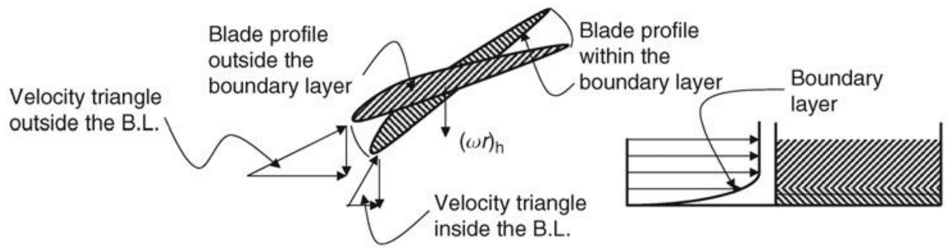
(a) The evolution of axial velocity profile with successive blade rows in a multistage compressor



(b) Total pressure loss distribution with blade height in a cascade

The boundary layer formation on the annulus is expected to grow with distance and in response to an adverse pressure gradient. The growth of the boundary layer with the number of stages is demonstrated by the data of Howell (1945), where the axial velocity distribution continually deforms and by the exit of the second stage there is no resemblance to a "potential core," that is, flat top profile, that the inlet flowfield characterizes. The formation of corner vortices and the associated total pressure loss may be discerned from Howell data, as shown in Figure 8.43. Note that the flow deflection angle near the hub and the tip of the cascade, in the boundary layer, is in excess of data outside the annulus boundary layer as seen in Figure 8.43b. Modern blade design accounts for this difference in the relative flow angle in the boundary layer and *bends* the blade *ends (end-benders)* to minimize incidence loss. A schematic drawing of an end-bender blade is shown in Figure 8.44. Howell (1945) proposes a simple model for end wall losses

■ **FIGURE 8.44** **Schematic drawing of a bent tip blade to account for a larger boundary layer incidence due to a streamwise momentum deficit**



that relates an annulus drag coefficient C_{Da} to the ratio of blade spacing to blade height, that is,

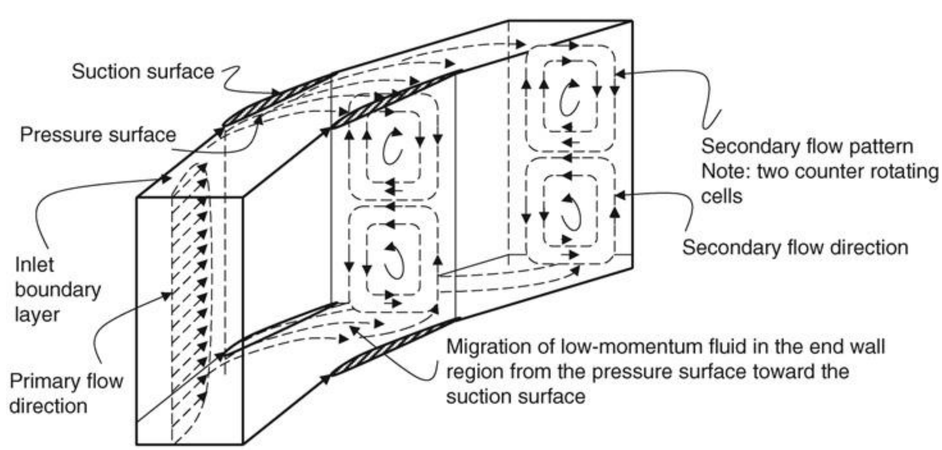
$$C_{Da} = 0.02(s/h) \quad \textbf{(8.163)}$$

A comparison paper by Howell (1945) on fluid dynamics of axial compressors should also be consulted. A secondary flow pattern is developed in a duct with a bend. The flow turning within the blade passages of a compressor then leads to a migration of the boundary layer fluid from the pressure surface toward the low-pressure side, that is, the suction surface. A flow pattern normal to the primary flow direction is then set up, which is called a secondary flow. A schematic drawing of a secondary flow generation and pattern is shown in Figure 8.45. Pioneering work on the formulation of secondary flows is due to Hawthorne (1951) and a simplified engineering formulation is due to Squire and Winter (1951). A simple secondary flow loss model is proposed by Howell (1945) and is expressed in terms of secondary (or induced) drag coefficient C_{Ds},

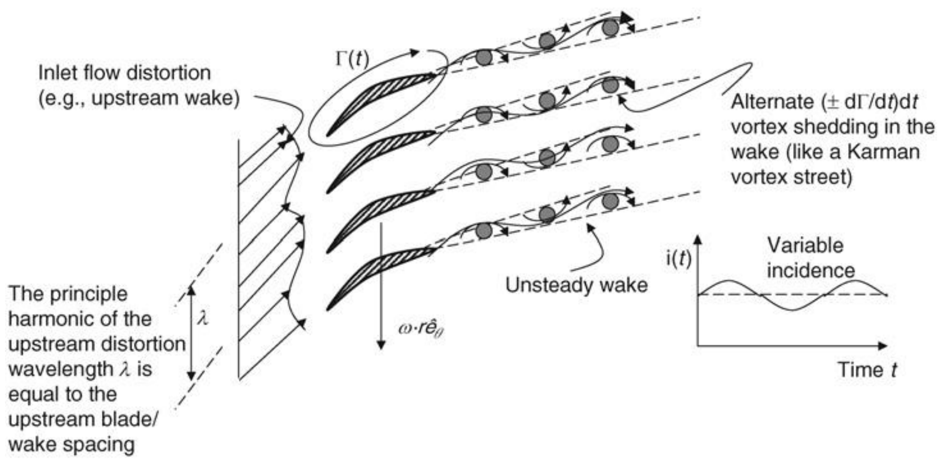
$$C_{Ds} = 0.018C_L^2 \quad \textbf{(8.164)}$$

The unsteadiness is an inherent mechanism for energy transfer between a rotating blade row and the fluid in turbomachinery. The upstream wake interaction with the following blade row is one source of unsteadiness (Kerrebrock and Mikolajczek, 1970). This *wake chopping* interaction leads to unsteady lift, which, following Kelvin's circulation theorem,

■ **FIGURE 8.45** **Viscous flow in a bend creates a pair of counterrotating streamwise vortices, which set up a secondary flow pattern**

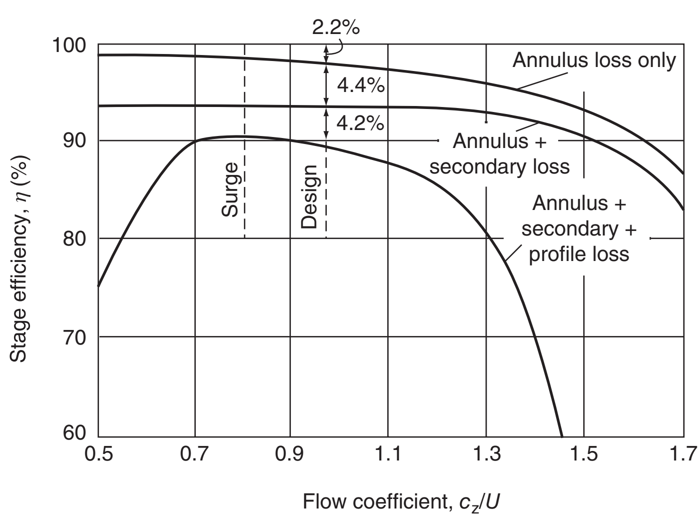


■ FIGURE 8.46
Unsteady flow interaction with a blade row causes periodic vortex shedding in the wake



results in a shed vortex of opposite spin in the wake. Another source of unsteadiness to a blade row is its relative motion with respect to a downstream blade row. Although, the blade is not engaged in a viscous wake chopping activity, as in the previous case, it is operating in an unsteady pressure field created by the downstream blade row. Operating in the unsteady pressure field is then referred to as unsteady *potential* interaction. Blade vibration in bending, twist or combined bending, and torsion is inevitable for elastic cantilevered structures. Therefore, blade vibration induces a spanwise variation of the incidence angle, hence an unsteady lift with a subsequent vortex shedding in the wake. The unsteady vortex shedding phenomenon is schematically shown in Figure 8.46. Kotidis and Epstein (1991) have measured unsteady radial transport in the rotor wake of a transonic fan. They have related unsteady losses to the radial transport in the spanwise vortex cores as well as turbulent mixing. The individual losses and their contributions to the overall loss are depicted in Figure 8.47 (from Howell, 1945). We note that at the design point the

■ FIGURE 8.47
Contributions of different losses in a compressor stage. Source: Howell 1945. Reproduced with permission from SAGE

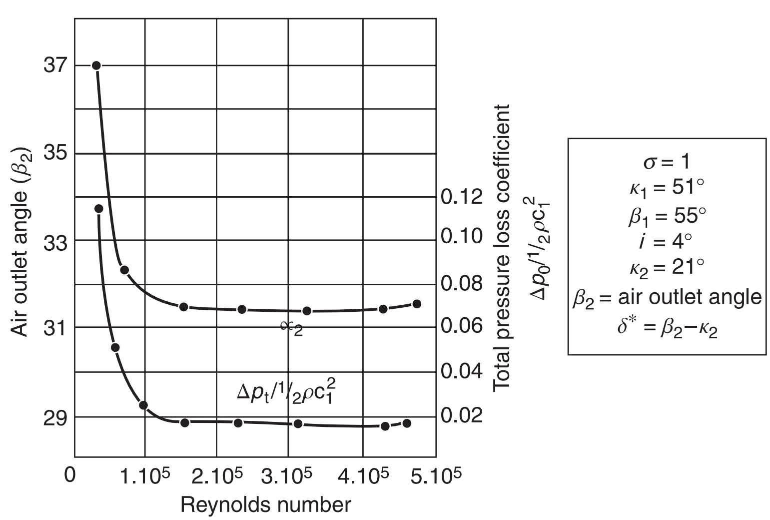


overall losses are near minimum and in low flow, the compressor enters surge and at the high flow rates profile losses dominate due to flow separation.

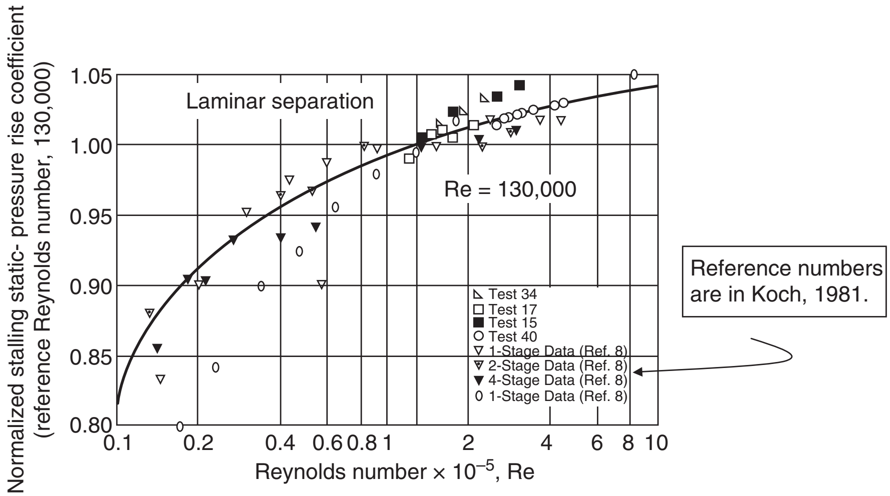
8.6.5.3 Reynolds Number Effect. The flow environment in a compressor, due to adverse pressure gradient, is sensitive to the state of a boundary layer. A laminar boundary layer readily separates in an adverse pressure gradient environment whereas a turbulent boundary layer may sustain a large pressure rise without separation or stall. The blade chord length, c, represents a suitable length scale for the Reynolds number evaluation as well as the relative velocity to the blade, W. Due to nearly flat compressor blade surfaces, we may borrow concepts from the flat plate boundary layer theory to predict the behavior of the boundary layer in adverse pressure gradient on a compressor blade. The effects of unsteadiness inherent in a turbomachinery as well as higher levels of turbulence in the core flow tend to provide for a higher mixing at the boundary layer and thus enhance its ability to withstand pressure rise. A more accurate approach should also include the effect of the blade curvature and its destabilizing (on a concave surface) or stabilizing (on a convex surface, such as the suction surface of a blade) effect on the boundary layer development. A simple statement can be made regarding the state of the boundary layer and its relation to compressor loss and that is to postpone/avoid flow separation, the boundary layer on a compressor blade has to be turbulent. This poses a rough minimum Reynolds number based on the blade chord of ~200,000 that serves as a rule of thumb, that is,

$$\mathrm{Re_c} \equiv \rho \cdot W \cdot c/\mu \geq 200{,}000$$

The Reynolds number trend on the cascade exit flow angle and total pressure loss may be seen from cascade test data of Rhoden (1956) in Figure 8.48. We note that below $\mathrm{Re_c} \sim 100{,}000$ the cascade suffers from a rapid increase in total pressure loss and large exit flow angle deviation, which is due to the phenomenon of laminar separation. This represents a *lower critical Reynolds number.* Koch (1981) examined the stalling pressure rise capability of axial-flow compressor stages, which indicated a much-reduced sensitivity to Reynolds



■ **FIGURE 8.48**
Cascade test results for the effect of Reynolds number on exit flow angle and the total pressure loss. Source: Rhoden 1956. Courtesy of U.K. Government

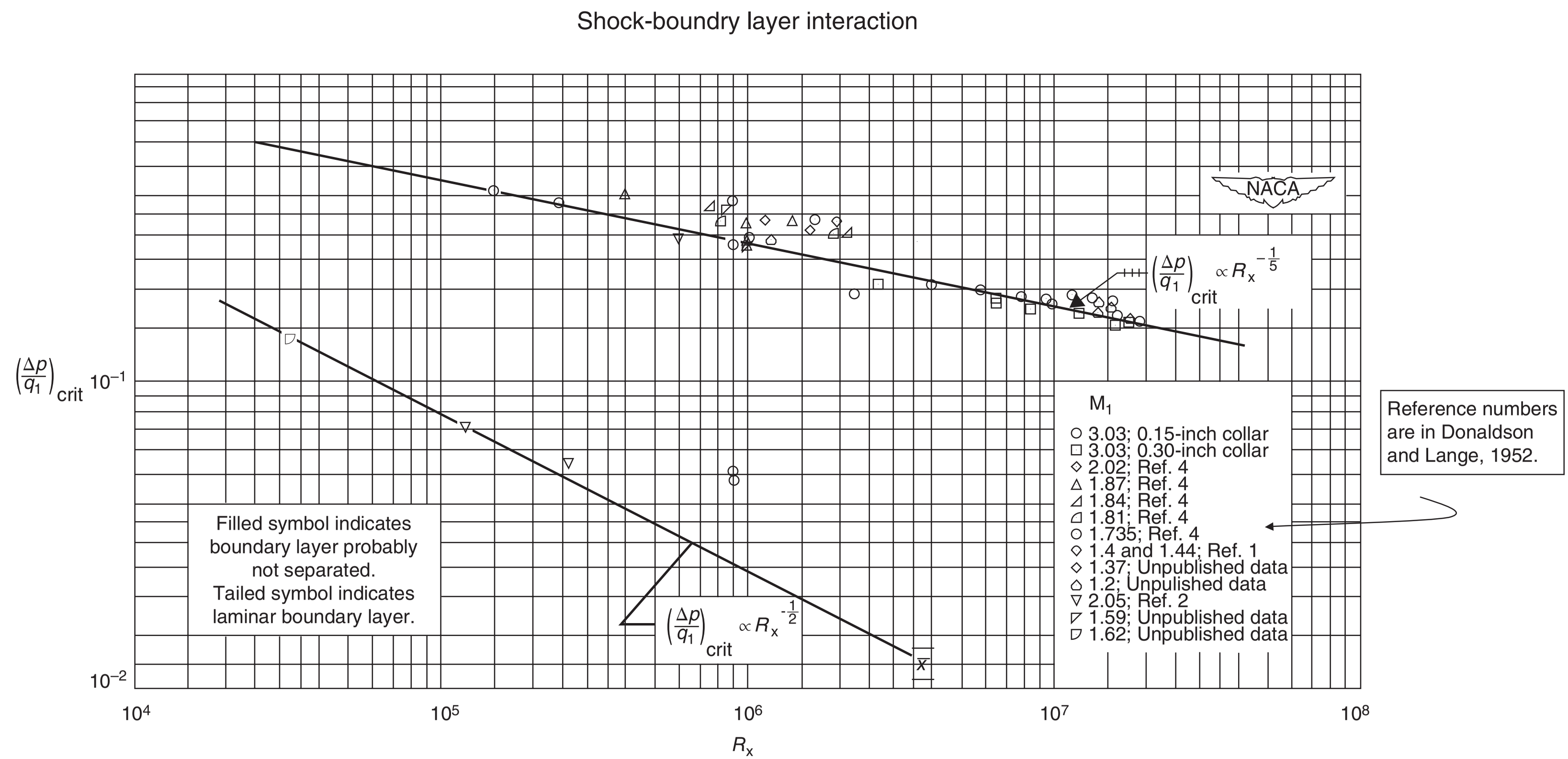


■ FIGURE 8.49 Variation of normalized stalling pressure rise coefficient with Reynolds number in single and multistage compressor tests. Source: Koch 1981. Reproduced with permission from ASME

number below 200,000. The source of reduced sensitivity is found in higher turbulence levels and the unsteadiness inherent in a compressor stage. Koch's data normalized to Re = 130,000 is shown in Figure 8.49.

The boundary layers in transonic compressors readily reach the turbulent regime due to a high relative velocity *W*. For example, the Reynolds number on a fan blade operating at (relative) Mach 1.5 at the tip and 0.75 at the hub with a 1-ft (30.5 cm) chord at standard sea level conditions ranges from $\sim 5 \times 10^6$ to $\sim 10^7$. For $Re_c \sim > 500{,}000$ the compressor efficiency remains nearly constant, which marks the *upper critical Reynolds number*. The blade surfaces behave as hydraulically rough and thus independent of Reynolds number for $Re_c > 500{,}000$. At high altitudes, where the air density is low, and for a slow aircraft (e.g., a long-endurance observation platform), a suitable compressor blade should be designed with a much wider chord to combat the perils of laminar separation.

The effect of Reynolds number is expected to be minimal on secondary flow losses as the secondary flows are predominantly pressure-driven. Conventional, that is, steady, low turbulence, cascade tests do not reproduce compressor test rig results due to higher turbulence levels that promote mixing and the effect of unsteadiness with similar impact on the boundary layer. Finally, for the effect of Reynolds number on shock-boundary layer interaction, we may examine the work of Donaldson and Lange (1952). Figure 8.50 is a log–log graph of critical pressure rise across a shock that leads to boundary layer separation as a function of Reynolds number (from Donaldson and Lange). The two branches that appear on the graph correspond to the familiar laminar and turbulent boundary layers on a flat plate. We note that a turbulent boundary layer sustains nearly an order of magnitude (~10 times) higher stalling pressure rise across the shock than a corresponding laminar boundary layer in a supersonic flow, as pointed out earlier in this section. Also note that the Reynolds number dependence for the stalling pressure rise in a turbulent flow $\sim Re^{-0.2}$ and for the laminar boundary layer is inversely proportional to the square root of Reynolds number, that is, $\sim Re^{-0.5}$, which is identical to the Reynolds number behavior of momentum deficit thickness or friction drag coefficient on flat plates.



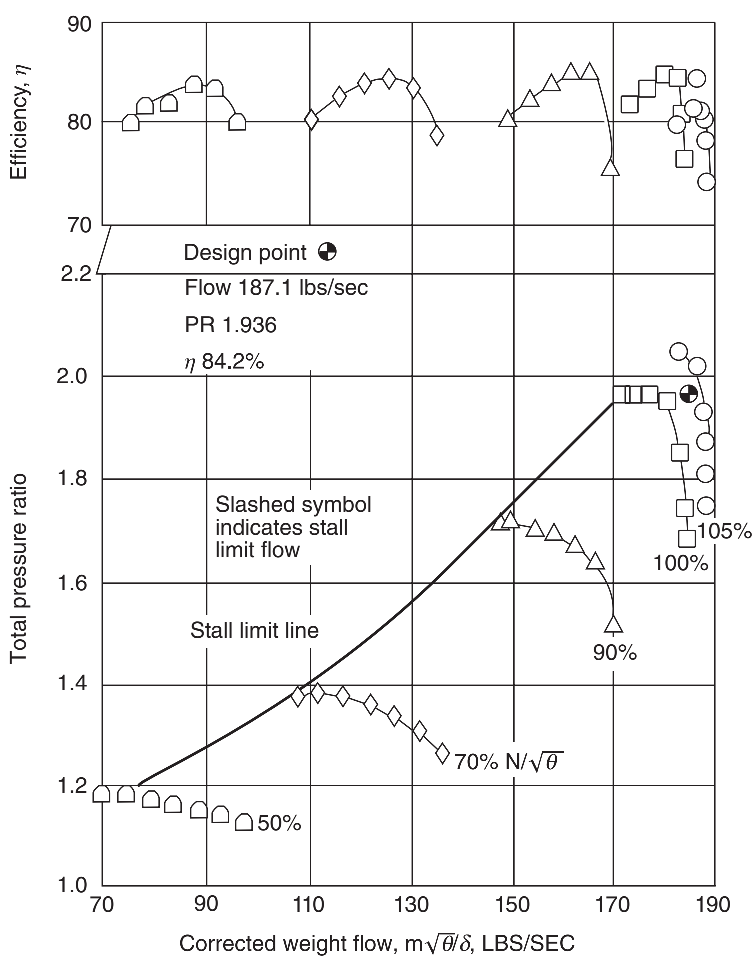
■ **F I G U R E 8 . 5 0 Reynolds number dependence of the critical pressure rise across a shock that leads to boundary layer separation. Source: Donaldson, Du, and Lange 1952. Courtesy of NASA**

8.7 Compressor Performance Map

Compressor pressure ratio plotted against the mass flow rate through the compressor is the compressor performance map. It is customary to graph the constant rpm lines on the performance chart as well as the adiabatic efficiency. The performance map of a single-stage transonic compressor is shown in Figure 8.51 (from Sulam, Keenan, and Flynn, 1970). The relative tip speed of the rotor is 1600 ft/s (488 m/s), and it represents a high-pressure ratio compressor stage ($\pi_s = 1.92$) with an adiabatic efficiency of 84.2% at the design point. The mass flow rate is corrected to the standard reference conditions, namely, the standard sea level pressure and temperature. The corrected mass flow rate is defined as

$$\dot{m}_{c_2} = \frac{\dot{m}_2\sqrt{\theta_2}}{\delta_2} \tag{8.165}$$

■ **FIGURE 8.51**
Performance map of a transonic compressor stage. Source: Sulam, Keenan, and Flynn 1970. Courtesy of NASA



where

$$\theta_2 \equiv \frac{T_{t2}}{T_{ref}} \tag{8.166}$$

$$\delta_2 \equiv \frac{p_{t2}}{p_{ref}} \tag{8.167}$$

The reference pressure and temperature are the standard sea level conditions, namely, p_{ref} = 1.01 bar (or 101.33 kPa) and T_{ref} = 288.2 K.

The corrected mass flow rate in a compressor is a pure function of the axial Mach number operating at the standard pressure and temperature. We can easily demonstrate this by writing the continuity equation of one-dimensional flow in terms of total pressure and temperature, the axial Mach number M_z, and the flow area A, namely,

$$\dot{m} = \sqrt{\frac{\gamma}{R}}\frac{p_t}{\sqrt{T_t}} \cdot A \cdot M_z \left(\frac{1}{1+(\gamma-1)M_z^2/2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{8.168}$$

$$\frac{\dot{m}\sqrt{T_t}}{p_t} = \sqrt{\frac{\gamma}{R}}A \cdot M_z \left(\frac{1}{1+(\gamma-1)M_z^2/2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{8.169}$$

Now, we may multiply both sides by the standard pressure and divide by the square root of the standard temperature to get the "corrected mass flow rate,"

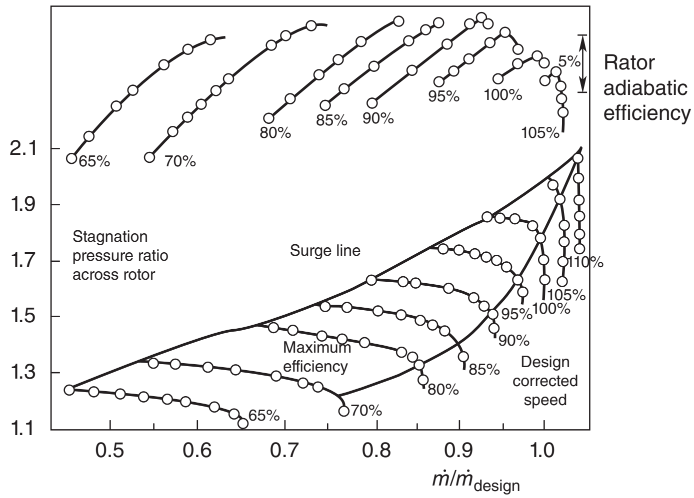
$$\frac{\dot{m}\sqrt{\theta}}{\delta} = \sqrt{\frac{\gamma}{R}}\frac{p_{ref}}{\sqrt{T_{ref}}}A \cdot M_z \left(\frac{1}{1+(\gamma-1)M_z^2/2}\right)^{\frac{\gamma+1}{2(\gamma-1)}} = f(\gamma, R, M_z) \tag{8.170}$$

The performance of a compressor depends on the axial and blade tangential Mach numbers M_z and M_T as discussed earlier. The corrected mass flow rate is a unique function of the axial Mach number and in addition it represents the mass flow rate into the compressor at the standard day pressure and temperature. By defining a corrected mass flow rate, we have basically taken out the effect of nonstandard atmospheric conditions of engine static testing or flight operation. The blade tangential Mach number M_T is proportional to the shaft rotational speed (or angular frequency N) divided by the local speed of sound or the square root of static temperature, namely,

$$M_T \propto \frac{N}{\sqrt{T}} \propto \frac{N}{\sqrt{\theta}} \tag{8.171}$$

The RHS of Equation 8.171 is called the *corrected shaft speed*,

$$N_c \equiv \frac{N}{\sqrt{\theta}}$$



■ **FIGURE 8.52**
Performance map of a modern fan rotor. Source: Cumpsty 1997. Reproduced with permission from Cambridge University Press

The corrected shaft speed is a unique function of the blade tangential Mach number, which along with the axial Mach number determines the performance of a compressor or fan, that is,

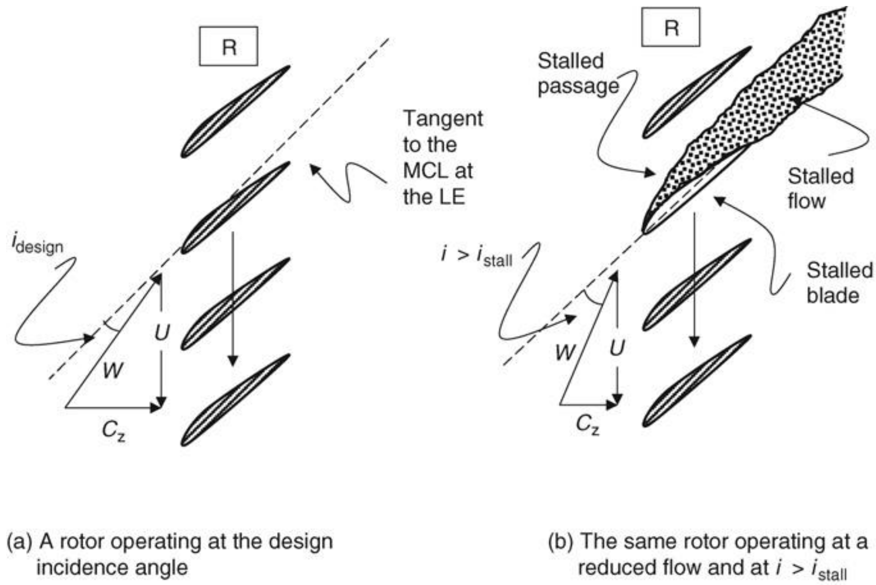
$$\pi_c = \pi_c(\gamma, R, M_z, M_T) = \pi_c(\gamma, R, \dot{m}_c, N_c) \tag{8.172}$$

$$\eta_c = \eta_c(\gamma, R, \dot{m}_c, N_c) \tag{8.173}$$

It is also customary to nondimensionalize the corrected mass flow rate by the "design mass flow rate," and the corrected shaft speed is nondimensionalized by the "design shaft speed" in the compressor performance map. An example of this is shown in Figure 8.52 (from Cumpsty, 1997).

8.8 Compressor Instability – Stall and Surge

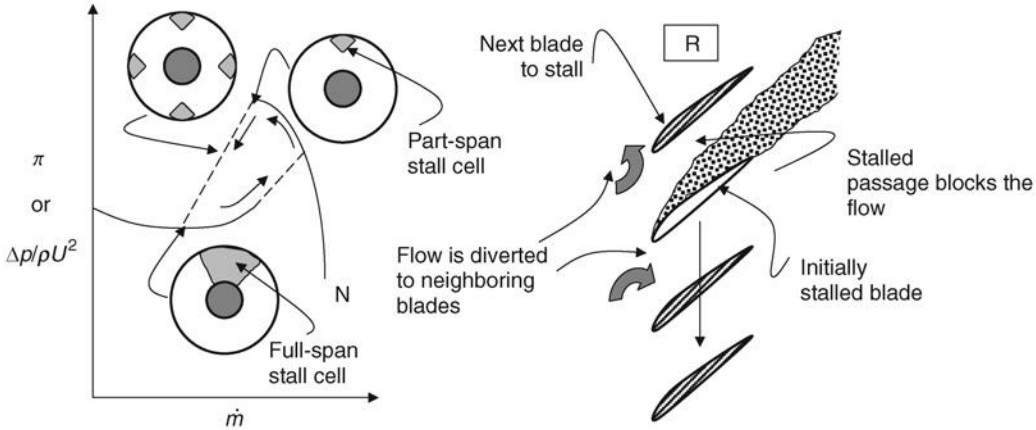
The phenomenon of stall in a compressor has its roots in the aerodynamics of lifting surfaces in a high angle of attack. Therefore, from aerodynamics we understand stall as flow separation from the suction or pressure surfaces of an airfoil with large positive and negative angles of attack, respectively. In a compressor operating at a constant rotational (shaft) speed when the mass flow rate drops, the axial velocity decreases, hence the incidence angle increases. We remember from cascade data a parameter that was called "optimum incidence" angle, at which the blade losses were at a minimum. Large deviation from this minimum-loss incidence angle causes a rapid rise in total pressure loss, which signifies boundary layer separation. The positive and negative stall boundaries were then defined for a cascade of a given blade profile shape, solidity, stagger angle, and Mach number. Therefore, when the flow rate in a compressor drops while operating at constant shaft speed, the danger of stall lurks in the background. A stalled compressor flow is unsteady and, hence, offers a means of driving the naturally occurring blade vibrations into resonance. This is the mechanism for energy flow from the fluid into the blade vibration



■ **FIGURE 8.53**
A rotor blade operating at its best incidence angle and at a reduced mass flow leading to a stalling incidence angle

that causes *flutter*. Flutter is the *self-excited* aeroelastic instability of an aerodynamic lifting surface. In contrast, *buffet* (or *buffeting*) is a *forced vibration* of an aerodynamic surface that is in the turbulent wake of an upstream wing/fuselage, for example, horizontal tail buffet. Figure 8.53 shows schematic drawing of a compressor rotor with two velocity triangles, one representing its design point operation and the second corresponding to a stalled flow.

The stalled flow in a compressor rotor may be initiated with a single blade at a certain radius. The stalled blade passage acts as a "blocker" and diverts the flow to the neighboring (as yet unstalled) blades. Following the drawing in Figure 8.53b, we note that the flow diversion from the stalled passage to the one above the stalled blade causes an increase in that blades incidence angle, pushing it toward stall. Similarly, the blade below the stalled passage gets a diverted flow, which causes a reduction of the flow incidence angle to the blade below. Therefore, it moves away from stall. We note that the stalled flow in a passage is moving in the opposite direction to the blade rotation, hence it is given the name *rotating stall*. The angular speed of rotating stall propagation is $\sim 1/2$ of rotor angular speed, that is, $\omega/2$, and in the opposite direction to the rotor rotation. Hence, in the laboratory (i.e., absolute) frame of reference, a rotating stall *patch* spins with the rotor (i.e., the same direction as the rotor) but at half the speed. To get a feel for the frequency of the rotating stall cells, for example, in a large turbofan engine with a shaft speed of ~50 Hz, a single stall cell has half the frequency or 25 Hz. Now for a two-to-four stall cells circumferentially arranged around the compressor rotor, the stall cell frequency becomes 50–100 Hz. The rotating stall starts with a single cell and develops into a number of cells with a partial span extent (known as the part-span stall) and may grow into a full-span stalled flow with subsequent reduction in the mass flow rate. This behavior is shown schematically in Figure 8.54. The first appearance of stall is limited to a single cell with subsequent reduction of the flow the average pressure rise drops and the cells multiply into a periodic arrangement, as shown in Figure 8.54. The recovery from



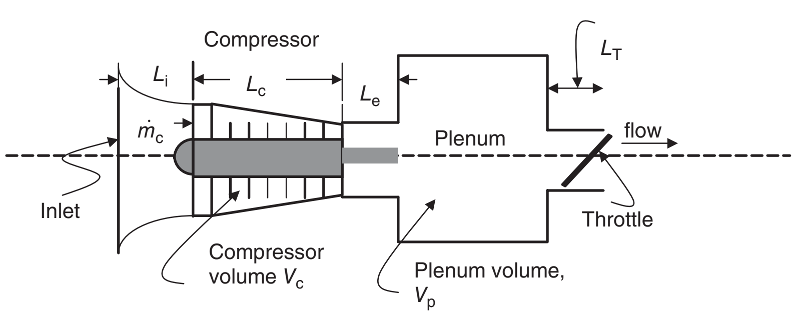
■ **FIGURE 8.54 Schematic drawing of the mechanism of rotating stall, part and full-span stall cells and a hysteresis loop in stall recovery**

the stalled operation in a compressor exhibits a hysteresis behavior, which accompanies systems governed by *nonlinear dynamics.*

The coupling between the compressor stall instability and the combustion chamber resonant characteristics could lead to an overall breakdown of the flow, or flow oscillation, in the compressor and combustor. The overall flow breakdown in the "system" composed of compressor and combustor is called *surge.* In a fully developed stage, surge is an axisymmetric oscillation of the flow with a characteristic timescale dictated by the plenum chamber time to empty and fill. In the initial transient stage, surge is asymmetric and thus creates transverse loads on the blades, which may rub on the casing and lead to structural damage and system breakdown. Early contributions to understanding compressor stall and surge and their distinction were made in the United States at NACA (e.g., Bullock and Finger, 1951) and by Emmons, Pearson, and Grant at Harvard (1955).

A unifying approach to treat the compression system instability was first presented by Greitzer (1976). Greitzer developed a successful one-dimensional theory for the onset of surge in a compressor coupled to a plenum chamber cavity, which may serve as a model of the combustor in a gas turbine engine. The elements of the compression system model are shown in Figure 8.55 to be composed of an inlet duct, a compressor, an exit duct connecting to a plenum chamber followed by a throttle that sets the backpressure, and the mass flow rate through the compressor.

■ **FIGURE 8.55 Compression system model that is used to study *local* (rotating stall) and *global* (surge) system instabilities**



Greitzer proposes the ratio of two characteristic timescales as the parameter that governs the dynamics of this system. One characteristic timescale is the compressor throughflow, which may be written as

$$\tau_{\text{throughflow}} \sim \frac{\rho \cdot V_c}{\dot{m}_c} \tag{8.174}$$

The second characteristic timescale is that of the plenum chamber, which is the time to charge the plenum chamber to a critical pressure rise condition for a stable compressor operation Δp_c,

$$\tau_{\text{charge}} \sim \frac{\left(\dfrac{\Delta p_c}{RT}\right) \cdot V_P}{\dot{m}_c} \tag{8.175}$$

Expressing the pressure rise in the plenum as the square of the wheel speed,

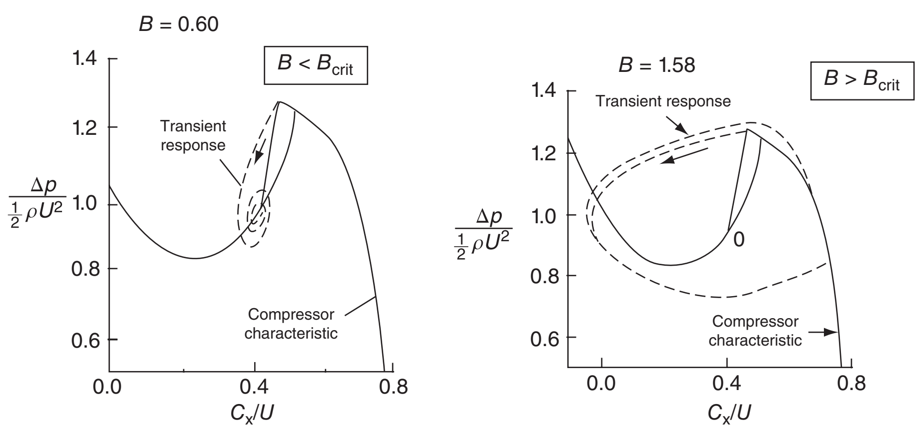
$$\Delta p_c \sim \rho\,(\omega \cdot r)^2 \tag{8.176}$$

$$T \sim a^2 \tag{8.177}$$

and the temperature as the speed of sound squared, Greitzer has shown that the ratio of the two timescales is the square of a parameter B that dictates the fate of disturbances in a compressor, that is,

$$B \equiv \frac{\omega \cdot r}{2a}\sqrt{\frac{V_P}{V_c}} \sim \left(\frac{\tau_{\text{charge}}}{\tau_{\text{throughflow}}}\right)^{0.5} \tag{8.178}$$

The critical value of the B-parameter, which causes compressor instability to grow into a surge instead of a localized rotating stall, is shown to be ~0.7–0.8 by Greitzer. Experimental investigations of compression system instability have supported the proposed model. Figure 8.56 (from Greitzer, 1976) shows the computed results of a transient behavior of



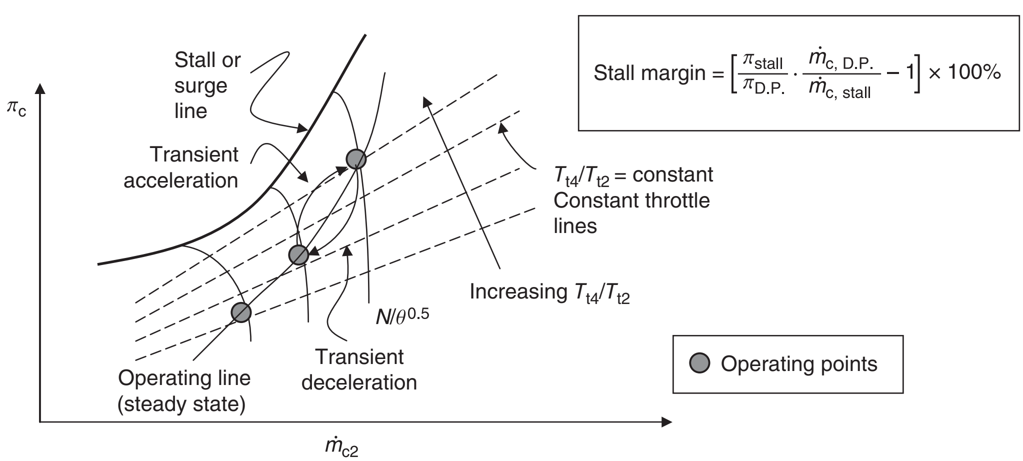
■ **FIGURE 8.56** **Transient response of a compressor to instabilities showing the appearance of a rotating stall (left) and the emergence of surge (right). Source: Greitzer 1976, Fig. 7.56, p. 190. Reproduced with permission from ASME** a compressor instability that settles into a rotating stall for low B-parameter and surge instability for a B-parameter of 0.6 and 1.58, respectively.

Current understanding of the compressor system instability has allowed strategies to be developed in its active control. Fast detection and cancellation of early oscillations are the keys to a successful active control system design. A review article by Paduano, Greitzer, and Epstein (2001) provides a rich exposition to the subject.

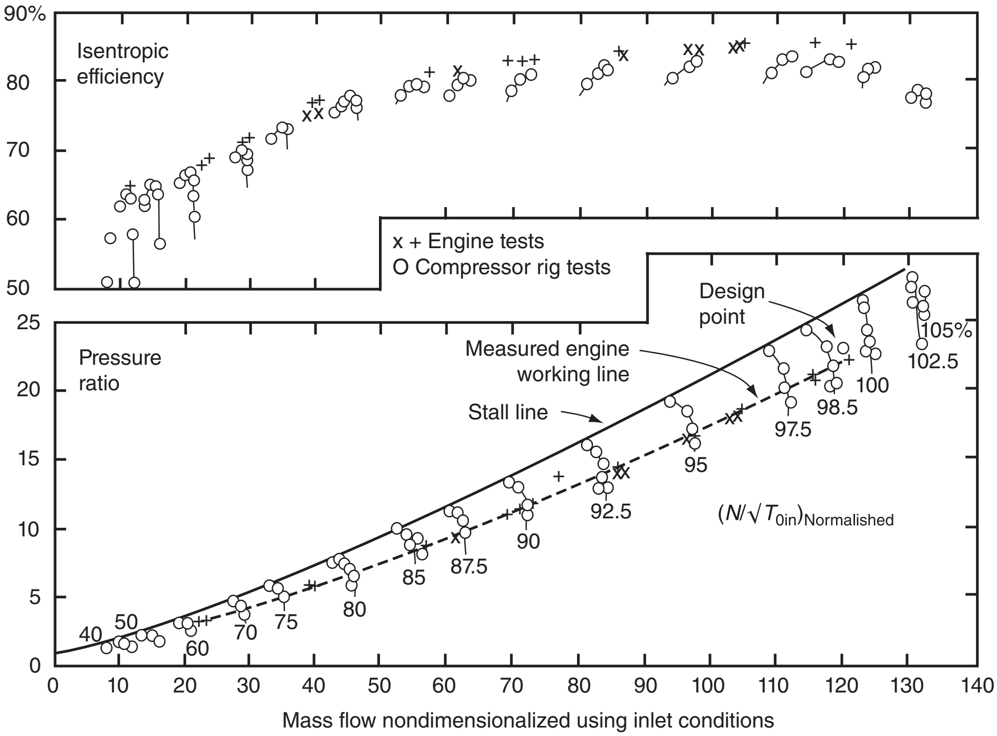
8.9 Multistage Compressors and Their Operating Line

The throttle in a gas turbine engine is the fuel flow control to the combustor. Constant throttle lines are, therefore, the lines of constant T_{t4}/T_{t2}. The path of steady-state operation of the compressor with the throttle setting is called the operating or working line in a compressor. It is often superimposed on the compressor performance map, as shown schematically in Figure 8.57. The transient behavior of "spool up" is shown in Figure 8.57 as well. In the context of the stall margin defined as the percentage stall pressure rise divided by the corresponding percentage drop in the mass flow rate, the transient engine operation deviates from the steady-state operating line and approaches the stall or surge line. The level and type of inlet flow distortion also affects the stall margin. We shall discuss inlet distortion and compressor stall later in this chapter.

Compressor performance map for the GE Energy Efficient Engine (E^3) is shown in Figure 8.58 (from Cumpsty, 1997). Compressor test rig data as well as the engine test data are superimposed for comparison. Engine measurements are limited to the data points along the engine working line, whereas the compressor rig offers the versatility to produce the entire map. As a point of reference, E^3 program was initiated by NASA in 1973 (sparked by the oil embargo) with the goal of 12% reduction in specific fuel consumption over then current commercial aircraft engines of the day. Both GE and Pratt & Whitney developed technology demonstrator engines under this program that exceeded NASA's goal. A book by Garvin (1998) presents a history of aircraft engine development in the United States with a particular emphasis on GE's achievements in the commercial



■ **FIGURE 8.57 Constant throttle lines and a steady-state operating line are superimposed on a typical compressor performance map (D.P. is the design point)**



■ **FIGURE 8.58** **Compressor performance map from GE E[3] program with a design point pressure ratio of 23 achieved in 10 stages with variable stators in the first six stages. Source: Cumpsty 1997. Reproduced with permission from Cambridge University Press**

engine market. The lead taken by the military needs and the lag of the commercial side in aircraft engine development are presented in the context of the cold war, and the impact of business and strategic teaming/partnerships that have led to a business success. Propulsion engineers need to look at the business side of engineering to appreciate issues of cost, markets, customer service, and product support. For additional reading on the history of aircraft gas turbine engine development in the United States, the book by James St. Peter (1999) is recommended.

Let us do a simple and quick calculation of this compressor's polytropic efficiency e_c based on its design point pressure ratio and adiabatic efficiency. We estimate the compressor adiabatic efficiency of ~0.86 from Figure 8.58 at the design point and for the compressor pressure ratio of 23 we substitute these in the modified form of Equation 4.22:

$$e_c = \frac{\gamma - 1}{\gamma}\left[\frac{\ln \pi_c}{\ln\left\{1 + \frac{1}{\eta_c}\left[\pi_c^{\frac{\gamma-1}{\gamma}} - 1\right]\right\}}\right]$$

to get an $e_c \approx 0.907$. An average stage pressure ratio in 10 stages is

$$(\pi_s)_{\text{Avg}} \approx (\pi_c)^{\frac{1}{N}} = (23)^{\frac{1}{10}} \cong 1.368$$

This represents just an *average* of the stage pressure ratios and we recognize that the front stages operate at a higher loading (front stage pressure ratio of ~1.6) than the aft stages (pressure ratio ~1.15) due to a higher blade tangential Mach number M_T.

Now, we shall demonstrate that a throttle position T_{t4}/T_{t2} establishes the compressor pressure ratio, the mass flow rate, and the corresponding shaft speed. From the chapter on cycle analysis and off-design considerations, we argued that the turbine nozzle and the exhaust nozzle throat remain choked over a wide operating range of the engine. As we recall the turbine expansion parameter,

$$\tau_{t-\text{off}-\text{design}} = \tau_{t-\text{design}} = \text{constant}$$

The compressor–turbine power balance demands

$$\dot{m}_0(h_{t3} - h_{t2}) = \dot{m}_0(1 + f)(h_{t4} - h_{t5})$$

To simplify the equation, assume the gas is calorically perfect and neglect the small contribution of fuel-to-air ratio in favor of 1 on the RHS of the above equation, to get

$$T_{t2}(\tau_c - 1) \approx T_{t4}(1 - \tau_t)$$

Therefore, the compressor temperature ratio is related to the throttle setting and a constant turbine expansion parameter τ_t via

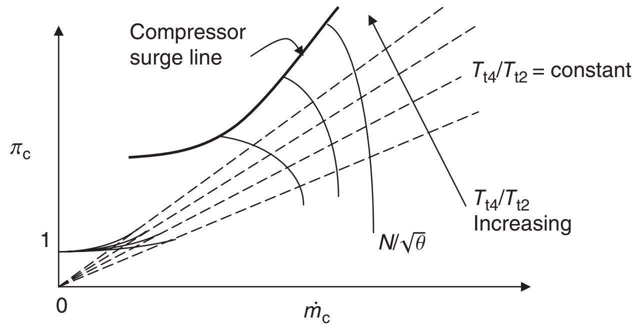
$$\tau_c \approx 1 + (1 - \tau_t)\frac{T_{t4}}{T_{t2}} \tag{8.179}$$

Here, we see that the compressor temperature ratio increases linearly with the throttle setting. The compressor pressure ratio is related to the temperature ratio via the polytropic efficiency; therefore, the compressor pressure ratio is established by the throttle setting as

$$\pi_c \approx \left[1 + (1 - \tau_t)\frac{T_{t4}}{T_{t2}}\right]^{\frac{\gamma e_c}{\gamma - 1}} \tag{8.180}$$

The continuity equation at stations 2 and 4, that is, the compressor face and the combustor/turbine nozzle exit connects the axial Mach number at the engine face to the throttle setting according to

$$\begin{aligned}\dot{m}_2 &= \sqrt{\frac{\gamma_2}{R_2}}\frac{p_{t2}}{\sqrt{T_{t2}}}A_2 \cdot M_{z2}\left(1 + \frac{\gamma_2 - 1}{2}M_{z2}^2\right)^{-\frac{(\gamma_2+1)}{2(\gamma_2-1)}} \\ &\approx \dot{m}_4 = \sqrt{\frac{\gamma_4}{R_4}}\frac{p_{t4}}{\sqrt{T_{t4}}}A_4\left(\frac{\gamma_4 + 1}{2}\right)^{-\frac{(\gamma_4+1)}{2(\gamma_4-1)}}\end{aligned} \tag{8.181}$$



■ **FIGURE 8.59**
Constant throttle lines on a compressor performance map

By neglecting the combustor total pressure loss, we may replace the combustor exit total pressure p_{t4} with the compressor exit pressure p_{t3} to rewrite Equation 8.181 as

$$M_{z2}\left(1+\frac{\gamma-1}{2}M_{z2}^2\right)^{-\frac{(\gamma+1)}{2(\gamma-1)}} \approx \text{constant}\frac{\pi_c}{\sqrt{T_{t4}/T_{t2}}} \tag{8.182}$$

We assumed the flow areas A_2 and A_4 remain constant in deriving Equation 8.182 as well as all the gas property variations included in the proportionality constant on the RHS of Equation 8.182. The left-hand side (LHS) of Equation 8.182 is proportional to the corrected mass flow rate $\dot{m}_{c2}$, hence

$$\dot{m}_{c2} \propto \frac{\pi_c}{\sqrt{T_{t4}/T_{t2}}} \tag{8.183}$$

Therefore, constant throttle lines, T_{t4}/T_{t2} = constant, are straight lines on the compressor performance map π_c versus $\dot{m}_{c2}$, as shown schematically in Figure 8.59. Also higher throttle constants attain higher slopes, as shown in Figure 8.59, which get the compressor operating point closer to the compressor surge line. Figure 8.59 shows the constant throttle lines and their convergence on compressor pressure ratio of 1, corresponding to a mass flow rate of zero.

The engine face axial Mach number is established by the throttle setting via Equation 8.182 and the compressor pressure ratio was related to the throttle setting via Equation 8.180, therefore,

$$M_{z2}\left(1+\frac{\gamma-1}{2}M_{z2}^2\right)^{-\frac{(\gamma+1)}{2(\gamma-1)}} \approx \text{constant}\frac{\left[1+(1-\tau_t)(T_{t4}/T_{t2})\right]^{\frac{\gamma.e_c}{\gamma-1}}}{\sqrt{T_{t4}/T_{t2}}} \tag{8.184}$$

We may solve Equation 8.184 for the engine face axial Mach number for a given throttle setting. We may also simplify the LHS somewhat by recognizing that the second term in the parenthesis involving axial Mach number is small compared to one, hence applying binomial expansion yields,

$$M_{z2}\left(1-\frac{\gamma+1}{4}M_{z2}^2\right) \approx \text{constant}\frac{\left[1+(1-\tau_t)(T_{t4}/T_{t2})\right]^{\frac{\gamma.e_c}{\gamma-1}}}{\sqrt{T_{t4}/T_{t2}}} \tag{8.185}$$

Now, let us demonstrate the dependence of wheel speed on the throttle setting. The compressor power is consumed by the rotor blades that experience a torque τ_r and spin at the angular rate of ω. The stator blades do not contribute to the power transfer. We need to sum overall the rotor blade rows in a multistage compressor to calculate the compressor power, namely,

$$\wp_c = \omega \cdot \sum_{j=1}^{N} \tau_{r_j} \tag{8.186}$$

where N is the number of stages. The rotor torque is the integral of angular momentum increase across the blade row and may be expressed as

$$\tau_r = \int_{r_{h2}}^{r_{t2}} r \cdot C_{\theta 2}(r) \cdot \rho_2(r) \cdot C_{z2}(r) \cdot 2\pi r \, dr - \int_{r_{h1}}^{r_{t1}} r \cdot C_{\theta 1}(r) \cdot \rho_1(r) \cdot C_{z1}(r) \cdot 2\pi r \, dr \tag{8.187}$$

The geometrical parameters are sketched in Figure 8.60. In their simplest form the integrals are expressed as an average angular momentum at the pitchline radius r_m namely,

$$\tau_r = \dot{m}\left[\left(r_{m2} C_{\theta 2m}\right) - \left(r_{m1} C_{\theta 1m}\right)\right] \tag{8.188}$$

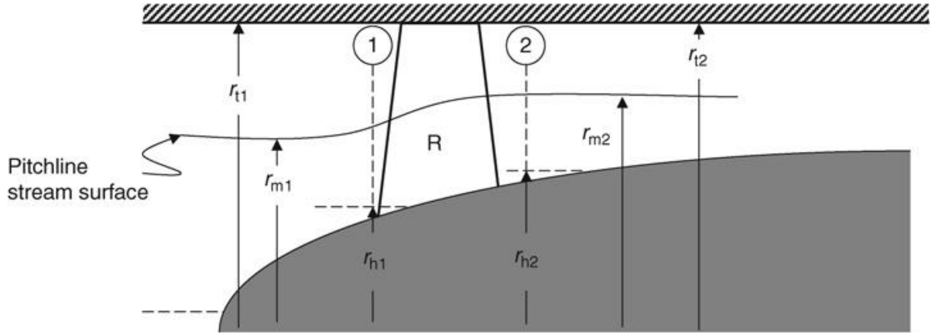
The rotor torque, however, is a function of the shaft speed as the absolute swirl downstream of the rotor changes with shaft speed according to

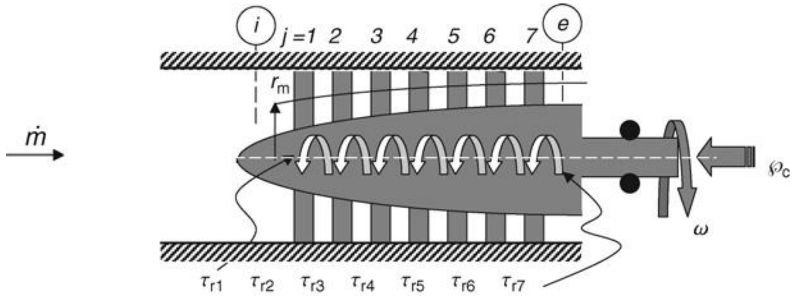
$$C_{\theta 2} = C_{z2} \tan \alpha_2 = U + W_{\theta 2} = U + C_{z2} \tan \beta_2 \tag{8.189}$$

The rotor relative exit flow angle β_2 remains nearly constant with the shaft speed over a wide operating range of the compressor, that is, for attached boundary layer flow to the blades. Substituting Equation 8.189 into Equation 8.188, we get

$$\tau_r = \dot{m}\left[r_{m2}\left(\omega \cdot r_{m2} + C_{z2} \tan \beta_2\right) - r_{m1} C_{z1} \tan \alpha_1\right] \tag{8.190}$$

■ **FIGURE 8.60**
Definition sketch of the geometrical parameters





■ **FIGURE 8.61** **Seven-stage compressor consuming power $\wp_c$ at the shaft speed of ω and the individual rotor torques $\tau_{r1}, \tau_{r2}, \ldots, \tau_{r7}$**

Expressing the shaft power in terms of the total enthalpy rise and the mass flow rate, we get

$$\wp_c = \dot{m}\left(h_{te} - h_{ti}\right) = \dot{m}h_{ti}(\tau_c - 1) = \omega \cdot \sum_{j=1}^{N} \tau_{rj} \tag{8.191a}$$

Figure 8.61 shows the parameters in Equation 8.191a.

By substituting for the rotor torque from Equation 8.190 to Equation 8.191a, for the compressor temperature ratio from Equation 8.179, and canceling the mass flow rate, we get

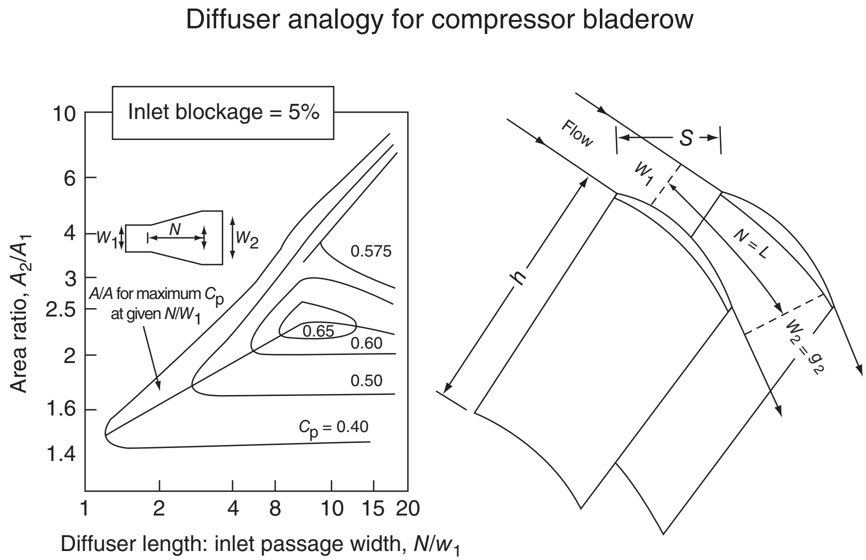
$$h_{t2}(1 - \tau_t)\frac{T_{t4}}{T_{t2}} = \omega \sum_{j=1}^{N} \left[r_{me}(\omega \cdot r_{me} + C_{ze} \tan \beta_e) - r_{mi} C_{zi} \tan \alpha_i\right]_j \tag{8.191b}$$

This is a quadratic equation in ω in terms of the throttle setting T_{t4}/T_{t2}, flow angles α_1 and β_2 that remain nearly constant, axial velocities across the rotor, and the radial position of the pitchline stream surface. In practice at a given throttle setting, the shaft finds its own speed consistent with the compressor power consumption and mass flow rate through the machine. We have thus established the interdependence of compressor pressure ratio, corrected mass flow rate, and shaft speed on the throttle setting.

The simplifications introduced in the above discussions may be removed one by one, if we are willing to iterate to find a consistent set of compressor performance parameters. For example, the choking condition in stations 4 and 8 may be relaxed in the case that the nozzle pressure ratio is below critical (i.e., necessary for choking). The subsonic throat Mach number may be found through application of continuity equation using a trial and error approach.

8.10 Multistage Compressor Stalling Pressure Rise and Stall Margin

Koch (1981) developed an analogy between the stalling pressure rise capability of an axial-flow compressor stage and two-dimensional diffusers. Classical diffuser data of Reneau, Johnston, and Kline (1966) and Sovran and Klomp (1967) for a straight centerline, 2D diffuser serve as the point of analogy. The pitchline radius of the compressor stage is taken as the reference point in developing the stall analogy with the diffuser. This approach is



■ FIGURE 8.62 (Left) Diffuser performance chart. Source: Reneau et al. 1967. Reproduced with permission from ASME. (Right) The analogy to a compressor blade row. Source: Wisler 2000.

useful in the preliminary design phase of an axial-flow compressor in estimating the maximum pressure rise potential as well as the stall margin for a given design point operation. Figure 8.62 shows the diffuser performance data of Reneau, Johnston, and Kline for an inlet boundary layer blockage of 5%. The length of the diffuser is N, which is analogous to L, as shown on the definition sketch on the right (from Wisler, 2000), to represent a diffusion length scale of a compressor blade passage at the pitchline radius.

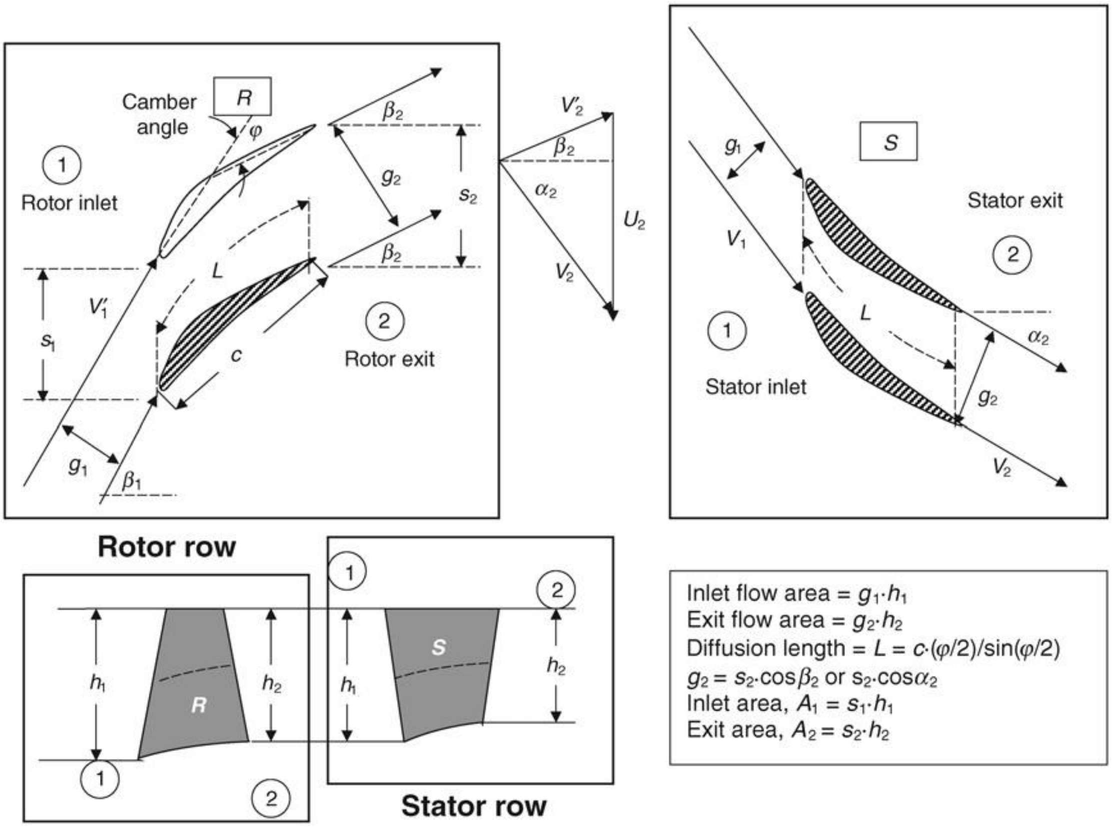
As a first approximation, the diffusion length L is represented by the arc length of the mean camber line of the airfoil at the pitchline. Assuming a circular arc for the mean camber line, the length L is related to the camber angle φ and the chord length c via

$$L \approx c \cdot \frac{\varphi/2}{\sin(\varphi/2)} \tag{8.192}$$

Although the area ratio of a diffuser, A_2/A_1, is fixed by the geometry of the diffuser, its counterpart in a cascade depends on the blade incidence, that is, the staggered spacing of the streamtube that enters and exits the blade row. The exit flow area of the blade channel is, however, fixed over a wide operating range of the compressor flow and, therefore, it serves as the reference area in the correlation development by Koch. The inlet flow area is a function of the operating point (i.e., the throttle setting) and decreases with an increasing incidence angle. Figure 8.63 shows the staggered spacing g_1 and g_2 and the flow areas at the inlet and outlet of a compressor blade row.

Here, we have adopted the terminology of Koch in representing the rotor relative velocity as V' and the absolute velocity as V. A nondimensional diffusion parameter is the ratio of the diffusion length L to the staggered spacing at the exit of the blade row, that is, L/g_2,

$$\frac{L}{g_2} = \sigma_1 \cdot \frac{\varphi/2}{\sin(\varphi/2)} \cdot \frac{s_1}{s_2}\frac{1}{\cos\beta_2} = \sigma_1 \cdot \frac{\varphi/2}{\sin(\varphi/2)} \cdot \frac{A_1}{A_2} \cdot \frac{h_2}{h_1 \cdot \cos\beta_2} \tag{8.193}$$



■ **FIGURE 8.63 Definition sketch of the staggered inlet and outlet spacing and area ratio for a compressor stage (V_1' and V_2' are relative velocities W_1 and W_2)**

where the camber angle φ is in radians and the blade inlet solidity is represented by σ_1 at the pitchline. The blade solidity, the camber angle, and the exit flow angle are all reflected in the nondimensional diffusion length ratio in a compressor blade row. The stator blade row diffusion length is calculated in a similar manner using the stator inlet solidity at the pitchline, camber angle, area ratios, and the exit absolute flow angle. The effect of higher camber angle is seen as an increase in diffusion, hence static pressure rise. The effects of blade aspect ratio, tip clearance gap, and Reynolds number are accounted for through the end wall boundary layer thickness. In a diffuser, the inlet boundary layer blockage is a key parameter in the performance and stalling characteristics of diffusers, and hence the end wall boundary layer thickness in a compressor stage serves the same principle.

A stage average approach is adopted in developing a correlation between the stalling pressure rise of a diffuser and a compressor stage. For example, the diffusion length ratio L/g_2 of the stage is the weighted average of the rotor and the stator values with blade row inlet dynamic head used as the weighting factor, that is,

$$\left.\frac{L}{g_2}\right|_{\text{Stage}} = \left[\frac{(L/g_2)_{\text{Rotor}} \cdot q_1' + (L/g_2)_{\text{Stator}} \cdot q_1}{q_1' + q_1}\right] \qquad \textbf{(8.194)}$$

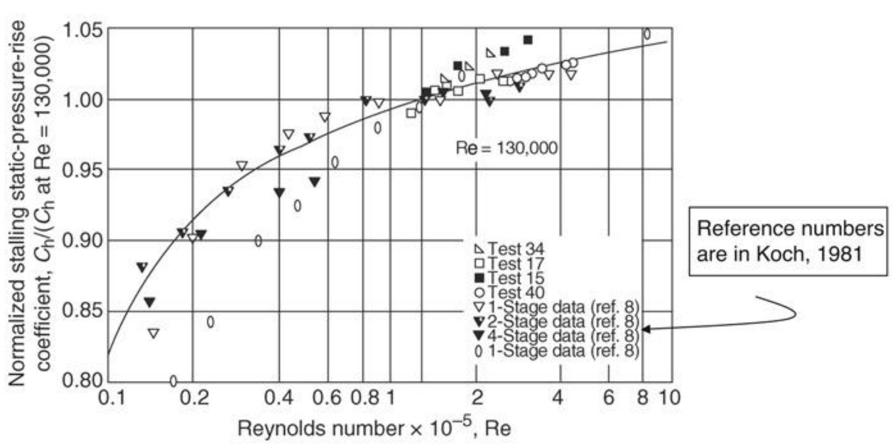
where q_1' is the rotor inlet dynamic head (relative) and q_1 is the stator inlet dynamic head. Koch (1981) defines an enthalpy equivalent of the static pressure rise in a compressor stage according to

$$C_h \equiv \frac{c_p T_1\left[\left(\left.\frac{p_2}{p_1}\right|_{\text{Stage}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right] - \frac{(U_2^2 - U_1^2)_{\text{Rotor}}}{2}}{\left[\frac{\left.V_1'^2\right|_{\text{Rotor}} + \left.V_1^2\right|_{\text{Stator}}}{2}\right]} \tag{8.195}$$

The numerator of Equation 8.195 is the stage static enthalpy rise based on the isentropic stage temperature ratio, which is corrected for the radial shift across the rotor at the pitchline radius. To be comparable, the rotor *free work contribution* associated with the radial shift needs to be corrected for in the pressure rise comparisons of diffusers and compressor stages. The denominator is the sum of the free stream dynamic heads to the rotor and stator. There are *two* dynamic heads in the denominator, corresponding to *two* static enthalpy rises across the rotor and stator in the numerator.

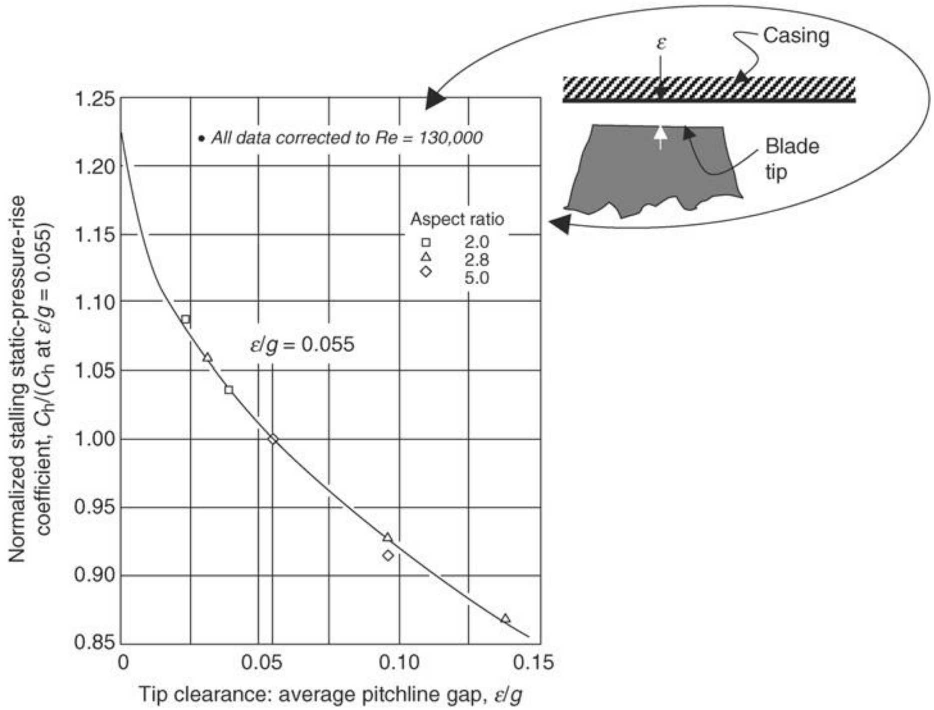
The contributions of Reynolds number, tip clearance, and the axial spacing between the rotor and the stator rows to the stalling pressure rise coefficient are represented in Figures 8.64 to 8.66, respectively (from Koch, 1981). Normalizing values for Reynolds number, ratio of tip clearance to gap, and the ratio of axial distance to the blade spacing are used in presenting the stalling pressure rise coefficient. These are Reynolds number = 130,000, the average tip (radial) clearance-to-gap ratio $\varepsilon/g = 0.055$, and the ratio of axial clearance to the blade spacing $\Delta z/s = 0.38$. The Reynolds number is calculated based on the blade row inlet relative velocity and the chord length at the pitchline radius. The gap g and the blade spacing s represent a stage average at the pitchline radius in Figures 8.65 and 8.66.

Note that a higher stalling pressure rise is achieved in a compressor stage with a decreasing tip clearance as well as a decreasing axial spacing between the blade rows. The effect of blade stagger was significant on the stalling pressure rise capability of axial-flow compressor stages according to Koch (1981). The effect of stagger is related to a recovery potential of a total pressure deficit region (e.g., upstream wake) interacting with the stator

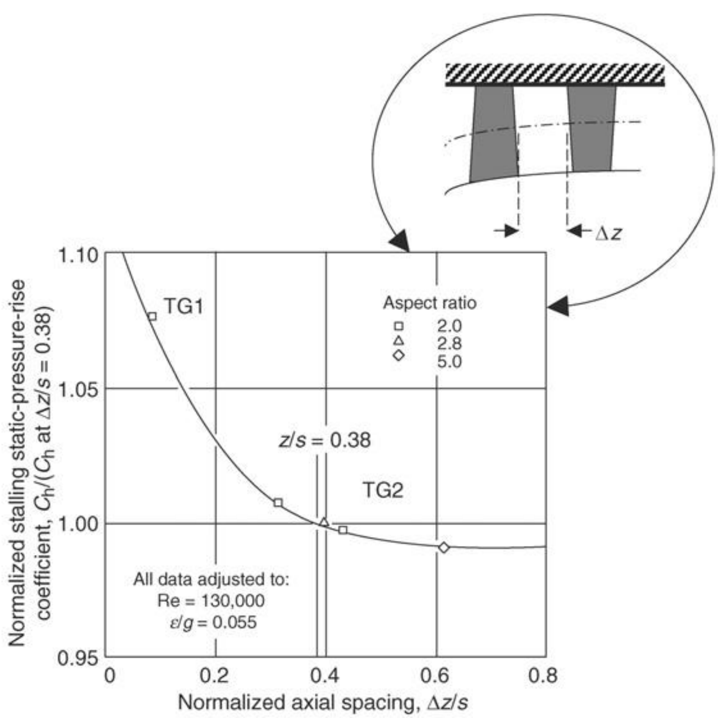


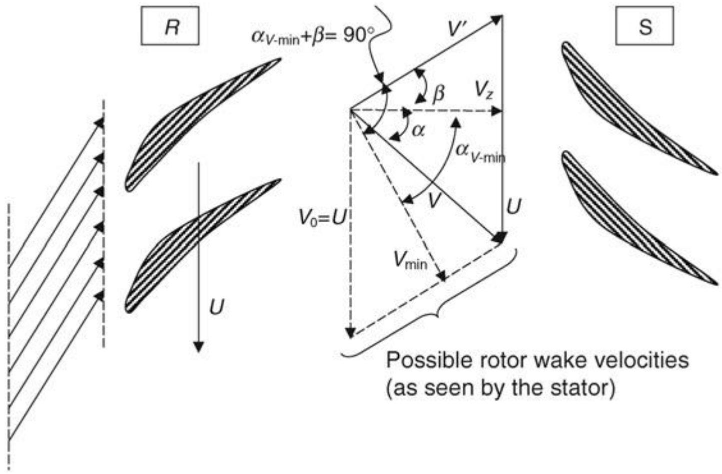
■ FIGURE 8.64 The effect of Reynolds number on stalling pressure rise coefficient of an axial-flow compressor stage (normalized by Reynolds number of 130,000). Source: Koch 1981. Reproduced with permission from ASME

■ **FIGURE 8.65**
The effect of tip clearance on the stalling pressure rise coefficient. Source: Koch 1981. Reproduced with permission from ASME



■ **FIGURE 8.66**
The effect of axial spacing between rotor and stator blade rows on stalling pressure rise coefficient. Source: Koch 1981. Reproduced with permission from ASME





■ **FIGURE 8.67 Velocity vector diagram upstream of the stator with possible upstream wake velocity vectors. Source: Adapted from Koch 1981. Reproduced with permission from ASME**

blade row. Leroy Smith first identified this phenomenon in 1958. The velocity vector diagrams upstream of the stator are shown in Figure 8.67. Note that the minimum wake velocity as seen by the stator occurs when the angle between the relative velocity vector V' and the wake velocity vector is 90°. In case of zero wake velocity in the relative frame, the wake as seen by the stator is equal to U and is in the tangential direction. This point is shown as $V_0 = U$ in Figure 8.67.

The effect of low stagger (i.e., high flow coefficient) is seen in the small included angle between the relative and absolute velocity vectors $\alpha + \beta$, which will result in a low momentum wake velocity $V_{\text{min}} \ll V$. With a low dynamic head associated with this flow, the low stagger stages seem more prone to stall than a high stagger counterpart. In a high stagger stage, that is, a low flow coefficient case, $\alpha + \beta > 90°$; therefore, the upstream wake enters the stator with $V_{\text{wake}} > V$, hence a higher dynamic pressure fluid is less prone to stall. The observations of $\alpha + \beta$ and 90° and their impact on the loss recovery were made by Ashby in a discussion of Smith's (1958) paper. Here, we are alerted to the importance of the vector diagram in a compressor stage and its impact on its stall margin. Koch devised an *effective dynamic pressure factor* F_{ef} that accounts for the rotor wake interaction with the stator, that is, the effect of stagger on pressure recovery. The formulation of F_{ef} is presented by Koch to be

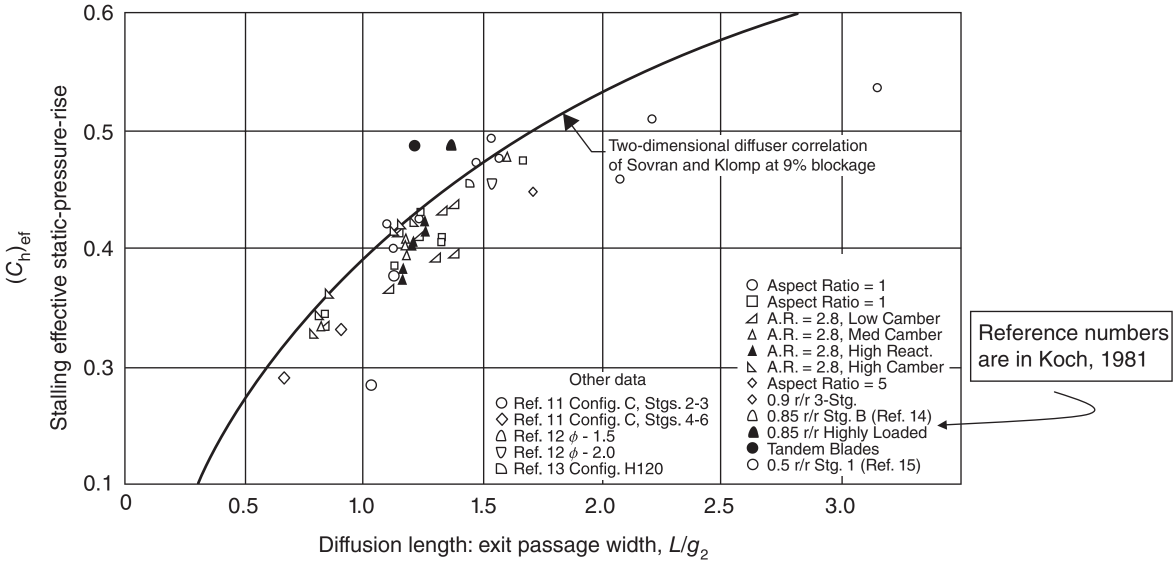
$$F_{\text{ef}} = \frac{V_{\text{ef}}^2}{V^2} = \frac{V^2 + 2.5V_{\text{min}}^2 + 0.5V_0^2}{4V^2} \tag{8.196}$$

The minimum dynamic head is related to the included angle $(\alpha + \beta)$ of the stage vector diagram according to

$$\frac{V_{\text{min}}^2}{V^2} = \sin^2(\alpha + \beta), \quad \text{if}(\alpha + \beta) \leq 90° \text{ and } \beta \geq 0° \tag{8.197a}$$

$$\frac{V_{\text{min}}^2}{V^2} = 1.0, \quad \text{if}(\alpha + \beta) > 90° \quad \text{High Stagger} \tag{8.197b}$$

$$\frac{V_{\text{min}}^2}{V^2} = \frac{V_0^2}{V^2}, \quad \text{if}\beta < 0° \qquad \text{Low Stagger} \tag{8.197c}$$

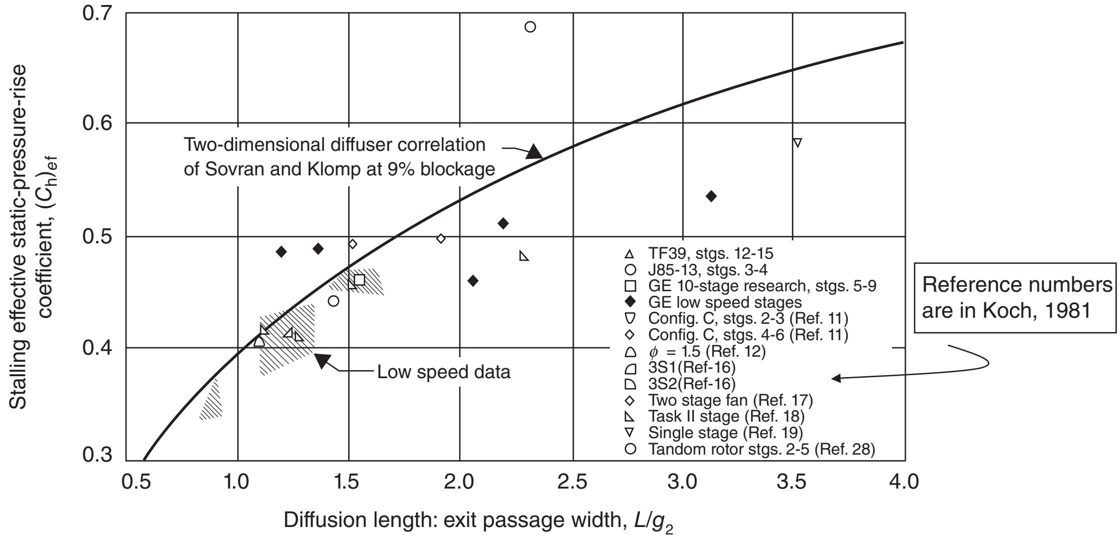


■ **FIGURE 8.68 Correlation of effective static pressure rise coefficient at stall for low-speed compressor stages. Source: Koch 1981. Reproduced with permission from ASME**

The procedure to account for the effects of Reynolds number, tip clearance, axial spacing, and stagger on stalling static pressure rise coefficient in a compressor stage is to first *adjust* the stalling pressure rise for Reynolds number other than 130,000 (via Figure 8.63), the tip clearance (to gap ratio) other than 5.5% (via Figure 8.64), and the axial spacing (to blade spacing) other than 0.38 (via Figure 8.65). Therefore, we first calculate the $(C_\mathrm{h})_\mathrm{adj}$. The effect of stagger and the velocity vector diagram is now introduced through the *effective dynamic head factor* F_ef, which is multiplied by the free stream dynamic head of the stator, that is,

$$(C_\mathrm{h})_\mathrm{ef} = (C_\mathrm{h})_\mathrm{adj}\left[\frac{V_1'^2\Big|_\mathrm{Rotor} + V_1^2\Big|_\mathrm{Stator}}{V_1'^2\Big|_\mathrm{Rotor} + F_\mathrm{ef}\cdot V_1^2\Big|_\mathrm{Stator}}\right] \qquad \textbf{(8.198)}$$

Koch's stalling effective pressure rise correlation for low- and high-speed axial-flow compressor stages is shown in Figures 8.68 and 8.69, respectively. The maximum recovery of the two-dimensional diffuser for an inlet blockage of 9% of Sovran and Klomp (1967) marks the upper limit in Figures 8.68 and 8.69, which serves as an approximate stalling boundary for axial-flow compressor stages. We note that the stall margin correlation for the low-speed compressor predicts typically below 0.5 of static pressure recovery at stall. The high-speed compressor stages produce a higher effective stalling static pressure rise coefficient of up to ~0.55. The case of a tandem rotor compressor with an L/g_2 of 2.33 poses the highest stalling static pressure recovery, that is, ~23% above the diffuser stall curve, as shown in Figure 8.69. The tandem blading serves as a means of flow control in a compressor rotor and the simple diffuser performance underpredicts the stall behavior of a compressor stage that employs flow control. An analogy of a tandem rotor compressor



■ **FIGURE 8.69 Correlation of effective static pressure rise coefficient at stall for high-speed compressor stages. Source: Koch 1981. Reproduced with permission from ASME**

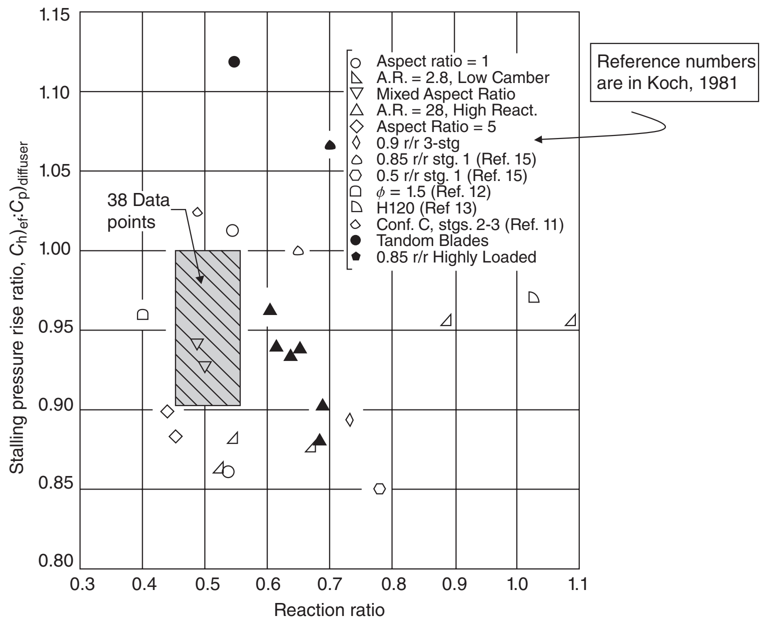
stage should be made with a wide-angle diffuser that employs splitter plates as a means of high static pressure recovery.

Koch demonstrates that effect of the stage reaction on the stalling pressure rise in axial-flow compressor is well predicted by the diffuser stall margin correlation of Figure 8.68 or 8.69. The range of stage reactions from 0.39 to 1.09 (where the stator accelerates the flow, like a turbine) is shown in Figure 8.70. It is surprising that over such a wide range of stage reaction, the stalling pressure rise in a compressor correlates with the diffuser predictions.

In a multistage high-speed compressor, the stagewise distribution of the static pressure rise at stall is compared with the stall margin correlation of Figure 8.71 (from Koch).

Figure 8.71 shows that in a multistage compressor, there is at least one or a group of stages that rises to the predicted level of stalling pressure according to the diffuser correlation. For example, the 10-stage research compressor operating at 95% speed at stall had its eighth stage reach/exceed the stall boundary as predicted by the correlation of Figure 8.69. The 16-stage TF39, at 97% speed, had stages 12 and 15 exceed the stall boundary and stages 13 and 14 operate within 96 percentile of the stall boundary. Note that we are not looking for a perfect match between the correlation and the stall pressure rise of every stage in a multistage compressor. We are rather interested in the trends and behavior of individual and groups of stages and their stall margin. In this context, the shaded symbols in Figure 8.71 represent the "critical" stages, which lie above 95 percentile of the stall boundary. These stages are likely to stall first and cause a breakdown of the flow in the rest of the compressor. The stalling pressure rise data in Figure 8.71 represent *actual* high-speed multistage compressor tests, which validate Koch's semiempirical correlation presented in this section. The stall margin correlation based on the diffuser analogy has proven to be a valuable tool in the preliminary design stage of axial-flow compressors.

■ **FIGURE 8.70**
Effect of stage reaction on the ratio of the stalling pressure rise to a 2D diffuser pressure recovery. Source: Koch 1981. Reproduced with permission from ASME



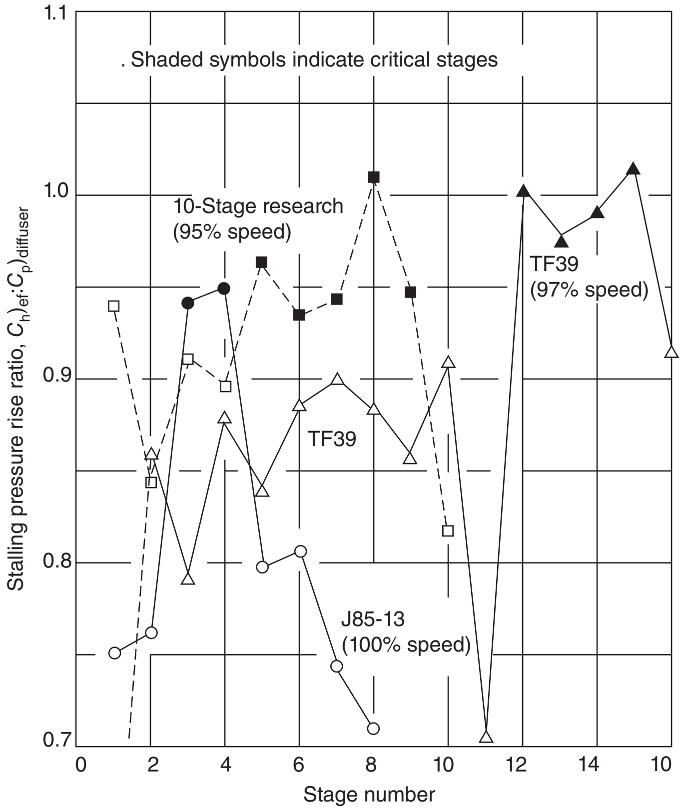
We have taken the stalling effective pressure rise coefficient of Figures 8.68 and 8.69 and graphed a family of curves with 95%, 90%, . . . , 70% of the stall correlation (Figure 8.72). We have thus created a chart with 5% stall margin increments as a function of the stage-averaged diffusion length ratio L/g_2. We note that increasing the diffusion length ratio L/g_2, which points to either higher solidity or lower aspect ratio blading, helps with the stall margin. Wide chord blades also operate at a higher Reynolds number, which is beneficial to boundary layer stability, as evidenced in Figure 8.64. The minimum operational tip clearance of ~1% blade height and the minimum axial spacing between blade rows of ~0.25 axial chord also create higher static pressure rise in a compressor stage. Stage vector diagram affects the loss recovery ($\alpha + \beta \geq 90°$) and thus stall static pressure rise.

8.11 Multistage Compressor Starting Problem

The annulus flow area in a multistage compressor is designed based on the calculated density rise along the axis of the machine operating at the design point. The assumption of constant axial velocity, for example, implies that the channel area shrinks inversely proportional to the density rise, to conserve mass, namely,

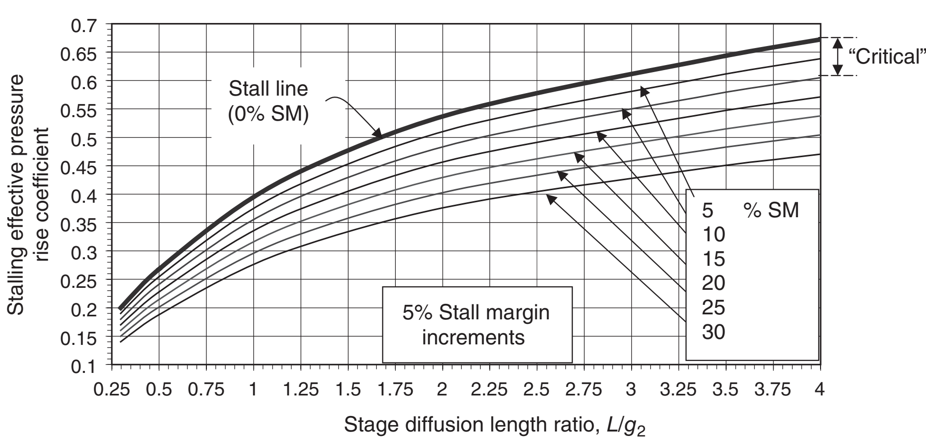
$$\frac{A(z)}{A_1} = \frac{\rho_1}{\rho(z)} \tag{8.199}$$

This behavior is schematically shown in Figure 8.73.

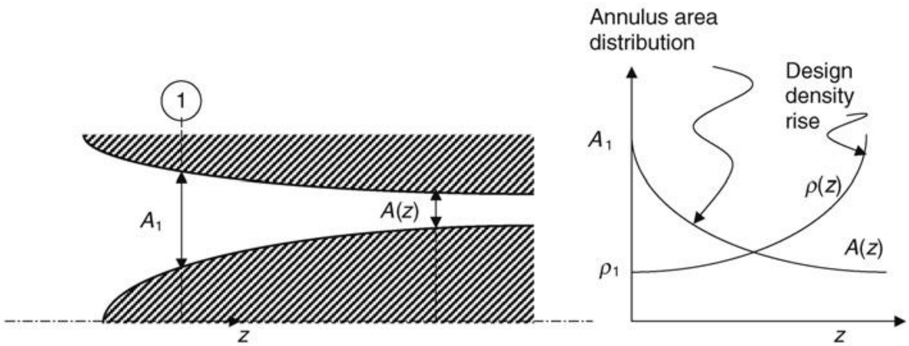


■ **FIGURE 8.71**
Stall pressure rise, per stage, in a multistage compressor at high speed. Source: Koch 1981. Reproduced with permission from ASME

In the starting phase of a compressor, the mass flow rate is initially small, which means a higher loading for the front stages. The picture for the aft high-pressure stages is just the opposite of the front low-pressure stages. Initially, the compressor does not develop the density rise that it was designed to produce, hence with lower than design densities in the aft stages the axial velocity is increased to satisfy the continuity equation. Higher than the design axial velocity leads to a lower loading of the aft stages. Actually



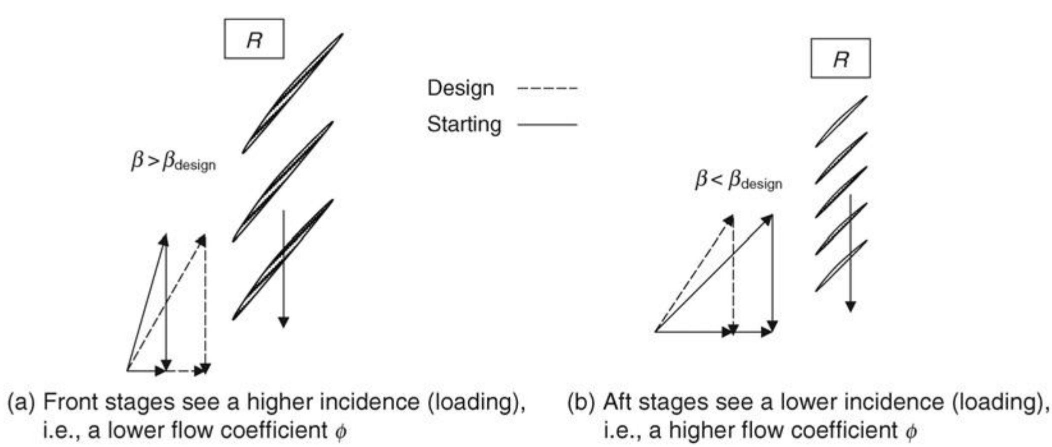
■ **FIGURE 8.72**
Axial-flow compressor stage stall margin (SM) chart with 5% stall margin increments



■ **FIGURE 8.73** **Flow annulus area and the density rise are inversely proportional in a multistage axial compressor**

the aft stages would be *windmilling* and their blades operating in a separated flow. A comparison of the velocity triangles in the starting phase and the design point will help us understand the (starting) problem in a high-pressure compressor. Figure 8.74 shows the velocity triangles for the front and aft stages of a compressor. Note that the flow coefficient is lower than the design value for the front stages, and the flow coefficient is higher for the aft stages. We remember from the compressor performance map that with a reduction of flow coefficient the compressor can go into stall for a given shaft speed. With front stages not producing the design density ratio, the axial flow in the aft stages accelerates to conserve mass. Hence, the two ends of the compressor see the opposite flow fields and are subjected to opposite flow coefficients, that is, lower and higher than the design, respectively. However, the tendency for both front and aft stages is to stall in the starting phase of a high-pressure compressor.

The loading mismatch between the front and aft stages in the starting phase of a high-pressure axial compressor may be solved in several ways. Here we introduce three distinct approaches to the starting problem.



■ **FIGURE 8.74** **Velocity triangles at design and starting phases of a high-pressure compressor (a) Front stages see a higher incidence (loading), i.e., a lower flow coefficient ϕ (b) Aft stages see a lower incidence (loading), i.e., a higher flow coefficient ϕ**

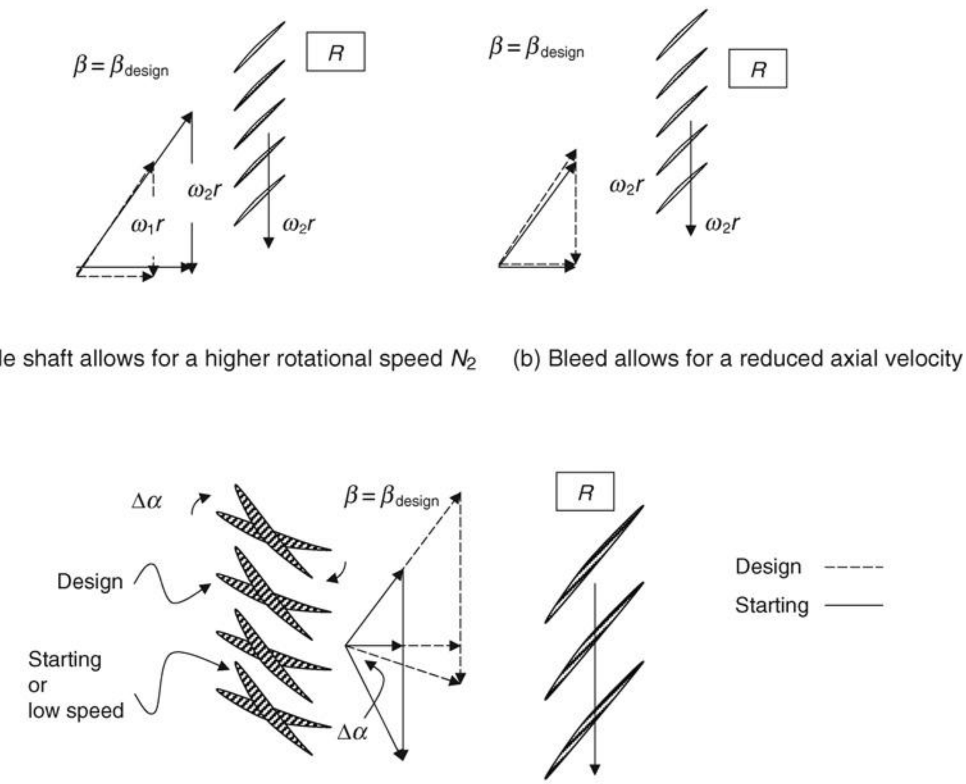
Proposition 1: Split shaft (or multispool shaft system)
Let the aft stages operate at a higher rotational speed than the front stages. This proposition aims at matching the relative flow angle to the rotor blades in the aft stages. Pratt & Whitney spearheaded the development of the split shaft concept in the United States in the 1950s. Today, all modern high-pressure compressors employ a two-to-three spool or shaft configuration to alleviate the problem of starting and improve compressor efficiency and stability. This method primarily attacks the aft stages, which would have been windmilling without a higher shaft speed, as in a single-shaft gas turbine engine. Rolls-Royce is the only engine company that has manufactured a production three-spool commercial engine, known as the RB 211, with several successful derivatives.

Proposition 2: Variable stators
This proposition aims at adjusting the absolute flow angle (α) through a stator variable setting to improve the mismatch between the flow and the rotor relative flow angle β. GE-Aircraft Engines spearheaded the development of high-pressure compressors with the variable stator approach in the United States, also in 1950s. Today, modern high-pressure compressors use variable stators for the front several stages to help with the starting/off-design operation efficiency. This method primarily attacks the front stages. Although a variable stator benefits all stages of a compressor, its use is limited to the front few (say six) low-pressure stages as the sealing of hot gases through the variable stator seal poses an operational and maintenance problem on the high-pressure end of the compressor.

Proposition 3: Intercompressor bleed
The windmilling operation of the aft stages is the result of high axial velocities through those stages. To cut down on excessive axial velocities, we may bleed off some mass flow in the intermediate stages of the compressor, the so-called intercompressor bleed. Therefore in the starting phase, bleed ports need to be opened to help the high-pressure end of the compressor to operate properly. This proposition represents a relatively low-cost method of starting compressors that does not employ either a split shaft or the variable stators. Stationary gas turbine power plants employ the intermediate bleed solution as a method of starting the compressor. This method attacks the aft as well as the front stages, as the bleed ports are located in the middle. Therefore, mass withdrawal causes a reduction in the axial velocity to the aft stages as well as lowering the backpressure for the front stages, hence increasing the flow speed in the machine. The ability to *tailor* the flow through the engine to improve overall efficiency, component stability, and provide cooling to accessories and engine components all speak in favor of intercompressor bleed. In fact, all modern gas turbine engines today employ intercompressor bleed/flow control. These methods are summarized in Figure 8.75.

8.12 The Effect of Inlet Flow Condition on Compressor Performance

Aircraft gas turbine engines operate downstream of an air intake system. The level of *distortion* that an inlet creates at the compressor face affects the performance and the stability of the compressor. First, what do we mean by *distortion?* In simple terms, distortion represents nonuniformity in the flow. The nonuniformity in total pressure as in boundary layers and wakes, the nonuniformity in temperature as in gun gas ingestion or thrust reverser flow ingestion and the nonuniformity in density, as created by hot gas ingestion are some of the different types of distortion. The common feature of all



(a) Multiple shaft allows for a higher rotational speed N_2 (b) Bleed allows for a reduced axial velocity

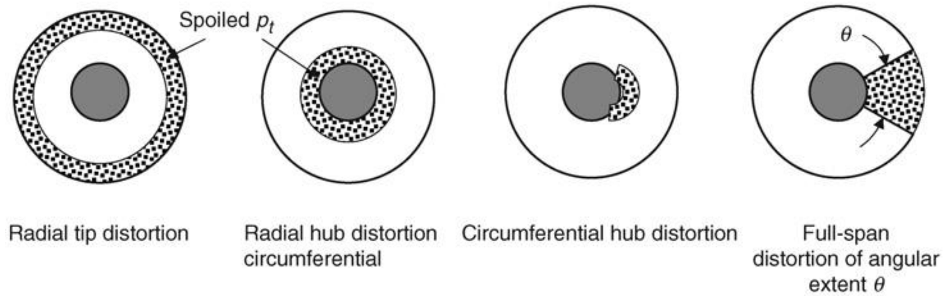
(c) A variable stator turns the flow in the direction of the rotor in the starting or low-speed operation

■ **FIGURE 8.75** **Three propositions on how to start a high-pressure compressor**

different types of distortion is found in their destabilizing impact on the compressor performance. This means that all distortions reduce the stability margin of a compressor or fan, potentially to the level of compressor stall or the engine surge. The types of distortion are

1. Total pressure distortion $p_t(r, \theta)$
2. Total temperature distortion $T_t(r, \theta)$
3. Flow angle distortion $\alpha(r, \theta)$
4. Secondary flow – swirl at the engine face $C_{\theta 1}(r, \theta)$
5. Entropy distortion $s(r, \theta)$
6. Combinations of some or all of the above distortions.

The most common type of inlet distortion is the total pressure distortion that is caused by separated boundary layers in the inlet. Under normal operating conditions, the boundary layers in the inlet are well behaved and remain attached. However, if the boundary layer



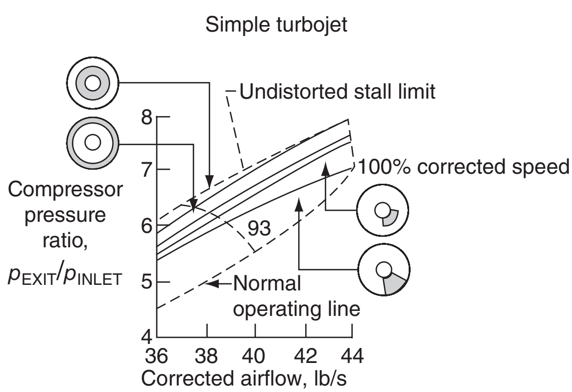
■ **FIGURE 8.76**
Different types of total pressure distortion

management system in a supersonic inlet, as in the bleed system, fails to react to an abrupt change in the flight operation (potentially due to a rapid combat maneuver), the flowfield at the engine face will contain large patches of low-energy, low-momentum flow that could cause flow separation in the front stage(s) of the fan or compressor. In describing the total pressure distortion and its impact on compressor performance, we divide the spatial extent of the spoiled flow according to its radial and circumferential extent, as shown in Figure 8.76.

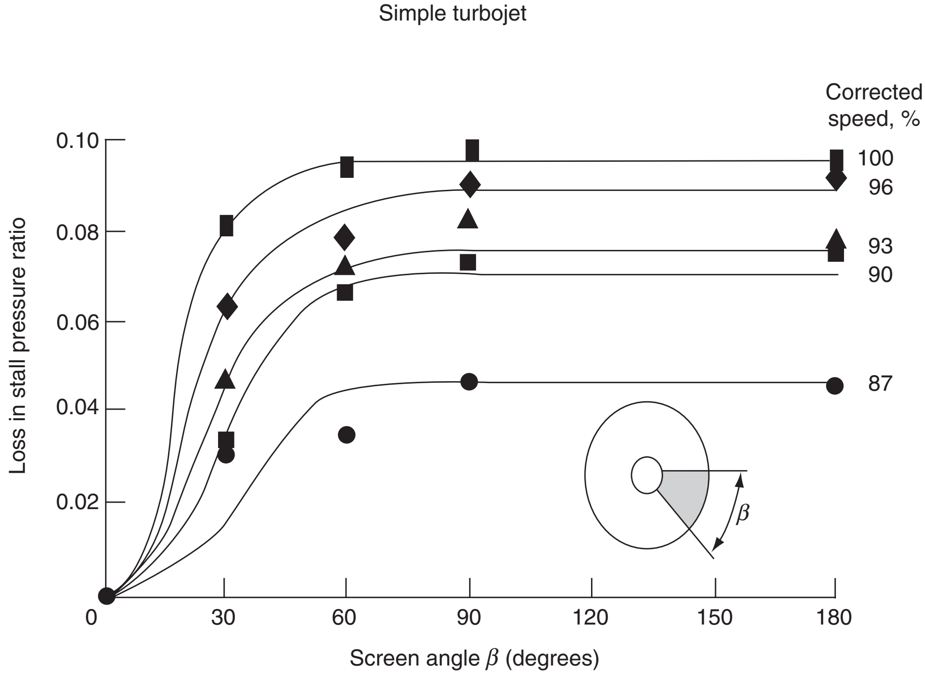
These inlet total pressure distortion patterns may be simulated by installing screens of varying porosity upstream of the compressor or fan (typically one diameter upstream) in propulsion system ground test facilities. The distortion patterns generated by screens in a test set up in ground facilities represent the "steady-state" component of the distortion and thus lack the "dynamic" or transitory nature of the distortion encountered in real flight environment. The full description of distortion requires both the steady-state and the dynamic components, as in the study of turbulent flow requiring a mean and an rms level of the fluctuation.

The results of a NASA-Glenn 10×10 supersonic wind tunnel study on the response of a simple turbojet engine (J-85) to steady-state inlet total pressure distortion are shown in Figure 8.77 (from Povolny *et al.*, 1970).

The normal operating line, the undistorted stall boundary, and two corrected shaft speeds of 100% and 93% design are shown in dashed lines. Four solid lines correspond to the stall boundaries of the four distortion patterns simulated at the compressor face



■ **FIGURE 8.77**
Effect of inlet total pressure distortion on compressor stability. Source: Povolny et al. 1970. Courtesy of NASA

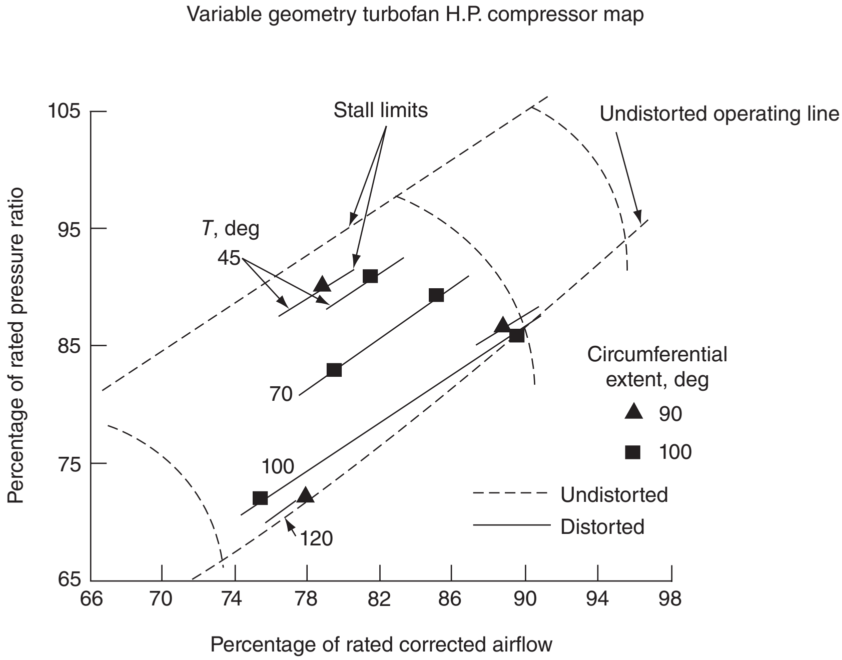


■ **FIGURE 8.78** **Effect of the extent of circumferential spoiling on compressor performance. Source: Povolny et al. 1970. Courtesy of NASA**

via screens. From having the least to the most impact on the stall margin deterioration, we identify the culprits as (1) radial hub, (2) radial tip, (3) circumferential hub, and (4) the full-span circumferential distortion, respectively. We also note that the full-span distortion at the 100% corrected speed operates at the stall boundary, that is, zero-stall margin! Further research identified a critical circumferential extent of the spoiled sector that causes the maximum loss in the stall pressure ratio of a compressor is at nearly 60°, as evidenced in Figure 8.78 (from Povolny *et al.*, 1970).

The loss in stall pressure ratio with circumferential inlet distortion reaches to ~10% at 100% corrected speed, as shown in Figure 8.78. At higher shaft speeds, the incidence angle in the spoiled sector is larger than at the lower shaft speed, thus the trend of higher loss of the stalling pressure ratio at higher shaft speeds becomes evident using simple velocity triangle arguments.

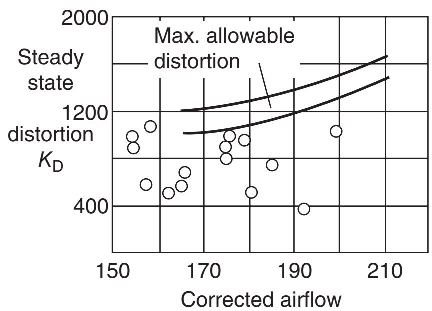
The temperature distortion also leads to a reduction in stall margin. In general, static temperature distortion in a flow brings about density nonuniformity, which creates a nonuniform velocity field. Consequently, it is impossible to create a static temperature distortion without creating other forms of nonuniformity, for example, density, velocity, total pressure, in the flow. To quantify the impact of a spatial temperature distortion on engine stall behavior, NASA researchers have conducted experiments with representative data shown in Figure 8.79 (from Povolny *et al.*, 1970). The undistorted operating line, stall limit, and different shaft speeds are shown in dashed lines. Data points corresponding to the effect of temperature distortions of 45–120°F on the stall behavior of a variable geometry turbofan engine high-pressure compressor are plotted in solid lines. The circumferential extents of the temperature distortions were 90 and 100° in different tests. A 100°F distortion of ~90–100° circumferential extent is seen to stall the high-pressure compressor operating at its 90% corrected flow.



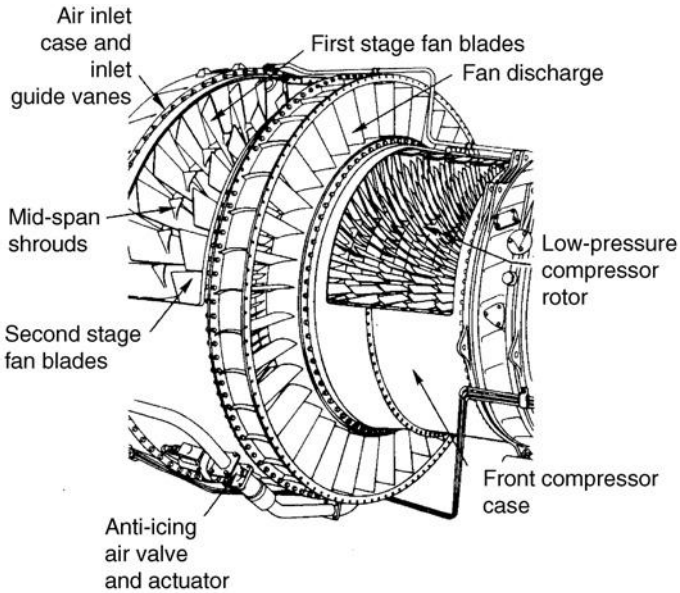
■ **FIGURE 8.79**
Effect of spatial temperature distortions on engine stall. Source: Povolny et al. 1970. Courtesy of NASA

The inadequacy of steady-state distortion simulation in a wind tunnel is best seen in Figure 8.80 showing F-111 flight test data of compressor stall with TF-30 engine. A steady-state distortion parameter K_D is graphed at different corrected airflows through the engine, and a band of maximum allowable distortion based on previous wind tunnel tests is shown to miss the mark of in-flight compressor stall data by a large margin (from Seddon and Goldsmith, 1985). The culprit is identified as the *dynamic distortion.*

The steady-state distortion parameter K_D expresses a weighted average of engine face total pressure distortion pattern (circumferential and radial) recorded by an engine face rake system composed of several radial pitot tube measurements. For a mathematical description of K_D and other distortion parameters, Hercock and Williams (1974) may be consulted. In addition, Farr and Schumacher (1974) present evaluation methods of dynamic distortion in F-15 aircraft. Schweikhand and Montoya (1974) treat research instrumentation and operational aspects in YF-12. These references are recommanded for further reading.



■ **FIGURE 8.80**
In-flight measurements of engine surge. Source: Seddon and Goldsmith 1985. Reproduced by permission from AIAA



■ **FIGURE 8.81** **Dual-compressor rotor with two front fan stages. Source: Reproduced by permission of United Technologies Corporation, Pratt & Whitney**

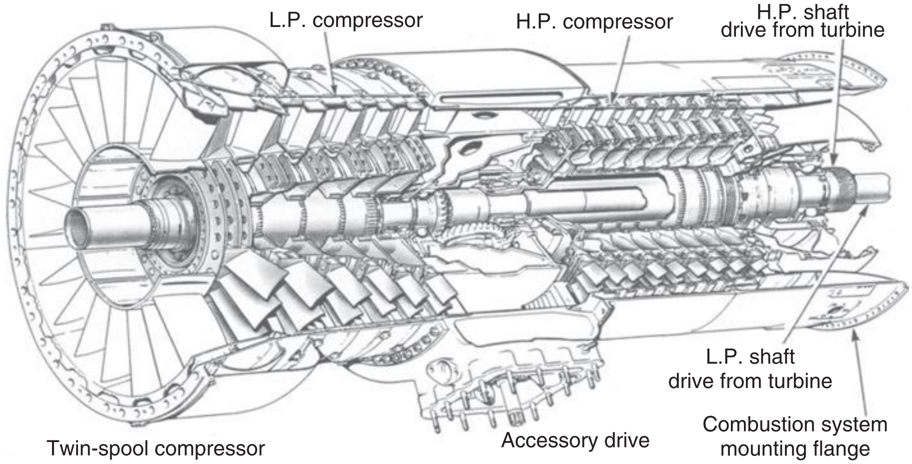
8.13 Isometric and Cutaway Views of Axial-Flow Compressor Hardware

To meet the real world, engineers need to spend time with the real hardware. Isometric and cutaway views of the hardware provide for a three-dimensional feel but may not be a substitute for the cutaway of a real engine (or its components) in the laboratory. The real engine, albeit old and surplus, shows real fasteners, components assembly, manufacturing tolerances; some operational degradation, for example, wear, on the rotor blade tips, shows labyrinth seals between rotating and stationary parts; it shows signs of erosion in the compressor and turbine and perhaps corrosion in the hot section. A visual inspection and a feel of the compressor and turbine blades show the turbine blades receive heavy deposits from the combustor (and its fuel additives) and give a new meaning to "surface roughness," which will not feel "hydraulically smooth."

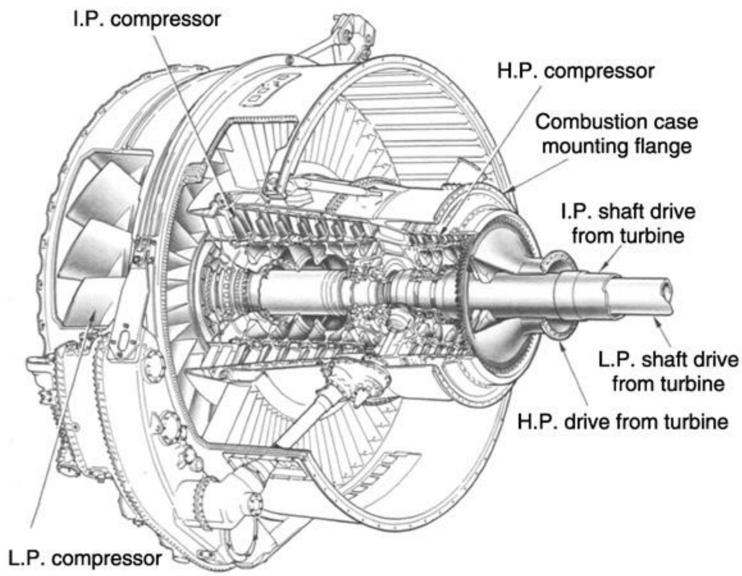
Having pointed out some of the advantages of real hardware, let us examine some isometric and cutaway views of axial-flow compressors, designed, built, tested, and flown on aircraft in this section. Figure 8.81 is taken from a manual by Pratt & Whitney Aircraft (1980), whereas Figures 8.82–8.84 are taken from Rolls-Royce's *The Jet Engine* (2005). Figure 8.81 shows a two-stage fan and the low-pressure compressor rotor. Mid-span shrouds prevent the first bending mode of the first fan rotor blades. The anti-icing air valve and the actuator point to an operational need of the aircraft (i.e., flight under icing condition) and demands on the compressor air.

8.14 Compressor Design Parameters and Principles

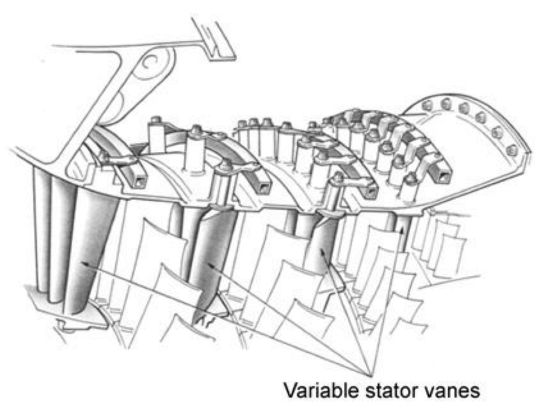
In this section, we present design guidelines that are useful in the preliminary design of axial-flow compressors. The approach to turbomachinery design in textbooks is neither unique nor exact. In general, we use the lessons learned from our predecessors and the



■ **FIGURE 8.82**
A twin-spool compressor. Source: The Jet Engine, 2005. Reproduced by permission from The Jet Engine, Copyright Rolls-Royce plc 2005



■ **FIGURE 8.83**
A triple-spool compressor. Source: The Jet Engine, 2005. Reproduced by permission from The Jet Engine, Copyright Rolls-Royce plc 2005



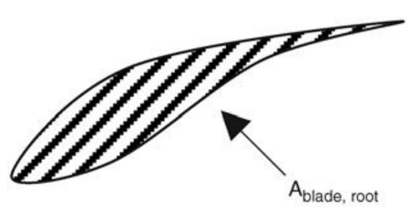
■ **FIGURE 8.84**
Typical variable-stator vanes. Source: The Jet Engine, 2005. Reproduced by permission from The Jet Engine, Copyright Rolls-Royce plc 2005

invaluable contributions of NASA and open literature, which often come from academia and industry. One possible approach to compressor design is outlined in steps that are particularly useful for students who want to learn and practice their turbomachinery design skills.

We propose to design an axial-flow compressor for a design-point mass flow rate and pressure ratio. Here, we first review the steps and in the next section we design an axial compressor at the pitchline. We start the design process by choosing

1. Axial Mach number at the compressor/fan face is M_{z1} ~0.5
2. Flow area at the compressor face is $A_1 = \pi r_{t1}^2 - (r_{h1}/r_{t1})^2$ that sizes the engine and thus establishes the mass flow rate
3. Typical hub-to-tip radius ratio, r_{h1}/r_{t1} is ~0.4, a little less or more (may be up to ~0.5 or as low as ~0.3). The relative tip Mach number M_{r1} establishes the tip radius
4. $M_{r1} = \sqrt{M_{z1}^2 + M_{T1}^2}$
   - M_{T1}, tip tangential Mach number, that is, $(\omega r_t)/a_1$ is in excess of sonic
   - M_{r1} is supersonic, ~1.2 to ~1.5 (remember that M_z ~0.5)
5. Thickness-to-chord ratio varies from ~10% in the subsonic hub region to ~3% in the supersonic tip region, linearly varying in between
6. The Reynolds number based on chord has to be > 300,000 (in relative frame), at altitude. This establishes the minimum chord length for turbulent boundary layer at all altitudes, the other parameter contributing to the chord length is the blade aspect ratio and bending stresses. Often chord length is several times bigger than that required for the 300,000 Reynolds number at altitude
7. Centrifugal stress at the blade root (i.e., the hub) is calculated based on the simple formula

$$\sigma_{c,root} \equiv \frac{F_{c,\,root}}{A_{blade\cdot root}} \tag{8.200}$$



The centrifugal force is the integral of mv^2/r, which is $m\omega^2 r$, that is,

$$F_{c,\,root} = \int_{r_h}^{r_t} \rho_{blade} \cdot A(r)\omega^2 r\mathrm{d}r \tag{8.201}$$

where ρ_{blade} is blade material density and $A(r)$ is the blade cross-sectional area as a function of span.

$$\sigma_c = \frac{1}{A_h}\int_{r_h}^{r_t} \rho_{\text{blade}} \cdot A_b(r)\omega^2 r\mathrm{d}r \tag{8.202}$$

$$\frac{\sigma_c}{\rho_{\text{blade}}} = \frac{\omega^2}{A_h}\int_{r_h}^{r_t} A_b(r) r\mathrm{d}r = \omega^2 \int_{r_h}^{r_t} \frac{A_b}{A_h} r\mathrm{d}r \tag{8.203}$$

$$\frac{\sigma_c}{\rho_{\text{blade}}} = \omega^2 \int_{r_h}^{r_t} \frac{A_b}{A_h} r\mathrm{d}r \tag{8.204}$$

The blade area distribution along the span, $A_b(r)/A_h$, is known as *taper* and is often approximated to be a linear function of the span. Therefore, it may be written as

$$A_b = A_h - \frac{r - r_h}{r_t - r_h}(A_h - A_t) \rightarrow \frac{A_b}{A_h} = 1 - \frac{r - r_h}{r_t - r_h}\left(1 - \frac{A_t}{A_h}\right) \tag{8.205}$$

We may substitute $A_b(r)/A$ in the integral and proceed to integrate; however, a customary approximation is often introduced that replaces the variable r by the pitchline radius r_m. The result is

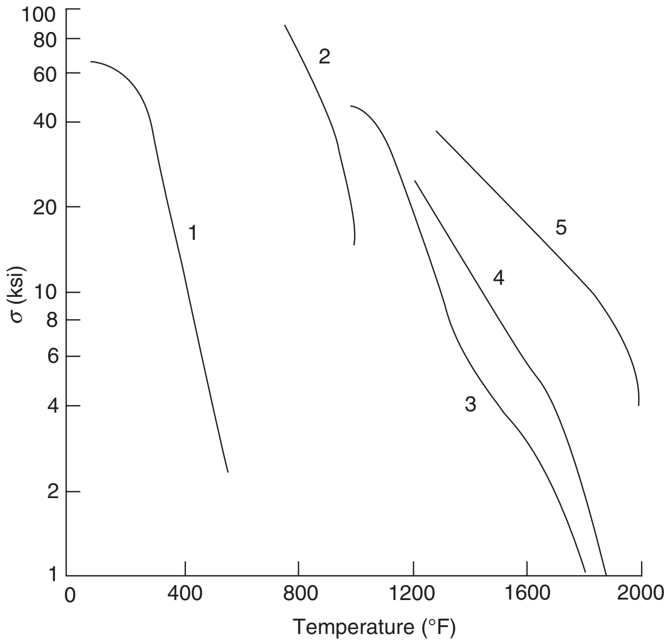
$$\frac{\sigma_c}{\rho_{\text{blade}}} = \frac{\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right) \tag{8.206}$$

The taper ratio A_t/A_r is ~0.8–1.0. Therefore, the ratio of centrifugal stress to the material density is related to the square of the angular speed, the taper ratio, and the flow area, $A = 2\pi r_m(r_t - r_h)$. Equation 8.206 is the basis of the so-called AN^2 rule, where the RHS is related to the size (i.e., A) of the machine and the square of the angular speed (i.e., N^2), and the LHS is related to material property known as specific strength.

The material parameter of interest in a rotor is the *creep ruputure strength*, which identifies the maximum tensile stress tolerated by the material for a given period of time at a specified operating temperature. Based on the 80% value of the allowable 0.2% creep in 1000 h for aluminum alloys and the 50% value of the allowable 0.1% creep in 1000 h for other materials, Mattingly, Heiser, and Pratt (2002) have graphed Figures 8.85 and 8.86. They show the allowable stress and the allowable specific strength of different engine materials as a function of temperature

8. The solidity at the pitchline $\sigma_m \sim 1$ to 2

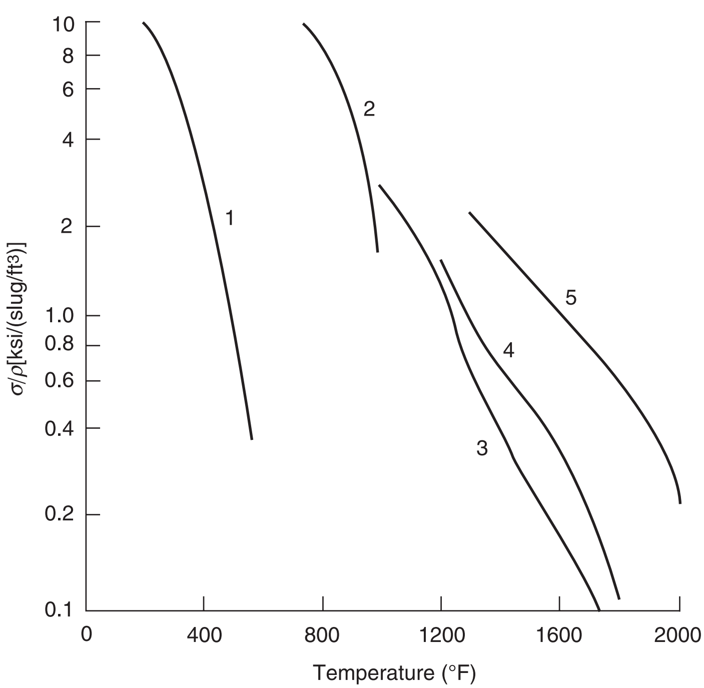
9. The degree of reaction at the pitchline may be chosen to be $°R_m \sim 0.5$ or 0.6 for subsonic sections and considerably higher (e.g., 0.8) for the supersonic portion of the blade. This choice specifies the $C_{\theta 2,m}$. We can use this information along with Euler turbine equation to establish the rotor-specific work at the pitchline



Material	No.
Aluminum alloy	1
Titanium alloy	2
Wrought nickel alloy	3
High-strength nickel alloy	4
Single-crystal superalloy	5

■ **FIGURE 8.85** **Allowable stress versus temperature for typical engine materials. Source: Mattingly, Heiser, and Pratt 2002. Reproduced with permission from AIAA**

■ **FIGURE 8.86** **Allowable strength-to-weight ratio for typical engine materials. Source: Mattingly, Heiser, and Pratt 2002. Reproduced with permission from AIAA**



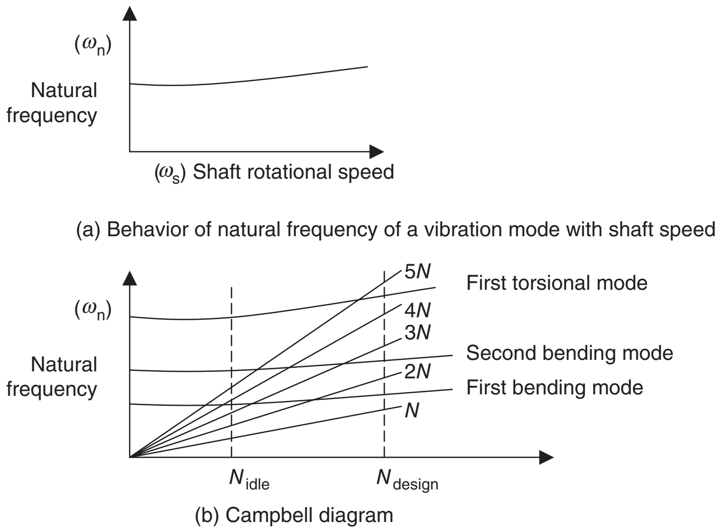
as well as the stagnation temperature rise, that is, we get T_{t2m}. The choice of $°R$ is less critical than D-factor or de Haller criterion

10. The D-factor at the pitchline should be ≤ 0.5 or 0.55, often a quick check on the acceptable level of diffusion is made using the de Haller criterion that limits diffusion to a level corresponding to $W_2/W_1 \geq 0.72$
11. The choice of the vortex design, that is, C_θ(r), establishes the degree of reaction and D-factor at other radii, which we examine closely. It is acceptable for the degree of reaction to deviate from 0.5, but we do not want it to become negative in a compressor (i.e., it then behaves like a turbine!). The D-factor has to remain below the critical value of ~0.5 (or 0.55). We may iterate on the vortex design choice in order to get these parameters right
12. The polytropic efficiency is assumed to be ~0.90, that is, e_c~0.90, which in combination with T_t ratio gives p_t ratio, that is, we get total pressure at the pitchline at station 2, downstream of the rotor
13. To get the flow parameters downstream of the stator, we preserve T_t and calculate p_{t3} based on an assumed ϖ_s of ~0.04–0.05; also the idea of repeated stage gives us the α_3
14. We assume the axial velocity remains constant throughout the compressor (at the design radius, i.e., the pitchline). This is a preliminary design choice
15. Bending stresses on a cantilevered blade, under aerodynamic loading, is estimated (by Kerrebrock, 1992) to be

$$\frac{\sigma_{\text{bending}}}{p} \approx \left(\frac{C_z}{U_t}\right)\left(\tau_s - 1\right)\left(\frac{s}{2c}\right)\left(\frac{r_t}{t_{\max}}\right)^2 \tag{8.207}$$

    Bending stress is, therefore, proportional to gas pressure, flow coefficient at the tip, stage total temperature ratio, and to the square of tip radius-to-thickness ratio. Bending stress is also inversely proportional to solidity
16. Thermal stresses are often small in axial-flow compressors and fans; however, they are significant in the turbine section. Also, the aft stages of high-pressure compressors in modern gas turbine engines operate at high gas temperatures, such as T_{gas} ~850–900 K, which require special attention to thermal strains and tip clearance, as well as total stress (i.e., the sum of centrifugal, bending, and thermal) at the hub
17. Blades as cantilevered structures attain diverse mode shapes, such as first bending, second bending, first torsional, coupled first bending and torsional, as well as higher modes of vibration. Each mode shape has its own natural frequency ω_n. The effect of blade rotation is to *stiffen* the structure and thus raise these natural frequencies. The shaft and the discs also exhibit their natural vibrational mode shapes and frequencies. To avoid resonance between these frequencies and the shaft rotational speed (and its multiples), a frequency diagram, known as *Campbell diagram* (Figure 8.87), is generated to examine possible match points between these frequencies inherent in the compressor, that is, in the natural modes and rotor shaft frequency. Kerrebrock (1992), Wilson and Korakianitis (1998), and Mattingley, Heiser, and Pratt (2002) provide a detailed account of engine structure.

■ **FIGURE 8.87**
Frequency diagram showing possible resonance condition between the dominant vibration modes and multiples of shaft frequency



8.14.1 Blade Design – Blade Selection

For blade design, we need to estimate the following angles:

1. Incidence angle i
2. Deviation angle δ^* $\quad \delta^* = m\varphi/\sqrt{\sigma}$

The incidence angle is estimated at the location of minimum loss, from cascade data that is known as i_{optimum}. Typically, optimum incidence angle is $\sim -5°$ to $+5°$.

Iteration loop
The deviation angle is first estimated from the simple Carter's rule:

$$\delta^* \approx \frac{\Delta\beta}{4\sqrt{\sigma}} \tag{8.208}$$

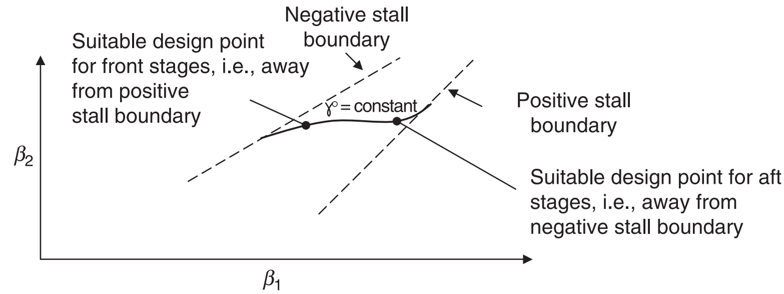
which is based on the net flow turning and the coefficient "m" in Carter's formula is taken to be 0.25.

Also, based on the inlet and exit flow angles, we may use the (NACA 65-series) cascade correlation data of Mellor (Appendix I) to identify a suitable 65-series cascade geometry that gives an adequate stall margin. Note that the front and aft stages of a multistage compressor or fan face different stall challenges; for example, the front stages are more susceptible to positive stall and the aft stages are more susceptible to negative stall. Therefore, we may choose the design point to be farther away from the two stall boundaries for front and aft stages. Figure 8.88 is a definition sketch for the design point selection.

The optimum incidence angle is defined (by Mellor) to be at the point of lowest profile loss, although some authors prefer the optimum incidence to correspond to the angle where lift-to-drag ratio is at its maximum. Here, we adopt Mellor's definition, due to its ease of use. The cascade loss data, as shown in Figure 8.22, can be used to arrive at the i_{opt}.

Johnsen and Bullock (1965) in NASA SP-36 outline optimum incidence estimation based on cascade data correlations for different profile types and profile thicknesses. This reference may be used for more accurate estimation of the incidence angle. From the inlet flow and an estimate of the optimum incidence angle, we calculate the leading-edge

■ **FIGURE 8.88** **Definition sketch for suitable design points for the front and aft stages of a multistage compressor or fan**



angle on the mean camber line (MCL), κ_1, and from the exit flow angle and deviation we calculate the blade angle at the T.E.

$$i \equiv \beta_1 - \kappa_1 \Longrightarrow \kappa_1 \equiv \beta_1 - i$$

$$\delta^* \equiv \beta_2 - \kappa_2 \Longrightarrow \kappa_2 \equiv \beta_2 - \delta^* \Longrightarrow \varphi \equiv \kappa_1 - \kappa_2$$

The iteration loop on the deviation angle may be initiated at this point, since we have calculated a camber angle and we may now use the expanded version of the Carter's rule:

$$\delta^* = m\varphi/\sqrt{\sigma}$$

Although we have introduced a possible iteration loop on the deviation angle, in the preliminary stage of the compressor design it is acceptable to use the simple formula of Carter for deviation.

The supersonic section is designed based on

- Double-circular arc (DCA) blades
- J-profile design
- Multiple-circular arc design (MCA).

These designs are described in Cumpsty (1989), Schobeiri (2004), and Wilson and Korakianitis (1998).

8.14.2 Compressor Annulus Design

The annulus flow area shrinks inversely proportional to the rising fluid density for a constant axial velocity, that is,

$$A(z)/A_1 = \rho_1/\rho(z) \quad \text{for} \quad C_z = \text{constant(say} \sim 500 \text{ ft/s or } 150 \text{ m/s)}$$

We can calculate density rise per stage (at the pitchline) from the pressure and temperature rise and the use of the perfect gas law. This is an approximate method, but is adequate for the preliminary design purposes.

Common practices are

- Keep the casing, that is, the tip, radius constant, therefore the information about the annulus area gives the hub radius
- Keep the hub radius constant and thus calculate the tip radius from the annulus area
- Keep the pitchline radius constant, which then shrinks the tip and hub equally, and we can calculate those from the annulus area

- Keep the axial gap between the rotor and stator blade rows to approximately 0.23–0.25 c_z (i.e., about one quarter axial chord length gap between blade rows).

8.14.3 Compressor Stall Margin

Compressor stall margin may be established based on the use of Koch equivalent diffuser model of the compressor blade passage. This is rather laborious, but produces good results. We start with the equivalent enthalpy rise coefficient C_h (for notation, see section 8.10).

Note that the difference between U_1 and U_2 across the rotor (in the numerator) is due to streamline shift in the radial direction. We further correct for Reynolds number, tip clearance gap, and the axial blade spacing other than the nominal values that are used in Koch's stall margin correlations. The correction factors are in Figures 8.64–8.66. Finally, we correct the adjusted enthalpy rise coefficient for the effect of stagger and the velocity vector diagram (wake effect) to arrive at the *effective* static enthalpy rise coefficient according to

$$(C_h)_{ef} = (C_h)_{adj}\left[\frac{V_1'^2\big|_{\text{Rotor}} + V_1^2\big|_{\text{Stator}}}{V_1'^2\big|_{\text{Rotor}} + F_{ef}\cdot V_1^2\big|_{\text{Stator}}}\right]$$

The critical stalling pressure rise is shown in Figure 8.72 as "stall line" or the case of 0% stall margin.

Guidelines on the Range of Compressor Parameters

Parameter	Range of values	Typical value
Flow coefficient ϕ	$0.3 \le \phi \le 0.9$	0.6
D-Factor	$D \le 0.6$	0.45
Axial Mach number M_z	$0.3 \le M_z \le 0.6$	0.55
Tip Tangential Mach Number, M_T	1.0–1.5	1.3
Degree of reaction	$0.1 \le {}^\circ R \le 0.90$	0.5 (for $M < 1$)
Reynolds number based on chord	$300{,}000 \le Re_c$	>500,000
Tip relative Mach number (1st Rotor)	$(M_{1r})_{tip} \le 1.7$	1.3–1.5
Stage average solidity	$1.0 \le \sigma \le 2.0$	1.4
Stage average aspect ratio	$1.0 \le AR \le 4.0$	<2.0
Polytropic efficiency	$0.85 \le e_c \le 0.92$	0.90
Hub rotational speed	$\omega r_h \le 380$ m/s	300 m/s
Tip rotational speed	ωr_t~450—550 m/s	500 m/s
Loading coefficient	$0.2 \le \psi \le 0.5$	0.35
DCA blade (range)	$0.8 \le M \le 1.2$	Same
NACA-65 series (range)	$M \le 0.8$	Same
De Haller criterion	$W_2/W_1 \ge 0.72$	0.75
Blade leading-edge radius	$r_{L.E.}$ ~5–10% of t_{max}	5% t_{max}
Compressor pressure ratio per spool	$\pi_c < 20$	up to 20
Axial gap between blade rows	0.23 c_z to 0.25 c_z	0.25 c_z
Aspect ratio, fan	~2–5	<1.5
Aspect ratio, compressor	~1–4	~2
Taper ratio	~0.8–1.0	0.8

EXAMPLE 8.8

Design an axial-flow compressor with the following design point parameters:

1. Takeoff at ambient condition

$$T_0 = 25°\text{C}\,(298\,\text{K}), \quad p_0 = 101\ \text{kPa},$$
$$\gamma = 1.4, \quad c_p = 1004\,\text{J/kg}\cdot\text{K}, \quad M_0 = 0$$

2. Overall compressor pressure ratio is 20
3. Polytropic efficiency is assumed constant along the compressor, $e_c = 0.90$
4. Mass flow rate is 100 kg/s

Design Choices

No IGV
Repeated stage
Constant axial velocity C_z

Calculate

Compressor
annulus geometry at the inlet and exit
number of stages and stage pressure ratio

First Stage
Reynolds number based on chord
blade section design at the pitchline

SOLUTION

The design axial Mach number at the compressor face is assumed to be

$$M_{z1} = 0.5$$

The static temperature at the engine face is

$$T_1 = T_{t1}/[1 + (\gamma - 1)M_1^2/2]$$
$$= 298\ \text{K}[1 + 0.2(0.5)^2] \cong 283.8\ \text{K}$$
$$a_1 = \sqrt{(\gamma - 1)c_{p1}T_1} = 337.6\ \text{m/s}$$

Therefore, the axial velocity is $C_{z1} \approx 168.8$ m/s.

The static pressure at the engine face is

$$p_1 = p_{t1}/[1 + (\gamma - 1)M_1^2/2]^{\gamma/(\gamma-1)}$$
$$= 101\ \text{kPa}/[1 + 0.2(0.5)^2]^{3.5} \cong 85.14\ \text{kPa}$$

The static density at engine face is

$$\rho_1 = p_1/RT_1$$

The gas constant $R = (\gamma - 1)c_p/\gamma$, which gives $R = 286.86$ J/kg · K and the static density at the engine face is

$$\rho_1 \approx 1.046\,\text{kg/m}^3$$

The continuity equation for steady, uniform flow is

$$\dot{m} = \rho AV = \rho_1 A_1 C_{z1}$$

We calculate the flow area at the engine face

$$A_1 = (100\,\text{kg/s})/(1.046\,\text{kg/m}^3)/(168.8\ \text{m/s})$$
$$\approx 0.56644\ \text{m}^2$$

The flow area at the engine face may be written in terms of the tip radius and hub-to-tip radius ratio, according to

$$A_1 = \pi r_{t1}^2\left[1 - \left(\frac{r_{h1}}{r_{t1}}\right)^2\right]$$

Here, we enter our next assumption, i.e., we choose the hub-to-tip radius ratio at the first stage to be

$$\frac{r_{h1}}{r_{t1}} = 0.5$$

This assumption along with the area A_1 gives the tip radius r_{t1} from the above equation, i.e., we calculate tip radius at the engine face to be

$$r_{t1} = \sqrt{A_1/\pi\left[1 - \left(\frac{r_{h1}}{r_{t1}}\right)^2\right]} \cong 0.49\ \text{m}$$

Obviously, our hub radius is half the tip radius by design choice, i.e.,

$$r_{h1} \cong 0.245\ \text{m}$$

Now, let us calculate the compressor discharge condition. The exit total pressure is 20 times the inlet total pressure therefore,

$$p_{\mathrm{t,exit}} = 20(101 \text{ kPa}) = 2{,}020 \text{ kPa}$$

The compressor temperature and pressure ratio are related via the polytropic efficiency following

$$\tau_{\mathrm{c}} = \pi_{\mathrm{c}}^{(\gamma-1)/e_{\mathrm{c}}\gamma}$$

Therefore, $\tau_{\mathrm{c}} \approx 2.5884$, which gives the discharge total temperature to be

$$T_{\mathrm{t,exit}} = 771.3\,\mathrm{K}$$

We assume the compressor discharge is swirl free and since we had also assumed the axial velocity to be constant along the compressor, we know that

$$C_{\mathrm{exit}} = C_{\mathrm{z,exit}} = \text{Constant} = 168.8 \text{ m/s}$$

We calculate the static temperature of the gas at compressor discharge, T_{exit}, according to

$$T_{\mathrm{exit}} = T_{\mathrm{t,exit}} - C_{\mathrm{exit}}^2/2c_{\mathrm{p,exit}}$$

Assuming a calorically perfect gas, $c_{\mathrm{p}} = 1004$ J/kg · K = constant and, therefore,

$$T_{\mathrm{exit}} = 757.14\,\mathrm{K}$$

The speed of sound at the exit is

$$a_{\mathrm{exit}} = \sqrt{(\gamma-1)c_{\mathrm{p}}T_{\mathrm{exit}}} \cong 551.4 \text{ m/s}$$

Thus, the exit Mach number at the compressor discharge is

$$M_{\mathrm{exit}} = 168.8/551.4 \approx 0.306$$

Note that our choice of constant axial velocity led to a reduction of flow Mach number at the compressor discharge, since the gas temperature and thus speed of sound increases along the compressor, which cause the Mach number to drop. This is a good situation for the burner, since we have to decelerate the compressor discharge gas to around Mach 0.2 for efficient combustion. The prediffuser in the burner has to decelerate the gas from Mach ~0.3 to ~0.2.

From M_{exit} and $p_{\mathrm{t,exit}}$ we get p_{exit}

$$p_{\mathrm{exit}} = p_{\mathrm{t,exit}}/[1 + (\gamma-1)M_{\mathrm{exit}}^2/2]^{\gamma/(\gamma-1)}$$
$$= 2020 \text{ kPa}/[1 + 0.2(0.306)^2]^{3.5} \cong 1{,}892.9 \text{ kPa}$$

We calculate the exit density ρ_{exit} from the exit temperature and pressure, from perfect gas law

$$\rho_{\mathrm{exit}} \approx 1892.9 \text{ kPa}/[(286.86\,\mathrm{J/kg\cdot K})(757.14\,\mathrm{K})]$$
$$\rho_{\mathrm{exit}} \approx 8.715\,\mathrm{kg/m^3}$$

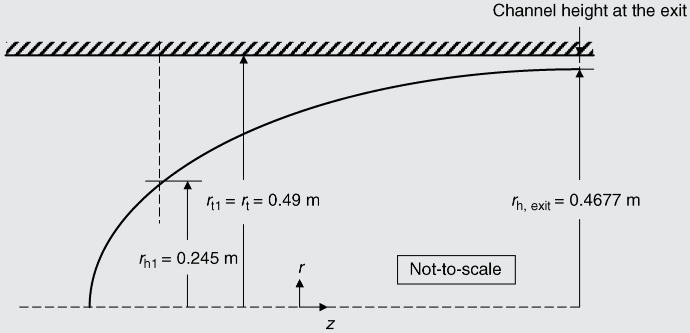
From continuity equation, we get the exit flow area,

$$A_{\mathrm{exit}} = \frac{\dot{m}}{\rho_{\mathrm{exit}}C_{\mathrm{z}}} = \frac{100 \text{ kg/s}}{8.715 \text{ kg/m}^3(168.8 \text{ m/s})} \cong 0.06797 \text{ m}^2$$

Assuming a constant tip radius r_{t} = constant, we calculate the hub radius at the compressor exit to be

$$r_{\mathrm{h,exit}} = \sqrt{r_{\mathrm{t}}^2 - A_{\mathrm{exit}}/\pi}$$
$$= \sqrt{(0.49)^2 - (0.06797)/3.14159265} \approx 0.4677 \text{ m}$$

Note that the channel height (or blade height) at the compressor exit is 0.02258 m ~ 2.2 cm. If we had shrunk the casing, i.e., the tip radius, instead of keeping it constant, the blade height at the exit would increase.



Here, we may do a quick trade study of the effect of outer casing shrinkage on the blade height in the last stage. The annulus area may be written in terms of the pitchline radius and the channel height as

$$A = 2\pi r_{\mathrm{m}}(r_{\mathrm{t}} - r_{\mathrm{h}})$$

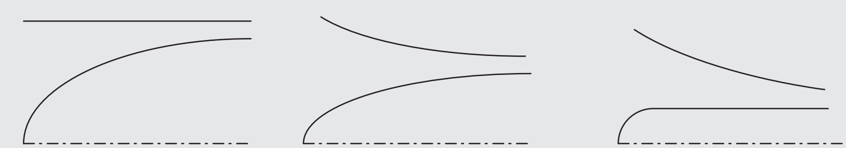
For the exit flow area of 0.06797 m^2, we have the product of pitchline radius and the blade height as

$$r_{\mathrm{m}} \cdot h_{\mathrm{blade}} \cong 0.010818\ \mathrm{m}^2$$
$$h_{\mathrm{blade}} \cong 0.010818 m^2/r_m$$

Therefore, the blade or channel height is inversely proportional to the pitchline radius. Here we examine three common choices:

(a) the constant tip radius, which resulted in h_{blade} of ~2.2 cm

(b) the constant pitchline radius $r_{\mathrm{m}} = (r_{\mathrm{h}} + r_{\mathrm{t}})/2 = 0.3677$ m, which gives h_{blade} of ~2.94 cm


(a) Constant tip radius (b) Constant pitchline radius (c) Constant hub radius

(c) the constant hub radius, gives an exit channel/blade height of $h_{\mathrm{blade}} \sim 4.07$ cm

We note that the increase in blade height between cases (a) and (b) is small, whereas the blade height nearly doubles in case (c) as compared to (a).

Pitchline calculations
Pitchline radius at the compressor face is at $r_{\mathrm{m1}} = (r_{\mathrm{h1}} + r_{\mathrm{t1}})/2 \cong 0.3677$ m

The shaft rpm is selected to give $U_{\mathrm{tip}} = 450$ m/s, which is very desirable for transonic compressor pressure ratio. The choice of shaft rpm and the blade geometry will be tested against the centrifugal stress calculations at the blade root.

But, for now we choose U_{tip} and then establish shaft rpm.

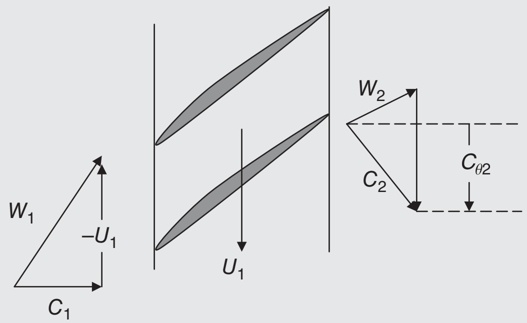
$$U_{\mathrm{tip}} = \omega r_{\mathrm{t}}, \text{ which gives}$$
$$\omega = 450\ \mathrm{m/s}/0.49\ \mathrm{m} = 918.4\ \mathrm{rad/s} \cong 8{,}770\ \mathrm{rpm}$$

Therefore, the rotor speed at pitchline is

$$U_{\mathrm{m}} = U_{\mathrm{tip}}(r_{\mathrm{m}}/r_{\mathrm{t}}) = (450\ \mathrm{m/s})(0.3677/0.49)$$
$$\approx 337.5\ \mathrm{m/s}.$$

The flow angles at the inlet to the rotor blade are

$$\alpha_1 = 0 \qquad \text{no IGV (design choice)}$$
$$\beta_1 = \tan^{-1}(337.5/168.8) \approx 63.43°$$



The relative velocity is $W_{\mathrm{m1}} \approx 377.4$ m/s, which also gives the relative inlet Mach number at the pitchline of

$$M_{1\mathrm{r,m}} = 377.4/337.6 = 1.12 \qquad \text{[pitchline is supersonic]}$$

In order to establish the flow angles at the rotor exit, we may choose the degree of reaction at the pitchline to be 0.5, i.e.,

$$°R_{\mathrm{m}} = 0.5 \quad 1^{\mathrm{st}} \text{ Design choice on } °R_{\mathrm{m}}$$

This choice immediately gives $C_{\theta 2\mathrm{m}}$, since

$$°R_{\mathrm{m}} = 1 - \frac{C_{\theta 1,\mathrm{m}} + C_{\theta 2,\mathrm{m}}}{2U}$$

The choice of degree of reaction of 1/2 will make $C_{\theta 2,\mathrm{m}} = U_{\mathrm{m}}$ and thus we are asking the rotor relative exit flow to be purely axial, i.e., $\beta_{2\mathrm{m}} = 0$.

The problem with this choice is that the net flow turning at the pitchline is ~63°, which means the blade needs too much camber and thus the excessive diffusion will cause boundary layer to separate. This should be evident from our calculation of the D-factor. Let us see if that fact is borne by our D-factor calculation.

$$D_{\text{r,m}} = 1 - \frac{W_{2\text{m}}}{W_{1\text{m}}} + \frac{|\Delta C_\theta|}{2\sigma_\text{m} W_{1\text{m}}}$$

$$W_{2\text{m}} = C_\text{z} = 168.8 \text{ m/s}$$

$$W_{1\text{m}} = [(168.8)^2 + (337.5)^2]^{1/2} = 377.4 \text{ m/s}$$

$$\Delta C_\theta = 337.5 \text{ m/s}$$

Let the solidity at the pitchline be $\sigma_\text{m} = 1$ (design choice), therefore we calculate the D-factor at the pitchline to be

$$D_{\text{r,m}} = 1.053 \qquad \text{[Unacceptable]}$$

As we suspected, the amount of turning is not tolerable by the boundary layer and it will stall. The D-factors above 0.55 indicate that the boundary layer is on the verge of stall. We are also guided by de Haller criterion that limits the amount of diffusion in the blade, i.e.,

$$\text{De Haller criterion is } \frac{W_2}{W_1} \geq 0.72$$

Using de Haller criterion, we arrive at the exit relative velocity at the pitchline of

$$W_{2\text{m}} = 271.7 \text{ m/s}$$

Therefore, $\beta_2 = \cos^{-1}(168.8/271.7) \cong 51.6°$

The relative swirl at the rotor exit is

$$W_{\theta 2\text{m}} = C_{z2} \tan\beta_{2\text{m}} \cong 212.9 \text{ m/s}$$

The loading parameter at the pitchline is

$$\psi_\text{m} = 124.6/337.5 \approx 0.369$$

The absolute swirl is $C_{\theta 2\text{m}} \approx (337.5 - 212.9)$ m/s = 124.6 m/s,

The absolute flow angle at the rotor exit at the pitchline radius is

$$\alpha_{2\text{m}} = \tan^{-1}(124.6/168.8) \approx 36.4°$$

The absolute velocity at the rotor exit is $C_{2\text{m}} = (C_z^2 + C_{\theta 2m}^2)^{1/2} = 209.8$ m/s

The stator thus has to turn the flow 36.4° and decelerate the flow to the axial velocity of 168.8 m/s if we are to achieve a repeated stage design. Where does this deceleration fall with respect to the de Haller criterion?

$$C_3/C_2 = 168.8/209.8 = 0.805$$

This clearly meets the De Haller criterion. Now, let us supplement our knowledge of rotor and stator diffusion by calculating their D-factors.

$$D_{\text{r,m}} = 1 - \frac{W_{2\text{m}}}{W_{1\text{m}}} + \frac{|\Delta C_\theta|}{2\sigma_{\text{r,m}} W_{1\text{m}}} \cong 1 - \frac{271.7}{377.4} + \frac{124.6}{2(1)(377.4)} \cong 0.445$$

$$D_{\text{s,m}} = 1 - \frac{C_{3\text{m}}}{C_{2\text{m}}} + \frac{|\Delta C_\theta|}{2\sigma_{\text{s,m}} C_{2\text{m}}} \cong 1 - \frac{168.8}{209.8} + \frac{124.6}{2(1.5)(209.8)} \cong 0.433$$

In the rotor calculation we chose the pitchline solidity of $\sigma_{\text{r,m}} = 1$, whereas for the stator we chose a pitchline solidity of $\sigma_{\text{s,m}} = 1.25$ (see the stator blade design section where this choice becomes clear). These are the kind of design choices that we can make in early stages of compressor preliminary design. We may revisit them at any point in our design process.

The pitchline degree of reaction is $°R_\text{m} = 1 - (C_{\theta 1\text{m}} + C_{\theta 2\text{m}})/2U_\text{m} = 1 - 124.6/2(337.5) = 0.815$

Rotor blade design at pitchline

We choose a double-circular arc blade for the supersonic section at the pitchline. The relative flow in supersonic blade sections is tangent to the upper surface. Also, we have to choose a blade thickness-to-chord ratio at the pitchline. The minimum t/c is at the tip, and for structural reasons it is limited to 3%. The hub, which operates in the subsonic relative flow, may be given a t/c of 10%. If we choose these t/cs as our design choices, and then impose a linear variation of the t/c along span, we arrive at the pitchline t/c of 6.5%.

Before we embark on the blade section design, we may establish the minimum chord length that is required for the turbulent boundary layer formation on the blade. The criterion of $\text{Re}_\text{c} > 300{,}000$ is clearly a useful guideline, but we have to satisfy that at all altitudes. In the low-density flight associated with the highest flight altitude in the flight envelope, we calculate the required chord length.

At 16-km altitude; the air density is ~0.136 times the sea level density. Therefore, at the engine face we will have

$$\rho_1 \approx (0.136)(1.046 \text{ kg/m}^3) \cong 0.142 \text{ kg/m}^3$$

Also, the kinematic viscosity ν at 16-km altitude is ~5.849 times the kinematic viscosity of air at sea level.

$$Re_c = \frac{W_{1m} \cdot c_m}{\nu_1} = \frac{377.4\,(m/s) \cdot c_m}{8.54 \times 10^{-5} m^2/s} \geq 300{,}000 \Longrightarrow c_m > 6.8\ \text{cm}$$

We may expand the chord length beyond this minimum for the following reasons:

- blade structural calculations (demanding a wider chord)
- blade aspect ratio (pointing to the advantages of a wide-chord design in stall margin)
- blade vibrational modes (from Campbell diagram)

The relative inlet flow is aligned with the upper surface, therefore, the incidence angle may be written as

$$i \approx \frac{\theta_{L.E.}}{2} \approx \left(\frac{t_{max}}{c}\right) \Longrightarrow i \approx 3.7°$$

The deviation angle may be estimated form the Carter's basic rule and then supplemented by 2°, since the effect of shock boundary layer interaction is to thicken the boundary layer on the suction surface, thus, causing a higher deviation angle, i.e.,

$$\delta^* \approx \frac{\Delta\beta}{4\sqrt{\sigma}} + 2^o \Longrightarrow \delta^* \approx 5°$$

The blade leading edge angle $\kappa_{1,m} = \beta_{1,m} - i$

$$\Longrightarrow \kappa_{1,m} \approx 59.7°$$

The blade trailing-edge angle $\kappa_{2,m} = \beta_{2,m} - \delta^*$

$$\Longrightarrow \kappa_{2,m} \approx 46.6°$$

Therefore, the camber angle is $\varphi_m = \kappa_{1,m} - \kappa_{2,m}$

$$\Longrightarrow \varphi_m \approx 13.1°$$

The stagger angle, for a double-circular arc blade, is related to the inlet flow angle, camber, and incidence angle according to

$$\beta_1 = \gamma^o + \frac{\varphi}{2} + i \Longrightarrow \gamma° \approx 60.6°$$

The number of rotor blades may now be calculated since we have chosen the solidity at the pitchline and selected the chord length. The blade spacing at the pitchline is

$$s_{r1,m} = \frac{2\pi r_{m1}}{N_{R1}}$$

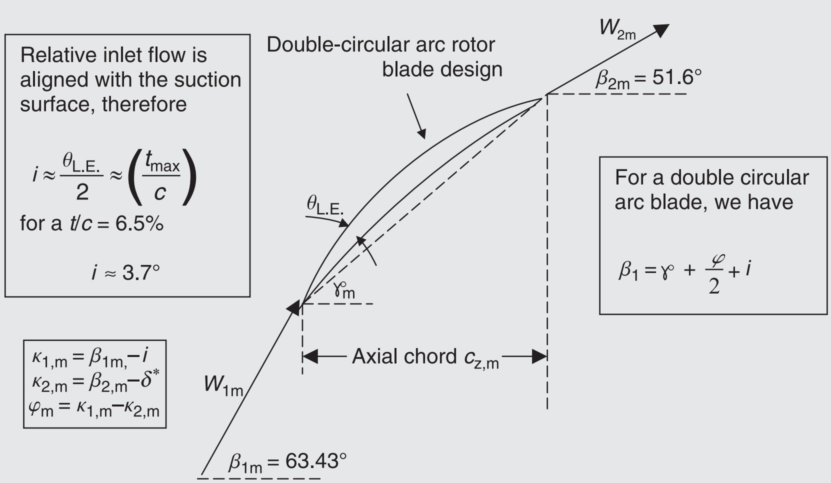
We also chose the rotor solidity of 1 at the pitchline, which gives

$$\sigma_{r1,m} = \frac{c_{r1,m}}{s_{r1,m}} = 1.0$$

Therefore, the number of rotor blades is $N_{R1} = 2\pi r_{m1}/c_{rm,1}$

For a rotor chord length of 7 cm (or 0.07 m) and the pitchline radius of 0.3677 m, we get the number of rotor blades in the first stage to be

$$N_{R1} = 33$$



Stator blade design at pitchline

The design of the stator blade at the pitchline follows the flow angles and Mach numbers in the absolute frame, i.e., we know that

$$\alpha_2 = 36.4°$$

$$\alpha_3 = 0° \quad \text{[repeated stage]}$$

To calculate the Mach number at the entrance to the stator, we need to calculate the static temperature and speed of sound at the pitchline radius downstream of the rotor.

From Euler turbine equation $T_{t2,m} = T_{t1} + U_m(C_{\theta 2} - C_{\theta 1})/c_p$

We made the assumption of no radial shift across the rotor and a calorically perfect gas. These assumptions or approximations are reasonable at the preliminary stage of design.

$$T_{t1} = 298\,\text{K}, U_m = 337.5\ \text{m/s}, C_{\theta 1} = 0,$$

$$C_{\theta 2} = 124.6\ \text{m/s}, c_p = 1{,}004\,\text{J/kg}\cdot\text{K}$$

Substitute these values in the Euler equation to get $T_{t2,m} = 339.9\,\text{K}$

Since the static temperature and total temperature are related via $T = T_t - C^2/2c_p$, we calculate the static temperature upstream of the stator at the pitchline radius

$$T_{2,m} \approx 318\,\text{K with a corresponding } a_{2,m} = 357.3\ \text{m/s}$$

$$M_{2,m} = 209.8/357.3 \approx 0.587\ \text{(subsonic stator flow)}$$

$$C_{2,m} = 209.8\ \text{m/s}$$

The stator blade at the pitchline is a subsonic section. Therefore, we may choose NACA-65 series cascade profiles from the data of Mellor (see Appendix I for a complete description) that

(a) best match the inlet and exit flow angles

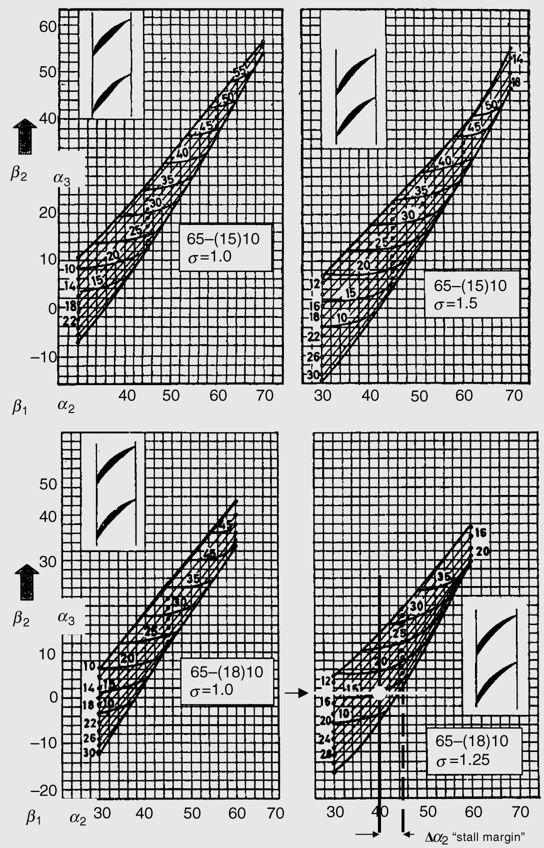
(b) provide for a reasonable "stall margin"

We choose the **65-(18)10** cascade with a solidity of $\sigma = 1.25$, where the intersection of the inlet and exit flow angles yield (note that the inlet and exit angles to the stator are α_2 and α_3 respectively)

$$\gamma° \approx 15°$$

Angle of attack, $\alpha_2 - \gamma° \approx 22°$ [which is consistent with our $\alpha_2 = 36.7°$ and the stagger of 15°]

We note a tolerance to inlet flow angle variation of ~6.5° margin to the positive stall boundary. We may use this range



of inlet flow angle variation to arrive at an approximate "stall margin" for the stator blade section at the pitchline in the first stage. A more accurate approach to stall margin calculation is based on Koch's method that we have detailed in this chapter.

Stage pressure ratio and number of stages

The total temperature rise across the first stage, at the pitchline, is

$$\Delta T_t = 41.9\,\text{K}$$

For a repeated stage and assuming the blade rotational speed U remains nearly constant at the pitchline of subsequent stages, we conclude that every stage achieves $\Delta T_t = 41.9$ K.

The total stagnation temperature rise for the compressor was calculated to be

$$\Delta T_{t,\,\text{compressor}} = 771.3\,\text{K} - 298\,\text{K} = 473.3\,\text{K}$$

(assuming a calorically-perfect gas)

If each stage produces 41.9 K total temperature rise, we need 11.29 stages to achieve the needed total temperature rise for the compressor (at the pitchline). Based on the approximations that we have made, e.g., calorically perfect gas and representing the 3D behavior of the compressor by its pitchline, we choose 12 stages for the compressor.

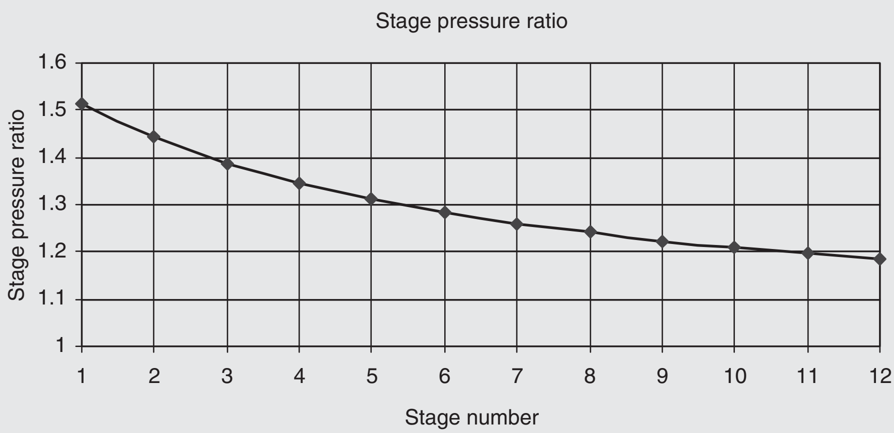
$$N_{\text{stages}} = 12$$

We calculate the stage pressure ratio π_s for each stage, and graph it in the following figure.

$$F_c = \int_{r_h}^{r_t} \rho_{\text{blade}} \cdot A_b(r)\omega^2 r\,dr$$

$$\sigma_c = \frac{1}{A_h}\int_{r_h}^{r_t} \rho_{\text{blade}} \cdot A_b(r)\omega^2 r\,dr$$

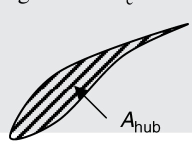
$$\frac{\sigma_c}{\rho_{\text{blade}}} = \frac{\omega^2}{A_h}\int_{r_h}^{r_t} A_b(r) r\,dr = \omega^2 \int_{r_h}^{r_t} \frac{A_b}{A_h} r\,dr$$



Note that the compressor pressure ratio at the pitchline, after 12 stages will be 22.5 (which is >20, the compressor design pressure ratio). The pitchline however does not represent the "average" of the stage pressure ratio. The main reason is that the pitchline is away from the end walls where losses are predominant. Therefore, we expect lower total pressure ratio per stage as that represented by our pitchline calculation. Another note is that the polytropic efficiency e_c should not be treated as constant along a multistage compressor. As compression progresses, the end wall losses grow and thus the slope of ideal-to-actual work (for small stage) drops. The polytropic efficiency may be as high as 0.90 (or even 0.92) for early stages, and as low as 0.85 in the aft stages.

Rotor blade stress calculations
The rotor blade is subject to centrifugal as well as bending, vibrational and thermal stresses. The dominant stress in a rotor however is the centrifugal stress σ_c.

$$\sigma_c \equiv \frac{F_c}{A_{\text{hub}}}$$



$$\frac{\sigma_c}{\rho_{\text{blade}}} = \omega^2 \int_{r_h}^{r_t} \frac{A_b}{A_h} r\,dr$$

The blade area distribution along the span, $A_b(r)/A_h$, is known as *taper* and is often approximated to be a linear function of the span. Therefore, it may be written as

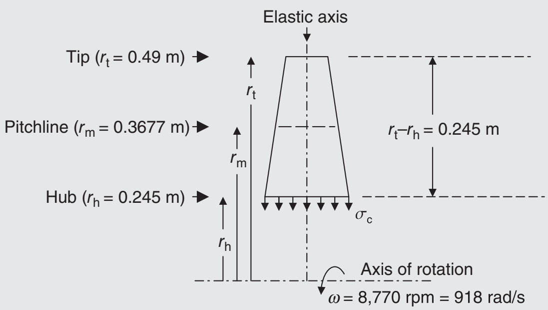
$$A_b = A_h - \frac{r - r_h}{r_t - r_h}(A_h - A_t)$$

$$\frac{A_b}{A_h} = 1 - \frac{r - r_h}{r_t - r_h}\left(1 - \frac{A_t}{A_h}\right)$$

We may substitute $A_b(r)/A$ in the integral and proceed to integrate; however a customary approximation is often introduced that replaces the variable r by the pitchline radius r_m. The result is

$$\frac{\sigma_c}{\rho_{\text{blade}}} = \frac{\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right)$$

Therefore, the ratio of centrifugal stress to the material density is related to the square of the angular speed, the taper ratio, and the flow area $A = 2\pi r_m(r_t - r_h)$.



Titanium alloy is suitable for compressor/fan rotor blades, due to its low density, high strength characteristics, which makes it a high strength-to-weight ratio, or high specific-strength material of choice for rotor blades in a compressor or fan. The density of Titanium alloy is

$$\rho_{\text{titanium}} = 4{,}680\,\text{kg/m}^3$$

The material parameter of interest in a rotor is the *creep rupture strength*, which identifies the maximum tensile stress tolerated by the material for a given period of time at a specified operating temperature. Based on the 80% value of the allowable 0.2% creep in 1000 h, for aluminum alloys and the 50% value of the allowable 0.1% creep in 1000 h for other materials, Mattingly, Heiser and Pratt (2002) have graphed Figures 8.85 and 8.86. They show the allowable specific strength of different engine materials as a function of temperature.

The allowable strength-to-weight ratio for the titanium alloy is noted to be ~9 ksi/slug/ft^3, which is equivalent to

Allowable creep rupture strength	120,560 m^2/s^2 (or ~120 kPa/kg/m^3)
Required strength	68,326 m^2/s^2 (or ~68 kPa/kg/m^3)

8.15 Summary

Axial-flow compressors are highly evolved machines that provide efficient mechanical compression (of the gas) in a gas turbine engine. The mechanical work is delivered to the medium by a set of rotating blade rows. The rotor blades impart swirl to the flow, thereby, increasing the total pressure of the fluid. In between the rotor blade rows are the stator blade rows that remove the swirl from the fluid, thereby increasing the static pressure of the fluid. The combination of one rotor row and one stator blade row is called a compressor stage. Since the primary flow direction is along the axis of the machine, staging the axial-flow compressor is easy. The flow in an axial-flow compressor is exposed to an *adverse pressure gradient* environment, that is, a climbing pressure hill that tends to stall the boundary layer. Therefore, the phenomenon of *stall* is experienced in a compressor. The stalled compressor flow is inherently unsteady and thus may cause a system-wide instability (between the compressor and combustor) known as *surge.*

In analyzing the flow in the rotor, it is most convenient to use the rotating frame of reference that spins with the rotor, which is known as the *relative frame of reference.* The flow in stator blade row is best analyzed by a stationary observer (i.e., a frame of reference that is fixed with respect to the casing) known as the *absolute frame of reference.* The velocity vectors in the two frames of reference form a triangle, described by

$$\vec{C} - \omega r \hat{e}_\theta = \vec{W}$$

The *Euler turbine equation* is known as the fundamental equation in turbomachinery that relates the change of angular momentum $\Delta\ (rC_\theta)$ across a rotor blade row to the rotor specific work via

$$w_c \equiv \frac{\wp_c}{\dot{m}} = \omega\Delta(rC_\theta) \tag{8.9a}$$

If we express the jump in swirl across a rotor in terms of the (rotor) inlet absolute and exit relative flow angles (that remain constant over a wide operating range of the compressor), we deduce from Equation 8.25a that

$$\tau_c - 1 = \left(\frac{U^2}{c_p T_{t1}}\right)\left[1 + \frac{C_{z_2}}{U}\tan\beta_2 - \frac{C_{z_1}}{U}\tan\alpha_1\right]$$

The significance of this equation is the appearance of U^2 in front of the bracket on the RHS, which indicates that the total temperature rise across a rotor is proportional to the square ofthe wheel speed U^2. Therefore a high blade Mach number is desirable, which is exactly the argument for the development of *transonic compressors.* A rule of thumb is that the blade tangential Mach number at the tip is slightly supersonic, $M_T \sim 1.2$–1.3. The structural limitations currently limit the blade tip Mach number to ~1.5 in conventional (subsonic throughflow) compressors. A high strength-to-weight ratio material, such as titanium, is desirable for fan blade construction.

Since we have to limit the level and extent of the flow diffusion in a compressor blade row, we define a *diffusion factor D* for the rotor and stator blade rows according to

$$D_r \equiv 1 - \frac{W_2}{W_1} + \frac{|W_{\theta2} - W_{\theta1}|}{2\sigma_r W_1} \qquad \text{(rotor } D\text{ -Factor)}$$

$$D_s \equiv 1 - \frac{C_3}{C_2} + \frac{|C_{\theta3} - C_{\theta2}|}{2\sigma_s C_2} \qquad \text{(stator } D\text{-Factor)}$$

The experience shows that diffusion factor is to be limited to ~.6 about the mid-span of a blade row and to ~0.4 near the hub or the tip. The lower diffusion imposed on the hub and the tip is due to a complex hub boundary layer/corner vortex formation and the tip clearance flows that dominate the two ends of a blade row. We may view the diffusion factor as a blade-row-specific figure of merit. We define a stage-based figure of merit as well, which is called the *degree of reaction* $°R$,

$$°R \equiv \frac{h_2 - h_1}{h_3 - h_1}$$

Degree of reaction is the ratio of the rise of the static enthalpy across the rotor to that of the stage. In essence, it speaks to the rotor's share of the static pressure rise to that of the stator. Therefore, a 50% degree of reaction stage equally divides the burden of the pressure rise to the rotor and the stator. Although 50% sounds desirable for equal burden, since the rotor blades spin their boundary layers are more stable and thus can withstand higher static pressure rise than the stationary stator blades. Consequently, a 60% degree of reaction may be more optimal than the 50%.

Cascade aerodynamics provides a rich background for a two-dimensional (subsonic) blade design. The minimum-loss incidence angle i_{opt} and the cascade (total pressure) loss bucket are used for the preliminary 2D design of compressor blade sections as well as identifying the *positive and negative stall boundaries.* The limited supersonic cascade data make the preliminary design of the supersonic sections of transonic fans less grounded. The general rules are based in gas dynamics, which indicate the selection of the thinnest possible sections in order to minimize the wave drag. The thinnest (structurally feasible) section is ~3% thick and the subsonic root is often ~10% thick.

Three-dimensional design of blades, in the preliminary stage, is achieved by the so-called *vortex design* approach. Here we introduce a catalog of swirl profiles, such as free-vortex, solid-body rotation, and others that will be anchored at the pitchline and then give us a swirl profile along the span. We applied radial equilibrium theory to calculate an axial velocity profile. Three-dimensional losses are due to secondary flows, tip clearance, and the blade junction corner vortex. The unsteady flow losses are related to upstream wake chopping (by the downstream blade row) and the subsequent vortex shedding in the wake. Losses due to compressibility are minimized by thin profile designs that form the sections of swept blades.

A single-spool compressor is of limited capability in high-pressure compressor applications. The primary reason for that is the differing rotational requirement of the front versus the aft stages. This is the crux of the classical *starting problem* for a high-pressure ratio compressor. Multistage compressors are often driven by different shafts in order to spin the high-pressure stages faster than the low-pressure compressor. Variable stators in the front stages are always adjustable to help with the starting problem as well as the off-design operation of the compressor. A fraction of airflow in the compressor is bled at the intermediate and exit sections to cool the HPT, the casing, and the exhaust nozzle. The cooling fraction is ~10–15% in modern engines.

As an upper bound, multispool, high-pressure ratio compressors can achieve a pressure ratio of ~45–50. The exit temperature of ~900 K is the limit of current materials for an uncooled compressor, which limit the compressor pressure ratio. The advances in computational fluid dynamics, parallel processing, and computer memory have elevated the design of compressors to be based on flow physics and with less reliance on empiricism and cascade data. The result has been the appearance of high-efficiency unconventional transonic blades and stages with *forward and aft swept blades* and *lean.* The advances in integrated manufacturing technology that involve super plastic forming and diffusion bonding (SPF/DB) are used for the manufacturing of modern (composite) wide-chord fan blades. New materials and manufacturing technology, e.g., BLISK, offer weight savings in compressors with the subsequent improvement on the engine thrust-to-weight ratio.

Dixon [7], Cheng *et al.* [2], Prince, Wisler and Hilvers [41], Whitcomb and Clark [53] are recommended for additional reading.

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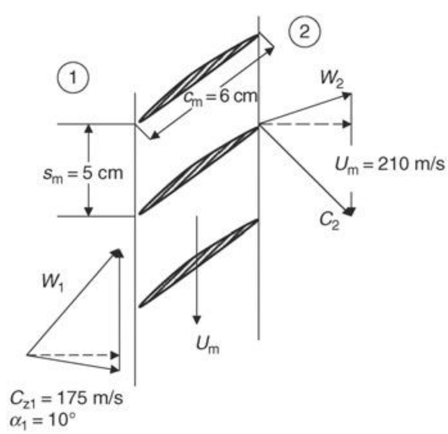
Problems

8.1 The absolute flow at the pitchline to a compressor rotor has a coswirl with $C_{\theta 1} = 78$ m/s. The exit flow from the rotor has a positive swirl, $C_{\theta 2} = 172$ m/s. The pitchline radius is at $r_m = 0.6$ m and the rotor angular speed is $\omega = 5220$ rpm. Calculate the specific work at the pitchline and the rotor torque per unit mass flow rate.

8.2 An axial-flow compressor stage has a pitchline radius of $r_m = 0.6$ m. The rotational speed of the rotor at pitchline is $U_m = 256$ m/s. The absolute inlet flow to the rotor is described by $C_{zm} = 155$ m/s and $C_{\theta 1m} = 28$ m/s. Assuming that the stage degree of reaction at pitchline is $°R_m = 0.50$, $\alpha_3 = \alpha_1$, and C_{zm} remains constant, calculate

- **(a)** rotor angular speed ω in rpm
- **(b)** rotor exit swirl $C_{\theta 2m}$
- **(c)** rotor specific work at pitchline, w_{cm}
- **(d)** relative velocity vector at the rotor exit
- **(e)** rotor and stator torques per unit mass flow rate
- **(f)** stage loading parameter at pitchline, ψ_m
- **(g)** flow coefficient φ_m

8.3 A rotor blade row is cut at pitchline, r_m. The velocity vectors at the inlet and exit of the rotor are shown.



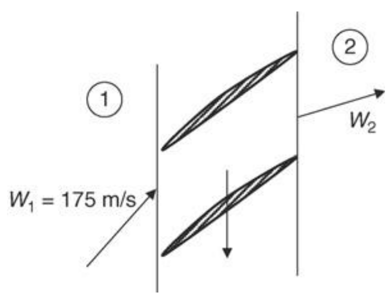
■ **FIGURE P8.3**

Assuming that $U_{m1} = U_{m2} = 210$ m/s and $C_{z1} = C_{z2} = 175$ m/s, $\rho_1 = 1$ g/m^3 $\beta_2 = -25°$, and $\varpi_r = 0.03$, calculate

(a) $W_{\theta 1}$ and $W_{\theta 2}$
(b) W_{1m} and W_{2m}
(c) D-factor D_{rm}
(d) Circulation Γ_m
(e) rotor lift at pitchline per unit span
(f) lift coefficient at pitchline
(g) rotor-specific work at r_m
(h) loading coefficient ψ_m
(i) degree of reaction $°R_m$

8.4 A rotor blade row at the hub radius is shown. The rotor total pressure loss coefficient at this radius is $\varpi_{rm} = 0.04$.

(a) How much deceleration is allowed in the rotor under de Haller criterion? i.e., What is the minimum W_2?
(b) What is the static pressure rise coefficient, assuming incompressible flow and W_{2min} from de Haller criterion?
(c) Compare the C_p in part (b) to the Arbitrary C_{pmax} shown in Figure 8.30 at the hub.



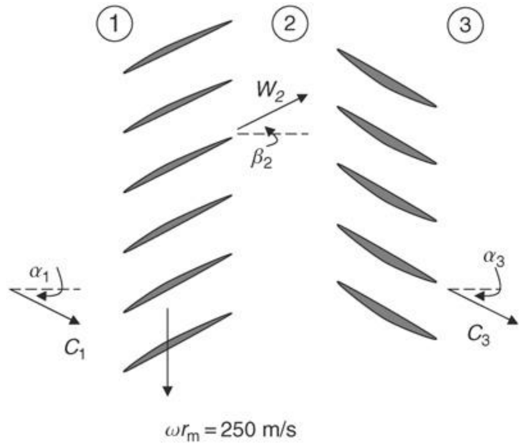
■ **FIGURE P8.4**

8.5 A compressor stage develops a pressure ratio of $\pi_s = 1.6$. Its polytropic efficiency is $e_c = 0.90$. Calculate the stage total temperature ratio τ_s and compressor stage adiabatic efficiency η_s. Assume $\gamma = 1.4$.

8.6 An axial-flow compressor stage is shown at the pitchline. Assuming

$\alpha_1 = \alpha_3$	
$\beta_2 = -30°$	$p_{t1} = 10^5$ Pa
$C_{z1} \cong C_{z2} \cong C_{z3}$	$T_{t1} = 290$ K
$r_1 \cong r_2 \cong r_3$	$M_1 = 0.5$
$\varpi_r = 0.03$	$\alpha_1 = 30°$
$\varpi_s = 0.02$	$\gamma = 1.4$
Calculate	$c_p = 1{,}004$ J/kg · K

(a) w_c (in kJ/kg)
(b) T_{t3}/T_{t1}
(c) M_{2r}
(d) p_{t2}/p_{t1}
(e) p_{t3}/p_{t2}
(f) η_s
(g) $°R_m$



■ **FIGURE P8.6**

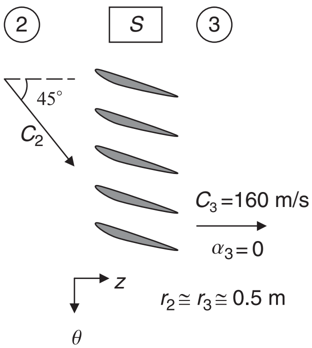
8.7 The flow at the entrance to an axial-flow compressor rotor has zero preswirl and an axial velocity of 175 m/s. The shaft angular speed is 5000 rpm. If at a radius of 0.5 m, the rotor exit flow has zero relative swirl, calculate at this radius

(a) rotor specific work w_c in kJ/kg
(b) degree of reaction $°R$

8.8 The absolute flow angle at the inlet of a stator blade in a compressor is $\alpha_2 = 45°$, as shown. The absolute total pressure and temperature in station 2 are $p_{t2} = 150$ kPa and $T_{t2} = 300$ K, respectively. The total pressure loss coefficient for this section of the the stator blade is $\varpi_s = 0.02$. Assuming the axial velocity remains constant and gas properties are $\gamma = 1.4$ and $c_p = 1.004$ kJ/kg · K, calculate

(a) entrance Mach number M_2
(b) exit total pressure p_{t3}
(c) exit Mach number M_3
(d) stator torque for a mass flow rate of $\dot{m} = 100$ kg/s
(e) static pressure rise, $\Delta p = p_3 - p_2$
(f) static temperature rise $\Delta T = T_3 - T_2$
(g) entropy rise $(s_3 - s_2)/R$

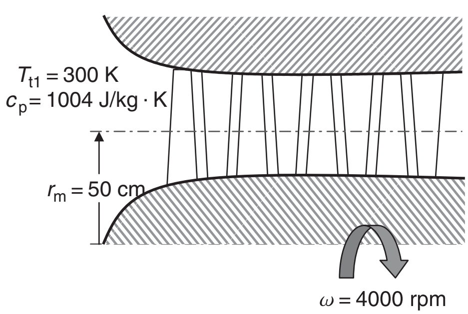
Note that the radius of this cut (section) is at $r \cong 0.5$ m from the axis of rotation, as shown in the diagram.



■ **FIGURE P8.8**

8.9 An axial-flow compressor with four stages is shown. Assuming a repeated stage design, with constant throughflow speed, C_z = 150 m/s, and 50% degree of reaction at the pitch-line with zero preswirl, calculate

(a) rotor specific work at the pitchline (kJ/kg)
(b) stage pressure ratio for an $\eta_s = 0.90$
(c) compressor pressure ratio π_c
(d) shaft power (in MW) for a mass flow rate of $\dot{m}_0$ = 100 kg/s
(e) *D*-factor for the first rotor at the pitchline for σ_r = 2.0



■ **FIGURE P8.9**

8.10 An axial-flow compressor stage is designed on the principle of constant through flow speed. The flow at the entrance to the rotor has 100 m/s of positive swirl and 180 m/s of axial velocity. Assuming we are at the pitchline radius r_m = 0.5 m, where the rotor rotational speed is U_m = 230 m/s, the degree of reaction $°R_m = 0.5$, the radial shift in the stream-tube is negligible, i.e., $r_{1m} \approx r_{2m} \approx r_{3m}$, and also assuming a repeated stage design principle is implemented, calculate

(a) α_{1m} and β_{1m}
(b) α_{2m} and β_{2m}
(c) rotor specific work at the pitchline w_{cm} in kJ/kg
(d) stator torque at the pitchline per unit mass flow rate, $\tau_s/\dot{m}$

8.11 In an axial-flow compressor test rig with no inlet guide vanes, a 1-m diameter fan rotor blade spins with a sonic tip speed, i.e., $U_{tip}/a_1 = 1.0$. If the speed of sound in the laboratory is a_0 = 300 m/s, and the axial velocity to the fan is C_{z1} = 150 m/s, calculate the fan rotational speed ω in rpm.

$$R = 287\ \text{J/kg} \cdot \text{K and } \gamma = 1.4$$

8.12 An axial-flow compressor rotor has an angular velocity of ω = 5000 rpm. The flow entering the compressor rotor has zero preswirl and an axial velocity of C_{z1} = 150 m/s. Assuming the axial velocity is constant throughout the stage, and the rotor specific work at the radius r = 0.5 m is w_c = 62 kJ/kg (γ = 1.4 and R = 287 J/kg · K) calculate

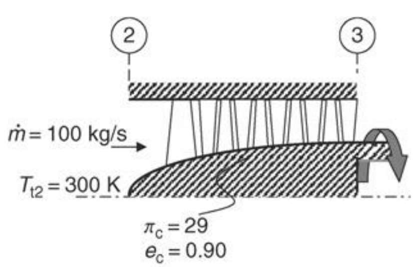
(a) stage degree of reaction, $°R$, at this radius
(b) total pressure ratio across the rotor, p_{t2}/p_{t1}, at this radius, assuming a polytropic efficiency of 90% and T_1 = 20 °C.

8.13 An axial-flow compressor rotor at the pitchline has a radius of r_m = 0.35 m. The shaft rotational speed is ω = 5000 rpm. The inlet flow to the rotor has zero preswirl and the axial velocity is $C_{z1} = C_{z2} = C_{z3}$ = 175 m/s. The rotor has a 50% degree of reaction at the pitchline. The stage adiabatic efficiency is nearly equal to the polytropic efficiency $\eta_s \cong e_c$ = 0.92. Assuming the inlet total temperature is T_{t1} = 288 K and c_p = 1.004 kJ/kg · K, calculate

(a) rotor specific work at $r = r_m$
(b) stage loading ψ at $r = r_m$
(c) flow coefficient at $r = r_m$
(d) rotor relative Mach number at the pitchline, $M_{1r,m}$
(e) stage total pressure ratio at the pitchline

8.14 For the multistage compressor, as shown, calculate

(a) compressor adiabatic efficiency η_c
(b) shaft power $\dot{W}_e$
(c) exit total temperature T_{t3}
(d) average π_s for ten stages, i.e., N = 10
Assume γ = 1.4 and c_p = 1004 J/kg · K.



■ **FIGURE P8.14**

8.15 A compressor test rig operates in a laboratory where θ_2 = 0.95 and δ_2 = 0.95. The mass flow rate is measured at the compressor face to be $\dot{m} = 100$ kg/s and the shaft power is measured to be $\wp_c = 50$ MW. Assuming compressor poly-tropic efficiency is $e_c = 0.90$, calculate

(a) compressor (total) pressure ratio π_c
(b) compressor corrected mass flow rate $\dot{m}_{c2}$
(c) compressor adiabatic efficiency η_c

8.16 A compressor adiabatic efficiency is measured to be $\eta_c = 0.85$ for a compressor total pressure ratio of $\pi_c = 20$. What is the "small-stage" efficiency for this compressor?

8.17 A compressor has a polytropic efficiency of $e_c = 0.92$ and a pressure ratio, $\pi_c = 25$ for an inlet condition of $T_{t1} = 520\ °R$ and $c_p = 0.24$ BTU/lbm · °R, calculate

(a) exit total temperature T_{t2}
(b) compressor adiabatic efficiency η_c
(c) compressor specific work w_c
(d) shaft power $\wp_s$ for a 100 lbm/s flow rate

8.18 A multistage compressor develops a total pressure ratio $\pi_c = 25$, and is designed with eight identical (i.e., "repeated") stages. The compressor polytropic efficiency is $e_c = 0.92$. Calculate

(a) *average* stage total pressure ratio π_s
(b) stage adiabatic efficiency η_s
(c) compressor total temperature ratio τ_c

8.19 Plot compressor adiabatic efficiency η_c versus π_c ranging from 1.0 to 50, for the following polytropic efficiencies $e_c = 0.95$, 0.90, and 0.85 as the running parameter.

8.20 A rotor blade row has a solidity of 1.0 at its pitchline. The absolute flow enters the rotor with no preswirl at $C_{z1m} = 500$ fps, and the rotor rotational speed is $U_m = 1200$ fps (at the pitchline). If the rotor exit flow angle at r_m is $\beta_{2m} = -30°$, calculate

(a) exit swirl velocities $W_{\theta 2m}$ and $C_{\theta 2m}$
(b) blade torque (per unit mass flow rate) at the pitchline assuming $r_m = 1.0$ ft
(c) the nondimensional total temperature rise across the rotor $\Delta T_t/T_{t1}$
(d) exit speed of sound a_2 assuming inlet speed of sound is $a_1 = 1100$ fps
(e) exit absolute Mach number M_{2m}
(f) inlet static pressure p_1 for an inlet total pressure of $p_{t1} = 14.7$ psia
(g) exit total pressure, assuming rotor adiabatic efficiency is 0.9
(h) rotor static pressure rise, $C_{Pm} = \Delta p_m/(1/2\ \rho W_{1m}^2)$
(i) shaft rotational speed ω (rpm)
(j) rotor torque at r_m per unit mass flow rate
(k) the axial force on the blades at r_m, assuming the mean chord $c_m = 4$ in.
(l) tangential force on the blade at r_m
(m) sectional lift-to-drag ratio, L'/D'
(n) rotor specific work at r_m

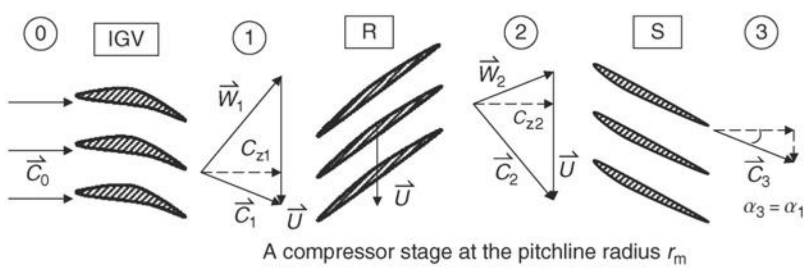
8.21 Calculate the Reynolds number based on chord at the pitchline for the rotor blade described in Problem 8.20, assuming fluid coefficient of viscosity is $\mu = 1.8 \times 10^{-5}$ kg/m · s. Compare the Reynolds number that you calculate to the upper critical Reynolds number in a compressor.

8.22 Calculate the circulation at the pitchline for the rotor blade row described in Problem 8.20. Also calculate the fraction of "ideal" lift that was destroyed by the total pressure losses in the blade row. Is there any indication of shock losses at the pitchline?

8.23 A rotor blade row has a hub-to-tip radius ratio of 0.5, solidity at the pitchline of 1.0, the axial velocity is 160 m/s, and zero preswirl. The mean section has a design diffusion factor of $D_m = 0.5$. Calculate and plot where appropriate

(a) exit swirl at the pitchline assuming the shaft rpm of 6000 and $r_m = 1.0$ ft (0.3 m)
(b) downstream swirl distribution $C_{\theta 2}$ (r) assuming a freevortex design rotor
(c) the radial distribution of degree of reaction °R along the blade span
(d) radial distribution of diffusion factor $D_r(r)$.

8.24 An axial-flow compressor stage is downstream of an IGV that turns the flow 15° in the direction of the rotor rotation, as shown. The axial velocity component remains constant throughout the stage at $C_z = 150$ m/s. The rotor rotational speed is $\omega = 3000$ rpm and the pitchline radius is $r_m = 0.5$ m. The rotor relative exit flow angle is $\beta_2 = -15°$.



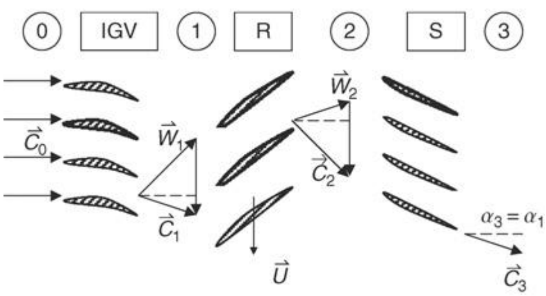
■ **FIGURE P8.24**

The static temperature and pressure of air upstream of the rotor are $T_1 = 20°C$ and $p_1 = 10^5$ Pa, respectively. Assuming $c_p = 1.004$ kJ/kg · K and $\gamma = 1.4$, calculate

(a) relative Mach number to the rotor, M_{1r}
(b) absolute total temperature T_{t1}
(c) rotor specific work w_c in kJ/kg
(d) total temperature downstream of the rotor, T_{t2}
(e) relative Mach number downstream of the rotor, M_{2r}
(f) stage total pressure ratio at the pitchline radius for $e_c = 0.92$
(g) stage degree of reaction $°R_m$ at the pitchline

8.25 A compressor stage with an inlet guide vane is shown at its pitchline radius $r_m = 0.5$ m. The rotor angular speed is $\omega = 4000$ rpm. The axial velocity is constant throughout at $C_z = 150$ m/s and the IGV imparts a preswirl of 75 m/s in the direction of rotor rotation, as shown. Assuming the inlet flow to IGV has $p_{t0} = 100$ kPa and $T_{t0} = 25°C$, calculate

(a) T_0, M_0, p_0
Assuming the IGV has a total pressure loss coefficient of $\varpi_{IGV} = 0.02$, calculate
(b) $p_{t1}, T_{t1}, T_1, M_1, p_1, M_{1r}$ and p_{t1r}
Knowing that the compressor stage has a degree of reaction of $°R = 0.5$ at the pitchline, calculate
(c) $C_{\theta 2}, T_{t2}, T_2, M_2, M_{2r}$
For a rotor total pressure loss coefficient of $\bar{\omega}_r = 0.03$ at the pitchline, calculate
(d) p_{t2}
Assume: $\gamma = 1.4$ and $c_p = 1004$ J/kg · K throughout the stage.



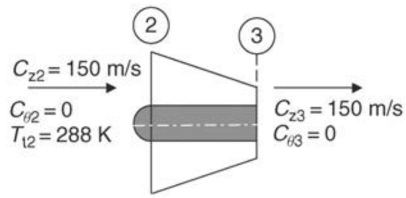
■ **FIGURE P8.25**

8.26 Apply Euler turbine equation to a streamtube that enters a compressor rotor blade row at $r = 2.5$ ft with zero preswirl and exits the row at $r = 2.7$ ft and attains 1000 ft/s of swirl velocity in the absolute frame. Assume that the shaft rotational speed is 5000 rpm.

8.27 Assume that we can analyze a 3D compressor rotor by stacking up its 2D flowfield, what is known as the "strip theory." Now, let us consider one such section. The inlet flow approaches the rotor blade at $\beta_1 = -45°$. The relative exit flow angle is $\beta_2 = -30°$ for a net 15° turning. The solidity of the rotor at this section is 1.5. Identify the most suitable NACA 65-series profile for this section that could produce the largest positive stall tolerance. Estimate the stagger angle that the blades need to be set at this section. To solve this problem, you have to use the cascade data. What is the safe operating range of the incidence angle (or equivalently the inlet flow angles), in degrees, for this section?

8.28 A simple method to establish the annulus geometry in a multistage compressor is to assume a constant through-flow (i.e., axial) speed Cz. Calculate

(a) the density ratio ρ_3/ρ_2
(b) the exit-to-inlet area ratio A_3/A_2



■ **FIGURE P8.28**

for

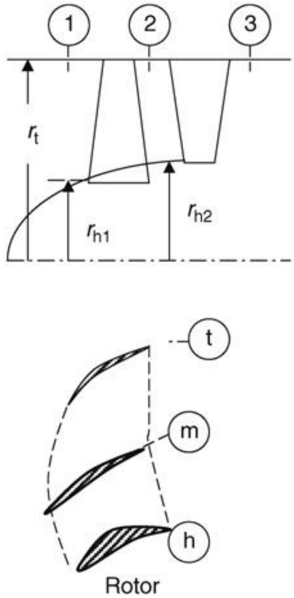
$$\pi_c = 35$$
$$e_c = 0.90$$
$$C_z = \text{constant}$$
$$\gamma = 1.4, c_p = 1{,}004\ \text{J/kg} \cdot \text{K}$$

Also, if the hub radius is constant and for a mass flow rate of 100 kg/s and the inlet total pressure of $p_{t2} = 100$ kPa, calculate the tip-to-tip radius ratio $(r_3/r_2)_{tip}$

8.29 Consider a compressor stage with no inlet guide vane. The hub-to-tip radius ratio for the rotor is $r_{h1}/r_{t1} = 0.5$ and the mass flow rate (of air) in the compressor is 100 kg/s, at the standard sea level condition, $p_{t1} = 100$ kPa and $T_{t1} = 288$ K. For an axial Mach number of $M_z = 0.5$, calculate

(a) the rotor hub and casing radii r_{h1} and r_{t1} (in meters)
To achieve a relative tip Mach number of $(M_{1r})_{tip} = 1.4$, calculate
(b) the rotor rotational speed ω (in radians per second)
For a design pitchline degree of reaction of $°R_m = 0.5$, and constant axial velocity, calculate
(c) the rotor exit swirl at the pitchline radius $C_{\theta 2m}$ (in m/s)
(d) the total temperature T_{t2m} (in K) using Euler equation
(e) the total pressure p_{t2m} (in kPa), assuming $\varpi_{rm} = 0.005$
(f) the fluid density ρ_{2m} in kg/m^3

Now, assume that we chose a free-vortex design for the rotor.



■ **FIGURE P8.29**

Calculate

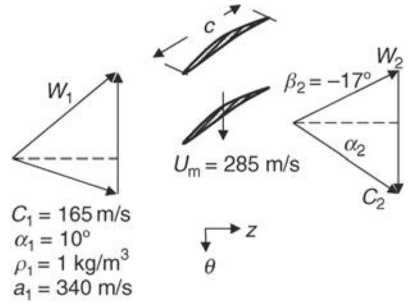
- **(g)** the degree ofreaction forthe rotorhub and tip $°R_h$ and $°R_t$
- **(h)** the diffusion factor at the pitchline D_m
- **(i)** the hub radius in station 2, r_{h2}, from continuity equation.

8.30 A compressor rotor velocity triangles at the pitchline are shown. The compressor hub-to-tip radius ratio is $r_h/r_t = 0.4$. The rotor solidity at the pitchline is $\sigma_m = 1.0$. Assuming constant axial velocity across the rotor at pitchline, calculate

- **(a)** solidity at the tip for $c_t = c_m$
- **(b)** solidity at the hub for $c_h = c_m$
- **(c)** degree of reaction $°R_m$
- **(d)** diffusion factor D_m
- **(e)** de Haller criterion
- **(f)** rotor loading coefficient ψ_m
- **(g)** flow coefficient φ_m

Assuming that the rotor has a free-vortex design, calculate

- **(h)** diffusion factor at the tip D_t
- **(i)** the degree of reaction at the hub, $°R_h$
- **(j)** the axial velocity distribution downstream of the rotor (assuming radial equilibrium)

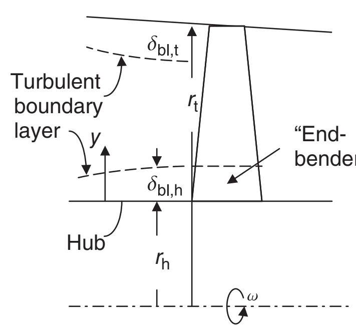


■ **FIGURE P8.30**

8.31 The flow coefficient to a rotor at pitchline is $\varphi_m = 0.8$, its loading coefficient is $C_m = 1.0$. The inlet flow to the rotor has zero swirl in the absolute frame of reference. Assuming axial velocity C_{zm} = constant across the rotor, calculate

- **(a)** the relative inlet flow angle β_{1m}
- **(b)** the relative exit flow angle β_{2m}
- **(c)** the degree of reaction $°R_m$

8.32 The end wall boundary layers and a rotor blade are shown. The hub radius is $r_h = 0.5$ m and the shaft speed is $\omega = 4000$ rpm. The rotor inlet flow has zero preswirl, i.e., $\alpha_1 = 0$. The rotor blade is twisted to meet the flow at zero incidence within the boundary layer. Calculate the blade stagger angle at the hub and its change within the boundary layer if $\delta_{bl,h} = 10$ cm and it has a 1/7th power law profile. $C_z(y)/C_{zm} = (y/\delta)^{1/7}$ and $C_{zm} = 150$ m/s. [Assume blade stagger angle is $\approx$ relative flow angle, β_1]



■ **FIGURE P8.32**

8.33 A multistage compressor has 12 repeated stages. Each stage produce a constant total temperature rise of $\Delta T_t = 25°C$. The inlet total temperature is $T_{t1} = 288$ K, and stage adiabatic efficiency is $\eta_s = 0.90$ and is assumed constant for the 12 stages. Calculate and graph the stage total pressure ratio for all 12 stages. What is the compressor overall total pressure ratio?

8.34 A compressor stage has 37 rotor blades and 41 stator blades. The shaft rotational speed is 5000 rpm. Calculate

(a) the rotor blade passing frequency as seen by the stator blades

(b) the stator blade passing frequency as seen by the rotor blades

8.35 The absolute flow to a compressor rotor has a coswirl with $\alpha_1 = 15°$. The exit flow from the rotor has an absolute flow angle $\alpha_2 = 35°$. The pitchline radius is at $r_m = 0.6$ m and the rotor angular speed is $\omega = 5220$ rpm. Assuming the axial velocity is $C_{zm} = 150$ m/s and is constant across the rotor, calculate

(a) the specific work at the pitchline

(b) the rotor torque per unit mass flow rate

(c) the degree of reaction

8.36 A rotor blade row has a hub-to-tip radius ratio of 0.4, solidity at the pitchline of 1.2, the axial Mach number of 0.6, and zero preswirl. The mean section has a design diffusion factor of $D_m = 0.5$. Assuming $a_1 = 330$ m/s, calculate and plot where appropriate

(a) exit swirl at the pitchline assuming the shaft rpm of 5000 and $r_m = 0.4$ m

(b) downstream swirl distribution $C_{\theta 2}(r)$, assuming a solid-body rotation vortex design rotor

(c) the radial distribution of degree of reaction $°R(r)$, along the blade span

(d) radial distribution of diffusion factor $D_r(r)$.

8.37 Bending stresses on a cantilevered blade, under aerodynamic loading, is estimated (by Kerrebrock, 1992) to be

$$\frac{\sigma_{\text{bending}}}{p} \approx \left(\frac{C_z}{U_t}\right)(\tau_s - 1)\left(\frac{s}{2c}\right)\left(\frac{r_t}{t_{\max}}\right)^2$$

Estimate the ratio of bending stress to inlet pressure ($\sigma_{\text{bending}}/p$) in a rotor with axial velocity 165 m/s, the rotor tip radius is $r_t = 0.75$ m, the mean solidity of 1.5, angular speed is $\omega = 4000$ rpm. The maximum blade thickness is $t_{\max} = 1$ cm. The stage total pressure ratio is 1.6 and the stage adiabatic efficiency is 0.88.

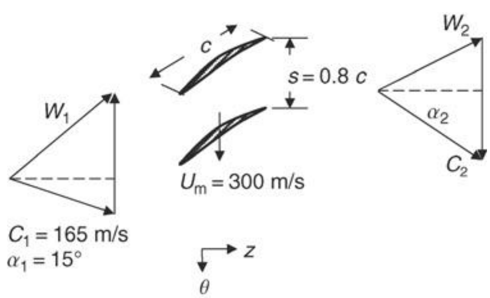
8.38 Centrifugal stress is proportional to AN^2 as we discussed in the compressor design section. It follows

$$\frac{\sigma_c}{\rho_{\text{blade}}} = \frac{\omega^2 A}{4\pi}\left(1 + \frac{A_t}{A_h}\right)$$

for a linear taper ratio. For a titanium rotor blade of $r_h = 0.4$ m, $r_t = 0.8$ m, and taper ratio $A_t/A_h = 0.8$, calculate the acceptable shaft speed ω if the allowable strength-to-weight ratio is ~9 ksi/slug/ft^3, which is equivalent to

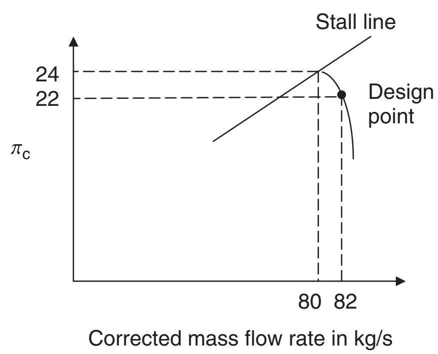
allowable creep rupture strength	120,560 m^2/s^2 (or ~120 kPa/kg/m^3)
required strength	68,326 m^2/s^2 (or ~68 kPa/kg/m^3)

8.39 A rotor section has de Haller criterion $W_2/W_1 = 0.75$. Assuming the axial velocity remains constant across the rotor and the velocity triangles are as shown, calculate the corresponding D-factor and degree of reaction for the rotor section.



■ **FIGURE P8.39**

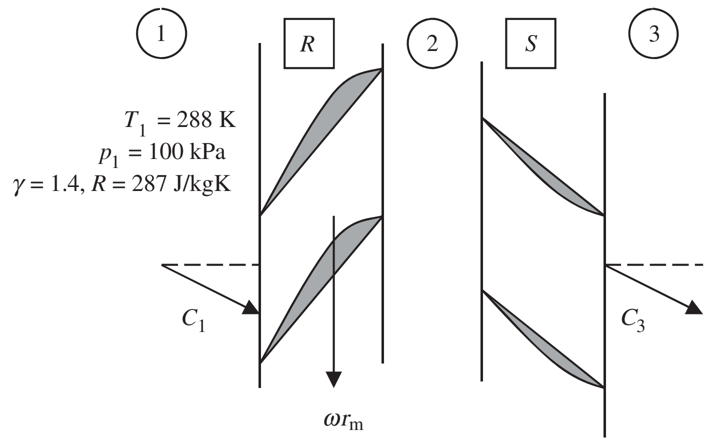
8.40 A portion of a compressor map surrounding the design point is shown.



■ **FIGURE P8.40**

Calculate this compressor's stall margin.

8.41 A compressor stage at the pitchline, $r_m = 0.5$ m, is shown. The inlet flow to the rotor has a preswirl with $\alpha_1 = 22°$. The axial velocity is $C_{z1} = C_{z2} = C_{z3} = 170$ m/s, i.e. constant throughout the stage. The stage is of repeated design, with $\alpha_3 = \alpha_1$. Rotor and stator solidities at pitchline are 1.2 and 1.0 respectively. The rotor inlet relative velocity at pitchline is sonic, i.e., $W_1 = a_1$ and the rotor relative exit velocity, following de Haller criterion, is $W_2 = 0.75\, W_1$.

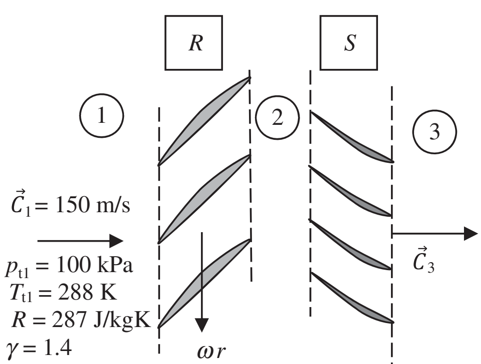


■ **FIGURE P8.41**

Calculate:

(a) shaft rotational speed, ω, in rpm
(b) rotor exit (absolute) swirl, $C_{\theta 2}$ in m/s
(c) stage degree of reaction, $°R_m$
(d) rotor D-factor, D_r
(e) stage total temperature ratio
(f) stage total pressure ratio for $e_c = 0.9$

8.42 An axial-flow compressor stage at its pitchline radius ($r_m = 0.50$ m) has zero preswirl, an axial velocity of 150 m/s. Its degree of reaction is $°R_m = 0.50$, and the shaft angular speed is 5,200 rpm. Assuming constant throughflow speed, i.e., C_z = const., and repeated stage design at the pitchline, calculate:

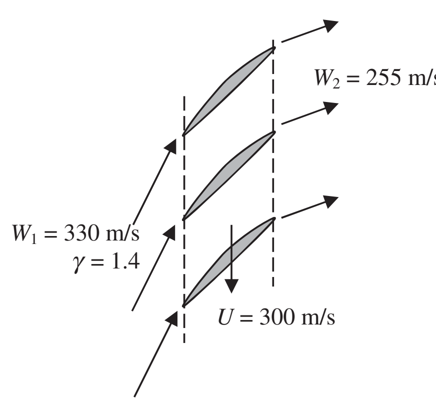


■ **FIGURE P8.42**

(a) relative Mach number at rotor inlet, M_{1r}
(b) rotor exit relative velocity, W_2 (m/s)
(c) static temperature downstream of the rotor, T_2 (K)
(d) absolute Mach number downstream of the rotor, M_2
(e) absolute Mach number downstream of the stator, M_3
(f) stage total temperature ratio, τ_s
(g) stage total pressure ratio if $\eta_s = 0.90$

8.43 The relative flow across a compressor rotor is shown. The rotor rotational speed is U = 300 m/s. The axial velocity is C_z = 165 m/s and it remains constant across the rotor. Assuming the rotor solidity is $\sigma_r = 0.8$, calculate:

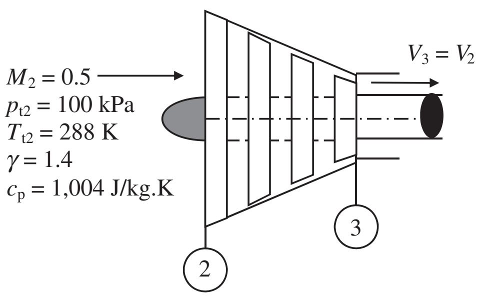
(a) de Haller criterion
(b) rotor degree of reaction (for a repeated stage)
(c) rotor specific work (in kJ/kg)
(d) rotor D-factor



■ **FIGURE P8.43**

8.44 A multistage compressor develops a total pressure ratio of $\pi_c = 35$ with a polytropic efficiency of $e_c = 0.90$. The air mass flow rate through the compressor is $\dot{m} = 200$ kg/s. Assuming γ and c_p remain constant, calculate:

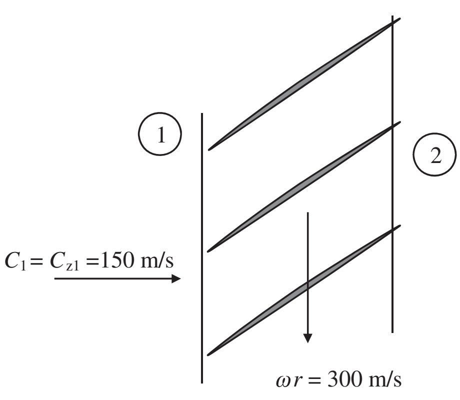
(a) compressor shaft power, $\wp_c$, in MW
(b) flow area in 2, i.e., A_2 in m^2
(c) density of air in station 3, ρ_3, in kg/m^3 [note: the axial velocity at the compressor exit, $V_3 = V_2$]
(d) The nondimensional entropy rise across the compressor, $\Delta s/R$



■ **FIGURE P8.44**

8.45 A cylindrical cut of the rotor in an axial flow compressor is shown. The rotor uses constant-axial velocity design, $C_{z1} = C_{z2}$. Assuming diffusion in the rotor, based on de Haller criterion is described by $W_2/W_1 = 0.75$, calculate:

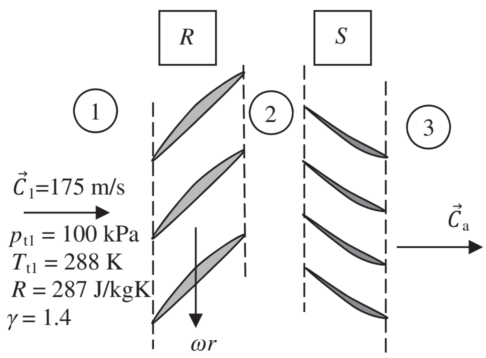
(a) relative flow angle at the rotor inlet, β_1 in degrees
(b) relative flow angle at the rotor exit, β_2 in degrees
(c) rotor exit swirl, $C_{\theta 2}$, in m/s
(d) stage loading parameter, ψ



■ **FIGURE P8.45**

8.46 An axial-flow compressor stage at its picthline radius ($r_m = 0.40$ m) has zero preswirl, an axial velocity of 175 m/s, a degree of reaction of $°R_m = 0.65$, with the shaft angular speed of 6450 rpm. Assuming constant throughflow speed, i.e., C_z = const., and a repeated-stage design at the pitchline, calculate:

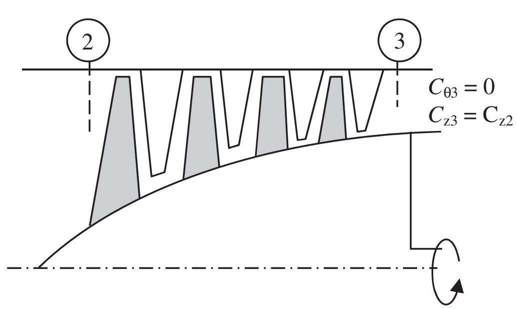
(a) relative Mach number at rotor inlet, M_{1r}
(b) rotor exit relative velocity, W_2 (m/s)
(c) static temperature downstream of the rotor, T_2 inK
(d) absolute Mach number downstream of the rotor, M_2
(e) absolute Mach number downstream of the stator, M_3
(f) stage total temperature ratio, τ_s
(g) stage total pressure ratio if $\eta_s = 0.88$



■ **FIGURE P8.46**

8.47 A multistage compressor has a total pressure ratio of $\pi_c = 15$, a polytropic efficiency of $e_c = 0.90$ and the inlet condition: $C_2 = C_{z2} = 160$ m/s, $T_{t2} = 288$ K, $p_{t2} = 100$ kPa. Assuming constant gas properties $\gamma = 1.4$ and $c_p = 1004$ J/kgK and C_z remains constant with compressor exit purely in the axial direction (i.e., no exit swirl), calculate:

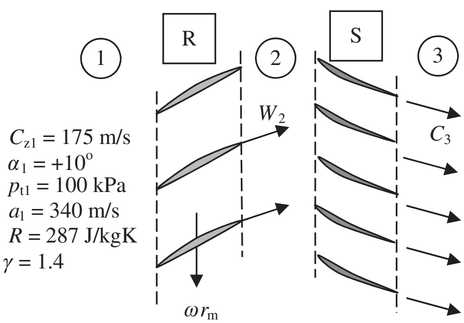
(a) compressor exit total temperature, T_{t3} in K
(b) compressor inlet Mach number, M_2
(c) compressor exit Mach number, $M_3 = M_{z3}$
(d) compressor Density ratio, ρ_3/ρ_2
(e) compressor area ratio, A_3/A_2



■ **FIGURE P8.47**

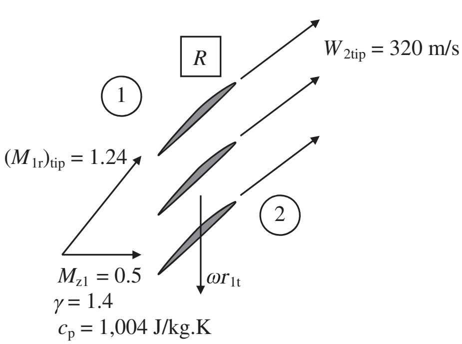
8.48 An axial-flow compressor stage is shown at its pitchline radius ($r_m = 0.30$ m). The axial velocity is 175 m/s and is constant. The rotor exit relative flow angle is $\beta_2 = -15°$. Assuming the inlet ***preswirl exists*** with $\alpha_1 = +10°$ and shaft angular speed is 7600 rpm, calculate:

- **(a)** absolute Mach number at the inlet, M_1
- **(b)** relative Mach number at the inlet, M_{1r}
- **(c)** static pressure at the inlet, p_1 (kPa)
- **(d)** total temperature at the inlet, T_{t1} (K)
- **(e)** rotor exit absolute swirl, $C_{\theta 2}$ (m/s)
- **(f)** stage degree of reaction, $°R_m$ (assume repeated stage)
- **(g)** rotor specific work at pitchline, w_m in kJ/kg
- **(h)** static temperature downstream of the rotor, T_2 (K)
- **(i)** absolute Mach number downstream of the rotor, M_2
- **(j)** speed of sound downstream of the stator, a_3 (m/s)
- **(k)** absolute Mach number downstream of the stator, M_3



■ **FIGURE P8.48**

8.49 An axial-flow compressor rotor has a tip relative Mach number (at the inlet) of 1.24, as shown. The tip radius is $r_{1t} = 0.73$ m and the inlet speed of sound is $a_1 = 337$ m/s. The rotor inlet flow has zero preswirl and has an axial Mach number of M_{z1} 0.5. The rotor exit has a relative velocity, $W_{2tip} = 320$ m/s. Assuming C_z is constant, calculate:



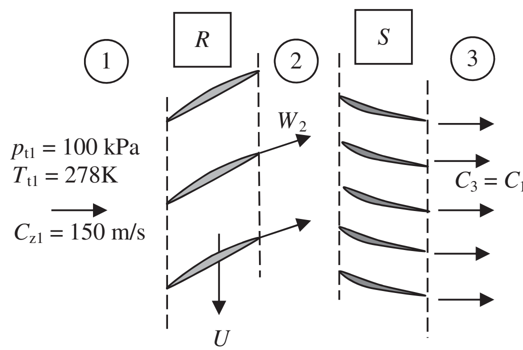
■ **FIGURE P8.49**

- **(a)** the shaft angular speed, ω in rpm
- **(b)** the rotor specific work, w_c, at the tip in kJ/kg
- **(c)** rotor exit absolute Mach number, M_2

8.50 An axial flow compressor stage at $r = 0.4$ m is shown. The inlet flow is purely axial with $C_{z1} = 150$ m/s, which remains constant across the rotor and stator. The rotor solidity at this radius is $\sigma_r = 1.5$. The rotor angular speed is $\omega = 7400$ rpm. The stage degree of reaction at this radius is $°R = 0.75$. The rotor total pressure loss parameter (in relative frame) is $\varpi_r = 0.05$.

The gas properties are: $\gamma = 1.4$ and $R = 287$ J/kgK. Calculate:

- **(a)** absolute swirl at rotor exit, $C_{\theta 2}$ in m/s
- **(b)** rotor exit (absolute) total temperature, T_{t2} in K
- **(c)** static temperature at the rotor exit, T_2 in K
- **(d)** rotor exit absolute Mach number, M_2
- **(e)** rotor exit relative Mach number, M_{2r}
- **(f)** relative dynamic pressure at rotor inlet, q_{1r}, in kPa
- **(g)** inlet relative total pressure, p_{t1r}, in kPa
- **(h)** static pressure at rotor exit, p_2, in kPa
- **(i)** rotor exit (absolute) total pressure, p_{t2}, in kPa
- **(j)** rotor D-factor, D_r, at this radius



■ **FIGURE P8.50**

8.51 The pitchline radius of a compressor rotor is at $r_m = 0.3$ m. The degree of reaction is $°R_m = 0.75$. The axial velocity to the rotor is $C_{z1} = 175$ m/s = constant across the blade row. The flow to the rotor has zero preswirl and the rotor angular speed is $\omega = 6500$ rpm. The solidity of the rotor at the pitchline is $\sigma_m = 1.0$. Calculate:

- **(a)** *absolute* swirl velocity downstream of the rotor, i.e., $C_{\theta 2m}$, in m/s
- **(b)** *relative* swirl upstream and downstream of the rotor, i.e., $W_{\theta 1}$ and $W_{\theta 2}$ in m/s
- **(c)** de Haller parameter, W_2/W_1
- **(d)** diffusion factor at the pitchline radius, D_{rm}

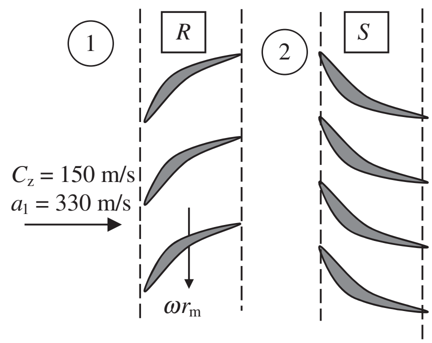
8.52 An axial-flow compressor rotor at the pitchline has a radius of $r_m = 0.5$ m. The shaft rotational speed is $\omega = 6000$ rpm. The inlet flow to the rotor has zero preswirl and the axial velocity is constant, with $C_z = 150$ m/s. The stage has a 80% degree-of-reaction at the pitchline where the solidity is $\sigma_m = 1.2$. The stage adiabatic efficiency is equal to the polytropic efficiency, $e_c = 0.92$.

Assuming that the inlet total temperature is $T_{t1} = 288$ K, $\gamma = 1.4$ and $c_p = 1.004$ kJ/kg.K, calculate:

(a) rotor specific work at $r = r_m$ in kJ/kg
(b) stage loading, ψ, at $r = r_m$
(c) flow coefficient, ϕ, at $r = r_m$
(d) rotor relative Mach number at the pitchline, $M_{1r,m}$
(e) stage total pressure ratio at the pitchline
(f) rotor diffusion factor at the pitchline
(g) is the de Haller criterion satisfied?

8.53 A compressor stage at its pitchline radius ($r_m = 0.25$) has $°R_m = 0.73$. The axial velocity is constant at $C_{zm} = 150$ m/s. The flow entering the compressor is swirl free, i.e., $\alpha_1 = 0$. The flow exiting the stage at pitchline is also swirl free, i.e., $\alpha_3 = 0$. The compressor angular speed is $\omega = 7500$ rpm. The speed of sound at the entrance to the stage is $a_1 = 330$ m/s and rotor solidity at the pitchline radius is $\sigma_{rm} = 0.80$. Assuming the gas constant is $R = 287$ J/kg·K and $\gamma = 1.4$, calculate:

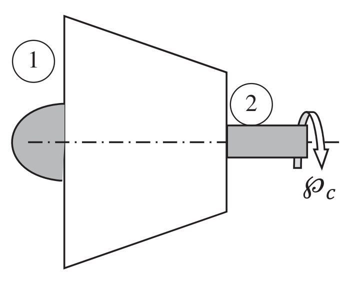
(a) absolute swirl downstream of the rotor, $C_{\theta 2m}$, in m/s
(b) rotor specific work at pitchline radius, w_c, in kJ/kg
(c) total temperature downstream of the rotor, T_{t2m}, in K
(d) de Haller criterion for the rotor at pitchline
(e) the rotor diffusion factor at the pitchline radius, D_{rm}



■ **FIGURE P8.53**

8.54 An axial-flow compressor has no IGV. Its inlet conditions are: $p_{t1} = 100$ kPa, $T_{t1} = 288$ K and the mass flow rate $\dot{m} = 100$ kg/s. The compressor pressure ratio is $\pi_c = 15$ and the polytropic efficiency is $e_c = 0.90$. The axial flow in the compressor is designed to be constant at $C_z = 166$ m/s. Assuming the gas constant is $R = 287$ J/kg · K and $\gamma = 1.4$, calculate:

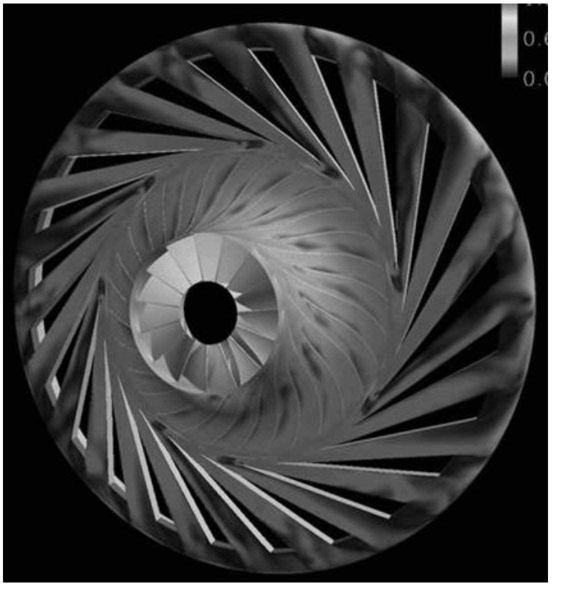
(a) absolute Mach number at the inlet to the compressor, M_1
(b) compressor shaft power, $\wp_c$, in MW
(c) compressor inlet flow area, A_1 in m^2
(d) exit Mach number, M_2 (assume the exit flow is swirl free)
(e) compressor exit area, A_2 in m^2



■ **FIGURE P8.54**

CHAPTER 9

Centrifugal Compressor Aerodynamics



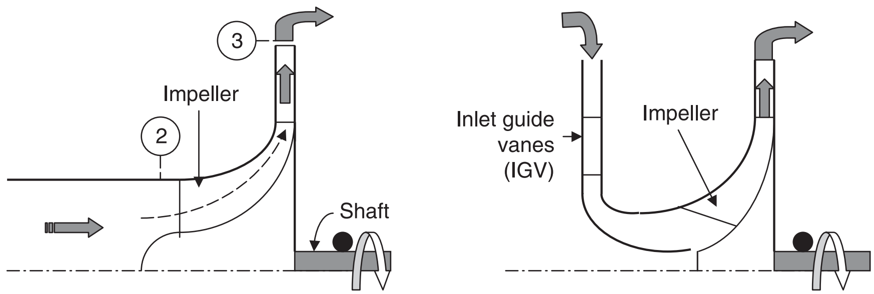
CFD results shown in a centrifugal compressor.
Source: Courtesy of NASA

9.1 Introduction

Centrifugal compressors belong to the general category of turbomachines. The flow may enter a centrifugal compressor in the axial direction. The rotor, which is known as the impeller, imparts energy to the fluid by the rotation of its aerodynamic surfaces (i.e., blades or impeller vanes) that are highly curved and twisted. The initial curvature and twist in the impeller, known as the inducer section, has the function of meeting the incoming flow at its relative flow angle. The second function of the inducer is to turn the relative flow toward the axial direction before it begins its journey in the radial direction. The third function of the inducer is to increase the fluid static pressure in the passage by decelerating the gas. The inducer exit flow is then further decelerated radially outward, by virtue of centrifugal force acting on the fluid in the spinning impeller (vaned) passages. Since the pressure rise in this type of configuration is primarily produced by centrifugal compression/force, this kind of turbomachinery is known as a centrifugal compressor. In contrast to axial-flow compressors where the flow deviation in the radial direction is negligible, the principle of operation of the centrifugal compressor is based on large radial shift between the inlet and exit of the impeller (Figure 9.1). Since the impeller exit radius is by design much larger then the impeller inlet radius, the centrifugal compressors have a lower mass flow rate per frontal area than the axial-flow compressors. A low mass flow per frontal area is of critical concern to aircraft propulsion system designers, since it translates into an increased drag count. However, in low-speed applications where the drag penalty is less significant, centrifugal compressors offer the advantage of high-pressure ratio per stage and robustness of construction that is less prone to structural failure. There are other

Aircraft Propulsion, Second Edition. Saeed Farokhi.

Companion Website: www.wiley.com/go/farokhi



■ **FIGURE 9.1**
Schematic drawing of a first (left) and a second stage (right) centrifugal compressor

applications for small gas generators on board aircraft that are suitable for centrifugal compressor application. For example, APUs (auxiliary power units) are used to start the engines and provide power to aircraft, which are entirely embedded in the fuselage; therefore, there are no concerns of external drag penalties for their use. Industrial and automotive turbocharging applications also use centrifugal compressors.

The radially pumped flow from the impeller first needs to be decelerated through a radial and vaned diffuser, and then it needs to turn back toward the axis of rotation for either the next compressor stage or the combustion chamber. Since any ducting that involves turns is a source of (total pressure) loss, multistage centrifugal compressors are considered cumbersome (and with lower efficiency) as compared with axial-flow compressors. However, the total pressure ratio of a single-stage centrifugal compressor may be as high as 10–12, whereas an axial-flow compressor stage produces a high of ~1.6–2.0 in advanced transonic fan stages.

9.2 Centrifugal Compressors

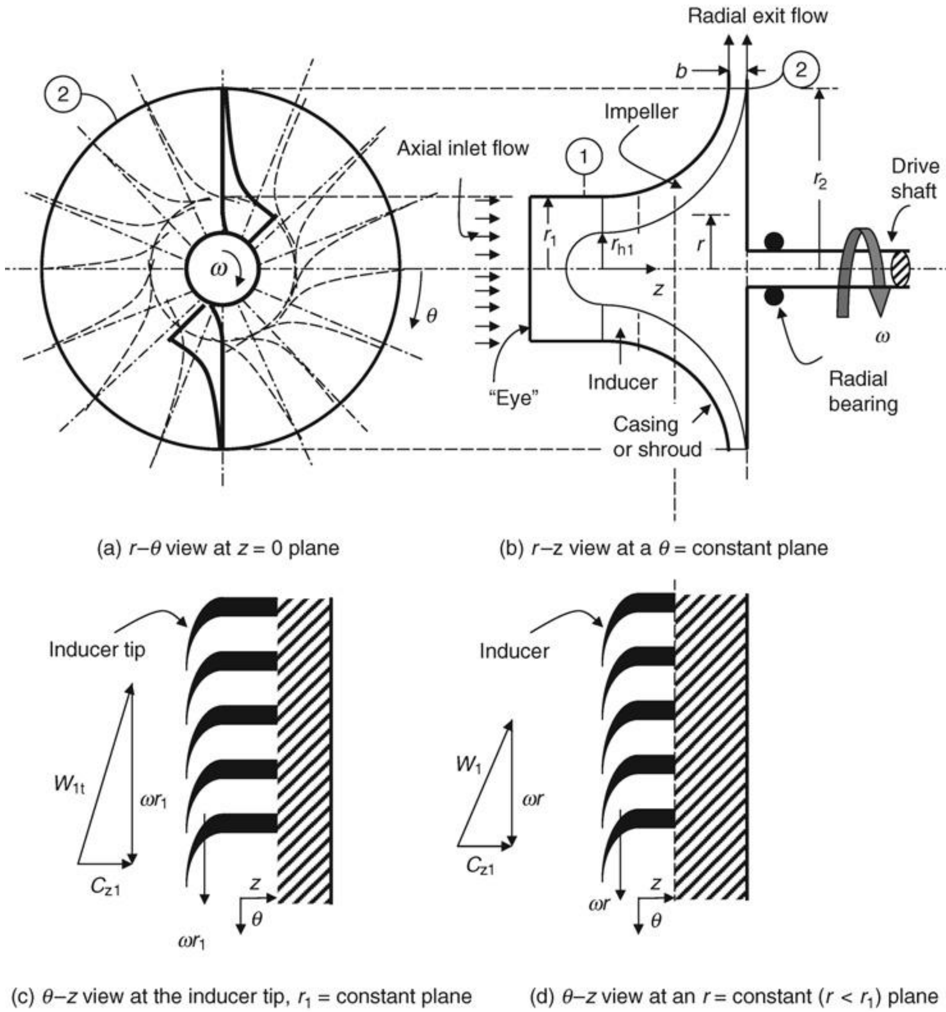
A centrifugal compressor is a robust mechanical compression system that pumps the gas from primarily an axial inlet condition to a radial exit direction. The elements of a centrifugal compressor rotor, known as the impeller, are shown in Figure 9.2.

The inlet section of an impeller is called the inducer, which is turned in the direction of impeller rotation to meet the flow at a small incidence angle. The radial displacement of the fluid in the inducer is small but the flow turning is rather large. The inducer turns the fluid from the inlet relative flow direction β_1 to the axial direction. The static pressure rise in the inducer is due to the conversion of relative swirl kinetic energy in a curved diffuser. The outer portion of the impeller is responsible for turning the flow in the radial direction and accelerating the fluid in the tangential direction. The torque acting on the fluid causes the angular momentum to change across the rotor blade following Euler turbine equation:

$$\tau_{\text{fluid}} = -\tau_{\text{blade}} = \dot{m}(r_2 C_{\theta 2} - r_1 C_{\theta 1})$$

As in the axial-flow compressors, we note that the torque acting on the fluid is positive across the rotor (as $r_2 C_{\theta 2} > r_1 C_{\theta 1}$) and thus the rotor torque being equal and opposite is negative. Then multiplying the rotor torque with the angular speed of the shaft, ω, we get a negative shaft power that is delivered to the rotor. This is consistent with the thermodynamic convention employed in the first law that considers the work done *on the surrounding* as positive. Therefore, the work done *by the surrounding* (as needed by

■ **FIGURE 9.2**
Definition sketch of a centrifugal compressor impeller with an inducer



compressors) is then negative. The torque acting on the fluid crossing the stator, as in the axial-flow compressors, is negative. Therefore, the stator vanes experience a positive torque. There will not be any contribution to the fluid power by the stator blades as they are stationary and thus perform no work on the medium. The shaft power delivered to the fluid is converted into total enthalpy rise according to

$$\wp = \dot{m}\omega(r_2C_{\theta 2} - r_1C_{\theta 1}) = \dot{m}(h_{t2} - h_{t1}) \tag{9.1}$$

The rotor total temperature ratio is deduced from Equation 9.1, that is,

$$\frac{T_{t2}}{T_{t1}} = 1 + \frac{\omega \cdot r_2}{c_pT_{t1}}\left[C_{\theta 2} - \left(\frac{r_1}{r_2}\right)C_{\theta 1}\right] \tag{9.2}$$

In Equation 9.1, the impeller radius ratio (r_1/r_2) is small, which makes the second term in the bracket negligible compared with the first term. In addition, either due to a lack of inlet guide vanes, which results in $C_{\theta 1}$ to be identically zero, or the fact that $C_{\theta 2} \gg C_{\theta 1}$ in a centrifugal compressor, we often neglect the second term in the bracket in Equation 9.2 in favor of the first term, that is,

$$\frac{T_{t2}}{T_{t1}} \cong 1 + \frac{\omega \cdot r_2 \cdot C_{\theta 2}}{c_p T_{t1}} = 1 + \frac{U_2^2}{c_p T_{t1}} \left(\frac{C_{\theta 2}}{U_2} \right) \tag{9.3}$$

However, note that we do not *have to* neglect the second term in Equation 9.2, as the inlet swirl $C_{\theta 1}$ may be treated as a known input to the problem.

There are three types of impeller geometry:

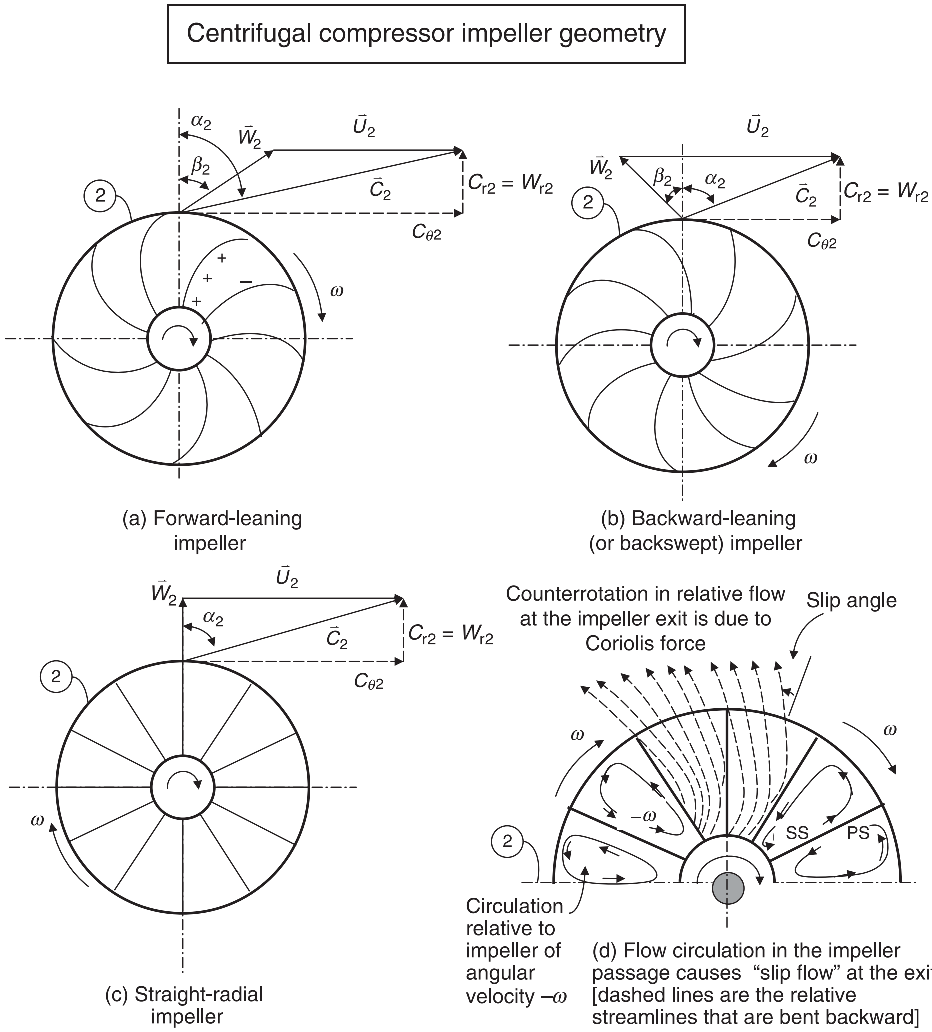
- Radial impeller
- Forward-leaning impeller
- Backward-leaning (or backswept) impeller.

A schematic drawing of impeller geometries and their corresponding velocity triangles are shown in Figure 9.3.

The radial impeller offers radial passages to accelerate the fluid in the rotor and ideally attain a radial exit flow. The forward-leaning impeller geometry accelerates the fluid in a spiral passage leaning in the direction of the rotor rotation. The backswept impeller is composed of spiral passages that direct the relative flow in the opposite direction to rotor rotation. The geometry of the impeller dictates the exit flow angle β_2 and thus strongly impacts the magnitude of the exit swirl velocity $C_{\theta 2}$. The three types of impeller geometry are shown in Figure 9.3, with the flow angles at the impeller exit now measured with respect to the radial direction, as shown. The velocity triangles at the impeller exit show that the absolute swirl is increased when the impeller geometry is forward leaning (with $\beta_2 < 0$). For the case of the straight radial impeller, the exit absolute swirl should theoretically be the same as the wheel speed U_2. The backward-leaning or backswept impeller geometry turns the relative flow in the opposite direction to the wheel rotation (with $\beta_2 > 0$), therefore the absolute swirl in the exit plane is reduced. All these arguments are the results of simple interpretation of the velocity triangles (Figure 9.3).

The Euler turbine equation suggests that the forward-leaning blades produce the highest total enthalpy rise and, hence, absorb the maximum shaft power. The impeller exit Mach number is the highest of the three designs and thus limits the static pressure recovery and efficiency of the radial diffuser. Also, the blades are under severe structural loads in bending due to their curvature. Similarly, the blades of a backward-leaning impeller are under high structural loading and the shaft power absorption is reduced. The exit Mach number of the backward-leaning impeller is the lowest of the three designs and will be explored as an alternative to radial impeller design. The lower exit Mach number of the backward-leaning impeller is attractive in achieving higher efficiency diffusion in the radial diffuser. The optimum geometry from structural standpoint seems to be the straight radial impeller with good power absorption, acceptable stability characteristics, and a low level of structural loading (in bending) due to purely radial design. However, the pressure rise in the compressor increases with the wheel speed, hence a backward-leaning impeller design could operate at higher U_T once the structural design issues are resolved.

In the rotor frame of reference, the flow in the impeller is primarily in the radial direction, which bends in the opposite direction to the wheel rotation due to a Coriolis



■ **FIGURE 9.3** **Definition sketch of the impeller blade shapes, flow angles, ideal velocity triangles at the impeller exit, and the phenomenon of slip**

force, that is, $\propto 2\omega W_r$. The presence of tangential pressure gradients in addition to the Coriolis effect combine to create a countercirculation in impeller blade passages toward the trailing edge. The counterrotation in the impeller exit flow causes a reduction in the absolute swirl, therefore a reduction in the power absorption and thus leads to a reduced pressure ratio. This phenomenon is known as *slip* in centrifugal compressors. An example of the slip in a straight radial impeller is shown in Figure 9.3d. The other two examples of impeller configurations, that is, the forward- and the backward-leaning types, are subject to the phenomenon of slip in their respective passages. We define a slip factor for radial impellers as the ratio of absolute swirl to the exit wheel speed, which represents the *ideal* swirl had the fluid attained the same tangential velocity as the disk, namely,

$$\varepsilon \equiv \frac{C_{\theta 2}}{U_2} \quad \text{(slip factor)} \tag{9.4}$$

Hence, the total temperature rise across the straight radial impeller is related to the wheel speed and the slip factor ε following the Euler turbine equation, that is,

$$T_{t2} = T_{t1} + \frac{\varepsilon \cdot U_2^2}{c_p} \tag{9.5a}$$

From Equation 9.5a, the impeller loading coefficient $\Delta h_t/U_2^2$ gives a simple expression for the radial impeller, which is independent of mass flow rate, assuming constant slip factor, that is,

$$\Delta h_t/U_2^2 = \varepsilon \tag{9.5b}$$

In general, the slip factor is defined as

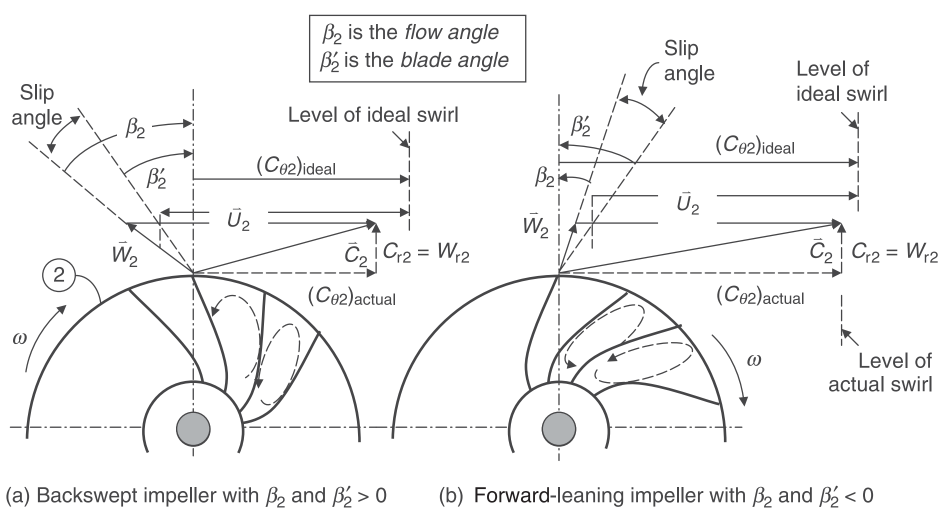
$$\varepsilon \equiv \frac{(C_{\theta 2})_{\text{actual}}}{(C_{\theta 2})_{\text{ideal}}} \quad \text{(slip factor)} \tag{9.6}$$

where the ideal exit swirl is calculated based on the *impeller blade exit angle* β_2', and the actual exit swirl is based on the *actual exit flow angle* β_2. Figure 9.4 shows a definition sketch of these angles for the backward and forward-leaning impellers.

If we adopt the sign convention that the impeller exit angle is measured from the vane position to the radial direction and is positive in the direction of the rotor rotation, we note that a backswept impeller has a positive β_2' and a forward-leaning impeller has a negative vane exit angle β_2'. With a similar sign convention for the exit flow angle, we may write the ideal and actual exit swirl velocities in terms of the impeller exit angle β_2', flow angle β_2 as

$$C_{\theta 2,\text{ideal}} = U_2 - C_{r2} \cdot \tan \beta_2' \tag{9.7a}$$

$$C_{\theta 2,\text{actual}} = U_2 - C_{r2} \cdot \tan \beta_2 \tag{9.7b}$$



■ **FIGURE 9.4** **Definition sketch for the slip flow in a backswept and forward-leaning impeller**

From Equation. 9.7a, we note that a backswept impeller, with a positive exit vane angle, reduces the absolute (ideal) swirl, whereas a forward-leaning impeller with a negative vane angle increases the exit absolute swirl. Similar argument can be made regarding the exit swirl velocity in the actual flow as described in Equation. 9.7b. We define the slip factor as the ratio of actual to the ideal exit swirl as

$$\varepsilon \equiv \frac{C_{\theta 2,\text{actual}}}{C_{\theta 2,\text{ideal}}} = \frac{U_2 - C_{r2} \cdot \tan \beta_2}{U_2 - C_{r2} \cdot \tan \beta_2'} \tag{9.8}$$

The impeller specific work and the loading are written as

$$\Delta h_t = U_2 C_{\theta 2} = U_2 (U_2 - C_{r2} \tan \beta_2) = U_2^2 \left[1 - \frac{C_{r2}}{U_2} \tan \beta_2\right] \tag{9.9a}$$

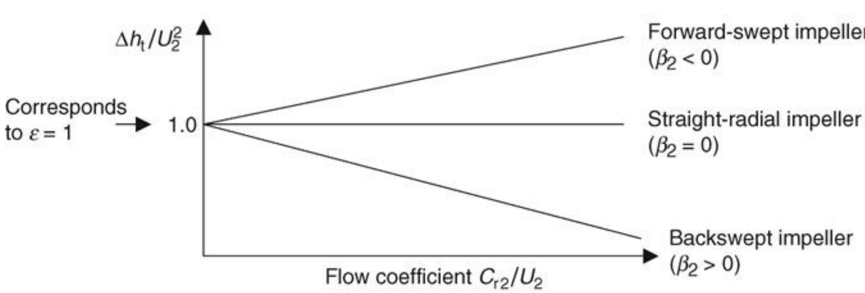
$$\Psi \equiv \frac{\Delta h_t}{U_2^2} = 1 - \left(\frac{C_{r2}}{U_2}\right) \tan \beta_2 \tag{9.9b}$$

We may graph Equation 9.7b, that is, the variation of the nondimensional work, or the impeller-loading coefficient $\Delta h_t / U_2^2$, with flow coefficient C_{r2}/U_2 (which is similar to C_z/U in axial-flow compressors), for different impellers. Figure 9.5 shows the impeller loading variation with flow coefficient for the three impeller types (with a constant slip factor ε).

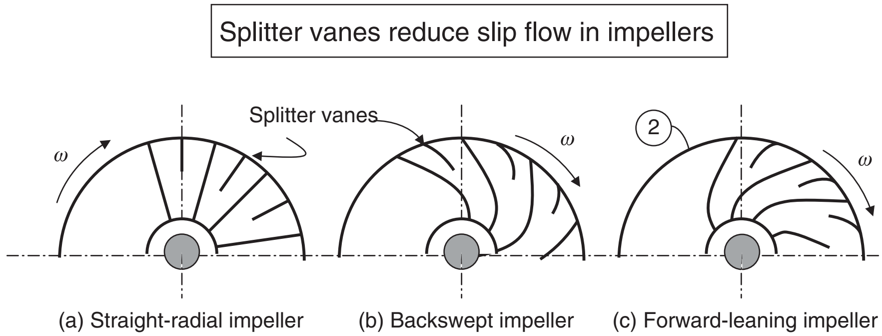
There are several correlations between the slip factor ε and the number of the impeller vanes N in radial impellers, which are proposed by Stodola (1927), Busemann (1928), and Stanitz–Ellis (1949). The Stanitz–Ellis model, which is similar to Stodola's, is considered a reasonable approximation for the slip factor in straight-radial impellers (and a first-order approximation for other impeller types),

$$\varepsilon = 1 - \frac{1.98}{N} \tag{9.10}$$

A 20-bladed impeller, for example, will experience a slip factor of ~0.9 in its energy transfer to the fluid. For radial impellers, the slip factor (as expressed in Equation 9.10) is independent of the mass flow rate. However, slip factor in reality is a function of the mass flow rate for all three configurations, that is, radial, forward- and backward-leaning impellers. Increasing the number of the impeller blades reduces the effect of slip, which in turn increases the weight. To achieve the effect of higher blade count without investing heavily in the system weight increase, a compromise solution is found by incorporating splitter vanes in the outer half of an impeller disk. The presence of splitter vanes acts as



■ **FIGURE 9.5** **Centrifugal compressor impeller loading variation with flow coefficient or mass flow rate**



■ FIGURE 9.6 **Schematic drawing of splitter vanes on the impeller disk of a centrifugal compressor**

an inhibitor of counterswirl produced in the outer portion of an impeller disk and possible reduction of flow separation in the impeller. Figure 9.6 shows the splitter vanes concept in a centrifugal turbomachinery. A parallel argument in the creation of slip is worthy of discussion. The flow at the exit of the inducer is turned parallel to the axis and is starting its journey in the radial direction. Therefore, the rotor flow is purely radial at the exit of the inducer. This resembles a source flow, which is irrotational. For the flow to remain irrotational (in the absence of external forces, i.e., in rotor frame of reference), counter eddies of the same angular speed need to be set up in the impeller blade passages. The counter eddies thus lead to an exit flow that is rotated in the opposite direction to the wheel rotation, by an angle known as the slip angle.

The stage total temperature ratio is the same as the impeller total temperature ratio, since the stationary blades are adiabatic and do no work on the fluid; therefore, we may write Equation 9.3 as

$$\tau_{\text{stage}} = \frac{T_{t2}}{T_{t1}} = 1 + \varepsilon \frac{U_2^2}{\left(\dfrac{\gamma R T_{t1}}{\gamma - 1}\right)} = 1 + (\gamma - 1)\,\varepsilon \frac{U_2^2}{a_{t1}^2} \tag{9.11}$$

The stagnation speed of sound at the inlet condition is a_{t1} in Equation 9.11. The ratio of wheel speed at the impeller exit to the stagnation speed of sound at the inlet is a non-dimensional parameter that is called the Mach index Π_M in the centrifugal compressor literature. In terms of the Mach index, we may write the stage total temperature ratio for a straight-radial impeller with zero preswirl as

$$\tau_s = 1 + (\gamma - 1)\varepsilon \cdot \Pi_M^2 \tag{9.12}$$

Upon closer examination of this parameter, that is, Mach index, we note that it is directly proportional to the impeller tangential Mach number at the exit and inversely proportional to the axial Mach number at the inlet of the compressor. This relation is shown in Equation 9.13.

$$\Pi_M^2 = \frac{U_2^2}{\gamma \cdot R \cdot T_{t1}} = \frac{U_2^2}{\gamma \cdot R \cdot T_1\left(1 + \dfrac{\gamma - 1}{2} M_1^2\right)} = \frac{M_{T2}^2}{1 + \dfrac{\gamma - 1}{2} M_1^2} \approx M_{T2}^2 \tag{9.13}$$

where $M_{T2} = \dfrac{U_2}{a_1}$.

EXAMPLE 9.1

Use a spreadsheet to calculate and graph the ratio of Mach index to the impeller tip tangential Mach number (from Equation 9.13) for a range of inlet Mach numbers from 0.2 to 0.6.

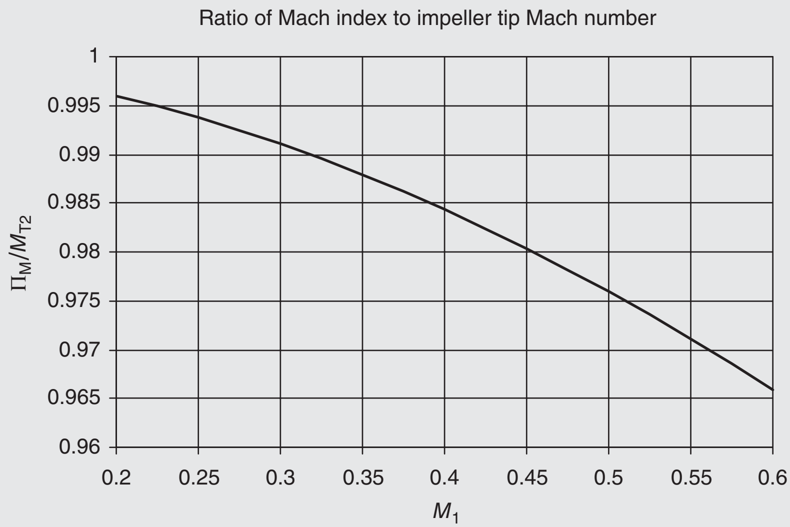
SOLUTION

The ratio of the Mach index and the impeller tip tangential Mach number M_{T2} is

$$\frac{\Pi_M}{M_{T2}} = \left(1 + \frac{\gamma - 1}{2} M_1^2\right)^{-0.5} \tag{9.14}$$

The result of spreadsheet calculation of Equation 9.14 is graphed in Figure 9.7.

At inlet axial Mach number of ~0.45, the Mach index is ~98% of the rotor tip tangential Mach number. The impeller tip tangential Mach number is thus the dominant parameter that determines the stage temperature ratio and pressure ratio, similar to the axial-flow compressor.

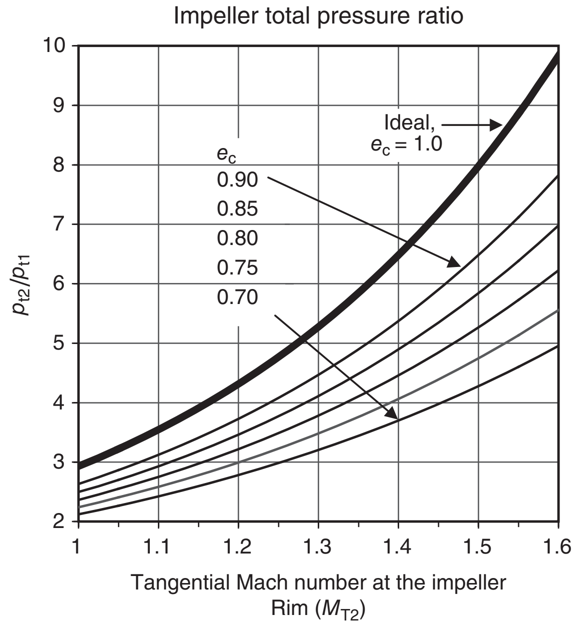


■ **FIGURE 9.7 Comparison between the Mach index and impeller tip Mach number M_{T2}**

The impeller total pressure ratio is related to the impeller total temperature ratio via polytropic efficiency according to

$$\frac{p_{t2}}{p_{t1}} = \left(\frac{T_{t2}}{T_{t1}}\right)^{\frac{\gamma \cdot e_c}{\gamma - 1}} \cong \left[1 + (\gamma - 1)\varepsilon \cdot M_{T2}^2\right]^{\frac{\gamma \cdot e_c}{\gamma - 1}} \tag{9.15}$$

where we replaced the Mach index with the tip tangential Mach number. Now, let us graph this equation for different tip tangential Mach numbers. Figure 9.8 shows the functional dependence of pressure ratio and tip Mach number. We note an exponential growth of the



■ FIGURE 9.8 Variation of impeller pressure ratio with tip tangential Mach number (based on the inlet speed of sound, a_1) for a straight-radial impeller with slip factor $\varepsilon = 0.9$

impeller total pressure ratio with the tangential Mach number at the impeller tip. There are two limitations to the exponential growth performance of centrifugal compressors with tip tangential Mach number depicted in Figure 9.8. One limiting factor is due to excessive centrifugal stresses and the structural loads; the second limiting factor is due to a drop in efficiency of the diffusion process in the radial diffuser at high inlet supersonic Mach numbers. To tackle the structural problem, a high strength-to-weight ratio material is desirable for the rotor application. Titanium alloys are thus suitable, and the lighter, lower cost aluminum is still considered a material of choice for small centrifugal compressor applications. The impeller rim speed of ~1700 ft/s (or ~520 m/s) represents the state-of-the art in centrifugal compressors. The corresponding tangential rim Mach number defined as the ratio of rim speed to the inlet speed of sound a_1 at standard sea level inlet condition, that is, a_1 of 1100 ft/s or 340 m/s, is limited to $M_{T2} \sim 1.55$. From the three possible impeller geometries, the forward-leaning design would create a higher exit Mach number than the rim tangential Mach number, based on the velocity triangle depicted in Figure 9.3a. Therefore the inlet Mach number to the diffuser will be even higher than 1.55, based on the inlet speed of sound (a_1). The straight-radial impeller would roughly achieve the same diffuser inlet Mach number as the impeller tangential rim Mach number. Note that the radial flow Mach number is small compared with a supersonic tip. To reduce the diffuser inlet Mach number to lower than 1.55 (based on a_1), we may lean the impeller blades backward, known as backsweep. The main disadvantage of curved impellers, that is, both the forward- and backward-leaning types, is in their higher (bending) stress levels than the radial vanes. The curved impellers experience large bending stresses due to the centrifugal force, whereas the radial impellers theoretically experience no bending stresses due to centrifugal loads. The higher cost of manufacturing attributed to curved impellers is considered their second disadvantage as compared with straight-radial impellers. Despite higher manufacturing costs and higher stresses, the trend is to maximize the wheel speed for efficient centrifugal compression while reducing the diffuser entrance Mach number by backward-leaning the impeller blades. The backswept impellers of ~45° exit sweep represent the state of the art in high performance centrifugal compressors.

So far we have referenced impeller exit velocity components to the inlet speed of sound. The behavior of actual flow in the diffuser depends, however, on the Mach number based on the local speed of sound at the impeller exit, a_2. The Euler turbine equation applied to an impeller with zero inlet swirl (or preswirl) is the starting point of our calculation of T_2, or a_2, that is,

$$\frac{T_{t2}}{T_{t1}} - 1 \cong \frac{U_2^2}{c_p T_{t1}} \left(\frac{C_{\theta 2}}{U_2}\right)$$

From the definition of slip factor, Equation 9.8, we will substitute for the ratio of exit absolute swirl to the wheel speed in the above equation to get

$$\frac{T_{t2}}{T_{t1}} - 1 \cong (\gamma - 1)\Pi_M^2 \cdot \varepsilon \cdot \left[1 - \left(\frac{C_{r2}}{U_2}\right) \tan \beta_2'\right] \tag{9.16}$$

Now, the total temperature ratio may be related to the static temperatures and the fluid kinetic energy following the definition of total enthalpy, that is,

$$\frac{T_{t2}}{T_{t1}} = \frac{T_2 + C_2^2/2c_p}{T_1 + C_1^2/2c_p} \tag{9.17}$$

The contribution of kinetic energy at the entrance to the centrifugal compressor may be neglected in favor of the static temperature and thus Equation 9.17 may be approximated as

$$\frac{T_{t2}}{T_{t1}} \cong \frac{T_2 + C_2^2/2c_p}{T_1} = \frac{T_2}{T_1} + \frac{C_2^2}{2c_p T_1} \tag{9.18}$$

The right-hand side of Equation 9.18 may be simplified if we replace the absolute kinetic energy with the swirl kinetic energy, which essentially neglects small contribution of the exit radial kinetic energy. We may solve for the static temperature ratio as

$$\frac{T_2}{T_1} \cong \frac{T_{t2}}{T_{t1}} - \frac{C_{\theta 2}^2}{2c_p T_1} = \frac{T_{t2}}{T_{t1}} - \frac{\gamma - 1}{2}\Pi_M^2 \left(\frac{C_{\theta 2}}{U_2}\right)^2 \tag{9.19}$$

By substituting for the total temperature ratio from Equation 9.16 and the ratio of exit swirl to the wheel speed from Equation 9.8, we get the impeller static temperature ratio as

$$\frac{T_2}{T_1} - 1 \cong (\gamma - 1)\,\Pi_M^2 \cdot \varepsilon \cdot \left[1 - \left(\frac{C_{r2}}{U_2}\right) \tan \beta_2'\right] \left\{1 - \frac{1}{2} \cdot \varepsilon \cdot \left[1 - \left(\frac{C_{r2}}{U_2}\right) \tan \beta_2'\right]\right\} \tag{9.20}$$

The parameters on the right-hand side of Equation 9.20 are all nondimensional design parameters. In its simplest form, where we set radial velocity nearly equal to zero for curved impellers or in the case of straight-radial blades, $\beta_2' = 0$, and neglect slip, that is, $\varepsilon = 1$, the static temperature ratio reduces to

$$\frac{T_2}{T_1} - 1 \approx \left(\frac{\gamma - 1}{2}\right)\Pi_M^2 = \frac{1}{2}\left(\frac{T_{t2}}{T_{t1}} - 1\right) \qquad \textbf{(9.21)}$$

Since the total temperature through the stator remains constant, the impeller static temperature rise is half of the stage total temperature rise, which for low speed at the exit of the diffuser becomes one half of the stage static temperature rise $(T_3/T_1 - 1)$. Therefore, under these conditions, the rotor and the stator produce the same static temperature rise, or what was called a 50% degree of reaction. The ratio of the speed of sound at the impeller exit to the inlet speed of sound is the square root of the static temperature ratio as

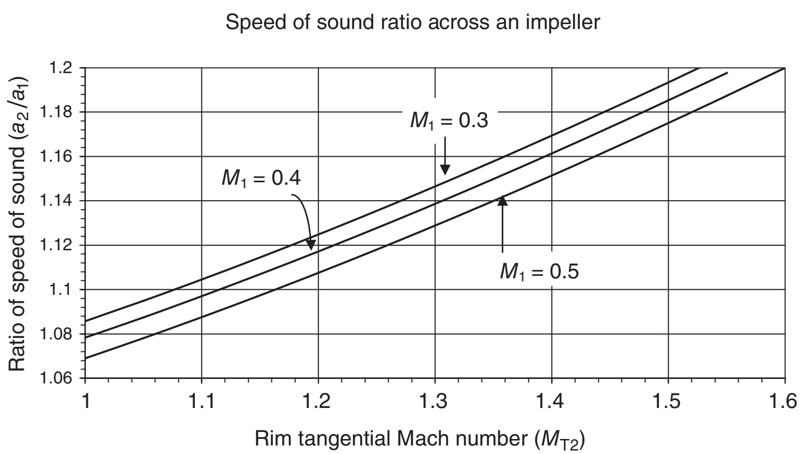
$$\frac{a_2}{a_1} \approx \sqrt{1 + \left(\frac{\gamma - 1}{2}\right)\Pi_M^2} \qquad \textbf{(9.22)}$$

From Equation 9.14, we may substitute for the Mach index in terms of the wheel tangential Mach number M_{T2} and the inlet Mach number M_1 to write the ratio of the speeds of sound across the impeller as

$$\frac{a_2}{a_1} \approx \sqrt{1 + \frac{\left(\frac{\gamma - 1}{2}\right)M_{T2}^2}{1 + \left(\frac{\gamma - 1}{2}\right)M_1^2}} \qquad \textbf{(9.23)}$$

We may now graph this expression to examine the rise of speed of sound across the impeller. Figure 9.9 shows the ratio of speed of sound across the impeller.

In the limiting case of $M_{T2} \sim 1.55$, the ratio of speed sounds across the impeller a_2/a_1 is ~1.2. Therefore, all the local Mach numbers at the impeller exit that we had referenced to the inlet speed of sound are ~20% smaller when we use local speed of sound a_2 instead of a_1. Now, let us recap, through an example, the impeller exit Mach numbers as viewed by a radial diffuser that follows the impeller.



■ **FIGURE 9.9** **The ratio of speed of sound across the impeller for three inlet Mach numbers of $M_1 = 0.3$, 0.4, and 0.5 and zero preswirl ($\gamma = 1.4$)**

EXAMPLE 9.2

Consider a centrifugal compressor impeller with the following inlet and design parameters:

Known/design inlet conditions

- Inlet absolute stagnation temperature and pressure are 288 K and 1.01×10^5 Pa, respectively, i.e., our design point operation is at the standard sea level static condition
- $C_1 = C_{z1} = 150$ m/s, this gives an axial inlet Mach number of ~0.5, no preswirl, which is our design choice
- Hub-to-tip radius ratio is $r_{h1}/r_1 = 0.5$, this too is a design choice
- Limit the relative Mach number at the eye radius to 1.0, to avoid shock losses in the inducer
- Compressor adiabatic efficiency $\eta_c = 0.72$ estimated based on similar machines
- Compressor mass flow rate is $\dot{m} = 5$ kg/s
- Gas properties are $c_p = 1004$ J/kg · K and $\gamma = 1.4$

Known/design exit conditions

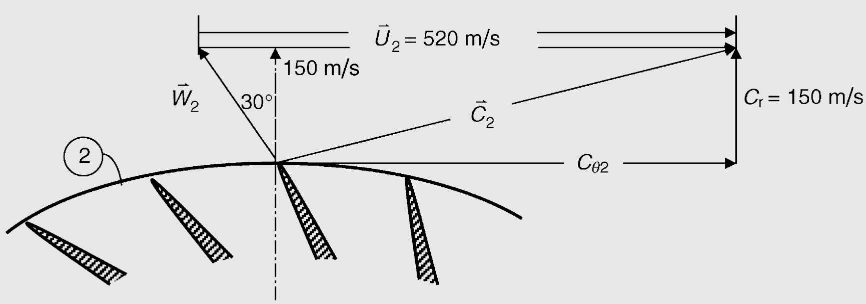
- $C_{r2} = C_{z1} = 150$ m/s, this could be our design choice, which sizes the impeller exit area
- Impeller rim speed is $U_2 = 520$ m/s, we chose the maximum rim speed in this example
- Impeller exit vane angle $\beta_2' = 20°$, i.e., we chose a backsweep design to reduce M_2
- Allow for a 10° slip angle, which makes the exit relative flow angle $\beta_2 = 30°$.
- Assume equal total pressure loss parameter ϖ in the impeller and the radial diffuser

Calculate

(a) the absolute Mach number entering the radial diffuser, M_2, and its components, i.e., radial and tangential M_{r2} and $M_{\theta 2}$
(b) the relative Mach number at the impeller exit, M_{2r}
(c) the compressor pressure ratio π_c and its polytropic efficiency e_c
(d) the inlet flow area and the "eye" radius r_1
(e) the shaft angular speed ω
(f) the impeller exit radius r_2
(g) the impeller exit width b
(h) the shaft power

SOLUTION

We first draw the impeller exit velocity triangle as an aid in calculating the velocity components.



The static temperature at the impeller inlet is

$$T_1 = T_{t1} - C_2^1/2c_p = 288\text{ K} - (150)^2/[2(1004)]$$

$$\text{K} = 276.8\text{ K} \Rightarrow \boxed{T_1 = 267.8\text{ K}}$$

$$a_1 = \sqrt{(\gamma - 1)\, c_p T_1} \Rightarrow a_1 = [0.4(1004)(276.8)]^{1/2}$$

$$= 333.4\text{ m/s} \Rightarrow \boxed{a_1 = 333.4\text{ m/s}}$$

The absolute Mach number in the inlet is $M_1 = 150/333.4 \cong 0.45 \Rightarrow \boxed{M_1 \cong 0.45}$

The absolute exit swirl velocity is the difference between the rim speed and the relative swirl velocity $W_{\theta 2}$, which from the velocity triangle, it is

$$C_{\theta 2} = 520(\text{m/s}) - 150.\tan 30°(\text{m/s})$$

$$= 433.4\text{ m/s} \Rightarrow \boxed{C_{\theta 2} = 433.4\text{ m/s}}$$

Therefore, the absolute velocity at the exit has a magnitude of

$$C_2 = [(433.4^2) + (150)^2]^{1/2} \text{ m/s}$$

$$= 458.6 \text{ m/s} \Rightarrow \boxed{C_2 = 458.6 \text{ m/s}}$$

We need the speed of sound at the exit, a_2, in order to establish the absolute exit Mach numbers. We may use the approximate form of the ratio of speeds of sound as expressed in Equation 9.22, i.e.,

$$\frac{a_2}{a_1} \approx \sqrt{1 + \frac{\left(\frac{\gamma-1}{2}\right)M_{T2}^2}{1+\left(\frac{\gamma-1}{2}\right)M_1^2}}$$

The impeller tip tangential Mach number is $M_{T2} = U_2/a_1 = 520/333.4 = 1.560$

The inlet absolute Mach number is $M_1 = C_1/a_1 = 150/333.4 = 0.450$

$$\frac{a_2}{a_1} \approx \sqrt{1 + \frac{(0.2)(1.560)^2}{1+(0.2)(0.45)^2}}$$

$$\cong 1.212 \Rightarrow \boxed{a_2 \cong 1.212(333.4) \text{ m/s} = 404.1 \text{ m/s}}$$

We may decide to use the more accurate Equation 9.16 to calculate the exit temperature and the speed of sound, i.e.,

$$\frac{T_{t2}}{T_{t1}} = \frac{T_2 + C_2^2/2c_p}{T_1 + C_1^2/2c_p}$$

We will use the Euler turbine equation to calculate the total temperature ratio across the impeller as

$$\frac{T_{t2}}{T_{t1}} \cong 1 + \frac{\omega \cdot r_2 \cdot C_{\theta 2}}{c_p T_{t1}} = 1 + \frac{U_2^2}{c_p T_{t1}}\left(\frac{C_{\theta 2}}{U_2}\right)$$

Substituting in the above equation for total temperature ratio, we get

$$\frac{T_{t2}}{T_{t1}} = 1 + \frac{(520)^2}{(1004)(288)}\left(\frac{433.4}{520}\right)$$

$$\cong 1.7794 \Rightarrow \boxed{T_{t2} \cong 512.5 \text{ K}}$$

The static temperature is related to total temperature via

$$T_2 = T_{t2} - C_2^2/2c_p \cong 512.47 \text{ K} - (458.6)^2/2(1004)$$

$$\text{K} \cong 407.7 \text{ K} \Rightarrow \boxed{T_2 \cong 407.7 \text{ K}}$$

The speed of sound at the impeller exit is

$$a_2 = \sqrt{\gamma \cdot RT_2} = \sqrt{(\gamma-1)c_p T_2}$$

$$= \sqrt{(0.4)(1004)(407.7)}$$

$$\cong 404.7 \text{ m/s} \Rightarrow \boxed{a_2 = 404.7 \text{ m/s}}$$

The difference between the approximate and the more exact formulation of the speed of sound at the impeller exit, as we calculated, is negligible (404.1 m/s versus 404.7 m/s).

Now we can calculate the impeller exit Mach numbers, based on the local speed of sound, namely,

$$M_{\theta 2} \equiv C_{\theta 2}/a_2 = 433.4/404.7 \cong 1.071$$

$$M_{r2} \equiv C_r/a_2 = 150/404.7 \cong 0.371 \Rightarrow \boxed{\begin{aligned} M_{\theta 2} &= 1.071 \\ M_{r2} &= 0.371 \\ M_2 &= 1.133 \end{aligned}}$$

$$M_2 \equiv C_2/a_2 = 458.6/404.7 \cong 1.133$$

The impeller exit Mach number is W_2/a_2 and $W_2 = 150 \text{ m/s}/\cos 30° = 173.2 \text{ m/s}$

Therefore, the impeller exit relative Mach number is $173.2/404.7 \cong 0.428$ $\boxed{M_{r2} \cong 0.428}$

Note that our design choices have led to a slightly supersonic inlet condition to the radial diffuser that follows the impeller. To calculate the compressor total pressure ratio, we use the adiabatic efficiency η_c and the compressor total temperature ratio T_{t2}/T_{t1} as we derived earlier

$$\pi_c = [1 + \eta_c(\tau_c - 1)]^{\frac{\gamma}{\gamma-1}} = [1 + 0.72(1.7794 - 1)]^{3.5}$$

$$\cong 4.75 \Rightarrow \boxed{\pi_c = p_{t2}/p_{t1} = 4.75}$$

Compressor polytropic and adiabatic efficiencies are related via

$$\eta_c = \left[\frac{\pi_c^{\frac{\gamma-1}{\gamma}} - 1}{\pi_c^{\frac{\gamma-1}{\gamma \cdot e_c}} - 1}\right]$$

Solving the above equation for e_c, we get

$$e_c = \frac{\left(\frac{\gamma-1}{\gamma}\right)\ln(\pi_c)}{\ln\left[1 + \frac{\pi_c^{\frac{\gamma-1}{\gamma}} - 1}{\eta_c}\right]} = \frac{0.2857\ln(4.75)}{\ln\left[1 + \frac{(4.75)^{0.2857} - 1}{0.72}\right]}$$

$$\cong 0.773 \Rightarrow \boxed{e_c = 0.773}$$

Now let us size the inlet flow area and the eye radius r_1 for the mass flow rate of 5 kg/s, under standard sea level static condition at the inlet. First, we need to calculate the fluid density at the inlet, ρ_1. We can calculate the density from the perfect gas law, which needs a gas constant R, or from the speed of sound, which requires γ. We calculate the static pressure at the inlet, i.e.,

$$p_1 = \frac{p_{t1}}{\left[1+\frac{\gamma-1}{2}M_1^2\right]^{\frac{\gamma}{\gamma-1}}} = \frac{1.01\times10^5\ \text{Pa}}{[1+0.2(0.45)^2]^{3.5}} \cong 8.7897\times10^4\ \text{Pa}$$

$$a_1^2 = \frac{\gamma\cdot p_1}{\rho_1} \Rightarrow$$

$$\rho_1 = (1.4)(8.7897\times10^4)/(333.4)^2\ \text{kg/m}^3 \cong 1.107\ \text{kg/m}^3$$

$$\dot{m} = \rho_1 A_1 C_{z1} \Rightarrow$$

$$A_1 = (5\ \text{kg/s})/[(1.107\ \text{kg/m}^3)(150\ \text{m/s})] = 0.03011\ \text{m}^2 = 301.1\ \text{cm}^2$$

$$A_1 = \pi\left(r_1^2 - r_{h1}^2\right) = \pi\cdot r_1^2\left[1-\left(\frac{r_{h1}}{r_1}\right)^2\right] \Rightarrow$$

$$r_1 = [(301.1\ \text{cm}^2)/\pi(1-0.25)]^{1/2} \cong 11.30\ \text{cm or } r_1 \sim 4.466\ \text{inches}$$

$$\boxed{r_e = r_1 \cong 11.30\ \text{cm}}$$

From the inlet relative Mach number of 1.0 at the eye radius r_e, we establish the angular speed of the shaft ω,

$$C_{z1}^2 + (\omega\cdot r_1)^2 = a_1^2\cdot M_{r1}^2 = (333.4)^2(1.0)^2 = 111{,}156\ \text{m}^2/\text{s}^2$$

Therefore,

$$(\omega r_1)^2 = [111{,}156 - (150)^2]\text{m}^2/\text{s}^2 = 88{,}656\ \text{m}^2/\text{s}^2,\ \text{or } \omega r_1 \cong 297.7\ \text{m/s},$$
$$\text{or } \omega \cong 2635\ \text{rad/s}$$
$$\text{or } \omega \cong 25{,}160\ \text{rpm}$$

$$\Rightarrow \boxed{\omega \cong 2635\ \text{rad/s} \cong 25{,}160\ \text{rpm}}$$

The impeller exit radius r_2 is established by the rim speed of 520 m/s specified in the problem according to

$$r_2 = U_2/\omega = [520\ \text{m/s}]/[2635\ \text{rad/s}]$$

$$\Rightarrow \boxed{r_2 \cong 19.73\ \text{cm}}$$

This gives a radius ratio of ~0.572 for the eye to exit, i.e., $r_e/r_2 \sim 0.572$. We shall examine the effect of this parameter on the inducer diffusion factor in the next section. The impeller exit width (or blade axial span at the exit) b is established from continuity equation and our design choice for the exit radial velocity $C_{r2} = W_{r2} = 150$ m/s, i.e.,

$$A_2 = \frac{\dot{m}}{\rho_2.C_{r2}} = 2\pi\cdot r_2\cdot b$$

In this preliminary stage, note that we made a first order approximation in the flow area by not accounting for the blade thickness multiplied by the number of blades that reduces the exit flow area. There are two unknowns in the above equation. One is the blade axial span at the exit, b, and the other is the gas density at the impeller exit, ρ_2. To calculate the gas density at the impeller exit, we first proceed to calculate the static pressure at the impeller exit p_2. Then by using a perfect gas law at station 2, we calculate the density of the gas, ρ_2. The exit static pressure p_2 is calculated from our assumption of equal total pressure loss parameter ϖ across the rotor and the diffuser. From the definition of $\varpi_{\text{imp.}}$, we have

$$\varpi_{\text{imp.}} \equiv \frac{p_{t1r}-p_{t2r}}{q_{1r}} = \frac{p_1\left[1+\frac{\gamma-1}{2}M_{r1}^2\right]^{\frac{\gamma}{\gamma-1}} - p_2\left[1+\frac{\gamma-1}{2}M_{2r}^2\right]^{\frac{\gamma}{\gamma-1}}}{\frac{\gamma}{2}p_1 M_{1r}^2}$$

Also, the diffuser total pressure loss parameter is

$$\varpi_{\text{dif.}} \equiv \frac{p_{t2}-p_{t3}}{q_2} = \frac{p_2\left[1+\frac{\gamma-1}{2}M_2^2\right]^{\frac{\gamma}{\gamma-1}} - \pi_s\cdot p_1\left[1+\frac{\gamma-1}{2}M_1^2\right]^{\frac{\gamma}{\gamma-1}}}{\frac{\gamma}{2}p_2 M_2^2}$$

We replaced the exit total pressure p_{t3} by the product of the stage total pressure ratio π_s and the inlet absolute total pressure. By setting the above two total pressure loss parameters equal to each other, we calculate the impeller exit static pressure p_2, namely,

$$p_2/p_1 \cong 1.795 \text{ or } p_2 \cong 1.578\times10^5\ \text{Pa}$$

$$\Rightarrow \boxed{p_2 \cong 1.578\times10^5\ \text{Pa}}$$

Therefore, the impeller exit density is $\rho_2 = \gamma p_2/(a_2)^2$, which is

$$\rho_2 = (1.4)\left(1.578 \times 10^5 \text{ Pa}\right) / (404.7 \text{ m/s})^2 \cong 1.3487 \text{ kg/m}^3$$

$$\Rightarrow \boxed{\rho_2 \cong 1.3487 \text{ kg/m}^3}$$

The impeller exit width b is now estimated from the continuity equation,

$$b \approx \frac{\dot{m}}{2\pi \cdot r_2 \rho_2 \cdot C_{r2}}$$

$$= \frac{5 \text{ kg/s}}{2\pi \cdot (0.1973 \text{ m})(1.3487 \text{ kg/m}^3)(150 \text{ m/s})}$$

$$\cong 0.01994 \text{ m} \Rightarrow \boxed{b \cong 2.0 \text{ cm}}$$

The shaft power is the product of the mass flow rate and the total enthalpy rise across the rotor,

$$\wp_s = \dot{m}c_p(T_{t2} - T_{t1}) = 5 \text{ kg/s}(1004 \text{ J/kg} \cdot \text{K})(512.5 - 288)$$

$$\text{K} \cong 1{,}127 \text{ kW} \Rightarrow \boxed{\wp_s \cong 1127 \text{ kW} = 1.127 \text{ MW}}$$

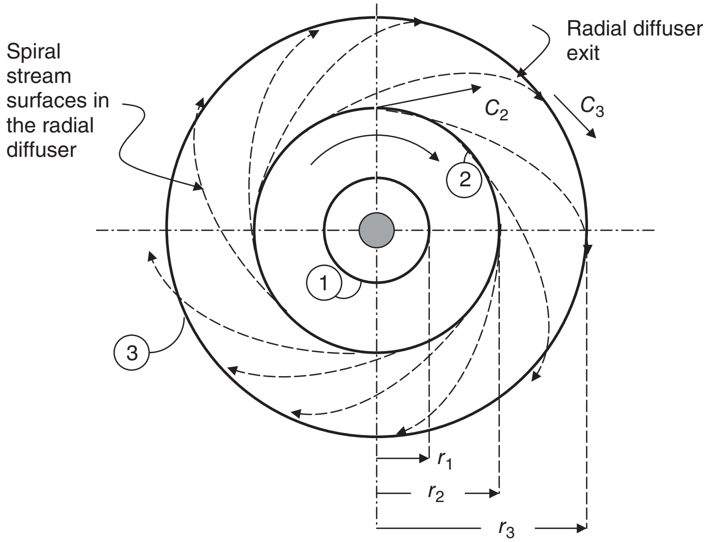
9.3 Radial Diffuser

The flow that leaves the impeller of a centrifugal compressor enters a radial diffuser. As the absolute flow at the impeller exit has swirl, the streamlines in the radial diffuser are spirals in shape. A definition sketch of a radial diffuser with its spiral streamlines and the station numbers is shown in Figure 9.10.

The discharge of the centrifugal impeller enters a vaneless radial diffuser at first, which may be followed by a vaned diffuser. The function of the radial diffuser, as in all diffusers, is to decelerate the flow and convert the kinetic energy to static pressure rise.

The diffuser in a centrifugal compressor lies between parallel walls with a spiraling flow that moves outward in the radial direction. Due to the parallel sidewalls, the diffuser area increases monotonically with radius according to

$$\frac{A(r)}{A_2} = \frac{r}{r_2} \tag{9.24}$$



■ **FIGURE 9.10**
Definition sketch of the flow in the radial diffuser

The swirl component of the flow in the radial diffuser decays as a result of

1. The conservation of angular momentum reduces the swirl inversely proportional to an increasing radius in the radial diffuser, and
2. The external torque on the fluid applied by the wall (due to fluid viscosity) is in the opposite direction to the fluid angular momentum, thus it acts to retard the fluid swirl as it moves outward in spirals.

The conservation of angular momentum, in the absence of external torque, requires

$$rC_\theta = r_2 C_{\theta 2} \tag{9.25a}$$

$$\frac{C_\theta(r)}{C_{\theta 2}} = \frac{r_2}{r} \tag{9.25b}$$

The ideal fluid path whose angular momentum is conserved results in a *logarithmic spiral, as* schematically shown in Figure 9.11. The effect of external retarding torque is

$$(\tau_{\text{ext}})_f = \dot{m}(r_3 C_{\theta 3} - r_2 C_{\theta 2}) = -\tau_w \tag{9.26a}$$

Therefore, in the presence of a retarding torque (due to friction), the angular momentum is reduced and swirl diminish even faster than a logarithmic spiral, namely,

$$\frac{rC_\theta}{r_2 C_{\theta 2}} = 1 - \frac{\tau_{\text{wall}}}{\dot{m}(r_2 C_{\theta 2})} \tag{9.26b}$$

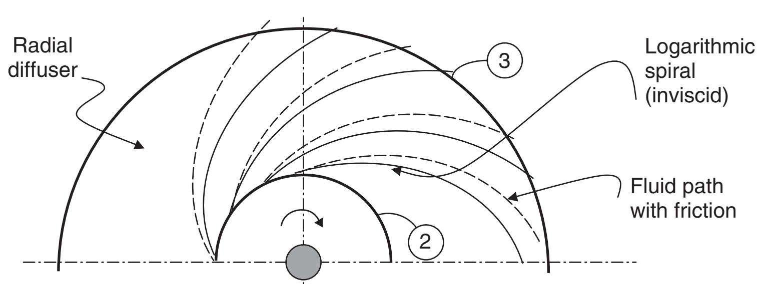
where the (external) wall torque is written as negative, which applies a retarding influence on the fluid angular momentum. The fluid path with friction *unwinds* faster than the inviscid solution. A schematic drawing of the two stream patterns in a radial diffuser with/without friction is shown in Figure 9.11.

The inlet Mach number M_2 to the radial diffuser may be calculated using the velocity triangle at the impeller exit.

$$C_2 = \sqrt{C_{r2}^2 + C_\theta^2} = \sqrt{C_{r2}^2 + (U_T - C_{r2}\tan\beta_2)^2} \tag{9.27}$$

The speed of sound at the impeller exit is based on T_2, for which we derived a simple expression,

$$\frac{T_2}{T_1} \approx 1 + \left(\frac{\gamma - 1}{2}\right) M_T^2 \tag{9.28a}$$



■ **FIGURE 9.11** **Schematic drawing of the fluid path in a radial diffuser with and without the effect of wall friction**

Therefore, the speed of sound at the impeller exit is related to the speed of sound at the inlet according to

$$\frac{a_2}{a_1} \approx \sqrt{1 + \left(\frac{\gamma - 1}{2}\right) M_{\mathrm{T}}^2} \tag{9.28b}$$

The combination of the exit speed to the local speed of sound is the Mach number, therefore,

$$M_2 \approx \frac{\sqrt{M_{z1}^2 + (M_{\mathrm{T}} - M_{z1} \tan \beta_2)^2}}{\sqrt{1 + \left(\frac{\gamma - 1}{2}\right) M_{\mathrm{T}}^2}} \tag{9.29}$$

In expression 9.29, we approximated the radial velocity at the impeller exit by the inlet axial velocity C_{z1}. We may use this as a design choice to approximate the blade axial span at the impeller exit b.

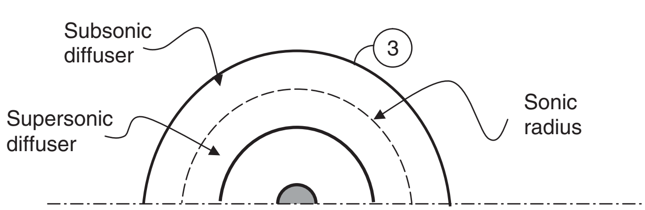
The effect of backward-leaning design (i.e., $\beta_2 > 0$) on the impeller exit Mach number is seen from Equation 9.28 to be reducing the exit velocity, thus the Mach number. A reduced impeller exit Mach number then relieves the requirements on the radial diffuser and improve its performance. Despite the choice of backward-leaning designs in reducing the exit velocity, our desire to achieve higher pressure ratios has pushed the exit Mach number of the impeller, that is, the diffuser inlet Mach number into supersonic regime. The radial diffuser is then divided into a supersonic diffuser followed by a subsonic diffuser. The dividing line between the two parts is the invisible sonic circle, as shown in Figure 9.12. The subsonic diffuser may be bladed (vanes) to improve the pressure recovery.

For the conversion of the entire kinetic energy to the fluid static enthalpy rise, we get

$$h_3 - h_2 = \frac{C_2^2}{2} \approx \frac{U_{\mathrm{T}}^2}{2} \tag{9.30}$$

Therefore, the static temperature ratio across the compressor stage is related to the square of the impeller exit Mach number, according to

$$\frac{T_3}{T_1} = \frac{T_2}{T_1} + \left(\frac{\gamma - 1}{2}\right) M_{\mathrm{T}}^2 \tag{9.31}$$



■ **FIGURE 9.12** **Radial diffuser with supersonic inlet flow and a sonic radius**

Now, substituting for the static temperature ratio for the rotor from Equation 9.28a, we get

$$\frac{T_3}{T_1} \approx 1 + (\gamma - 1)M_T^2 \tag{9.32}$$

The static pressure ratio in the limit of reversible and adiabatic flow for the stage is

$$\frac{p_3}{p_1} \approx \left[1 + (\gamma - 1)M_T^2\right]^{\frac{\gamma}{\gamma-1}} \tag{9.33}$$

In this model, we note that the static temperature rise across the diffuser is the same as the rotor, which produces the other half of the static temperature rise. In the context of the degree of reaction, the stage that we defined is a 50% degree of reaction type.

EXAMPLE 9.3

For an impeller tip Mach number of 1.2, and equal static pressure rise across the rotor and diffuser, calculate

- the static pressure ratio across the rotor and diffuser, p_3/p_1
- the static pressure ratio across the diffuser, p_3/p_2
- estimate diffuser static pressure rise (assume $M_2 \approx M_T$)

SOLUTION

In the limit of reversible adiabatic flow, we may use Equation 9.33 to get

$$\frac{p_3}{p_1} \approx [1 + 0.4(1.2)^2]^{3.5} \cong 4.9$$

The static pressure ratio across the diffuser is square root of the stage pressure ratio, namely,

$$\frac{p_3}{p_2} \approx \sqrt{4.9} \approx 2.2$$

The static pressure rise coefficient for the diffuser is

$$C_p \equiv \frac{p_3 - p_2}{q_2} = \frac{p_3 - p_2}{\gamma \cdot p_2 M_2^2/2} = \frac{2}{\gamma \cdot M_2^2}\left(\frac{p_3}{p_2} - 1\right) \tag{9.34}$$

The Mach number at the impeller exit is

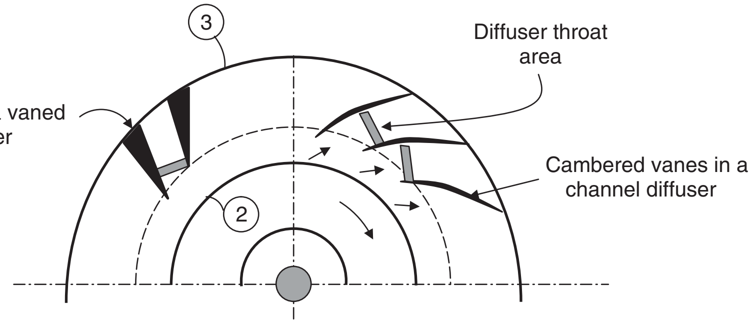
$$M_2^2 = M_{r2}^2 + M_{\theta 2}^2 \tag{9.35}$$

We need to analyze the velocity triangle at the impeller tip to evaluate these terms, i.e., the radial and tangential Mach numbers. For the sake of discussion, let us take M_2 to be nearly the same as M_T, which forces the static pressure rise coefficient in the radial diffuser to be

$$C_p \approx \frac{2}{(1.4)(1.2)^2}(2.2 - 1) \approx 1.19$$

From our earlier diffuser studies we know that this level of static pressure rise is not tolerable in a practical diffuser. In the axial-flow compressor Chapter 8, we set a maximum value of $C_{p,\max}$ of ~0.6, which in this case is nearly twice the maximum value. We conclude that the diffuser section needs to be more lightly loaded than the rotor if the boundary layer is to remain attached in the diffuser. This is not surprising that in centrifugal compressors the rotor tip Mach number plays such a dominant role in establishing the performance of the machine. We experienced a similar behavior in axial-flow machinery.

The placement of cambered vanes in radial diffusers similar to the splitter plates in axial diffusers significantly improves the static pressure recovery of radial diffusers. The vane leading edge needs to be aligned with the local relative flow in order to avoid lip



■ **FIGURE 9.13**
Schematic drawing of cambered vanes or wedges in a radial diffuser

separation. A schematic drawing of cambered vanes orwedges in a centrifugal compressor diffuser is shown in Figure 9.13.

The guiding hands of the vanes' sidewalls assist the flow in remaining attached to diffuser walls. This is another example of three-dimensional flow separation being delayed by taking one spatial degree of freedom away from the flow, that is, turning a 3D flow into an effective 2D flow. The diffuser throat blockage, as presented earlier in the inlet chapter, dominates the performance of the cambered vane diffusers. Pratt & Whitney has introduced "pipe" diffusers, which have produced a superior performance to the cambered vane or channel diffusers (Kenny, 1984).

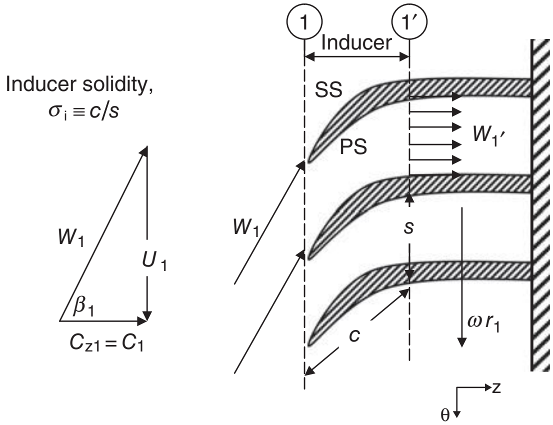
9.4 Inducer

The inducer is the part of the impeller that meets the flow at the compressor inlet at the relative flow angle β_1. Its primary function is to turn the flow toward the axial direction, as shown in Figure 9.13, thereby increasing its static pressure. In this context, the function of inducer is very similar to the rotor blades in an axial-flow compressor. There are some differences, though, that we need to consider. One is that the rotor blades in an axial-flow compressor do not need to turn the relative flow completely in the axial direction, whereas the inducer has to turn the flow completely in the axial direction. This causes a higher loading, that is, turning demand, on the inducer. The second difference is subtler than the first. The difference arises from the trailing edge of the blades. Namely, the inducer trailing edge is not free, that is, it is connected to the rest of the impeller. The axial-flow compressor rotor trailing edge is free and thus obeys the Kutta condition, which influences the trailing edge flow and the wake pattern. For example, the axial-flow compressor rotor immediately adjusts to the incoming flow disturbances by vortex shedding in the wake or adjusting the wake vortex strength in the spanwise direction. Despite these differences, we still proceed to analyze the inducer blades using the same tools we developed in the axial-flow compressors.

Let us identify station $1'$ as the exit of the inducer, as shown in Figure 9.14. We may define the diffusion factor for the inducer as

$$D_{\text{inducer}} \equiv 1 - \frac{W_{1'}}{W_1} + \frac{|\Delta C_\theta|}{2\sigma_{\text{i}} W_1} \quad \text{[inducer D – factor]} \tag{9.36}$$

The change of swirl across the inducer is approximately the wheel speed U_1, since the inlet flow to the inducer is swirl free (in the absolute frame) and the exit flow is nearly



■ **FIGURE 9.14** **Definition sketch of the inducer geometry and relative flow (in θ–z plane)**

axial in the relative frame. Now, if we assume that the axial velocity in the inducer remains constant, i.e.,

$$C_{z1} \cong W_{1'} \tag{9.37}$$

We may write the diffusion factor in terms of axial flow Mach number at the inlet and the impeller tip Mach number M_T, according to

$$D_{\text{inducer}} \cong 1 - \frac{C_{z1}}{\sqrt{C_{z1}^2 + U_1^2}} + \frac{U_1}{2\sigma_i\sqrt{C_{z1}^2 + U_1^2}} \tag{9.38}$$

$$D_{\text{inducer}} \cong 1 - \frac{1}{\sqrt{1 + \left(\frac{U_1}{U_2}\right)^2 \left(\frac{U_2}{C_{z1}}\right)^2}} + \frac{U_1/C_{z1}}{2\sigma_i\sqrt{1 + \left(\frac{U_1}{U_2}\right)^2 \left(\frac{U_2}{C_{z1}}\right)^2}} \tag{9.39a}$$

$$D_{\text{inducer}} \cong 1 - \frac{1}{\sqrt{1 + \left(\frac{r_1}{r_2}\right)^2 \left(\frac{M_T}{M_{z1}}\right)^2}} + \frac{(M_T/M_{z1})(r_1/r_2)}{2\sigma_i\sqrt{1 + \left(\frac{r_1}{r_2}\right)^2 \left(\frac{M_T}{M_{z1}}\right)^2}} \tag{9.39b}$$

Here, we note that the geometric parameters such as the impeller radius ratio and the inducer solidity are tied together with the choices of the tangential and axial flow Mach numbers to arrive at a reasonable inducer diffusion factor D_{inducer}, that is, $D_{\text{inducer}} < 0.6$. The axial Mach number varies in the range of $M_{z1} \sim 0.4$–0.6 and the impeller tip Mach number $M_t \sim 1$–1.5, the radius ratio, and the solidity may be plotted as a function of inducer diffusion factor.

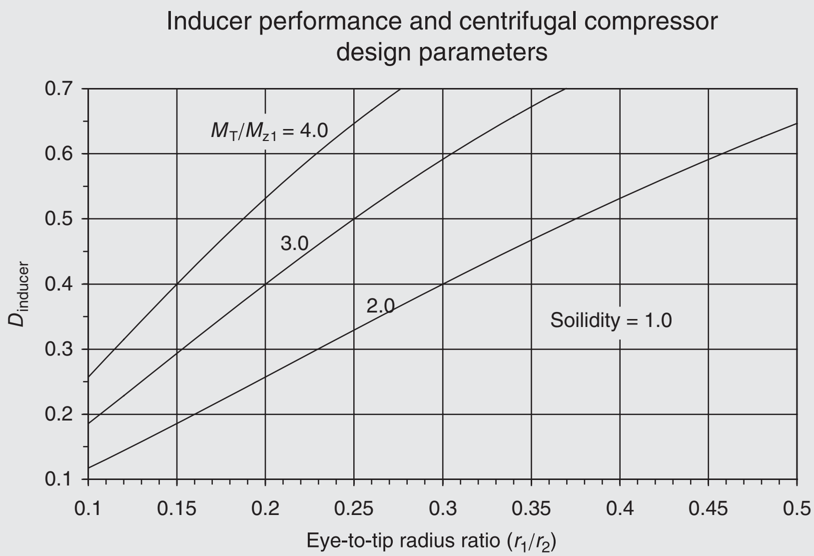
EXAMPLE 9.4

Calculate and graph the inducer D-factor for solidity of one and over a range of impeller tip Mach numbers and radius ratios.

SOLUTION

A spreadsheet is produced based on Equation 9.39b for solidity of 1. The eye-to-tip radius ratio varied from 0.1 to 0.5, and the tip-to-axial Mach number ratio varied from 2 to 4.0. The graph of the tabulated spreadsheet is shown in Figure 9.15. The parameter M_T/M_{z1} is chosen as a running parameter in Figure 9.15, as it determines the compressor pressure ratio. We observe that for a given solidity, e.g., $\sigma_i - 1$, and a maximum diffusion factor, of say 0.6, a high-pressure ratio compressor impeller requires a low eye-to-tip radius ratio (~0.22). But since the mass flow rate through the machine is proportional to the inlet area, for a high-pressure ratio impeller, the mass flow per unit frontal area (i.e., A_2 or A_3) drops. This inverse relationship between the radius ratio and the pressure ratio limits the practical designs of centrifugal compressors to a compromise between these conflicting requirements.

The advances in transonic rotor design in axial-flow compressors have prompted maximum relative Mach numbers approaching the inducer tip to be as high as ~1.5. The supersonic relative tip Mach number creates shocks, which demand extensive inducer geometric tailoring via viscous CFD analysis to minimize shock boundary layer interaction losses.



■ **FIGURE 9.15** **Inducer design parameters (for an inducer solidity of one)**

EXAMPLE 9.5

In Example 9.1, we calculated an eye-to-tip radius ratio of 0.57, an axial Mach number of $M_{z1} \sim 0.45$ and $M_t \sim 1.56$. Calculate the inducer diffusion factor in terms of its solidity. Is there any solidity that allows acceptable diffusion in the inducer?

SOLUTION

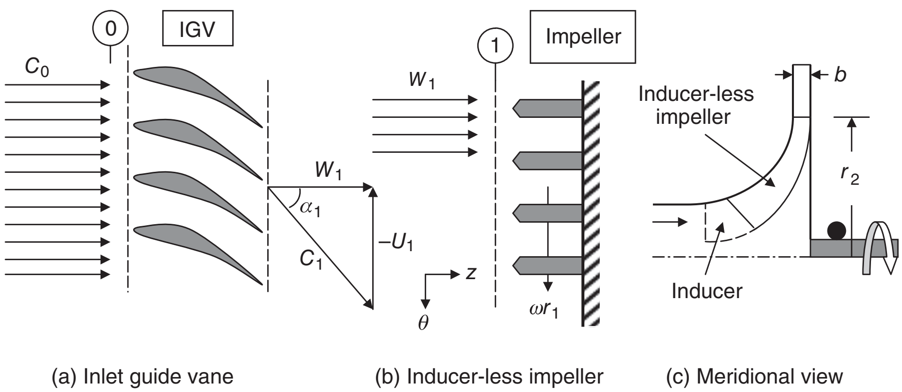
Upon substitution of the Mach numbers and radius ratio in Equation 9.39b that relates the diffusion factor to the impeller solidity, we get

$$D_i \cong 0.79 + \frac{0.446}{\sigma_i}$$

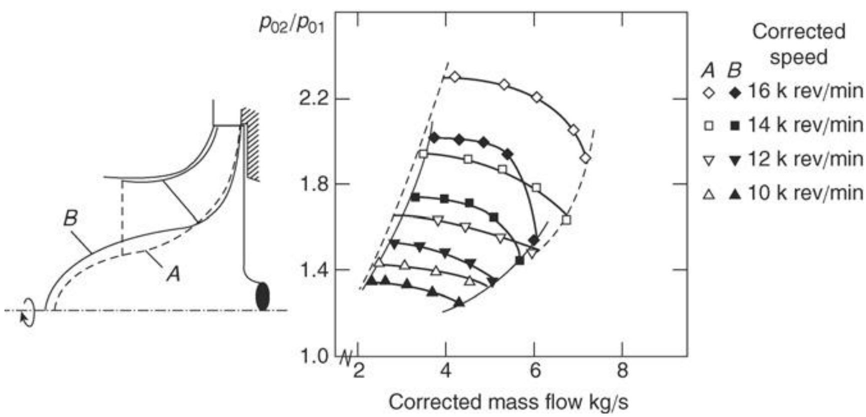
As suspected, the diffusion is too large and there is no choice of solidity (σ_i) that can decrease the inducer diffusion factor to below 0.79. Therefore, we are prompted to change our design parameters in order to reduce the inducer (diffusion) loading.

9.5 Inlet Guide Vanes (IGVs) and Inducer-Less Impellers

By incorporating inlet guide vanes in the incoming flow, we may impart a preswirl to the fluid, which is in the direction of impeller rotation. Therefore, it is possible to eliminate the need for an inducer if the relative flow to the impeller is already in the axial direction. Figure 9.16 shows the IGV exit flow and the impeller inlet relative flow, which is purely in the axial direction. In Figure 9.16c, we are comparing an impeller with inducer to an inducer-less impeller in the meridional plane. The presence of inducer offers a larger throat area than the inducer-less impeller. This geometrical influence causes the centrifugal compressor with an inducer-less impeller reach choking condition before a geometrically comparable centrifugal compressor with an inducer section. This may be seen from the data of Eckardt (1977). Figure 9.17 shows Eckardt's comparison of two centrifugal compressors with the same exit radius r_2 and width b. The "A" compressor has inducer and backsweep of 40°, whereas the "B" impeller is inducer-less and has 30° backsweep blades. The "A" impeller reaches a higher mass flow rate before choking.



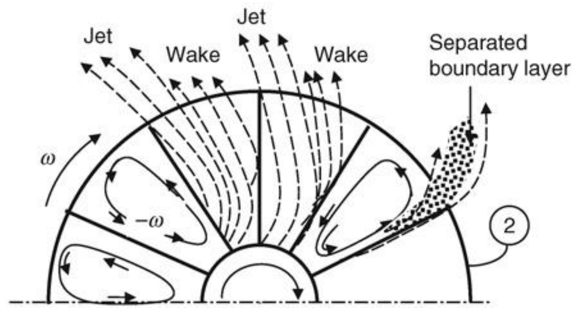
■ **FIGURE 9.16** **IGV-coupled inducer-less impeller and geometrical comparison of two impellers**



■ **FIGURE 9.17** **Performance maps of two impellers (A) with and (B) without inducer. Source: Eckhardt 1977. Courtesy of German Government**

9.6 Impeller Exit Flow and Blockage Effects

We have introduced the phenomenon of slip at the impeller exit. The turning of the relative flow in the opposite direction to that of the rotor starts inside the impeller passage.



■ **FIGURE 9.18**
The exit flow pattern from an impeller shows a "jetwake" profile

We have graphically depicted the relative streamlines in the impeller in Figure 9.3d. The accumulation of the flow on the pressure side results in a jet-like behavior and the sparse low-momentum flow on the suction side of the impeller vanes gives an appearance of the wake (or deficit momentum) flow. This behavior at the exit of the centrifugal compressor impellers is known as the *jet-wake flow,* first described by Dean and Senoo (1960). Figure 9.18 shows the jet-wake exit flow behavior in a radial impeller. The first observation is that the impeller exit flow is highly nonuniform. Therefore, the one-dimensional approximations that we often make need to be modified to reflect the nonuniform flow behavior. The second implication is that the boundary layer on the suction surface near the impeller exit is likely separated, as schematically shown in Figure 9.18. The combined effects of jet-wake velocity profile and separated flow gives rise to a high level of *blockage* at the exit.

Through the concept of blockage, we define an effective flow area $A_{2\text{eff}}$. The definition of blockage from diffuser studies in Chapter 6 is repeated here, where effective and geometrical areas are applied to the impeller exit:

$$B_2 \equiv 1 - \frac{A_{2\text{eff}}}{A_2}$$

where A_2 is the geometric flow area that may be written as

$$A_2 = (2\pi r_2 b - Ntb)\cos\beta_2' \qquad \textbf{(9.40)}$$

In Equation 9.40, N is the number of the impeller vanes, t is thickness of the blades at the exit, b is the axial span of vanes at 2, and β'_2 is the blade sweep angle at the exit (measured from the radial direction). The cosine term in Equation 9.40 gives a projection of the exit area that is in the radial direction. The conclusions from this section are

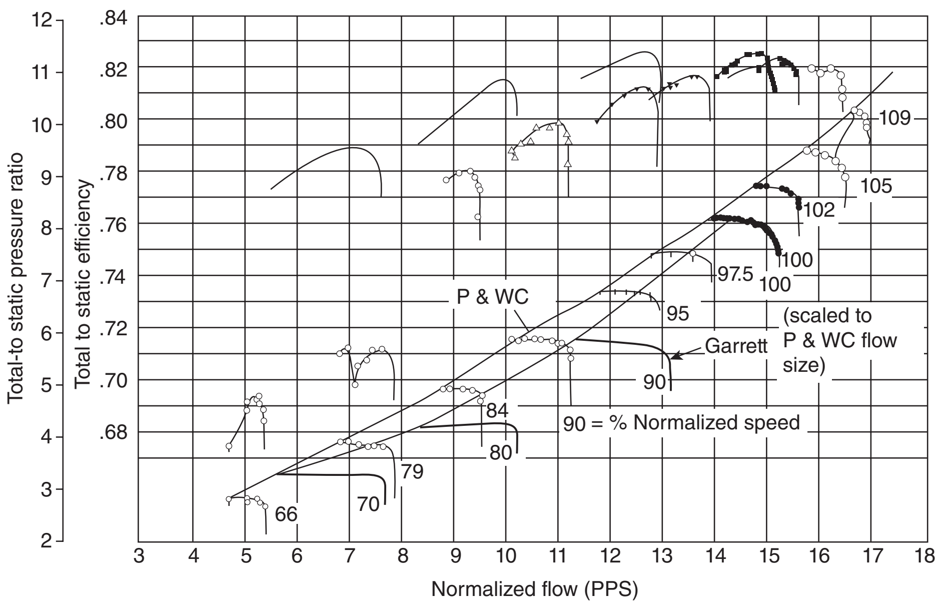
(a) One-dimensional flow analysis is totally inadequate and gives erroneous results
(b) Jet-wake profile and flow separation have to be accounted for by a suitable averaging technique and an estimate of blockage B_2, respectively. In either case, more experimental data are needed.

9.7 Efficiency and Performance

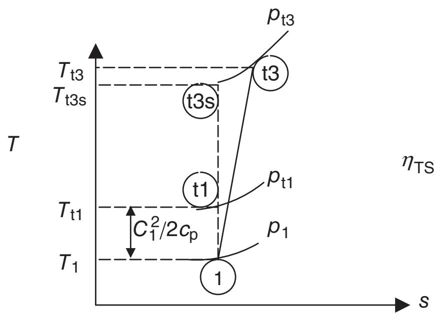
Centrifugal compressor efficiency has always lagged its counterpart, the axial-flow compressor, for the same pressure ratio. The flow path is rather torturous as it twists from one plane to another in a centrifugal compressor, thus massive secondary flow and separation losses become imminent. In addition, for high-pressure ratio centrifugal compressors, the flow in the diffuser becomes supersonic and its efficient diffusion rather complex. In a historical perspective presented by Kenny (1984), different loss sources, future design trends, and limitations of centrifugal compressors are discussed. Although it is very difficult, if not impossible, to separate losses in turbomachinery into their components without overlapping, which is double/triple accounting, it is still instructive to expose general loss sources in turbomachinery. In a centrifugal compressor, the losses are attributed to

1. Tip clearance loss in an unshrouded impeller
2. Secondary flow losses due to flow turning
3. Disk friction losses
4. Losses due to compressibility effects in the inducer due to shocks and flow separation
5. Jet-wake mixing flow losses in the diffuser
6. Diffuser compressibility losses due to high absolute Mach numbers
7. Flow unsteadiness and vortex shedding in the impeller wake.

Also a comparison of two centrifugal compressors designed for an 8:1 total-to-static pressure ratio and their efficiencies at off-designare shown in Figure 9.19 (also from Kenny). Note that the efficiencies are listed for total exit-to-static inlet state of the gas, as well as the pressure ratio. For a Mach 0.5 (absolute inlet) flow, the ratio of total-to-static pressure is $[1 + (0.2)(0.5)^2]^{3.5} \cong 1.186$, therefore an 8:1 design total-static pressure ratio means a total-to-total pressure ratio of $\pi_s \cong 6.74$. Although centrifugal compressors are less efficient; they can produce stage pressure ratios outside the reach of the axial-flow compressors. The centrifugal compressors represent the compressor of choice for small

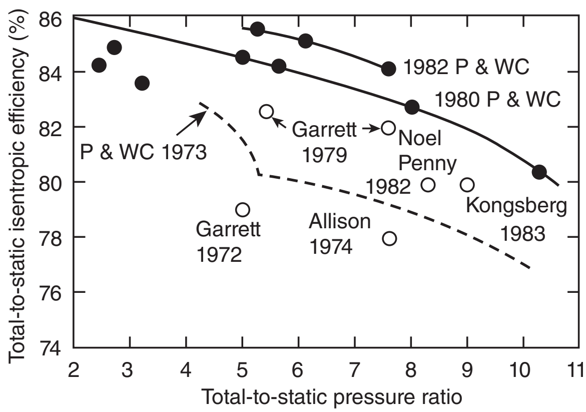


■ FIGURE 9.19 The performance map for two centrifugal compressors with an 8:1 total-static design pressure ratio. Source: Kenny 1984. Reproduced with permission from SAE International



$$\eta_{TS} \equiv \frac{h_{t3s} - h_1}{h_{t3} - h_1}$$

■ **FIGURE 9.20**
***T–s* diagram used in defining the total-to-static efficiency of a centrifugal compressor**

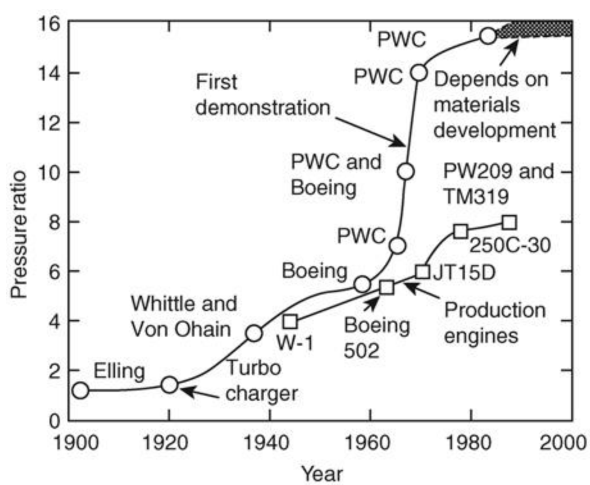


Note: All efficiencies are at 11% surge margin

■ **FIGURE 9.21**
Advances in centrifugal compressor performance. Source: Kenny 1984. Reproduced with permission from SAE International

aircraft engines, auxiliary power units, industrial applications, as well as automotive turbochargers.

Figure 9.20 shows a *T–s* diagram that is used in defining total-to-static efficiency of centrifugal compressors. Advances in centrifugal compressor design are demonstrated by an increase in total-to-static pressure ratio and adiabatic efficiency as shown in Figures 9.21 and 9.22 (from Kenny, 1984).



■ **FIGURE 9.22**
Historic perspective on centrifugal compressor (total to static) pressure ratio. Source: Kenny 1984. Reproduced with permission from SAE International

EXAMPLE 9.6

The inlet flow to a centrifugal compressor is described by

$$C_1 = 200\ \mathrm{m/s}$$
$$p_{t1} = 101\ \mathrm{kPa}$$
$$T_{t1} = 288\ \mathrm{K}$$

The compressor pressure ratio at the design point is $p_{t3}/p_{t1} = 6.8$ with a total-to-total adiabatic efficiency of $\eta_{TT} = 0.80$. Calculate the compressor total-to-static efficiency. Assume $\gamma = 1.4$ and $c_p = 1004\ \mathrm{J/Kg \cdot K}$.

SOLUTION

From the definition of total-to-total adiabatic efficiency, we calculate T_{t3} following

$$T_{t3} = T_{t1}\left\{1 + \frac{1}{\eta_{TT}}\left[\left(\frac{p_{t3}}{p_{t1}}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]\right\} \approx 550.5\ \mathrm{K}$$

Also, T_{t2s} is related to T_{t1} and compressor pressure ratio isentropically, i.e.,

$$T_{t2s} = T_{t1}\left(\frac{p_{t3}}{p_{t1}}\right)^{\frac{\gamma-1}{\gamma}} \approx 498\ \mathrm{K}$$

The static temperature of the inlet gas is calculated from

$$T_1 = T_{t1} - \frac{C_1^2}{2c_p} \approx 288\ \mathrm{K} - (200)^2/2/1004 \approx 268.1\ \mathrm{K}$$

The definition of total-to-static efficiency is $\eta_{TS} \equiv \dfrac{h_{t3s} - h_1}{h_{t3} - h_1}$ and we have calculated all the parameters in this equation, therefore, $\eta_{TS} \approx 0.814$

9.8 Summary

Centrifugal compressors achieve high compression per stage that is predominantly produced by centrifugal force with a large exit-to-inlet impeller radius ratio. The flow area through the inlet is thus a small fraction of the frontal area of the machine. Therefore, the mass flow rate per frontal area is lower in centrifugal than the axial-flow compressors.

The pressure ratio per stage is high, however, with lower adiabatic efficiency than the axial counterpart. The flow path from the inlet to the exit twists and turns, thus secondary flow losses are significant. The phenomenon of slip creates a jet-wake nonuniform velocity profile at the impeller exit. The geometry of the impeller blade passages is characterized by a large wetted perimeter and a small flow area, which makes for an equivalently small hydraulic diameter with large frictional losses. The radial flow from the impeller is decelerated in radial and vaned diffusers. The diffuser exit flow is turned 90° for a follow up combustor or even 180° degrees for the next centrifugal compressor stage. Therefore, staging centrifugal compressors involve turnaround ducts, which make them more cumbersome than axial compressors.

The robust construction of the centrifugal compressors and their high pressure ratio per stage capability make them ideal for small gas turbine engines, auxiliary power units, automotive turbo-chargers, and other industrial uses. The inducer-less and shrouded impellers are used in industrial applications, whereas the unshrouded impellers with inducers that offer lower weight and higher mass flow capability are suitable for aerospace applications.

Classical literature on turbomachinery is found in Stodola's text (1924, 1927) as well as Traupel's book (1977). Taylor (1964), Marble (1964), Kerrebrock (1992), Cumpsty (2004) and Dixon (1975) have treated compressors in detail and are recommended for further reading. Historical perspectives presented by Garvin (1998) and St. Peter (1999) compliment the subject. Hill and Peterson's book (1992) as well as Paduano, Greitzer and Epstein's review paper (2001) on compression system stability are recommended to the reader.

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Problems

9.1 A centrifugal compressor has 22 radial impellers with an (impeller) exit radius of $r_2 = 0.25$ m. Assuming the mass flow rate through the compressor is $\dot{m} = 10$ kg/s, the shaft speed is $\omega = 10{,}000$ rpm, and the inlet flow is swirl free, calculate

(a) the shaft power, $\wp_s$, in kW

(b) the total temperature rise in the compressor (assume $\gamma = 1.4$ and $c_p = 1004$ J/kg · K)

9.2 Size the exit radius of a centrifugal compressor impeller, r_2, that is to reach a tangential Mach number of $M_T = 1.5$. The shaft rotational speed is 25,000 rpm and the inlet flow condition is characterized by

$$p_1 = 100 \text{ kPa}$$
$$T_1 = 288 \text{ K}$$
$$M_1 = 0.5$$
$$\gamma = 1.4 \text{ and } c_p = 1{,}004 \text{ J/kg} \cdot \text{K}$$

9.3 We are interested in a parametric study of the diffusion factor for an inducer section of an impeller. We know from Equation 9.39b that

$$D_{\text{inducer}} \cong 1 - \frac{1}{\sqrt{1+\left(\frac{r_1}{r_2}\right)^2\left(\frac{M_{T2}}{M_{z1}}\right)^2}} + \frac{(M_{T2}/M_{z1})(r_1/r_2)}{2\sigma_i\sqrt{1+\left(\frac{r_1}{r_2}\right)^2\left(\frac{M_{T2}}{M_{z1}}\right)^2}}$$

There are three nondimensional groups that appear in the above equation.

(a) impeller radius ratio r_1/r_2
(b) ratio of impeller tip Mach umber to the inlet axial Mach number M_{T2}/M_{z1}
(c) inducer solidity σ_i

Graph D_{inducer} versus one of the three nondimensional groups while keeping the other two groups constant.

9.4 A centrifugal compressor has no inlet guide vanes and thus zero preswirl. The axial Mach number is $M_1 = 0.5$ and is uniform at the impeller face. The inlet condition is entirely known, $p_1 = 100$ kPa, $T_1 = 288$ K, and the geometry of the inlet, $r_{1h} = 10$ cm and $r_{1t} = 25$ cm. The exit radius is $r_2 = 0.35$ cm and shaft rotational speed is 10,000 rpm. The centrifugal compressor has 20 radial impeller blades.

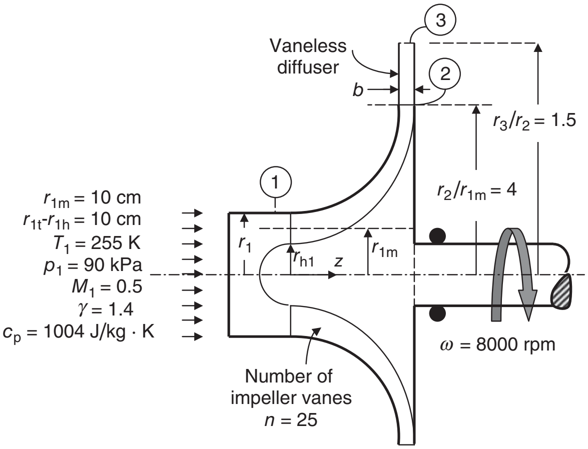
Assuming the compressor polytropic efficiency $e_c = 0.85$, calculate

(a) mass flow rate $\dot{m}_1$ (in kg/s)
(b) absolute swirl velocity at the impeller exit, $C_{\theta 2}$, in m/s
(c) compressor specific work, at the pitchline, w_c, in kJ/kg
(d) compressor shaft power in kW
(e) compressor pressure ratio π_s
(f) exit (absolute) Mach number at 2, M_2, assuming $C_{r2} = C_{z1}$
(g) swirl velocity at the exit of the radial vaneless diffuser

(assuming zero frictional losses in the diffuser) and radius of 45 cm

9.5 A centrifugal compressor has 25 impeller blades of radial design. The rotor exit diameter is $r_2 = 0.4$ m, and its rotational speed is $\omega = 8000$ rpm. Assuming the inlet to rotor flow is purely axial and the air mass flow rate is $\dot{m} = 25$ kg/s through the compressor, calculate

(a) compressor shaft power, $\wp_c$ in, MW
(b) (time rate of change of) angular momentum at the rotor exit



■ **FIGURE P9.5**

9.6 The inlet flow to a radial (vaneless) diffuser has a swirl component of $C_{\theta 2} = 200$ m/s and a radial component of $C_{r2} = 100$ m/s. The impeller exit radius $r_2 = 25$ cm and the flow static pressure and temperature are

$$p_2 = 300 \text{ kPa}$$
$$T_2 = 425 \text{ K}$$

Assuming inviscid flow in the radial diffuser, with $\gamma = 1.4$ and $R = 287$ J/Kg · K, calculate

(a) C_r(r)
(b) C_θ(r)

9.7 A centrifugal compressor discharge pressure and temperature are measured to be

$$P_{t3} = 303 \text{ kPa}$$
$$T_{t3} = 408 \text{ K}$$

The inlet flow is purely axial with $C_{z1} = 100$ m/s and the inlet total pressure and temperature are

$$p_{t1} = 101 \text{ kPa}$$
$$T_{t1} = 288 \text{ K}$$

Calculate

(a) total-to-static pressure ratio p_{t3}/p_1
(b) total-to-total pressure ratio
(c) total-to-static adiabatic efficiency η_{TS}
(d) total-to-total adiabatic efficiency η_{TT}
(e) polytropic efficiency e_c for total-static and total-total processes

9.8 To eliminate the inducer from a centrifugal compressor impeller, we need to induce a swirl profile in the incoming flow (known as preswirl) that is in the direction of the rotor and its relative flow to the impeller is purely axial. This task is done through an IGV.

For an inducer-less impeller of

$$r_{1h} = 5 \text{ cm}$$
$$r_{1t} = 10 \text{ cm}$$
$$\omega = 11{,}500 \text{ rpm}$$
$$C_{z1} = 100 \text{ m/s}$$

Calculate

(a) swirl profile $C_{\theta 1}$ (r)
(b) absolute flow angle α_1 (r)
(c) the IGV twist from hub-to-tip, assuming a constant deviation angle with span

9.9 The inlet static pressure to an impeller is $p_1 = 100$ kPa and its exit static pressure is $p_2 = 200$ kPa with inlet relative dynamic pressure $q_{1r} = 140$ kPa. Treating the impeller as a diffuser, assuming $\gamma = 1.4$, calculate

(a) inlet relative Mach number M_{1r}
(b) static pressure recovery coefficient C_{PR}

9.10 A radial diffuser has an exit-to-inlet radius ratio r_3/r_2 of 4. Assume the flow is incompressible and the fluid is inviscid. Calculate

(a) the diffuser area ratio A_3/A_2
(b) exit-to-inlet radial velocity ratio C_{r3}/C_{r2}
(c) exit-to-inlet swirl ratio $C_{\theta 3}/C_{\theta 2}$
(d) static pressure rise coefficient or static pressure recovery C_{PR}

9.11 A centrifugal compressor has a total pressure ratio of $p_{t3}/p_{t1} = 8$. Calculate and graph this compressor's total-to-static pressure ratio p_{t3}/p_1 for a range of inlet Mach numbers M_1 from 0.25 to 0.75 in steps of 0.05. Assume $\gamma = 1.4$.

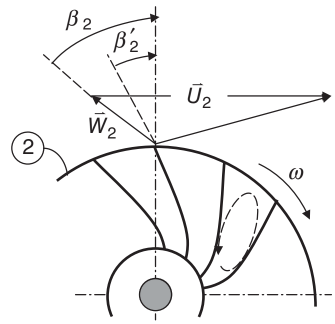
9.12 The total-to-total compressor adiabatic efficiency for a centrifugal compressor is $\eta_{TT} = 0.78$. The compressor total pressure ratio is $p_{t3}/p_{t1} = 7$. Assuming the inlet Mach number is $M_1 = 0.4$ and $\gamma = 1.4$, calculate

(a) total-to-static pressure ratio
(b) total temperature ratio T_{t3}/T_{t1}
(c) total-to-static adiabatic efficiency η_{TS}.

9.13 A backswept impeller has a tip speed of $U_2 = 366$ m/s and the impeller blade exit angle is $\beta_2' = 30°$. The radial velocity at the impeller exit is $C_{r2} = 125$ m/s. Stagnation speed of sound at the impeller inlet is $a_{t1} = 340$ m/s, assuming the inlet flow is purely axial, and the slip angle at the impeller exit is 6°, i.e., $\beta_2 - \beta_2' = 6°$, calculate

(a) slip factor ε
(b) Mach index Π_M
(c) Impeller specific work, w_c, in kJ/kg
(d) stage total temperature ratio τ_{stage}
(e) stage total pressure ratio if $e_c = 0.89$

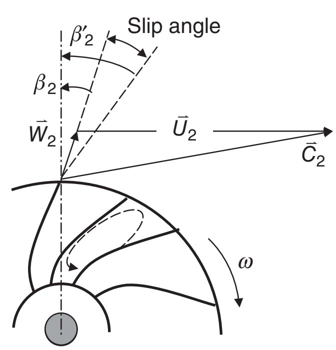
Assume $\gamma = 1.4$ and $c_p = 1004$ J/Kg · K.



■ **FIGURE P9.13**

9.14 A forward-swept impeller has a tip speed of $U_2 = 400$ m/s and the impeller blade exit angle is $\beta_2' = -30°$. The radial velocity at the impeller exit is $C_{r2} = 100$ m/s. Stagnation speed of sound at the impeller inlet is $a_{t1} = 330$ m/s, assuming the inlet flow is purely axial and the slip angle at the impeller exit is $-6°$, i.e., $\beta_2' - \beta_2 = -6°$, calculate

(a) slip factor ε
(b) Mach index Π_M
(c) impeller specific work, w_c, in kJ/kg
(d) stage total temperature ratio τ_{stage}
(e) stage total pressure ratio if $e_c = 0.89$.



■ **FIGURE P9.14**

9.15 A 19-bladed radial impeller has an exit radius $r_2 = 25$ cm. The impeller width at the exit is $b = 1$ cm. We wish to calculate and compare some impeller exit flow areas with successive levels of detail.

Case 1
Assume the individual trailing edge thickness of impeller blades is zero, i.e., $t = 0$, and calculate the geometric flow area.

Case 2
Assume the individual trailing edge thickness is $t = 2$ mm (per blade), and calculate the geometric flow area at the impeller exit.

Case 3
Assume the impeller exit blockage is 10% and now calculate the effective flow area.

Assuming the same slip factor between the three cases, relate the average radial velocity C_{r2} in cases 1, 2, and 3.

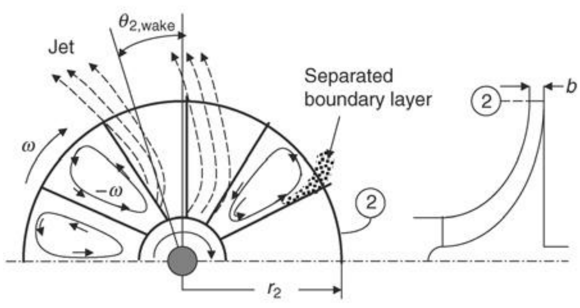
9.16 Air enters a centrifugal compressor at $p_{t1} = 101$ kPa, $T_{t1} = 288$ K, and purely in an axial direction with $M_z = 0.4$. Assuming the impeller radius at the exit is $r_2 = 25$ cm and the impeller is of radial design with slip factor $\varepsilon = 0.9$ and $M_{T2} = 1.5$, calculate

- **(a)** U_2 in m/s
- **(b)** specific work, w_c, in kJ/kg
- **(c)** impeller exit total temperature, T_{t2}, in K
- **(d)** shaft rotational speed, ω, in rpm

9.17 The flow pattern at the exit of a radial centrifugal compressor impeller shows a jet-wake behavior, as shown in Figure 9.18, which is reproduced here with additional descriptions. In order to estimate the blockage, we assume that the wake region has zero momentum and is of angular extent $\theta_{2,\,\text{wake}}$ (per passage) at the impeller exit, as shown. The number of impeller blades is N, and the blade thickness at the exit is t (note the blade sweep angle is zero).

Calculate

- **(a)** the geometric flow area at 2 in terms of r_2, t, N, and b
- **(b)** the effective flow area in terms of $\theta_{2,\,\text{wake}}$
- **(c)** the blockage B_2

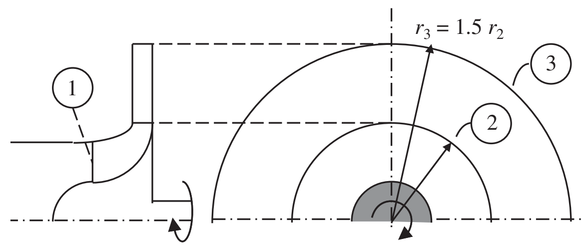


■ **FIGURE P9.17**

9.18 Consider a radial diffuser in a centrifugal compressor with straight radial impeller, as shown. The radial velocity at the impeller exit is $C_{r2} = 150$ m/s, the tangential velocity $C_{\theta 2} = 250$ m/s and the slip factor is $\varepsilon = 0.90$. The stagnation temperature at the inlet to the impeller is $T_{t1} = 298$ K.

Assuming inlet flow to the centrifugal compressor is swirl free, i.e., $C_{\theta 1} = 0$, calculate

- **(a)** absolute Mach number, M_2
- **(b)** swirl at the exit of the radial diffuser, $C_{\theta 3}$, assuming inviscid flow
- **(c)** exit swirl, $C_{\theta 3}$, if the average retarding torque acting on the fluid by the wall is 5% of (time rate of change of the) fluid angular momentum at 2.

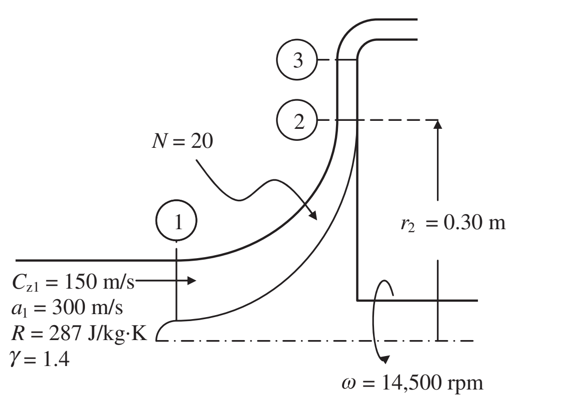


■ **FIGURE P9.18**

9.19 A centrifugal compressor impeller has 20 straight blades. The exit radius of the impeller is $r_2 = 0.30$ m and the angular speed is $\omega = 14{,}500$ rpm. The inlet flow to the compressor has zero preswirl and the axial velocity is $C_{z1} = 150$ m/s. The impeller exit has the radial velocity component C_{r2} that is equal to C_{z1}.

Assuming the speed of sound at the inlet to compressor is $a_1 = 300$ m/s, calculate

- **(a)** Mach index, Π_M
- **(b)** impeller exit absolute swirl, $C_{\theta 2}$, in m/s



■ **FIGURE P9.19**

(c) specific work of the rotor, w_c, in kJ/kg
(d) impeller exit absolute Mach number, M_2
(e) compressor total temperature ratio, τ_c

9.20 A centrifugal compressor impeller has 30 backswept blades with $\beta_2' = 20°$. The inlet flow condition to the compressor is purely axial with $C_{z1} = 200$ m/s and its total pressure and temperature are: $p_{t1} = 100$ kPa and $T_{t1} = 288$ K, respectively. The impeller rim is at a radius of $r_2 = 0.2$ m and the shaft rotational speed is $\omega = 26{,}500$ rpm. The radial velocity at the impeller exit is equal to the inlet axial velocity, as shown. Assuming the vaneless radial diffuser has an exit radius of $r_3 = 0.35$ m and further assuming that the flow in the radial diffuser is inviscid with gas properties $\gamma = 1.4$ and $R = 287$ J/kgK, calculate

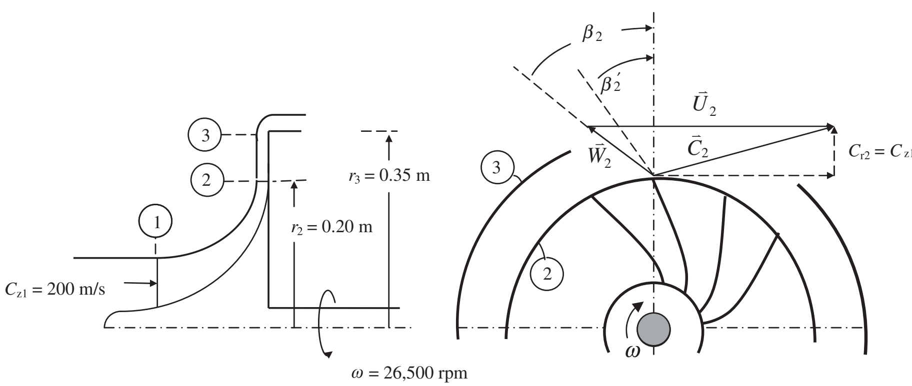
(a) impeller rim speed, U_2, in m/s
(b) impeller (actual) exit swirl, $C_{\theta 2}$, in m/s
(c) specific work of the compressor, w_c, in kJ/kg
(d) impeller absolute exit Mach number, M_2
(e) swirl velocity at the diffuser exit, $C_{\theta 3}$, in m/s
(f) radial velocity at the diffuser exit, C_{r3}, in m/s (neglecting density variations in the diffuser)

9.21 A centrifugal compressor impeller is of radial design and has 18 blades. The impeller exit radius is at $r_2 = 0.30$ m. The wheel rim speed is $U_2 = 400$ m/s. Assuming that flow in compressor inlet is swirl free, the axial velocity at the inlet and the radial velocity at the exit of the impeller are equal, i.e., $C_{z1} = C_{r2} = 150$ m/s, with $T_{t1} = 288$ K, $p_{t1} = 100$ kPa, $\gamma = 1.4$ and $c_p = 1004$ J/kg · K, calculate

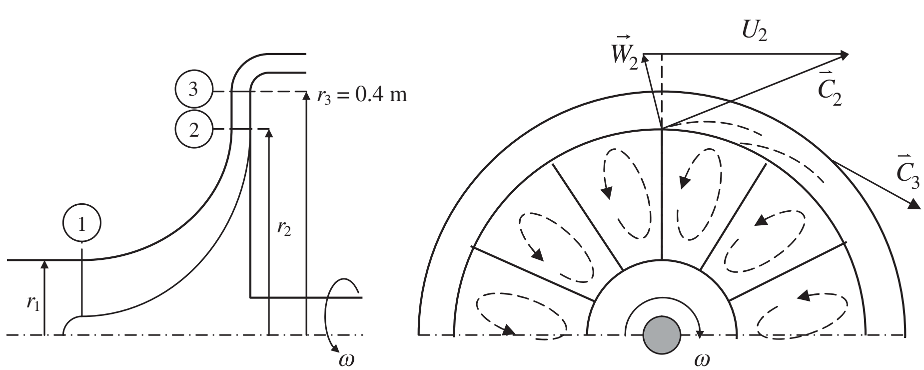
(a) absolute swirl at the impeller exit, $C_{\theta 2}$, in m/s
(b) absolute Mach number at the impeller exit, M_2
(c) absolute radial and tangential velocities at the radial diffuser exit, C_{r3} and $C_{\theta 3}$, in m/s
(Assume the fluid is incompressible and inviscid in the radial diffuser)

9.22 A centrifugal compressor has 20 radial impeller blades that rotate at $\omega = 18{,}500$ rpm. The impeller exit radius is $r_2 = 0.20$ m. Assuming that the inlet flow is swirl free, calculate

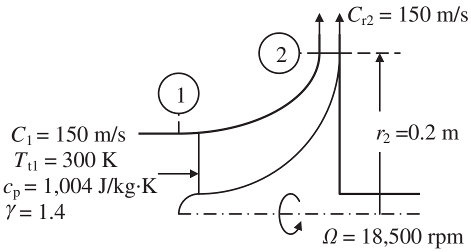
(a) impeller exit (absolute) swirl, $C_{\theta 2}$, in m/s
(b) impeller rim tangential Mach number, $M_T = \omega r_2 / a_1$
(c) speed of sound, a_2, in m/s at the impeller exit



■ **FIGURE P9.20**



■ **FIGURE P9.21**



■ **FIGURE P9.22**

9.23 A centrifugal compressor impeller has 20 straight impeller blades. The inlet flow condition to the compressor is purely axial with $M_1 = 0.4$. The geometric parameters of the centrifugal compressor are: $r_{h1} = 0.05$ m, $r_{t1} = 0.1$ m, $r_2 = 0.25$ m and $r_3 = 0.375$ m. The shaft rotational speed is $\omega = 14{,}000$ rpm. The static pressure and temperature at the inlet are $p_1 = 90$ kPa and $T_1 = 255$ K, respectively. The radial velocity at the impeller exit is equal to the inlet axial velocity, i.e., $C_{r2} = C_{z1}$ and the compressor polytropic efficiency is $e_c = 0.85$. Assuming the radial diffuser is vaneless and the flow in the radial diffuser is inviscid with gas properties $\gamma = 1.4$ and $R = 287$ J/kg · K, calculate

(a) flow area at the inlet to the centrifugal compressor, A, in m^2
(b) inlet axial velocity, C_{z1}, in m/s
(c) inlet density, ρ_1, in kg/m^3
(d) the mass flow rate, $\dot{m}$, in kg/s
(e) impeller rim speed, U_2, in m/s
(f) impeller (actual) exit swirl, $C_{\theta 2}$, in m/s
(g) compressor shaft power, $\wp_c$, in kW
(h) impeller absolute exit Mach number, M_2
(i) swirl velocity at the diffuser exit, $C_{\theta 3}$, in m/s
(j) stage total pressure ratio, π_c

9.24 The corrected mass flow rate at the inlet to a centrifugal compressor is $\dot{m}_{c1} = 20\frac{kg}{s}$ and the axial Mach number at the compressor face is $M_{z1} = 0.5$. For the hub-to-tip radius ratio of the impeller equal to 0.1, i.e., $r_{h1}/r_{t1} = 0.1$, calculate

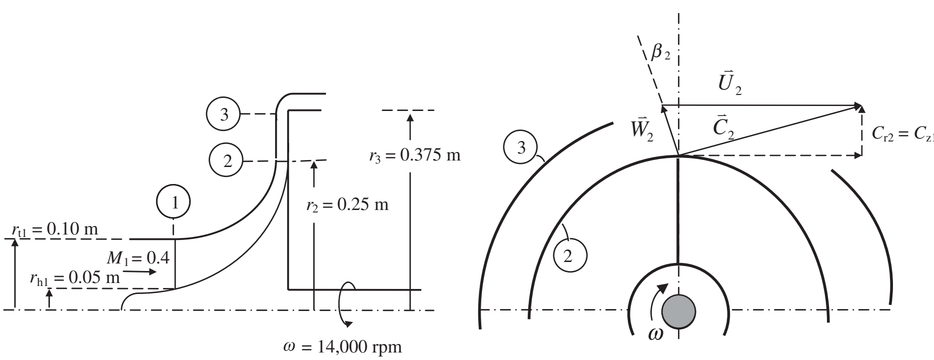
(a) inlet area, A_1, in m^2
(b) hub and tip radii, r_{h1} and r_{t1}, in cm
(c) shaft speed, ω, for the impeller inlet relative tip Mach number to be 0.8, i.e., $(M_{1r})_{tip}$

Assume gas total temperature at the impeller inlet is 288 K, $\gamma = 1.4$ and $R = 287$ J/kgK

9.25 A centrifugal compressor discharge static pressure is $p_3 = 352$ kPa and its total temperature is $T_{t3} = 444$ K. The flow at the impeller inlet has an axial component with $C_{z1} = 150$ m/s and a preswirl component with $C_{\theta 1} = 56$ m/s. The inlet total pressure and temperature are the standard sea-level conditions, i.e., $p_{t1} = 101$ kPa and $T_{t1} = 288$ K, respectively. Assuming the compressor polytropic efficiency is $e_c = 0.88$, calculate

(a) compressor total pressure ratio, p_{t3}/p_{t1}
(b) static pressure ratio, p_3/p_1, across the compressor
(c) exit Mach number, M_3
(d) compressor specific work, w_c, in kJ/kg

Assume fluid properties are: $\gamma = 1.4$ and $c_p = 1004$ J/kgK



■ **FIGURE P9.23**